

Cyclic Linearization of Syntactic Structure

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Abstract

This paper proposes an architecture for the mapping between syntax and phonology – in particular, that aspect of phonology that determines the linear ordering of words. We propose that linearization is restricted in two key ways. (1) the relative ordering of words is fixed at the end of each phase, or “Spell-out domain”; and (2) ordering established in an earlier phase may not be revised or contradicted in a later phase. As a consequence, overt extraction out of a phase P may apply only if the result leaves unchanged the precedence relations established in P. We argue first that this architecture (“cyclic linearization”) gives us a means of understanding the reasons for successive-cyclic movement. We then turn our attention to more specific predictions of the proposal: in particular, the effects of Holmberg’s Generalization on Scandinavian Object Shift; and also the Inverse Holmberg Effects found in Scandinavian “Quantifier Movement” constructions (Rögnvaldsson (1987); Jónsson (1996); Svenonius (2000)) and in Korean scrambling configurations (Ko (2003, 2004)). The cyclic linearization proposal makes predictions that cross-cut the details of particular syntactic configurations. For example, whether an apparent case of verb fronting results from V-to-C movement or from “remnant movement” of a VP whose complements have been removed by other processes, the verb should still be required to precede its complements after fronting if it preceded them before fronting according to an ordering established at an earlier phase. We argue that “cross-construction” consistency of this sort is in fact found.

1. Introduction¹

This paper proposes an architecture for the mapping between syntax and phonology – in particular, that aspect of phonology that determines

¹ The present paper is the first installment in a larger project (Fox and Pesetsky (in prep.)). The inspiration for this work can be traced to a conversation with John

ordering. In Fox and Pesetsky (in prep.), we will argue that this architecture, when combined with a general theory of syntactic domains (“phases”), provides a new understanding of a variety of phenomena that have received diverse accounts in the literature. This shorter paper focuses on two processes, both drawn from Scandinavian: the familiar process of *Object Shift* and the less well-known process of *Quantifier Movement*. We will argue that constraints on these operations can be seen as instances of the same property of grammar that explains the fact that movement is local and successive cyclic. We begin by sketching a model in which locality and successive cyclicity are consequences of the architecture that we propose, rather than specific facts about movement itself. We next present our proposal in somewhat greater detail, and show how it can account for a wide range of apparent limitations on movement – in particular, superficially contradictory restrictions on Object Shift and Quantifier Movement. The restrictions on Object Shift include those grouped under the rubric of *Holmberg’s Generalization*, which Quantifier Movement does not seem to obey. We will argue that Quantifier Movement instead obeys a near mirror-image of Holmberg’s Generalization (an “Inverse Holmberg Effect”), but that both Holmberg’s Generalization and its mirror image are expected if our proposed architecture is correct.

Our discussion will be for the most part informal, but we will conclude by offering a more formal implementation of our proposals. This

Frampton, and to a fruitful additional meeting with John Frampton and Sam Guttman. We are grateful to both of them, and also to the students and visitors who attended our Spring 2003 seminar at MIT, where much of this work was first presented and developed. We also want to express our gratitude for the contributions to our work of Elena Anagnostopoulou, Jonathan Bobaljik, Misi Brody, Noam Chomsky, Anders Holmberg, Sabine Iatridou, Kyle Johnson, Noriko Kawasaki, Heejeong Ko, Alec Marantz, Andrew Nevins, Øystein Nilsen, Jon Nissenbaum, Christer Platzack, Norvin Richards, Joey Sabbagh, Peter Svenonius, Shoichi Takahashi, and Sophia Tapio, among others. We also wish to thank for their comments and questions audiences at Sophia University (Tokyo); Université de Paris VII; Indiana University; Kyoto University (Kaken Ellipsis Workshop); University of Vittoria (Basque Country); University of Milan-Bicocca; the University of Massachusetts/Amherst; the Workshop on Interfaces (Château de la Bretèche); the 2003 LSA Summer Institute (Michigan State University); GLOW 2003 (Lund); and the First North American Syntax Conference (Concordia University, Montreal).

implementation will belong to a family of possibilities consistent with our overall model. We will leave for the fuller presentation a comparison of these alternatives, along with a range of other empirical consequences.

2. Successive-cyclicity and linear order

Much attention has been devoted to movement relations that appear to be long-distance or unbounded. A large body of work has converged on the conclusion that such relations actually represent the result of the compounding of a series of more local relations. For example, the apparent long-distance property of *wh*-movement in a sentence such as (1) seems to be a consequence of a series of shorter movement relations – at least through intermediate specifiers of CP and specifiers of VP² as shown in (1):

(1) **Successive-cyclic wh-movement**

I wonder [which book he [_{CP} _{CP} Mary [_{CP} _{CP} read _{CP}]]]

Evidence for the possibility of successive cyclic movement through Spec,CP and Spec,VP includes a variety of reconstruction phenomena (see Barss (1986), Lebeaux (1991) for CP; Fox (1999) for VP) – as well as the possibility of stranding *wh*-related elements (McCloskey (2000)) and perhaps *wh*-phrases themselves (McDaniel (1989); Gračanin (2004)) in at least some intermediate Spec,CP and Spec,VP positions.

Additional evidence supports an even stronger conclusion. If Chomsky (1973) and others are correct in linking island effects to the existence of intermediate landing sites, it must be the case that long-distance movement does not merely permit, but actually *requires* the use of these intermediate positions. We must then ask why. This question has figured prominently in much discussion over the past quarter century, and a variety of answers have been suggested.

² To keep the discussion simple, we will not make a distinction between *v*P and VP (as argued for by Hale and Keyser (1993) and Hale and Keyser (2002), among others) in the first part of this paper. We return to the *v*P/VP distinction below and in Fox and Pesetsky (in prep.).

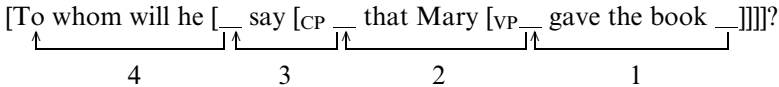
All proposals known to us attribute a special status to particular syntactic domains, variously called “bounding nodes”, “barriers” and “phases”. Our proposal will not differ in this respect. For now, we will follow Chomsky (2000) in using the term “phase” for the domains in question, to highlight a similarity between our proposal and his, to which we return later. Traditionally, phases play two distinct roles – and it is on this point that our proposal will crucially differ from others, including Chomsky’s. On the one hand, a phase is taken to determine an “impenetrable domain” from which movement is generally excluded. On the other hand, it is also taken to determine a specific position as an “escape hatch”. The escape hatch provides the one and only legal route for movement out of the phase. Thus, for example, if CP is a phase, it determines that movement out of CP is impossible from any position except Spec,CP – its designated escape hatch. This forces all apparent long-distance movement from CP to be analyzed as involving movement through its escape hatch. More generally, apparent movement from any phase, on traditional accounts, must involve the escape hatch of that phase as an intermediate step.

Our approach will have in common with traditional proposals the requirement that movement from a phase proceeds through certain specific types of positions within that phase. Unlike traditional proposals, however, our approach does not stipulate particular positions that provide escape hatches for movement – nor is there any concept of “escape hatch” built into the theory. “Escape hatch effects” are, if we are correct, an artifact of the role played by phases in *linearization* of the terminal elements of syntactic structure. These effects are thus a consequence of the mapping between syntax and phonology.³

Consider the derivation in (2), in which *wh*-movement proceeds through intermediate specifiers of VP and CP:⁴

³ For other proposals that relate island effects (and requirements of successive cyclicity) to syntax-phonology interactions, see Pesetsky (1998) (with an antecedent in Perlmutter (1972)); Uriagereka (1999); among others.

⁴ It is important to our discussion that the intermediate positions indicated in (2) are on the left side of VP and CP. This proposal is independently supported (for CP) by the observations of McCloskey (2000). Our paper as a whole will constitute a sustained argument for this position. See Nissenbaum (2000) for possibly conflicting evidence.

(2) **Successive-cyclic *wh*-movement through Spec,CP and Spec,vP**

Let us focus on a particular property of the movements seen in (2). Consider first step 1, which takes place entirely within the lowest VP phase. This step can be seen as “revising” the word order of VP insofar as *wh*-movement to Spec,VP causes the *wh*-phrase to precede the words *gave the book*, which it would otherwise follow. Now consider step 2. This step also “revises” word order, causing the *wh*-phrase to precede the words *that Mary* which, once more, it would otherwise have followed. Notice, however, that step 2 only revises word order with respect to these words – not with respect to the words contained within the previous phase. The *wh*-phrase preceded the words *gave the book* at the end of the lowest VP phase, and still precedes them at the end of the next phase.

This type of observation can be repeated for each subsequent step. Thus step 3 revises the order of the *wh*-phrase with respect to the verb *say*, but not with respect to the words *that Mary gave the book*. Likewise, step 4 revises the order of the *wh*-phrase with respect to the words *will he*, but not with respect to any of the words dominated by lower phases.

The generalization is clear. Each time the *wh*-phrase moves, it leaves unchanged its linear order with respect to words dominated by previous phases, but may revise linear order with respect to words introduced in the current phase. We propose that this property of (2) is not a coincidence. As we will argue, if this property did not obtain, the sentence could not be linearized. This proposal will constitute the core of our explanation for the obligatoriness of successive-cyclic movement and will be the key component of our account of Holmberg’s Generalization and of Inverse Holmberg Effects.⁵

⁵ Our account of Holmberg’s Generalization has significant properties in common with earlier work that views the phenomenon as a consequence of ordering generalizations that constrain the independent working of syntactic operations, e.g. Sells (2001), Williams (2003); Müller (2000); Kathol (2000). Our proposal differs from these works in a number of respects – most crucially in the fact that linearization is established on a phase-by-phase basis. This property of our proposal yields a structural distinction between aspects of linear order that may be affected by a syntactic operation and those

3. General proposal

We assume, with Chomsky (2000, 2001a), that syntactic derivations involve the operations Merge and Move (“remerge”), with the consequence that structure is built from “bottom to top”. We also take from Chomsky the suggestion that the mapping between syntax and phonology (*Spell-out*) takes place at various points in the course of the derivation – rather than at a single point, as in traditional models. In this work, we focus on one key aspect of Spell-out: *linearization*. When Spell-out applies, it is possible that other relations among syntactic units (besides linear order) are established – for example, relations relevant to the determination of phonological phrasing. We will not be discussing these other relations, but will restrict our attention to linear order.

We will use the term *Spell-out domain* to refer to the constituents that are mapped to phonology by Spell-out. (We return shortly to the relation between the notion *Spell-out domain* and the notion *phase*.) The list of Spell-out domains includes at least CP, VP and DP.⁶ Each time the derivation constructs a Spell-out domain D, Spell-out applies, linearizing D. The first time this happens, Spell-out takes D as input and yields straightforwardly a linearization of D. Each time a new Spell-out domain D' is constructed, Spell-out linearizes the new material in D' and adds information about its linearization to the information cumulatively produced by previous applications of Spell-out.

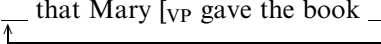
The key property of our proposal that contributes to an account of successive-cyclic movement is the following claim: *information about linearization, once established at the end of a given Spell-out domain, is never deleted in the course of a derivation*. The sole function of Spell-out is to *add* information. We will call this monotonicity property of Spell-out *Order Preservation*.

that may not, and consequently allows us to relate Holmberg’s Generalization to phenomena such as successive-cyclicity and Inverse Holmberg Effects. See Fox and Pesetsky (in prep.) for more discussion.

⁶ In Fox and Pesetsky (in prep.), we will have important qualifications about the type of verb-headed phrase that counts as a Spell-out Domain. For now, we continue to use the designation “VP”, ignoring the *v*P/VP distinction. We touch on this distinction briefly in section 5 below. See also Sabbagh (2003) for arguments that PP is a Spell-out domain and for a discussion of certain problems concerning the status of DP.

3.1. Successive-cyclicity as a consequence of Order Preservation

Let us examine how Order Preservation might enforce successive-cyclicity. Imagine that the phrase *to whom* in (2) had moved directly to the specifier of the lower CP – without first stopping in the specifier of VP, as sketched in (3):

- (3) ***Wh*-movement skipping Spec,VP – yielding an ordering contradiction**
 [To whom will he [__ say [CP __ that Mary [VP gave the book __]]]]?
- 

If the proposal above is correct, movement to Spec,CP must follow Spell-out of VP, since VP is a Spell-out domain. In English, heads precede their complements, and PP objects follow direct objects. As a consequence, the process of Spell-out applying to VP linearizes elements of VP, as represented informally in (4). The expression “A<B” should be read (for now) as “A precedes B”:

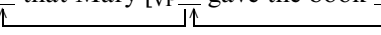
- (4) **Spell-out applies to VP (non-successive-cyclic derivation)**
gave < the book < to whom

After *to whom* moves to Spec,CP, Spell-out of CP yields the following ordering information – assuming, as is the general case in English, that the specifier of a phrase XP precedes its sister X’:

- (5) **Spell-out applies to CP (non-successive-cyclic derivation)**
to whom < that < Mary < VP

The information in (4) and the information in (5) impose conflicting requirements on the phonology. By (4), *to whom* must be pronounced after the words *gave*, *the* and *book*, but (5) requires *to whom* to be pronounced before these words (since they are contained in VP). Given Order Preservation, we suggest that it is this phonological problem – this *ordering contradiction* – that rules out the non-successive-cyclic derivation.

Consider, by contrast, the successive-cyclic derivation in which *to whom* first moves to Spec,VP before moving on to Spec,CP.

- (6) ***Wh*-movement stopping in Spec,VP – avoiding an ordering contradiction**
 [To whom will he [__ say [CP __ that Mary [VP __ gave the book __]]]]?
- 

In this derivation, when Spell-out applies to VP, the phrase *to whom* occupies Spec,VP, and is thus ordered to the left of the words *gave the book*

– rather than to the right of these words, as was the case in the non-successive-cyclic derivation above:

(7) **Spell-out applies to VP (successive-cyclic derivation)**

to whom < *gave* < *the book*

Once CP has been built, and *to whom* has moved to Spec,CP, Spell-out applies to CP just as it did in the non-successive-cyclic derivation:

(8) **Spell-out applies to CP (successive-cyclic derivation)**

to whom < *that* < *Mary* < VP

Continuing to read “A < B” as “A precedes B”, we can see that the information in (7) and (8) (unlike the information in (4) and (5)) does not yield an ordering contradiction. There is no conflict between the constraints imposed on the phonology by (7), which require that *to whom* precede the VP elements *gave the book*, and the constraints imposed by (8), which require that *to whom* precede the words *that Mary* in addition to preceding VP elements. The successive-cyclic derivation succeeds where the non-successive-cyclic derivation fails, for exactly the reason discussed at the end of the previous section. When the *wh*-phrase moved to Spec,CP in the successive-cyclic derivation, it left unchanged its linear order with respect to the words that had been linearized in the previous Spell-out domain, even though new orderings were established with respect to the words *that* and *Mary* that were merged in the course of building CP.

An architecture of this sort will in general force successive cyclicity when movement crosses a Spell-out domain boundary. Our proposal does, however, have certain unique features (both conceptual and empirical) which distinguish it from alternative hypotheses about the roots of successive cyclicity. These unique aspects of our proposal will become clear once we have spelled it out in somewhat greater detail. This presentation will still be somewhat informal. A more precise presentation will follow in section 7.

3.2. *Derivational scenarios*

Suppose a bottom-to-top derivation has created the syntactic domain D in (9), where D is also a Spell-out domain. Assume that the Spell-out operation establishes the ordering statements given in (9).

(9) **Spell-out of D**[_D X Y Z]

Ordering: X<Y
 Y<Z

Before proceeding, we should comment briefly on the status of the “ordering statements” provided in (9) (and many similar presentations to follow). Throughout most of this paper, the lists of ordering statements will be informal and will generally contain only information relevant to the point being made. Ultimately, we will specify the exact format in which actual ordering statements produced by Spell-out are presented to the phonetic systems. At that point, further questions will arise. For example, does the set of actual ordering statements in a situation such as (9) include a redundant statement such as “X<Z”? What are the actual elements mentioned in the ordering statements: phrases? words? combinations of the two? The answer to questions of this sort will turn out to have important consequences, which we will present in a fuller version of this work. For now, however, we can keep the discussion relatively simple by providing only relevant, non-redundant ordering statements in our informal presentations and using conventional abbreviations like “VP” whose exact role in these statements will ultimately demand clarification.

As an initial step towards a more precise formulation of our proposals, let us understand the significance of the symbol “<” as follows:

(10) **The relation “<” [first statement]**

An ordering statement of the form $\alpha < \beta$ is understood by PF as meaning that the last element dominated⁷ by α precedes the first element dominated by β .

A statement of the form “DP<VP”, for example, means that the last element dominated by DP precedes the first element dominated by VP. For now, we will generally assume that the relevant “elements” are words (although one could also imagine a variant in which the elements relevant to (10) are phonological segments). When presenting actual examples, we

⁷ “Dominate” throughout this work means “reflexively dominate”. This interpretation of the term is important for the case in which α or β is a word.

will indicate, where necessary, the words relevant to the interpretation of “<”.

It is also important to specify the way in which movement can affect phonology. Consider a traditional view of movement as an operation that creates a “chain” containing two elements. On this view, one element is new, and constitutes the “head” of the chain. The other element was previously merged, and constitutes a “trace”. With this view in mind, we will need to stipulate that Spell-out pays attention only to the head of a chain, and ignores traces. Thus, under a traditional view of movement, the statement of (10) should be modified as follows:

(11) **The relation “<” [modified to fit with copy theory of movement]**

An ordering statement of the form $\alpha < \beta$ is understood by PF as meaning that the last element dominated by α *and not dominated by a trace* precedes the first element dominated by β *and not dominated by a trace*.

In section 7, we will adopt a view of movement as a second merger of a single element rather than a copying procedure that constructs a chain (Blevins (1990); Chomsky (2001a, b); Epstein et al. (1998)). At that point in the discussion, we will no longer need a distinction between traces and non-traces, and will be able to revert to a simpler theory of how PF interprets “<” – essentially the statement in (10), coupled with a notion of “dominates” that is defined for structures in which a single element merges twice.

Of course, the system (under either view of movement) must already “know” what is first and last in X and Y in order to interpret statements like “ $X < Y$ ”. How the system acquires this information will also be a topic of discussion in our more formal presentation in section 7. Once again, however, it will be possible to see that the formal clarifications discussed in that section leave our previously discussed results unchanged.

Let us now return to (9). Imagine that after Spell-out applies to D, the syntax merges α with D, yielding (12):

(12) **α merges with D**

α [_D X Y Z]

Consider a situation in which, after merger of α , some element in D undergoes leftward movement into the higher Spell-out domain. As our

discussion of successive-cyclic *wh*-movement made clear, we will want to consider two possible scenarios, depending on whether or not the element that moved out of D previously occupied the left edge of D.

In the first scenario, it is X that moves leftward, as part of the construction of the next Spell-out domain D'. When D' is spelled out, the linearization of D' will add (to the ordering statements from D) the new (boldfaced) ordering statements listed in (13):

(13) **Scenario 1 (leftward movement from a left-edge position)**

$[_{D'} \dots X \alpha [_{D} t_X Y Z]]$



Ordering: $X < \alpha$ $X < Y$
 $\alpha < D \rightarrow \alpha < Y$ $Y < Z$

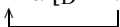
As we noted above, “ $X < \alpha$ ” means that the last element of X (not dominated by a trace) precedes the first element of α (not dominated by a trace). Likewise, “ $\alpha < D$ ” means that the last relevant element of α precedes the first relevant element of D – in this case, Y (i.e. the first element of Y, if Y itself is complex). The arrow after “ $\alpha < D$ ” shows this consequence of the ordering statement, namely that PF pronounces α before Y. We will not mark such consequences consistently – only when it helps make things clearer.

The key point, as we stressed when discussing successive-cyclic *wh*-movement, is the following. Because X was at the left edge of D when D underwent Spell-out, *the ordering statements added by Spell-out of D' are consistent with the ordering statements previously added in D*. Leftward movement from the left edge of a Spell-out Domain thus yields no ordering contradictions.

Now consider the alternative scenario, in which an element not at the left edge of D – for example, Y – undergoes movement as part of the formation of D'. In this scenario, when D' is spelled out, it will include all the ordering statements from D as well as the new ordering statements in (14):

(14) **Scenario 2 (leftward movement from a non-left-edge position)**

$*[_{D'} \dots Y \alpha [_{D} X t_Y Z]]$



Ordering: $Y < \alpha$ $X < Y$
 $\alpha < D \rightarrow \alpha < X$ $Y < Z$

In (14), Y was *not* at the left edge of D. Consequently, the ordering statements added when D' undergoes Spell-out are *not* consistent with the ordering statements previously added when D underwent Spell-out.

Let us see this in detail. The statement “ $\alpha < D$ ” is interpreted as “ α precedes X” (see (11)). The ordering statements of the previous Spell-out domain included “ $X < Y$ ”. The ordering statements of the new Spell-out domain indicate that Y in turn must precede α . This yields a contradiction when translated in the obvious manner into instructions for pronunciation: α must precede X, X must precede Y, but Y must precede α . There is no way out of this loop, since the property of Order Preservation makes it impossible to disregard (e.g. delete) any of the ordering statements produced by Spell-out.

The consequence of this system should be clear. All things being equal, leftward movement from a Spell-out domain D must take place from the left edge of D (and conversely for rightward movement). This, as we have seen, is a property that can explain successive-cyclicity.

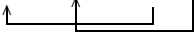
All things, however, are not always equal. In particular, we can now begin to discern some characteristics that distinguish our account of successive-cyclicity requirements from others. For one thing, our proposal accounts for “escape hatch phenomena” without invoking the concept of an escape hatch, as we shall illustrate below. This is a conceptual advantage that also has significant empirical consequences (but see note 9). Under particular circumstances, as we shall see, movement can extract an element from Spell-out domain D that did not occupy the relevant edge of D. Our proposal also differs from many others in tying successive-cyclicity requirements to properties of the syntax-phonology interface. We thus might expect that certain variations in pronunciation might obviate the need for successive-cyclicity. We will suggest that the phenomenon of “salvation by deletion” directly reflects this property of our system. We turn to these matters in the next section.

3.3. *Novel predictions*

Consider again the abstract derivation sketched in (9) and extended in (12). The first new case to examine is one in which *both* X and Y in (12) move out of D to positions to the left of α . In standard approaches,

movement of this sort should be impossible in principle, except under particular circumstances (e.g. multiple specifier constructions) in which both X and Y occupy “escape hatch” positions. In the system developed here, by contrast, such movement should be possible quite generally – so long as X and Y preserve their original order. This consequence of our system can be seen in Scenario 3 sketched in (15) below. Since X and Y preserve their original order, the new ordering statements added by Spell-out of D' are consistent with the ordering statements already established in D.

(15) **Scenario 3 (Movement from edge and non-edge positions)**

$$[D' \dots X \dots Y \alpha [D \ t_X \ t_Y \ Z]] \quad \text{Ordering:} \quad \begin{array}{ll} X < Y & X < Y \\ Y < \alpha & Y < Z \\ \alpha < D \rightarrow \alpha < Z \end{array}$$


Under this scenario no ordering contradiction arises, despite the fact that Y is not moving from the left edge of D. In Scenario 2, in which Y moved alone, an ordering contradiction arose from the distinct orderings of X and Y before and after movement. Scenario 3 contrasts with Scenario 2 precisely because the relative ordering of X and Y remains the same before and after movement. We will see shortly that this contrast between the two scenarios is empirically confirmed by a number of phenomena.

A different variation on Scenario 2, which we mention only in passing here, yields another novel prediction. Suppose D in (14) undergoes ellipsis, as shown in (16), with the consequence that X (and Z) in D are not pronounced at all. On such a scenario, any ordering statement that makes reference to X (or Z) has no impact on pronunciation. For ease of exposition, we can assume that these ordering statements are generated, but are deleted as a by-product of ellipsis. We indicate such deleted ordering statements with shading.

(16) **Scenario 4 (leftward movement from a non-left-edge position + ellipsis)**

$$[D' \dots Y \alpha [D \ X \ t_Y \ Z]] \quad \text{Ordering:} \quad \begin{array}{ll} Y < \alpha & X < Y \\ \alpha < D \rightarrow \alpha < X & Y < Z \end{array}$$


ellipsis

As result of deleting the shaded statements in (16), movement of Y produces no ordering contradiction. We thus expect that ellipsis of a domain such as D will allow certain otherwise illicit instances of movement. In

Fox and Pesetsky (in prep.), we argue that this prediction is true, and explains the phenomena often grouped under the rubric of “Salvation by Deletion” (Ross (1969); Chomsky (1972); Lasnik (2001); Merchant (to appear); Fox and Lasnik (2003)).⁸

We are now in a position to contrast our proposal with others, including other attempts to relate island phenomena to the syntax-phonology interface. As we have just seen, our proposal has the following consequences for a structure such as (12), repeated below as (17):

(17) α [_D X Y Z]

1. X may freely move leftward out of D.
2. Y may move leftward out of D so long as it does not phonologically cross X. Y moves out of D without phonologically crossing X if:
 - a. X has itself moved further to the left (Scenario 3), or
 - b. X is unpronounced as a consequence of ellipsis (Scenario 4).

Consequence 1 is what our proposal shares with theories that designate X as a specific escape hatch from D. Consequence 2 is unique to our proposal. Though our proposal, like others, designates certain categories (the Spell-out domains) as special in a way that interacts with island phenomena, these categories are in no sense syntactically opaque.

Earlier accounts such as the Subjacency Condition stipulate the syntactic opacity of particular domains, and therefore are compelled to designate specific peripheral positions as exceptions to this opacity. Certain more recent proposals (e.g. Chomsky (2000, 2001a); Uriagereka (1999); Nissenbaum (2000)) share with our proposal a logical connection between island phenomena and the timing of Spell-out. These proposals differ from their predecessors in not stipulating the opacity of various domains – deriving it instead as a by-product of Spell-out. They do not differ from their predecessors, however, in the factual assumption of syntactic opacity, with the necessary consequence that designated syntactic escape hatches need to be postulated. The proposals of Chomsky and Nissenbaum express this conclusion by distinguishing the categories whose construction triggers Spell-out (*phases*) from the smaller categories

⁸ This argument is extended by Takahashi (2004) to Object Shift, thus drawing a connection between the two scenarios. Takahashi’s work explores the syntax of pseudo-gapping and its consequences for linearization.

that are actually spelled out (the *Spell-out domains*). Thus, for example, when the CP phase is constructed, it is a subconstituent of CP that is actually spelled out (IP or ν P in different versions of this proposal). The spelled out domain – not the entire phase – thus becomes syntactically opaque. The phase minus its Spell-out domain constitutes the escape hatch. If Consequence 2 is correct, it is wrong to identify certain constituents as syntactically opaque, and there is no need to distinguish phases from Spell-out domains.⁹

The sections that follow constitute an extended argument that Consequence 2 is empirically correct. If these arguments are successful, we can conclude that the correct theory of “escape hatch effects” does not invoke the notion “escape hatch” in any form; and that the correct theory of “syntactic opacity effects” does not invoke the notion of a syntactically opaque domain.¹⁰

4. Holmberg’s Generalization

Let us begin by considering how the abstract discussion above applies to the specific case of movement out of VP in a head-initial language. (We continue to ignore the base position of the external argument, consistent with the assumption that VP, rather than ν P, is the relevant Spell-out

⁹ Chomsky (2001a) has offered a conceptual argument in favor of a distinction between phases and Spell-out domains. In particular, he suggests that in an optimal architecture for language, domains previously constructed are inaccessible to later operations. As a consequence, later operations need not look into these earlier domains – thus (in many cases) shortening computations. At the same time, since operations do appear to cross domain boundaries, there must be an “edge zone” that belongs to the earlier domain but is accessible to operations in the higher domain. Our proposals are simpler in that they do not need to stipulate an “edge zone”, though they might be viewed as weaker in allowing the syntax access to greater expanses of structure. The conceptual motivation for phases/Spell-out domains nonetheless remains similar to Chomsky’s. Though we do permit the syntax to examine the contents of earlier domains, our proposals share with Chomsky’s conception a certain “encapsulation” of these domains. The system never recomputes the linearization of earlier domains and thus allows communication with the phonological component, at least, to have a strictly local character. See section 7, as well as Fox and Pesetsky (in prep.) for further discussion.

¹⁰ It is interesting to note a prediction of our proposals not shared by most other theories: string-vacuous movement should not require successive-cyclicality, and should thus be immune to certain islands. Work by Sabbagh (2003) provides strong support for this conclusion from a detailed study of Right Node Raising and related constructions.

domain; see footnote 2.) We are thus considering the situation in (9), with $X = V$ and $D = VP$, as shown in (18a). This VP undergoes Spell-out, establishing the ordering statements shown. In the next Spell-out domain, various elements may merge with the structure in (18). Consider the case in which an adverb merges with VP, as shown in (18b):

(18) **Adv + VP**

- a. *Spell-out of VP*

[_{VP} V Y Z]

Ordering: V < Y
 Y < Z

- b. *Adverb merges with VP*

Adv [_{VP} V Y Z]

Suppose Y now needs to move leftward to a VP-external position. As we have seen, this would be unproblematic if Y had moved to the left edge of VP before Spell-out of VP, as we assume can happen in A-bar constructions. In the derivation sketched in (18), however, movement of Y into the higher Spell-out domain should be impossible unless the verb itself also moves leftward, landing in a position that preserves $V < Y$ order. This, we suggest, is the account of various constraints on Object Shift in Scandinavian languages that have been grouped under the rubric of “Holmberg’s Generalization” (Holmberg (1986, 1998)).

As is known, Scandinavian languages such as Swedish show the phenomenon of verb-second, which we will assume involves movement to C. The main verb of the sentence does not move to C in sentences that contain an auxiliary verb. In addition, embedded clauses in the mainland Scandinavian languages do not generally show verb-second.

When the main verb (V) moves to C, a DP object with certain semantic (or morphological) properties may (sometimes must) move leftward out of VP. The landing site of this movement is to the left of adverbs that can be shown to precede VP. Crucially, when V does not move to C, Object Shift is impossible. This restriction is the most discussed case of Holmberg’s Generalization, exemplified by the Swedish examples in (19):¹¹

¹¹ The examples in this section are taken from Holmberg (1998).

(19) **Object Shift blocked by (unmoved) verb intervener**¹²

- a. Jag kysste henne inte [_{VP} t_v t_o]
 I kissed her not
- b. *... att jag henne inte [_{VP} kysste t_o].
 ... that I her not kissed
- c. *Jag har henne inte [_{VP} kysst t_o].
 I have her not kissed

These facts follow from Order Preservation under one key assumption: that Object Shift, unlike certain other instances of movement, does not proceed successive-cyclically through Spec,VP. When an object is preceded within VP by other material, Object Shift should instantiate the unacceptable scenario 2. If material that precedes the object also moves out of VP, it should instantiate the acceptable scenario 3, which (as noted above) is a novel prediction of our account of successive-cyclicity. Let us examine the facts.

Consider first the case in which Object Shift is accompanied by V-to-C movement, as in (19a). The sketch in (20) below shows that when the verb raises, the ordering of elements in the VP Spell-out domain is not contradicted by ordering established in the CP domain:

(20) **Object Shift + V-movement to C**

- a. **VP:** [_{VP} V O]
 Ordering: V < O
- b. **CP:** [_{CP} S V [_{TP} t_s O adv [_{VP} t_v t_o]]]
 Ordering: S < V V < O
 V < O
 O < adv
 adv < VP → ∅

¹² When Object Shift of a pronoun is possible, it is obligatory. When Object Shift is blocked, as in (19b–c), the pronoun appears in its unshifted position to the right of the verb and negation. Our system is in general compatible with a variety of approaches to this “obligatory movement up to Holmberg’s Generalization”, e.g. covert movement when overt movement is blocked (Bobaljik (2002)). Bobaljik’s particular implementation of this proposal, which relies on a view of movement as copying, is not straightforwardly compatible with the view of movement that we adopt in section 7. We leave this issue open.

When VP is Spelled out, it is established that V precedes O. V continues to precede O in the CP spell-out domain. Since VP dominates only traces, the statement “adv<VP” has no consequences for pronunciation (as indicated by “ $\rightarrow \emptyset$ ”) and creates no contradictions.¹³ (If VP contained non-null post-object material, the statement “adv<VP” would, of course, order the adverb before this material.) This instantiates scenario 3.

Consider now derivations in which Object Shift is not accompanied by V-to-C movement. As (19b–c) showed, the result is unacceptable. In (21), we illustrate how this is accounted for by Order Preservation in the case of (19b), where the presence of a complementizer blocks V-to-C movement.

(21) ***Object Shift without V-movement to C (embedded clause)**

- a. VP: [VP V O]
Ordering: V<O
- b. CP: [CP C [TP S O adv [VP V t_o]]]
Ordering: C<S V<O
S<O
O<adv
adv<VP $\rightarrow$ adv<V

In (21), the ordering statements produced at Spell-out of CP do contradict the statement “V<O” produced at Spell-out of VP. Given “O<adv” and “adv<V”, O must precede V, which contradicts “V<O”, thus instantiating the unacceptable scenario 2.

When Aux moves to C and V remains in situ, as in matrix clauses like (19c), Object Shift yields the same contradiction, as we can see in (22):

(22) ***Object Shift without V-movement to C (matrix clause with auxiliary verb)**

- a. VP: [VP V O]
Ordering: V<O

¹³ We show the subject moving to Spec,CP, as in analyses of Verb-second that follow the original proposal of den Besten (1983). There are alternative proposals, stemming from Travis (1984) and Zwart (1996), according to which the verb in an SVO sentence moves only as far as T and the subject only as far as Spec,TP. This analysis would involve the same ordering as that shown in (20).

- b. **CP:** $\left[_{CP} S \text{ aux } \left[_{TP} t_S O \text{ adv } t_{aux} \left[_{VP} V t_o \right] \right] \right]$
Ordering: $S < \text{aux}$ $V < O$
 $\text{aux} < O$
 $O < \text{adv}$
 $\text{adv} < VP \rightarrow \text{adv} < V$

Alternative proposals have been advanced that also account for the facts in (19). These proposals focus on syntactic or morphological consequences of verb movement per se. An example of a verb-centered syntactic alternative is the proposal of Chomsky (1993). This proposal assumed a clause structure in which Object Shift and the obligatory raising of the subject create locality problems that can only be resolved by movement of the verb (under the assumption that verb movement expands the domains of locality). A verb-centered morphological alternative was offered by Bobaljik (1994, 1995, 2002), who proposed that a verb must be adjacent to an inflectional head from which it receives its morphology. Object Shift disrupts this adjacency unless the verb raises over the object.

Verb-centered accounts of the facts in (19) have been called into serious question, however, by Holmberg (1998), who made the following observation: “Less often mentioned, but no less true, is the fact that not just an unmoved verb, but any phonologically visible category inside VP preceding the object position will block Object Shift.” The examples in (23) substantiate this observation. Note that Object Shift is blocked in each of these examples despite the fact that V has moved to C. Object Shift in each case is blocked by a different intervener, marked with an underline:

(23) **OS blocked by non-verb interveners**¹⁴

a. **First-object intervener**

*Jag gav den inte [_{VP} t_v Elsa t_o].

I gave it not Elsa

b. **Particle intervener**

*Dom kastade mej inte [_{VP} t_v ut t_o].

They threw me not out

¹⁴ Comparable examples without Object Shift are of course acceptable.

These facts do not follow from verb-centered accounts of (19). They suggest that the ability of an element to block Object Shift is not related to its categorial status. Verbs and other elements act alike. Instead, as stressed by Holmberg, it is the linear position of an element that is responsible for the intervention effect. (Holmberg develops this observation as part of an argument that a phonological version of “Relativized Minimality” applies to certain instances of movement. His proposal has some important points of contact with ours, to which we return below.) It should be clear that Holmberg’s observations are an automatic consequence of our theory of linearization. They are simply another instantiation of “scenario 2”. In each example in (23), Spell-out of VP yields an ordering statement of the form “intervener<O” which is contradicted by the ordering statements produced at Spell-out of CP, once the object has moved leftward over the intervener. This is sketched in (24), where “XP” is the first object or particle intervener in (23):

- (24) OS blocked by non-verb interveners (scenario 2)

VP: [VP V XP O]
Ordering: V<XP
 XP<O

CP: [CP S V [TP t O adv [VP tv XP to]]]
Ordering: S<V V<XP
 V<O XP<O
 O<adv
 adv<VP → adv<XP

prediction. Ordering contradictions produced by Object Shift over the underlined interveners in (23) should be circumvented whenever the original order is restored – i.e. by movement into the higher Spell-out domain of not only the shifted object, but also the intervener. Holmberg’s (1998) observations in (25) corroborate this prediction. When the interveners in (23) undergo movement to a position to the left of the landing site for Object Shift, the impediment to Object Shift is removed:

(25) **OS not blocked when non-verb intervener moves further to the left**¹⁵

a. **First object intervener moves**

Vem_{IO} gav_V du den_O inte [_{VP} t_V t_{IO} t_O].
who gave you it not

b. **Particle intervener moves**

UT_P kastade dom mej_O inte [_{VP} t_V t_P t_O]
out threw they me not
(bara ned för trappan).
(only down the stairs)

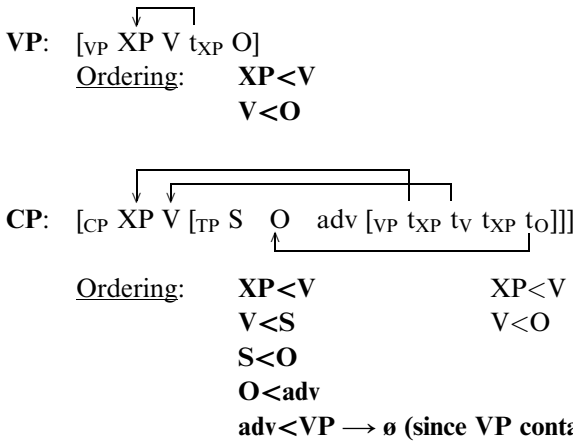
In (26), we can see why no ordering contradiction arises. We assume that the XP intervener (unlike the shifted object) moves first to the left edge of VP (as discussed in section 2) before undergoing further A-bar movement to Spec,CP. This initial step also serves to place the intervener to the left of V, an ordering preserved at the CP level:

¹⁵ A preposition in PP may also act as an intervener for Object Shift of its object, as noted by Holmberg:

*Jag talade henne inte [_{VP} t_V med t_O].
I spoke her not with

We do not have instances of P-topicalization or *wh*-movement comparable to the examples in (25) that allow us to see whether this effect too may be circumvented.

- (26) **OS not blocked when non-verb intervener moves further to the left (scenario 3)**



The relative ordering of XP, V and O, established at Spell-out of VP, is not contradicted by any statements added when CP is spelled out.

An additional observation by Holmberg (1998) is of particular interest. Holmberg pointed out that Object Shift is possible not only when V moves to C but also when V appears by itself in Spec,CP due to a process that we will call “Bare V topicalization”, exemplified in (27).

- (27) **Bare V topicalization**

- a. Kysst har jag henne inte (bara hållit henne i handen).
 kissed have I her not (only held her by the hand)
- b. Sett_v har han me kanske t_o ...
 seen has he me perhaps
 (men han vet inte vad jag heter).
 (but he knows not what I am called)

It is immediately obvious that our proposal predicts the acceptability of Object Shift in these examples, even though V has not moved to C. Thanks to the fronting of the bare V, the object follows V when CP is spelled out (after Object Shift), just as it did when VP was spelled out. What is relevant to the acceptability of Object Shift is the relative ordering of V and O at the various points of Spell-out, not the nature of the syntactic processes that establish these orderings.

Nonetheless, the proper syntactic analysis of Bare V Topicalization turns out to be of interest in the present context. As noted by Holmberg, one might analyze this construction in one of two ways. Bare V Topicalization might involve raising of V to Spec,CP – an instance of head movement to a specifier position, raising issues for the Head Movement Constraint:¹⁶

(28) **Bare V Topicalization analysis #1: Object Shift + V-movement to Spec,CP**

- a. **VP:** [_{VP} V O]
 Ordering: **V<O**
- b. **CP:** [_{CP} V aux [_{TP} S O adv t_{aux} [_{VP} t_V t_O]]]
Ordering: **V<aux** **V<O**
aux<S
S<O
O<adv
adv<VP → ∅

Alternatively, one might analyze Bare V Topicalization as movement of a VP from which non-head constituents have been previously extracted – an instance of VP “remnant movement”. If Bare V Topicalization is an instance of head raising, then the examples in (27) simply instantiate Scenario 3. If Bare V Topicalization is an instance of VP remnant movement, it instantiates an interesting variant of Scenario 3 in which X ends up to the left of Y not because X itself moves, but because a category that contains X (and formerly contained Y) moves. This possible derivation is sketched in (29):

¹⁶ It is possible that all head movement, including V-to-C movement, forms a specifier, with the moved head amalgamating with its host by a separate process. Such an analysis has been proposed by Matushansky (to appear) (see also Fukui and Takano (1998); Toyoshima (2001); Pesetsky and Torrego (2004)). This possibility does not eliminate the question raised in the text about the Head Movement Constraint. Note also that the topicalized verb in the Bare V Topicalization construction does not amalgamate with the auxiliary verb that occupies C.

(29) **Bare V Topicalization analysis #2: Object Shift + remnant VP topicalization**

VP: [VP V O]

Ordering: V<O2. *Remnant VP Topicalization*CP: [CP [VP V t_O] aux [TP S t_{aux} O adv t_[VP V t_O]]]]1. *Object Shift*

Ordering: VP<aux → V<aux V<O
 aux<S
 S<O
 O<adv

As indicated above, when VP is spelled out, it is established that V precedes O. In the process of building CP, Object Shift applies, followed by topicalization of the VP that the object was extracted from. (The auxiliary verb also raises to C, in keeping with the verb-second character of the language.) When CP is spelled out, new ordering statements are added, including “VP<aux”. Crucially, the last element of VP (not dominated by a trace) is now the verb itself, so by (11), the statement “VP<aux” means that V precedes the auxiliary verb. Since the V precedes the auxiliary verb, and the auxiliary verb precedes the object, the resulting ordering is fully consistent with the statement “V<O” established in the earlier spell-out domain.

A combination of Object Shift and VP-remnant movement must in fact be possible. When VP contains material besides the verb and a direct object, this material may front along with the verb even when the direct object is first extracted by Object Shift. Crucially, since material that is fronted along with the verb ends up preceding the shifted object, it must already have preceded the object when the VP was spelled out. Thus, for example, the fact (seen in (23a)) that the indirect object is ordered before the direct object within VP means that this ordering must be preserved if both Object Shift and VP-remnant movement apply.¹⁷ As a consequence,

¹⁷ This prediction of our proposal was brought to our attention by Alec Marantz (personal communication). Anders Holmberg offered the facts in (30a–b) as confirmation of this prediction.

the indirect object may be topicalized with the verb to the exclusion of the direct object, but the direct object may not be topicalized with the verb to the exclusion of the indirect object. This contrast may be seen in (30a–b):

- (30) a. ?[Gett henne t_{DO}] har jag den $_{DO}$ inte ...
 given her have I it not
 b. *[Gett t_{IO} den] har jag henne $_{IO}$ inte ...
 given it have I her not

Likewise, Object Shift applied to the subject of an ECM complement followed by remnant movement of the higher VP should be impossible, since the remainder of the ECM complement followed the subject when the higher VP was spelled out. This prediction is correct, as can be seen in the following example from Holmberg (1998):¹⁸

- (31) *[v_P Hört t_o hålla föredrag] har jag henne $_o$ inte t_{v_P} .
 heard t give talk have I her not

It is instructive at this point to consider the conclusions that Holmberg himself drew from the ill-formedness of (31). A consideration of Holmberg's analysis of such examples brings out another way in which our proposal differs from certain predecessors. Holmberg's analysis anticipated ours in attributing significant constraints on Object Shift to phonological factors related to linear precedence. It was Holmberg who first argued that Object Shift may not apply so as to alter linear precedence relations. Crucially, however, Holmberg proposed that this constraint is a condition on the application of Object Shift itself: Object Shift may not cross a phonological intervener. The fact that leftward movement of an intervener may save a derivation in which Object Shift applies both supported Holmberg's proposal and posed a problem for it at the same time. In (25), for example, movement to Spec,CP of an indirect object or particle permits an otherwise blocked instance of Object Shift. This type of fact was taken as a demonstration that phonologically null interveners (e.g. traces) do not block Object Shift, thus arguing for the phonological character of the phenomenon. On the other hand, since the phonological

¹⁸ See Nilsen (2003) for additional arguments for VP remnant movement in Norwegian, in the context of a proposal substantially different from ours.

constraint that governs Object Shift was viewed as a constraint on the operation itself, Holmberg had to conclude that Object Shift in such cases applies after the rules that move the intervener, and thus must apply countercyclically. Holmberg suggested that Object Shift applies in a post-syntactic PF component, thus simultaneously making sense of its apparent countercyclic character and its sensitivity to linear precedence.

What is particularly interesting about Holmberg's proposal in the present context is its incompatibility with a remnant movement analysis of Bare V Topicalization. When movement of the object via Object Shift interacts with movement of an independent constituent, it is possible to posit an ordering of the two operations such that Object Shift applies after the independent intervener has moved. Thus, in (25), Object Shift can be viewed as applying after movement of an indirect object or particle to Spec,CP. Likewise, if Bare V Topicalization is analyzed as in (28), Object Shift can be viewed as applying after V-movement. By contrast, if what looks like V-movement in the Bare V Topicalization construction is actually movement of a remnant VP from which the object has been extracted, it must obviously be the case that Object Shift *precedes* the movement that allows it to escape the phonological intervention constraint – and thus it cannot be claimed that the phonological condition on Object Shift is a condition on the rule itself, as assumed by Holmberg.

Our proposal is similar to Holmberg's in the phonological nature of its explanation for constraints on Object Shift. Our proposal differs from Holmberg's in regarding the phonological constraint on Object Shift not as a constraint on the operation of the rule itself, but as a domain-final constraint on *output*.¹⁹ It is quite expected under our proposal that Object Shift may cross elements which, if they did not later move, would create an ordering contradiction – so long as these elements do move before the next application of Spell-out. Thus, the remnant movement analysis of Bare V Topicalization is available to us, and to the extent that it can be supported, would provide further evidence for our proposal.

¹⁹ In this sense, our version of Holmberg's phonological constraint operates as a filter, rather than as a condition on a rule (thus resembling the models of predecessors mentioned in footnote 5). Note, however, that it is not a stipulated filter. Instead, it is simply the process of linearization in a model within which linearization statements, once made, cannot be unmade.

We have already seen in (30) some evidence that remnant VP topicalization does exist and interacts with Object Shift as predicted by our proposal. Holmberg himself notes potential further evidence provided by ECM constructions like those in (32). If it were possible to raise V to Spec,CP, we would incorrectly expect (32a–b) to be acceptable with the analysis indicated:

- (32) a. *Hört_v har jag henne_o inte t_v [t_o hålla föredrag].
 heard have I her not give talk
 b. *Hört_v har jag inte t_v [Per hålla föredrag].
 heard have I not Peter give talk

If V movement to Spec,CP is impossible, the analyses shown in (32) are of course unavailable. The word order in (32a–b) would have to be derived by extracting the ECM infinitival from VP to a position to the right of *inte* (in addition to applying Object Shift in (32a)) before fronting the remnant VP. Given the general, independently observable impossibility of fronting ECM infinitivals in languages such as Swedish, this initial step can be excluded, as Holmberg himself points out – thus explaining the ungrammaticality of the examples.²⁰ Under a V-movement analysis, there does not appear to be a simple alternative, though Holmberg (1998) does sketch a tentative proposal related to the assignment of case.

If V-movement to Spec,CP does not exist and all Bare V Topicalization is remnant VP movement, we must analyze examples like those in (33) from Holmberg (1998) as involving (short) extraction of PP or DP followed by VP remnant topicalization. The relevant short extraction cannot be Object Shift (as Holmberg notes) since Object Shift is largely limited to pronouns.

- (33) a. Sag gjorde han på henne (but he said nothing).
 looked did he at her

²⁰ In English, Heavy Shift may perhaps be followed by remnant VP topicalization, e.g. *Read we did the books that she recommended*. Heavy Shift of an ECM infinitive seems unacceptable, especially with remnant VP topicalization:

- (i) ??Mary saw with her binoculars Bill give that talk we were expecting.
 (ii) *See she did Bill give that talk.

The VP of an ECM infinitive also may not be stranded:

- (iii) *See Bill she did give that talk.

- b. Sett har jag inte den idioten.
 seen have I not the idiot
 (but I have talked with him on the telephone)

In all examples of this sort, our proposal makes a key prediction that is (as far as we can tell) confirmed by the available data. Whatever material fails to front under remnant VP topicalization in examples like (33) must be drawn from the right periphery of the fronted VP, so that the original order is retained.²¹

We have just seen some reasons to view Holmberg's Generalization as a consequence of an Order Preservation property of domain-final syntax-phonology mapping rather than as a constraint on particular movement operations.²² Once one views Holmberg's Generalization as a con-

²¹ Note also that if Bare V Topicalization is remnant VP fronting, as we have argued, then what needs to be preserved from phase to phase is clearly linear order, not some other relation, e.g. c-command. The preservation of V-O order after remnant VP movement, for example, satisfies the requirements of our system, even though the combination of Object Shift and remnant VP fronting produces a c-command configuration quite different from that which preceded these movement processes. See Williams (2003), 17–21 and Müller (2000) for some relevant discussion.

²² Other alternatives need to be carefully considered as well. Bobaljik (2002) and Anagnostopoulou (2003) have argued that the various blocking effects for Object Shift should not receive a uniform explanation. The blocking effect of an unmoved verb is attributed by Anagnostopoulou to Relativized Minimality interactions involving the external argument, along the lines of Chomsky (1993) – with verb movement to T creating an “equidistance” exception. The same phenomenon is attributed by Bobaljik to a morpho-phonological adjacency condition on V and a higher functional head, disrupted by Object Shift when V fails to raise.

The blocking effect of an indirect object on Object Shift of the direct object is attributed by both Anagnostopoulou and Bobaljik to Relativized Minimality interactions between the two objects. Anagnostopoulou suggests that the ameliorating effect of A-bar movement of the indirect object might follow from special properties of traces, as in (Chomsky (2000); Chomsky (2001a)).

The blocking effect of a particle in Swedish is related (by Bobaljik) to the general incompatibility of independently occurring particles in the Swedish passive (Svenonius 1996). Svenonius notes, however, that incorporation of the particle into the passive participle permits the passive, which suggests to us that the incompatibility of Swedish passive with particles is an independent phenomenon, related to the morphological interactions between particle incorporation and finiteness found elsewhere in Germanic – not a fact to be considered in parallel with the blocking effect of particles on OS with finite verbs.

sequence of domain-final syntax-phonology interaction, one expects the precise size of the relevant domains to have empirical consequences. Consider, for example, the fact that our proposal in effect distinguishes two types of movement that interact with the phonology in distinct ways. When a pair of elements have been ordered in a previous Spell-out domain, movement in a later Spell-out domain may not apply so as to reorder these elements; but a pair of elements that have not yet been ordered may undergo whatever movement the laws of syntax require, without fear of producing an ordering contradiction. It is the identity of the Spell-out domains that determines which movements belong to which category. This property of our proposal is directly reflected in the data discussed in this section. Because the lower Spell-out domain in Swedish contains V and the direct object O, but does not contain adverbs or negation, Object Shift is free to move across adverbs without any danger of an ordering contradiction, but is not free to move across V. In a system such as Holmberg's, where the relevant condition is not domain-limited, this difference between VP-external elements (such as adverbs) and VP-internal elements (such as V and O) is not expected.

5. An Inverse Holmberg Effect

We began our discussion with the observation that apparent long-distance leftward movement (e.g. *wh*-movement or topicalization) conforms to Order Preservation only if it proceeds via the left edge of each Spell-out domain that it crosses. In this way, we explained its successive-cyclic character. We then noted a further prediction of Order Preservation:

The obviation of the verb blocking effect by V-topicalization is not explained by Bobaljik or by Anagnostopoulou.

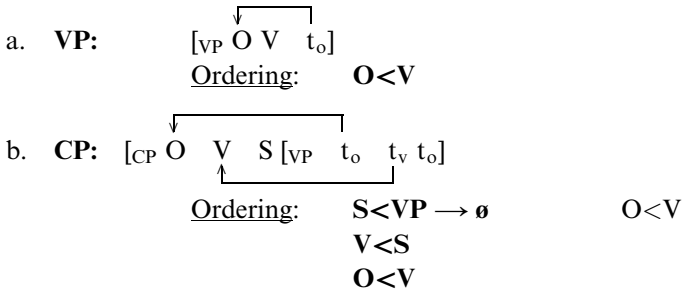
We suspect that the uniformity of our proposal across these various constructions (a property inherited from Holmberg 1999) argues in its favor, though there might be possible elaborations of the alternative views that would pose challenges. Anagnostopoulou documents a wide variety of cases in which constraints on A-movement are obviated by other movement operations, some of them not obviously order-preserving. We leave a fuller comparison for future work. For some comparative comments in a different context, see Ko (2004).

that even an element that does not occupy the left edge of a Spell-out domain may be moved leftward from that domain so long as elements to its left also move in a fashion that preserves the original linear order. Our example of this scenario was Object Shift, where an object preceded by V within VP could be extracted from VP so long as V also moved to a position preceding the landing site of the object.

Our proposal makes further important predictions (for VO languages) about constructions that combine movement of an element to the left edge of VP (as in *wh*-movement or topicalization) with verb movement. Consider the ordering consequences of movement of a postverbal element such as a direct object to the left edge of VP. Such movement will establish $O < V$ order when VP is spelled out. If V later raises to C, and O moves to Spec,CP, as in *wh*-movement, $O < V$ order is preserved, as seen in (34) and (35):

- (34) Ven kysste jag?
 who kissed I
 ‘Who did I kiss?’

(35) **Successive-cyclic *wh*-movement + V-movement to C**



The second step of object movement is crucial in (35). If, instead of moving to Spec,CP, the object had either remained at the left edge of VP or moved to a position lower than C, V-to-C movement would yield an ordering contradiction. Movement of this sort would thus display an “Inverse Holmberg Effect” – a local movement of a VP-internal element that is permitted only when V-raising does *not* take place. In fact, this effect appears to be attested.

Several Scandinavian languages, including Icelandic, show a phenomenon known as “Quantifier Movement” (QM) (Rögnvaldsson

(1987); Jónsson (1996); Svenonius (2000)).²³ A quantified expression that undergoes QM moves leftward across VP-internal material, landing in a VP-external position slightly lower than the landing site of Object Shift. QM thus fails to obey Holmberg's Generalization, as can be seen in the Icelandic QM sentences in (36).

(36) **Quantifier Movement (Icelandic)**²⁴

- a. Jón hefur ekkert [sagt Sveini ____].
Jon has nothing said Svein-DAT
'Jon has told Svein nothing.'
- b. Äg hef mörgum bókum skilað Jóni.
I have many books returned Jon-DAT
'I have returned many books to Jon.'
- c. Hann mun mikið hafa vilja [lesa ____].
he will much have wanted read
'He has wanted to read much.'

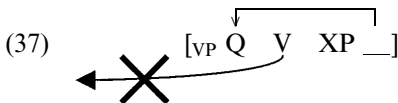
If our proposals are correct, the quantified expression in these examples must have stopped at the left edge of VP before landing in its final position – a VP-external position, as shown clearly in (36c). This intermediate landing site is necessary because in (36) the phrase moved by QM precedes VP-internal material. This is material that it would have ended up following if QM had not applied. If QM first targets the left edge of VP, the order found in (36a–c) is already established when VP is spelled out, and no contradiction arises.²⁵

²³ QM in Norwegian behaves differently from its counterparts in the other Scandinavian languages, and actually appears to obey Holmberg's Generalization. Our remarks about QM should be understood as excluding Norwegian, where QM presumably does not proceed via the left edge of VP as it does in the other languages under discussion (Christensen (1986, 1995); Svenonius (2000)).

²⁴ We gloss over another factor relevant to the proper analysis of QM. Negative quantified expressions in Scandinavian appear to undergo QM obligatorily, at least where QM makes a difference to word order (Jónsson (1996, 94); Christensen (1995); Svenonius (2000)).

²⁵ It is crucial either that the subject is outside the domain spelled out when VP is linearized or else that the quantified expression "tucks in" below the subject in Spec,vP. The fact that V-to-C movement may place the verb to the left of the subject in certain verb-second constructions suggests the former conclusion. For further discussion of the status of subjects with respect to the VP Spell-out domain, see Fox and Pesetsky (in progress) and Ko (2004).

If QM involves an intermediate landing site at the left edge of VP, we predict an interesting interaction between QM and V-movement. Linearization of VP orders the moved quantifier not only to the left of elements such as the indirect object, but also to the left of V itself. As a consequence, V-to-C movement should be blocked when QM occurs:²⁶



This expectation is borne out and represents exactly the situation just discussed under the rubric of an “Inverse Holmberg Effect”. Example (38) is a minimal variation on (36a), differing only in the absence of an auxiliary verb, which forces main verb movement to C. As indicated, once this minimal variation is introduced, QM is no longer possible:²⁷

(38) **QM to left edge of VP incompatible with V-to-C movement**

*Jón sagði ekkert Sveini ____.

Jon said nothing Svein-DAT

(Rögnvaldsson (1987), quoted by Svenonius (2000))

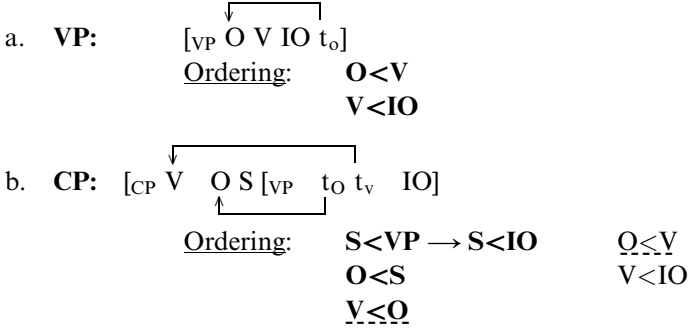
²⁶ It is not crucial to assume that V in Icelandic subject-first main clauses actually moves to C. What is important is simply the fact that V moves to a position higher (and to the left of) the landing site of QM – the same position in (38) as that occupied by the (highest) auxiliary verb in (36). This position might be lower than C, in light of the possibility of embedded verb-second constructions in Icelandic (and in light of Travis (1984) and Zwart (1996)’s more general proposal that V in Germanic SVO constructions occupies T, rather than C). See Iatridou and Kroch (1992), among others, for more discussion.

²⁷ One might argue that examples like (38a) are actually excluded because the indirect object (*Sveini*) must undergo Object Shift when Object Shift is not independently impossible. Given that the Object Shift position appears to be slightly higher than the QM position, obligatory Object Shift would also enforce IO<DO order. This view would be consistent with the acceptability of (36), given that OS is in fact impossible in such sentences (Holmberg’s Generalization). In fact, however, though some speakers have a preference for OS of full DPs when OS is possible, this judgment appears to represent nothing more than a preference. Jónsson (1996), for example, notes the acceptability of such examples as (i), in which the position of the adverb makes it clear that OS has not applied to the full DP *Mariu*:

Jón sagði hreinlega Mariu alla söguna.
 John told simply Mary whole story-the
 ‘John simply told Mary the whole story.’

The ordering contradiction produced by the combination of QM and V-to-C movement in (38) is shown schematically in (39):

(39) *QM + V-movement to C



The impossibility of V-movement when QM applies, seen in the contrast between (36) and (38), is a general phenomenon, not limited to double-object constructions (as noted by Svenonius (2000)). In (40a–b), in which V remains in situ, QM is seen crossing P and a direct object, in addition to V (with a slightly awkward outcome, as noted). In (40c–d), in which V moves to C, QM is seen to be impossible, as predicted.

- (40) a. Ég hef sett eitur í mörg glös.
 I have set poison in many glasses
 ‘I have placed poison in many glasses’
 b. ?Ég hef mörg glös sett eitur í.
 I have many glasses set poison in
 c. Ég setti eitur í mörg glös.
 I set poison in many glasses
 d. *Ég setti mörg glös eitur í.
 I set many glasses poison in
 [examples from Svenonius (2000)]

Likewise, (41a) shows long-distance QM from a controlled infinitival CP into a higher clause, which is once again incompatible with movement of the higher verb *skipaði*, as seen in (41b).

- (41) a. ?Hann hefur engum skipaði henni að giftast.
 he has nobody ordered her C to-marry
 ‘He hasn’t ordered her to marry anybody’
 b. *Hann skipaði henni engum [_{VP} *t_V* [_{CP} að giftast ____].
 he ordered her nobody C to-marry

The contrast in (41a–b) has an additional feature of interest. Johnson and Vikner (1994) argue that infinitival verbs in control clauses raise to T in Icelandic, as demonstrated by the relative position of infinitival *lesa* and negation in (42).

- (42) a. María lofaði að lesa ekki bókina.
 Maria promised C to-read not book-the
 b. María lofaði að ekki lesa bókina.
 Maria promised C not to-read book-the
 (Sigurðsson (1989))

If the verb in examples like (41a) has also moved to T, we may conclude that QM is not in general incompatible with V-raising. The fact that the moved quantifier *engum* ends up preceding the embedded verb means that it must have moved through the left edge of the lowest VP as an initial step of movement. Subsequent raising of the embedded verb *giftast* to T of the embedded clause would create an ordering contradiction – but for the fact that *engum* itself has moved leftward once again (to Spec of the embedded CP, a position to the left of complementizer *að*), restoring *engum* < *giftast* order. QM is thus compatible with V-raising for the same reason *wh*-movement in (34) is compatible with V-raising: movement to Spec,CP restores the order established when VP was spelled out.

This observation brings out a general characteristic of our proposals. In this section, we have discussed interactions among specific instances of movement: V-movement, QM and *wh*-movement. If our proposals are correct, however, we should not be surprised to discover very similar interactions in many other syntactic contexts. This is because, for us, these interactions arise exclusively from the linearization consequences of movement within and across Spell-out domains. Thus, though the ordering statements sketched in (35) and (39) included specific labels such as “V” and “O”, they could just as well have been given more abstractly.

For example, the role of “V” could be played by any element X that would have occupied the left edge of VP if some other element Y had not moved to its left. Such movement of Y in general takes away the privileged status of X as an “edge” element – i.e. an element that can move leftward in the next Spell-out domain without fear of ordering contradiction. We predict that *whenever* a situation of this sort arises, movement of Y should block movement of X (as in (39)) – with the now-familiar exception that arises when further instances of movement restore the original order of X and Y (as was the case with V-to-C and wh-movement in (35)).

Recent work by Ko (2003, 2004) allows us to demonstrate this point quite nicely. She shows that certain long-standing problems in the analysis of scrambling have a straightforward explanation as generalizations of the paradigms in (35) and (39). In scrambling languages such as Korean and Japanese, as studied by Kuroda (1983) and Miyagawa (1989), among others, a quantifier may be stranded by movement of the associated nominal expression, as seen in (43):

(43) **Object scrambling over subject. stranding quantifier**

↓				
[O	S	t_{obj}	NQ_{obj}	V]
Maykcwu-lul _i	John-i	t_i	sey-pyeng	masi-ess-ta
Beer-Acc	John-Nom		three-Cl _{bottle}	drink-Past-Dec
‘John drank three bottles of beer.’				

Kuroda and Miyagawa argue that before scrambling, the quantifier and associated nominal expression are structurally close (e.g. sisters²⁸) – and that it is movement that separates them under certain circumstances.

Given that scrambling seems to be free (and multiple scrambling is independently possible), it is interesting that object scrambling over a subject may not be followed by subject scrambling over the object, where the

²⁸ Miyagawa’s condition is slightly looser in ways that we will not discuss here. See also Ko (2004) for some discussion of a Korean construction in which the closeness condition does not obtain: a floating quantifier construction in which the quantifier agrees in case-marking with its associated nominal. This construction displays none of the key properties discussed in this section, for reasons discussed by Ko.

*[S O t_{subj} NQ_{subj} t_{obj}]

*Haksaying-tul-i₁ maykcwu-lul₂ t₁ sey-myeng t₂

Student-Pl-Nom beer-Acc three-CL_{person}

V]

masi-ess-ta

drink-Past-Dec

‘Three students drank beer.’

John- <i>i</i>	[na-nun [t _i	Mary-lul	ttayli-ess-ta-ko]]
John-Nom	I-Top	t	Mary-Acc
sayngkakha-n-ta			hit-Past-Dec-C
think-Pres-Dec			
'John, I think that __ hit Mary.'			

John- <i>i</i>	[_{CP} <i>way</i>	[<i>t_i</i>	Mary-lul	ttayli-ess-ni]]?
John-Nom	why	t	Mary-Acc	hit-Past-Q
'Why did John hit Mary?'				

²⁹ We present Korean examples only, but Japanese shows the same phenomena.

b. ... stranding a quantifier

Hakpumo-tul-ika_i way t_i **sey-meyng** hakkyo-lul

Parents-Pl-Nom why t_i 3-CL school-Acc

pangmwunha-yess-ni?³⁰

visit-Past-Q

‘Why did three parents visit the school?’

Ko concludes that examples like (44) are not excluded by a general ban on subject scrambling. She argues instead that such examples are excluded by the linearization consequences of the various movements.

Central to Ko’s analysis is an issue that we have sidestepped so far. If the subject originates within some type of VP, at what point in the derivation is it linearized with respect to other members of this phrase?³¹ Since the Scandinavian languages freely raise the main verb to C over the subject, the subject is presumably not linearized in Scandinavian with respect to the main verb before CP is constructed. That is one reason why we have referred so far to “VP” (rather than “vP”) as a Spell-out domain. If vP were a Spell-out domain in Scandinavian, verb-raising would yield an ordering contradiction. In Korean, however, Ko argues that vP is a Spell-out domain (possibly in addition to VP), and that the subject is therefore linearized with respect to VP-internal material before CP is built. We return to this apparent variation among languages in Fox and Pesetsky (in prep.), where we relate it to another source of variation (the overt/covert distinction). For now, we concentrate on the consequences of Ko’s assumption for scrambling.

Ko’s solution to the Subject Puzzle is based on the following observation. If vP is a Spell-out domain, an object that ends up preceding the external argument must first have scrambled to the edge of vP so as to establish “O<S” order. For this reason, any subsequent scrambling of

³⁰ Ko (personal communication) notes that this example is slightly awkward as a matrix question, for unclear reasons, but improves when embedded under a verb such as *anta* ‘know’.

³¹ In this section, we discuss only subjects which are external arguments. Miyagawa (1989) observed that the subject of passive and unaccusative constructions behaves like an object with respect to quantifier stranding, an unsurprising observation in the context of Ko’s proposals, as she discusses.

the subject that would yield “S<O” order is blocked as an ordering contradiction. This consequence is, of course, simply an instance of the Inverse Holmberg Effect in (39), with the scrambled object playing the role of the Icelandic quantified expression, and the subject playing the role of the verb.

Given that a quantifier and its associated nominal start out as sisters, (44) is excluded because its linearization imposes conflicting requirements on the derivation. For the object to precede the subject floated quantifier, it must scramble above the subject before *vP* is spelled out. In this position, however, it will not only precede the subject’s quantifier, but will also precede the rest of the subject phrase. The nominal part of the subject might then be separated from its quantifier by leftward scrambling – but this instance of movement yields an Ordering Contradiction. Having established (when *vP* was spelled out) that the object precedes this material, later movement cannot reverse that order.³²

Ko provides many additional examples of this effect. Among many other things, she shows that her analysis predicts a distinction between the effects on linearization of low and high adverbs. Low adverbs, by virtue of belonging to the *vP* Spell-out domain, pattern with objects in blocking stranding of a subject quantifier. High adverbs, as we have seen in (46), do not. Ko provides evidence for this generalization from a wide variety of adverbs. Her discussion thus demonstrates (internal to Korean) the same point that we have already stressed in comparing Korean quantifier stranding to Icelandic QM: that it is not the details of specific structures and operations that yield Holmberg and Inverse Holmberg effects, but the consequences of these structures and operations for linearization.

³² It is necessary to assume that the external argument in Korean, having been merged as Spec,*vP*, may not merge a second time to *vP* after object scrambling, thus establishing “S<O” order within the lower Spell-out domain, despite the object scrambling. Ko notes that the impossibility of *vP*-internal external-argument scrambling of this sort follows from Chomsky’s proposal that probe-goal relations (a precursor to movement) are established under *c*-command. (The head *v* does not *c*-command its specifiers.) Likewise, for our account of QM to be complete, we must ensure that the main verb in Icelandic cannot adjoin to its own maximal projection after QM has moved a quantified expression to its specifier.

6. Conclusions

The final sections of this paper bring into relief the central argument for our architecture: the fact that a heterogeneous set of syntactic constructions appear to be constrained by a single factor – the avoidance of ordering contradiction. The choice seems to be between a heterogeneous set of syntactic constraints (directly reflecting the heterogeneity of the data) or a homogeneous account in terms of the communication between syntax and phonology.

If our proposal is correct, the overall picture of syntax itself is simpler. For example, there is no need to seek a syntax-internal account of constraints on subject scrambling in Korean, nor an account of the variety of elements that block (and fail to block) Object Shift in Scandinavian.

On the other hand, certain effects continue to arise from the inner workings of syntax itself. For example, if our proposal is right, it is phonology that blocks overt long-distance *wh*-movement unless it proceeds successive-cyclically via the left edge of Spell-out domains. Our proposals say nothing in themselves, however, about the circumstances under which movement to these left-edge positions is allowed or prohibited. If, for example, movement to Spec,CP occurs only when triggered by the presence on C of a particular feature, then the absence of such a feature will create an ordering contradiction – but one should not look to our system for an explanation of the absence of the relevant feature.

This observation, of course, raises interesting questions about the overall division of labor between syntax and phonology. Our proposals provide some new ways of investigating such questions. For example, if we are correct, one can distinguish constraints on movement that are “syntax-internal” from those that reflect properties of Linearization by looking for environments where problems for Linearization might be overcome by other instances of movement (or by processes such as ellipsis). In this paper, we focused on cases in which multiple instances of movement eliminate problems for linearization. In Fox and Pesetsky (in prep.), we extend the discussion to consider ellipsis as well – an investigation with further implications for a variety of constructions.

7. Appendix

Our presentation has been informal. In the fuller version of this material, we sketch in greater detail how this approach might be formalized. This involves a choice among various possibilities, and we attempt to justify our choice on the basis of several empirical and conceptual considerations. Though space does not permit us to cover this material here, it may be useful nonetheless to present the rudiments of the implementation that we end up with. We do so, with a minimum of commentary, in this Appendix.

When a domain is spelled out, there is an *Ordering Table* that contains the output of any previous applications of Spellout. Spellout adds new ordering statements to the Ordering Table.

Laws of Precedence: The new ordering statements will be determined by *Laws of Precedence*.³³

(47) **Format for the Laws of Precedence**

Where X is the mother of α and β ; and α has properties φ (e.g. is a head, complement, specifier, etc.); and β has properties φ' , α *precedes* β .

(48) **Some Possible Laws of Precedence for English**

- a. Where X is the mother of α and β and α is a specifier of β , α *precedes* β .
- b. Where X is the mother of α and β and β is a complement of α , α *precedes* β .

Movement: We assume that Movement is a “remerge” operation (Blevins (1990); Chomsky (2001a); Epstein et al. (1998); among others) rather than a copying procedure. The consequence of movement is the merger to X of a single element α previously merged to Y .

³³ We are agnostic as to the balance between language-specific and UG-determined aspects of these Laws; e.g. whether UG encompasses a “Head Parameter” Travis (1984, 1991) or not (Haider (1992); Kayne (1994)).

This view is helpful to us for the following reasons. If movement of a constituent α with phonological properties P is viewed as a process that produces a copy of α and merges the copy in a new position, then the grammar must include some statement that prevents one of the two copies from being pronounced, i.e. that prevents P from being copied along with the other properties of α . Various proposals have been offered that have this effect (see Groat and O’Neil (1996); Pesetsky (1998); Fox and Nissenbaum (1999); Bošković (2001); Bobaljik (2002); Landau (2003); among others). On the other hand, if we view movement as a process that takes a single instance of α and remerges it, the issue of multiple realizations of P does not arise (since nothing has been copied). Furthermore, the notion that an ordering contradiction might arise from movement makes better sense under a “remerge” view than under a “copy” view. Under a copy view, it is not obvious why “ $\alpha < \beta$ ” and “ $\beta < \text{copy}(\alpha)$ ” should necessarily produce a problem for Spell-out. In contrast, under the remerge view, the contradiction is self-evident.

The fact that “traces do not count” is now a fact about what counts as the mother of a moved constituent, at least as far as ordering is concerned:

(49) **Mother**

The *mother* of α is the constituent formed by the most recent Merge of α .

(50) **Dominates (transitive closure of “mother”)**

A node X *dominates* α iff

- (i) X is the mother of α ; or
- (ii) X is the mother of β and β dominates α .

Spellout: Spellout of a domain K consists of the following two steps:

1. The precedence relations that are given by relevant Laws of Precedence are gathered into a set $LP(K)$.
2. An operation *Linearize* maps $LP(K)$ into a set $L(K)$, which contains ordering statements for terminal elements of K , and adds $L(K)$ to the Ordering Table.

(51) **Ordering statements provided by Laws of Precedence**

$LP(X) :=$

- $\{\alpha < \beta: \alpha \text{ and } \beta \text{ are dominated by } X;$
- $\alpha \text{ precedes } \beta \text{ by a Law of Precedence;}$
- $\text{and neither } \alpha \text{ nor } \beta \text{ is dominated by a Spell-out Domain}$
- $\text{other than } X\}$

(52) **Linearize**

- (i) Form the Linearization Set $L(K) :=$
 $\{\alpha' < \beta': \alpha < \beta \in LP(K),$
 $\text{and } \alpha' \text{ is an End of } \alpha \text{ and}$
 $\beta' \text{ is a Beginning of } \beta\}.$
- (ii) Update the Ordering Table by adding the members of $L(K)$
[that is, $\text{Ordering-Table} \rightarrow (\text{Ordering-Table} \cup L(K))$].

In Fox and Pesetsky (in prep.), we argue for the particular technical notion of “End” and “Beginning” (terms mentioned in (52)) given below in (53).

(53) **End and Beginning**

- (i) x is an End of α iff
 x is a terminal element reflexively dominated by α and
 $\neg \exists y \text{ s.t. } y \text{ is dominated by } \alpha \text{ and } x < y \in \text{Ordering Table}.$ ³⁴
- (ii) x is a Beginning of α iff
 x is a terminal element reflexively dominated by α and
 $\neg \exists y \text{ s.t. } y \text{ is dominated by } \alpha \text{ and } y < x \in \text{Ordering Table}.$

The choice of these particular notions is dictated by a consideration of the degree of redundancy that is allowed to exist in the Ordering Table. Intuitively, an “End of α ” according to (53) is a constituent that (at a given point in the derivation) still has a chance of being the last thing pronounced in α . We reserve for the fuller presentation a comparison of this view with other natural notions of End and Beginning.

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³⁴ “Reflexively dominates” is the union of “dominates” and identity. If lexical items always dominate subcomponents (e.g. *dog* dominates its phonological segments), then we can eliminate the word *reflexively* here, since it will be the /d/ or /g/ of *dog* that counts as the beginning or end of the lexical item.

References

- Anagnostopoulou, E. (2003). *The Syntax of Ditransitives: Evidence from Clitics*. Berlin, New York: Mouton de Gruyter.
- Barss, A. (1986). *Chains and Anaphoric Dependence: On Reconstruction and Its Implications*, Massachusetts Institute of Technology: Doctoral Dissertation.
- Blevins, J. (1990). *Syntactic Complexity: Evidence for Discontinuity and Multidomination*, University of Massachusetts at Amherst: Doctoral dissertation.
- Bobaljik, J. (1994). What Does Adjacency Do? In: Harley, H. & Phillips, C. (eds.) *The Morphology-Syntax Connection* (pp. 1–32). MIT.
- (1995). *Morphosyntax: The Syntax of Verbal Inflection*, Massachusetts Institute of Technology: Doctoral Dissertation.
- (2002). A-Chains at the PF-Interface: Copies and ‘Covert’ Movement. *Natural Language and Linguistic Theory*, 20, 197–267.
- Bošković, Ž. (2001). *On the Nature of the Syntax-Phonology Interface: Cliticization and Related Phenomena*. Oxford: Elsevier Science.
- Chomsky, N. (1972). Some Empirical Issues in the Theory of Transformational Grammar. In: Peters, P. S. (ed.) *Goals of Linguistic Theory*. Englewood Cliffs, NJ: Prentice-Hall.
- (1973). Conditions on Transformations. In: Anderson, S. & Kiparsky, P. (eds.) *A Festschrift for Morris Halle* (pp. 232–286). New York: Holt Rinehart & Winston.
- (1993). A Minimalist Program for Linguistic Theory. In: Hale, K. & Keyser, J. (eds.) *The View from Building 20* (pp. 1–52). Cambridge, Mass.: MIT Press.
- (2000). Minimalist Inquiries. In: Martin, R., Michaels, D. & Uriagereka, J. (eds.) *Step by Step: Essays on Minimalist Syntax in Honor of Howard Lasnik* (pp. 89–156). Cambridge, Mass.: MIT Press.
- (2001a). Derivation by Phase. In: Kenstowicz, M. (ed.) *Ken Hale: A Life in Language* (pp. 1–52). Cambridge, Mass.: MIT Press.
- (2001b). *Beyond Explanatory Adequacy*. Cambridge, Mass.: MIT Working Papers in Linguistics.
- Christensen, K. K. (1986). Norwegian *Ingen*: A Case of Post-Syntactic Lexicalization. In: Östen, D. & Holmberg, A. (eds.) *Workshop at the Ninth Scandinavian Conference* (pp. 21–35). Stockholm: University of Stockholm.
- (1995). Quantifier-Movement and Its Syntactic Reflexes. Ms., University of Bergen.
- Den Besten, H. (1983). On the Interaction of Root Transformations and Lexical Deletive Rules. In: Abraham, W. (ed.) *On the Formal Syntax of the Westgermania*. Amsterdam: John Benjamins.
- Epstein, S. D., Groat, E. M., Kawashima, R. & Kitahara, H. (1998). *A Derivational Approach to Syntactic Relations*. Oxford: Oxford University Press.
- Fox, D. (1999). Reconstruction, Binding Theory, and the Interpretation of Chains. *Linguistic Inquiry*, 30, 157–196.
- Fox, D. & Lasnik, H. (2003). Successive Cyclic Movement and Island Repair: The Difference between Sluicing and VP Ellipsis. *Linguistic Inquiry*, 34, 143–154.
- Fox, D. & Nissenbaum, J. (1999). Extraposition and Scope: A Case for Overt QR. In: *Proceedings of the West Coast Conference on Formal Linguistics 18* (pp. 132–144). Somerville, MA: Casadilla Press.
- Fox, D. & Pesetsky, D. (in prep.). *Cyclic Linearization and the Typology of Movement*.

- Fujita, N. (1994). *On the Nature of Modification: A Study of Floating Quantifiers and Related Constructions*, University of Rochester.: PhD.
- Fukui, N. & Takano, Y. (1998). Symmetry in Syntax: Merge and Demerge. *Journal of East Asian Linguistics*, 7, 27–86.
- Gračanin, M. (2004). V-N Compounds in Italian and Croatian. Ms., MIT.
- Groat, E. & O’Neil, J. (1996). Spell-out at the LF Interface. In: Abraham, W., Epstein, S. D., Thráinsson, H. & Zwart, C. J.-W. (eds.) *Minimal Ideas* (pp. 113–139). Amsterdam: John Benjamins Publishing Company.
- Haider, H. (1992). Branching and Discharge. vol. 23. Working Papers of the SF13 340. University of Stuttgart.
- Haig, J. H. (1980). Some Observations on Quantifier Float in Japanese. *Linguistics*, 18, 1065–1083.
- Hale, K. L. & Keyser, S. J. (1993). On Argument Structure and the Lexical Expression of Syntactic Relations. In: Hale, K. & Keyser, S. J. (eds.) *The View from Building 20* (pp. 53–110). Cambridge, Mass.: MIT Press.
- (2002). *Prolegomenon to a Theory of Argument Structure*. Cambridge, Mass.: MIT Press.
- Holmberg, A. (1986). *Word Order and Syntactic Features*, Stockholm: Doctoral Dissertation.
- (1998). Remarks on Holmberg’s Generalization. *Studia Linguistica*, 53, 1–39.
- Iatridou, S. & Kroch, A. (1992). The Licensing of CP-Recursion and Its Relevance to the Germanic Verb-Second Phenomenon. *Working Papers in Scandinavian Syntax*, 50, 1–24.
- Johnson, K. & Vikner, S. (1994). The Position of the Verb in Scandinavian Infinitives: In V or C but Not in I. *Working Papers in Scandinavian Syntax*, 53, 61–84.
- Jónsson, J. G. (1996). *Clausal Architecture and Case in Icelandic*, Linguistics, University of Massachusetts: Doctoral Dissertation.
- Kathol, A. (2000). *Linear Syntax*. Oxford: Oxford University Press.
- Kayne, R. S. (1994). *The Antisymmetry of Syntax*. Cambridge, Mass.: MIT Press.
- Ko, H. (2003). Asymmetry in Scrambling and Cyclic Spell-Out. Paper presented at Approaching Asymmetry at the Interface Conference, Université du Québec à Montréal.
- (2004). Asymmetries in Scrambling and Cyclic Linearization. Ms., MIT.
- Kuroda, S.-Y. (1983). What Can Japanese Say About Government and Binding. In: Barlow, M. (ed.) *Proceedings of the West Coast Conference on Formal Linguistics* (pp. 153–164). Stanford: Stanford Linguistics Association.
- Landau, I. (2003). Modular Recoverability: Chain Resolution in Hebrew V(P)-Fronting. Ms., Ben Gurion University.
- Lasnik, H. (2001). When Can You Save a Structure by Destroying It? Paper presented at Proceedings of the North East Linguistic Society, Georgetown University.
- Lebeaux, D. (1991). Relative Clauses, Licensing, and the Nature of the Derivation. In: Rothstein, S. D. (ed.) *Perspectives on Phrase Structure: Heads and Licensing*. San Diego: Academic Press.
- Matushansky, O. (to appear). Head Movement in Linguistic Theory. In: Sabbagh, J. & Hiraiwa, K. (eds.) *MIT Working Papers in Linguistics*. Cambridge, Mass.: MIT.
- McCloskey, J. (2000). Quantifier Float and *Wh*-Movement in an Irish English. *Linguistic Inquiry*, 31, 57–84.
- McDaniel, D. (1989). Partial and Multiple *Wh*-Movement. *Natural Language & Linguistic Theory*, 7, 565–604.

- Merchant, J. (to appear). Variable Island Repair under Ellipsis. In: Johnson, K. (ed.) *Topics in Ellipsis*. Cambridge, UK: Cambridge University Press.
- Miyagawa, S. (1989). *Structure and Case Marking in Japanese*. San Diego: Academic Press.
- Müller, G. (2000). Shape Conservation and Remnant Movement. Paper presented at Proceedings of the North East Linguistic Society, Rutgers University.
- Nilsen, Ø. (2003). *Eliminating Positions: Syntax and Semantics of Sentence Modification*, University of Utrecht: Doctoral dissertation.
- Nissenbaum, J. (2000). *Investigations of Covert Phrase Movement*, Linguistics, Massachusetts Institute of Technology: Doctoral Dissertation.
- Perlmutter, D. (1972). Evidence for Shadow Pronouns in French Relativization. In: Peranteau, P. M., Levi, J. N. & Phares, G. C. (eds.) *The Chicago Which Hunt: Papers from the Relative Clause Festival, April 13, 1972*. Chicago: Chicago Linguistic Society.
- Pesetsky, D. (1998). Some Optimality Principles of Sentence Pronunciation. In: Barbosa, P., Fox, D., Hagstrom, P., McGinnis, M. & Pesetsky, D. (eds.) *Is the Best Good Enough?* (pp. 337–384). Cambridge, Mass.: MIT Press.
- Pesetsky, D. & Torrego, E. (2004). Tense, Case, and the Nature of Syntactic Categories. In: Guéron, J. & Lecarme, J. (eds.) *The Syntax of Time*. Cambridge, Mass.: MIT Press.
- Rögnvaldsson, E. (1987). On Word Order in Icelandic. In: *Proceedings of the Seventh Biennial Conference of Teachers of Scandinavian Studies in Great Britain and Northern Ireland* (pp. 33–49). London: University College London.
- Ross, J. R. (1969). Guess Who? In: Binnick, R., Davison, A., Green, G. and Morgan, J. (eds.) *Papers from the Fifth Regional Meeting of the Chicago Linguistic Society*. Chicago: University of Chicago.
- Sabbagh, J. (2003). Ordering and Linearizing Rightward Movement. Ms., MIT.
- Saito, M. (1985). *Some Asymmetries in Japanese and Their Theoretical Implications*, Massachusetts Institute of Technology: Doctoral Dissertation.
- Sells, P. (2001). *Structure, Alignment and Optimality in Swedish*. Stanford, CA: CSLI Publications.
- Sigurdsson, H. Á. (1989). *Verbal Syntax and Case in Icelandic*, University of Lund: Doctoral Dissertation.
- Svenonius, P. (2000). Quantifier Movement in Icelandic. In: P. Svenonius (ed.) *The Derivation of VO and OV*. Amsterdam: John Benjamins.
- Takahashi, S. (2004). Pseudo-Gapping and Cyclic Linearization. In: Moulton, K. and Wolf, M. (eds.) *Proceedings of the North East Linguistic Society 34*.
- Toyoshima, T. (2001). Head-to-Spec Movement. In: Alexandrova, G. M. & Arnaudova, O. (eds.) *The Minimalist Parameter: Selected Papers from the Open Linguistics Forum, Ottawa, 12–23 March 1997*. Amsterdam: John Benjamins.
- Travis, L. D. (1984). *Parameters and Effects of Word Order Variation*, Massachusetts Institute of Technology: Doctoral Dissertation.
- (1991). Parameters of Phrase Structure and Verb-Second Phenomena. In: Freidin, R. (ed.) *Principles and Parameters in Comparative Grammar* (pp. 339–364). Cambridge, Mass.: MIT Press.
- Uriagereka, J. (1999). Multiple Spell-Out. In: Epstein, S. & Hornstein, N. (eds.) *Working Minimalism* (pp. 251–282). Cambridge, Mass.: MIT Press.
- Williams, E. (2003). *Representation Theory*. Cambridge, Mass.: MIT Press.
- Zwart, C. J.-W. (1996). *Morphosyntax of Verb Movement: A Minimalist Approach to the Syntax of Dutch*. vol. 39. Dordrecht: Kluwer.

Sound Patterns of Syntax: Object Shift*

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Abstract

In this paper, I explore what a purely phonological account of object shift (OS) involves and what research questions it leads to, in particular what it means for word order to be phonologically motivated and what morpho-phonological primitives are involved. I pursue the possibility that what licenses OS of full DPs in Icelandic is phonological properties, not found in other Scandinavian languages, together with overt case marking. Although the position of the object is determined phonologically, the architecture I propose, in which topic and focus play a central role, allows for an account of the semantic effect associated with OS in Icelandic.

1. Introduction

The phenomenon of Object Shift (OS) in general and Holmberg's Generalization in particular have been intriguing to linguists working within the Minimalist Program since Holmberg, 1986 in view of the restriction (in Scandinavian languages) of OS to structures that have undergone verb-movement. This type of restriction is problematic since there is no obvious way of linking the occurrence of one rule to the occurrence of another. Another puzzle raised by this phenomenon is its differing prop-

* I am grateful to Sofie Raviv, Lisa Rochman, and Idan Landau and especially to Tova Rapoport for their input and suggestions at various stages of the work on this paper. Thanks for their useful comments to the audience of the Glow workshop on Information structure in generative theory vs. pragmatics at Lund University in April 2003 and to the participants in the department of Linguistics, UCSD colloquium in August 2004. Special thanks go to Katalin É. Kiss. My research on the division of labor between Syntax and Phonology is supported by the Israel Science Foundation.

erties across Scandinavian languages: In Mainland Scandinavian (MSc) and Faroese, only weak pronouns shift, yet in Icelandic, full DPs shift as well. Finally, OS has interpretive effects: only topics or non-foci shift. Holmberg, 1999 argues that OS applies in a post-spell-out component of stylistic rules, containing rules dependent on focus structure. Movement is triggered by the need of –foc constituents to be governed by a +foc feature and is only licensed when all phonologically visible non-adjunct material has been removed from between the launching site and the landing site of the movement. Holmberg derives the linkage between verb movement and OS from the assignment of an inherent +foc feature to verbs. The restriction on OS in terms of phonological visibility calls for a phonological account, yet Holmberg rejects the idea that OS is a phonological or a morphological operation since it does not make reference to phonological or morphological primitives. Chomsky, 2001b, although tempted by a phonological analysis of OS, rejects it at least in part precisely because it is driven by the semantic interpretation of the shifted object. Bobaljik, 2002 proposes a copy theory of movement which allows for either copy to be pronounced. OS occurs obligatorily in the syntax yet morphological adjacency constraints determine which copy is pronounced at PF. This allows for a syntactic account of OS which is sensitive to PF properties such as adjacency.

In this paper, I explore what a purely phonological account of OS involves and what research questions it leads to, in particular what it means for word order to be phonologically motivated and what morpho-phonological primitives are involved. I pursue the possibility that it is the special properties of Icelandic word and compound stress, not found in the other Scandinavian languages, together with overt case marking, which licenses OS of full DPs. I also appeal to the grammatical model proposed in Erteschik-Shir, 1997, which allows for the association of phonological processes with semantic effects. I start by examining the free ride approach to OS in which the object prosodically incorporates into the verb and moves along with it (section 2.1) and in so doing uncover some of the restrictions which phonological displacement must be subject to. Although Holmberg's generalization falls out automatically from this approach, it must nevertheless be rejected because it leaves cases of OS in which the object is not attached to the verb unaccounted for. Section 2.2 introduces a processing constraint which requires that

argument roles be overtly identified and shows how adverb placement is one way of overtly identifying the subject. Since processing constrains the position in which the adverb can occur, positioning of the adverb must be phonological if we take PF to be concerned with both production and processing. Yet section 2.2 shows that adverb displacement does not adhere to the phonological principles set out in section 2.1. Section 2.3, inspired by Áfarli 1997, proposes a mechanism of adverb linearization which is compatible with these phonological principles and takes into account that not only must object pronouns prosodically incorporate, but so must these adverbs. Since prosodic incorporation plays an important role in the account of OS, section 2.4 explores this phonological process more carefully. Section 3 outlines how a phonological approach has the potential to explain language variation, including fine distinctions between dialects. Section 4 argues that it is Icelandic phonology which accounts for full DP OS in this language. Section 5 provides a brief comparison with Bobaljik, 2002; his proposal is designed to allow for phonological restrictions and in this way comes closest to my own. The last section outlines an architecture which can accommodate a phonological account of word order.

2. The phonological basis of Object Shift

2.1. First Pass

The linkage of OS to verb-movement is illustrated in (1) and (2) (all examples are in Danish unless otherwise specified):

- (1) Peter mødte ham ikke.
 Peter met him not
 ‘Peter didn’t meet him.’
- (2) a. Peter har ikke mødt ham.
 Peter has not met him
- b. ... at Peter ikke mødte ham.
 that Peter not met him

In (1) both the verb and the object have moved. In the sentences in (2), neither the verb nor the object have moved. In (2)a, the presence of the

auxiliary blocks verb-movement and (2)b illustrates the lack of verb movement in subordinate clauses.

In Erteschik-Shir, 2001; Erteschik-Shir and Strahov, 2004, it was argued that the object piggy-backs on the verb and therefore moves only when the verb moves. OS as such does therefore not occur.¹ This approach (also known as the ‘free ride’ approach; see Hellan, 1994), automatically captures Holmberg’s generalization since if the object attaches to the verb, it will only move when the verb moves. In order to see how the free-ride approach does in fact offer a solution to examples such as this one, it is necessary to examine the implementation of this approach. An obvious tack to take is to view the weak pronouns as undergoing cliticization (as in Hellan and many others especially Josefs-son, 1992; 1993, and Deprez, 1994). Such an account is however not extendable to Icelandic, in which full DPs shift and so cannot serve as an explanation for OS as a whole. In the papers cited above, I therefore proposed a different form of attachment, prosodic incorporation (PI), a phonological process which forms a single prosodic unit out of the prosodically incorporated element and its host. What was proposed there was an extension of cliticization processes not only to weak pronouns but also to other constituent types. This more general incorporation process is what I refer to as PI. I outline the phonology of PI in section 2.4.

If prosodic incorporation is a phonological process, and if, as I propose, it feeds verb movement, then verb movement must also be a phonological process. (This is consistent with Chomsky, 2001b, who views head movement as post-syntactic.) Assume that strings such as (3) are, very roughly, the output of syntax.²

- (3) [ikke Peter mødte ham]
not Peter met him

I do not take a position here on whether the syntax feeds phonology in chunks or phases (Chomsky, 1999) and whether these chunks are

¹ Under the proposed account, the object does not shift. For convenience I still refer to the phenomenon as OS.

² I assume late merger of inflectional morphology, but nothing here depends on exactly how it is implemented. For convenience, I list the inflected forms in the phonological string.

syntactically or phonologically motivated. I limit discussion to the larger VP domain within which all arguments are merged. Together with most researchers, I also assume (for the moment) that negation and other VP adverbials are merged at the (left) boundary of the VP.³ For the time being, I take it that only the linear string and its boundaries are visible to phonological processes.

The phonological derivation of (1) is shown in (4).

- (4) a. [ikke Peter mødte ham]
 b. PI: [ikke Peter mødte+ham]
 c. V-2: [mødte+ham ikke Peter]
 d. top: [Peter_{top} mødte+ham ikke]

(4)a is the output of PI, where PI is indicated by ‘+’. PI applies to weak functional elements such as pronouns and auxiliaries and is constrained by language particular prosodic rules.⁴ In the system envisaged here, PI is optional, yet non-PI-ed weak elements are ruled out as being unpronounceable, i.e., uninterpretable by the PF interface. Once PI has occurred, the host and the PI-ed element together form a phonological constituent and will therefore move as a unit as shown in (4)c. Here the movement of the verb (or auxiliary) to second position (V-2) is implemented as phonological displacement to the left edge (indicated by the bracket).⁵ Topicalization applies to the subject, moving it to initial position resulting in (4)d.⁶ Disregarding details the derivation is disarmingly simple. And the examples in (2) also follow since there the pronouns incorporate in situ.

A potential problem for the free ride analysis of OS is posed by the seeming need for rule ordering. This would make the approach extremely unattractive. Rule ordering, however, need not be imposed. The

³ Section 2.3 presents a different account of adverb positions.

⁴ See Selkirk, 1996 for an analysis of the incorporation of function words.

⁵ For the moment I disregard the possible motivation for phonological dislocations and hypothesize that they are triggered by pronunciation and processing needs. I return to this issue in section 6.2.

⁶ See Erteschik-Shir, 2003, to appear for an architecture in which top and foc features are introduced lexically.

I discuss the motivation for fronting the subject in section 2.3 and conclude that the subject in fact stays in situ. The word order of (1) is achieved, I propose, by implementing constraints on adverb pronunciation.

derivations in (5) show that other potential orders result in unpronounceable strings.

- (5) a. [ikke Peter mødte ham]
- b. V-2: [mødte ikke Peter ham]
- c. top: [Peter mødte ikke ham]
- d. PI: does not apply
- (6) a. [ikke Peter mødte ham]
- b. V-2: [mødte ikke Peter ham]
- c. PI: [mødte ikke Peter+ham]
- d. top: [Peter+ham mødte ikke]
- (7) a. [ikke Peter mødte ham]
- b. PI: [ikke Peter mødte+ham]
- c. top: [Peter ikke mødte+ham]
- d. V-2: does not apply

In (5), V-2 precedes PI leaving the pronoun unpronounceable: weak pronouns cannot be pronounced on their own and in Danish, adverbials do not provide a host for incorporation. (Note that the array of hosts is language specific and may vary even across dialects. Also, as we shall see below in section 3, some dialects of Swedish, and even Danish, do allow the pronunciation of weak pronouns following negation.)

In (6), PI precedes topicalization and the object incorporates into the subject, a possible host as we shall see below. Topicalization of the subject will drag along the incorporated object resulting in (6)d. This string violates a processing constraint which blocks elements with $\varnothing$ -features from intervening between the subject and its verb.⁷

(7) illustrates the case in which topicalization precedes V-2. Here V-2 is blocked from applying since the target of a P-syntactic process must be a string edge and therefore the verb could only move to first position, and not to second position as required.

So far, the proposed phonological implementation can be made to work without imposing rule ordering. We have also proposed the following ground rules for phonological displacement (D):

⁷ This constraint is subsumed under a more general processing constraint on identification of arguments (ID) introduced in (16).

(8) Properties of D

- a. D is to an edge.
- b. Ds are unordered.

I also adopt the hypothesis that motivation for D is limited to easing pronunciation (production) and parsing (processing) as stated in (9):⁸

(9) Motivation for D

- a. production
- b. processing

Restrictions on pronunciation will vary vastly across languages and depend on the general phonological properties of the language. These, as will be shown below, include stress and prosodic and intonational phrasing. Restrictions on processing will depend on disambiguating factors. Among the disambiguating factors to be discussed are overt case marking, overt agreement and the fronting of topics. Since the motivation for phonological displacement is ease of pronunciation and parsing, and neither is an absolute value, at least in some cases displacement will be optional.

2.2. *Second Pass*

An immediate problem with this approach (and indeed the reason it was rejected by Hellan), is the fact that the free ride approach leaves unaccounted for cases of OS in which the object is not attached to the verb.⁹ (10) illustrates such a case.

- (10) I går mødte Peter ham ikke
 Yesterday met Peter him not
 'Yesterday Peter didn't meet him.'

In (10) the object has shifted, but clearly not with the verb since the subject intervenes between the verb and the shifted object. (11) shows that the derivation of such sentences requires additional assumptions.

⁸ I ignore morpho-phonological constraints requiring adjacency between roots and affixes. In the late insertion model I envisage, affixation does not require movement.

⁹ For this point see also Bobaljik, 2002, p. 26.

- (11) a. [ikke Peter mødte ham i går]
 b. V-2: [mødte ikke Peter ham i går]
 c. PI: [mødte ikke Peter+ham i går]
 d. top: [i går mødte ikke Peter+ham]
 e. subj?: [i går mødte Peter+ham ikke]

Up to and including (11)d the derivation is very much as before. Note that if PI precedes V-2 (as in (4)), the output will be (12) in which an element with φ -features intervenes between the subject and its verb.

- (12) [i går mødte+ham Peter ikke]

This constraint holds whether the subject is to the left of the verb as in (6)d or whether it is to its right as in (12). Yet the movement of the subject in (11)e poses a problem since it looks like the subject (with its incorporated object) moves to a non-edge position. Such movement is excluded in the phonology according to the ground rules in (8) and it is not clear what motivation there would be for such a move even if it were not excluded. An obvious alternative is moving the adverb to the right. Phonological displacement is not directionally restricted and movement of the adverb to the right would be to an edge position. Yet what would be the motivation for moving the adverb? Since the unmoved adverb is positioned between the verb and its subject, the motivation could be to enable adjacency between the two, yet such adjacency is not required in subordinate clauses such as (13).

- (13) ... at Peter ikke mødte ham.
 ... that Peter not met him.
 '... that Peter didn't meet him.'

Here negation intervenes between the subject and the verb to no ill effect. It follows that there is no adjacency requirement between subject and verb and that adverbs (but not elements with φ -features) can intervene. There are two types of motivation for phonological movement, either parsing ease or pronounceability. Let us examine each one in turn.

According to Raviv, 2005 adverbials, auxiliaries and modals make a difference in the potential for topicalization of full DPs in Danish:

- (14) a. *Peter mødte Sara
 Peter met Sara
 'Sara met Peter.'

- b. Peter mødte Sara ikke.
Peter met Sara not
'Sara didn't meet Peter.'
 - c. Peter mødte ikke Sara.
Peter met not Sara'
'Peter didn't meet Sara.'
- (15) a. *Peter kender Sara fra fjernsynet
Peter knows Sara from TV-the
'Sara knows Peter from TV.'
- b. Peter burde Sara kende fra fjernsynet.
Peter ought to Sara know from TV-the
'Sara ought to know Peter from TV.'

Raviv argues that the presence of the adverbial and the modal in the b. sentences prevents ambiguity as follows: The subject is identified (as a subject) due to its position preceding the adverb whereas arguments following the adverb are identified as objects. The adverb therefore functions to delineate the boundary between subjects and other arguments.¹⁰ Similarly the word order in (15)b is disambiguated by the presence of the modal (an auxiliary would have the same effect). If the first argument, *Peter*, were a subject, then the modal would be followed by the verb. Since it is in fact followed by another argument, *Sara*, this latter argument must be the subject.

Raviv bases her analysis on Erteschik-Shir, 2003, to appear which introduces the following parsing constraint on the identification of arguments. This identification constraint applies at the PF interface with the articulatory-perceptual system:

(16) *ID*:

In a string, $X \dots Y$, where X and Y are arguments, ID X as subject and Y as object if neither is marked otherwise.

Thus in a string consisting of $\text{arg}_1\text{-V-arg}_2$, arg_1 would be identified as a subject and arg_2 as an object, unless the case marking on arg_1 is that of

¹⁰ This is of course not the case with OS in which the object pronoun precedes the adverb. Object pronouns are identified by their overt case marking and therefore need not be identified by their linear order. The same holds for full DPs in Icelandic as demonstrated in section 4.

an object or, alternatively, the case marking on arg_2 is that of a subject (or both). In these cases, arg_1 would be parsed as a topicalized object. This explains why the sentences in (17) can be understood with a topicalized object whereas (14)a cannot.

- (17) a. Ham mødte Sara.
 him met Sara
 ‘Sara met him.’
 b. Peter mødte hun.
 Peter met she
 ‘She met Peter.’

The verb also functions as an identifier. It identifies one of the arguments adjacent to it, as the subject. Since the subject and object can both be adjacent to the verb, the verb only identifies the subject unambiguously if the subject and the verb agree overtly. The ID of the subject is therefore inhibited if an object intervenes between it and the verb, explaining the restriction on intervening elements with φ -features.

ID would identify the topicalized object in (14)a and (15)a as a subject and the postverbal argument as an object, indeed the only interpretation of the string. In (14)b and (15)b the added adverbial and modal force a different identification thus licensing topicalization in these cases. It seems then, that in spite of the fact that the adverb seemingly should play no role in the assignment of syntactic roles to the arguments, it in fact plays a disambiguating role. Is this, then, the motivation for the movement of the adverb to the right? The answer is negative since both adverb orders render acceptable sentences, as seen in (14)b and (14)c.

We now turn to an examination of the phonology of Scandinavian adverbials, in order to see if the motivation for the placement of adverbs is to be found in this domain. Although some adverbials also have a free-standing (strong) form as in (18), many have only a weak form which must prosodically incorporate as in (19).¹¹

- (18) a. Sandsynligvis mødte Peter ham.
 probably met Peter him
 ‘Peter probably met him.’

¹¹ It is likely that the distinction here is between sentence adverbials which have both a strong form and a weak form and VP adverbials which have only a weak form.

- b. Peter mødte ham, sandsynligvis.
Peter met him, probably.
- c. Peter mødte+ham+sandsynligvis.
- (19) a. *Ikke/*Ofte mødte Peter ham
not/often met Peter him
- b. *Peter mødte ham, ikke/ofte.
Peter met him, not/often
- c. Peter mødte+ham+ikke/ofte.

The strong types of adverbs occur sentence initially and sentence finally with comma intonation as shown in (18)a and (18)b. These positions are not available for the weak adverbs as illustrated in (19)a and (19)b. (18)c and (19)c show that both forms incorporate elsewhere, i.e., in the cases under examination.

Although the requirement that adverbs incorporate in (18)c and (19)c does constrain the possible locations of adverbs, it does not lead us to a solution to the problem of motivating the shift of the adverb to the right. Compare the final two strings in (11), repeated in (20), with (21).

- (20) a. top: [i går mødte ikke Peter+ham]
- b. subj?: [i går mødte Peter+ham ikke]
- (21) Peter mødte+ikke Sara.
PETER met not Sara

(21) shows that the adverb can incorporate into the verb. It follows that neither the adverb nor the pronoun are stranded in (20)a which therefore cannot be ruled out for reasons of pronounceability. Motivation for the movement of the adverb to the right of the subject is therefore still lacking.

2.3. *Third Pass*

Although the position of the subject to the left of the adverb in (20) is motivated in terms of processing, there is no obvious way of formulating it as motivation for the rightward movement of the adverb. The problem disappears, however, if we abandon the common assumption that adverbs are merged at the left VP edge. Instead, following Åfarli, 1997 and

Chomsky, 2001a, Bobaljik, 2002¹², adjuncts are assumed to be merged on a separate plane. Åfarli's approach is designed to explain the possible positions of adverbials, among them the order of weak pronouns and adverbials. He analyzes weak pronouns as clitics which "must form a prosodic unit together with some other phonetically realized host adjacent (typically left-adjacent) to it, where that host may be an element belonging to any category (verb, noun, adjective, preposition, complementizer, etc.) as long as it is not an adverbial" (p. 102). Although the intuition behind Åfarli's approach is similar to the one presented here, it does not allow for the prosodic incorporation of nonclitics and so does not generalize this account to OS of full DPs. This is because Åfarli's implementation is syntactic. I present it here, and examine the properties of a phonological implementation.¹³

According to Åfarli, adverbials are attached in a third dimension to AgrP (either Agr_sP or Agr_oP). They are then linearized in one of three ways: Either the adverbial is "bent" down, to the left, or to the right, deriving a structure in which the adverbial ends up adjoined to Agr', or to the left or right of AgrP, respectively. In the latter case, the adverbial is sentence final and is pronounced with comma intonation. Since multiple adverbials can attach and each can bend independently, many adverb orders can be derived. Excluded, however, is the following in Swedish (from Åfarli, p. 96, his (9b)):

- (22) *Har nogon student last möjligen boken?
 has any student read possibly the book

Bending the adverb down positions it after the subject; bending to the left positions it to the left of the subject, and bending to the right positions it at the end of the sentence (with comma intonation). In (22) the adverb is positioned inside the VP, not a bending option in Åfarli's system.

Bending, according to Åfarli, must take place before Spell-Out since it is necessary for both PF and LF interpretation. In the present approach, interpretation follows from the adjunction site of the adverbials together

¹² The idea originates with Lebeaux, 1988; 1992 who argues for the insertion of adjuncts at s-structure. The need for deriving adjuncts in this way follows from their invisibility with respect to adjacency requirements, among them those pertaining to OS.

¹³ For an initial account of linearization from a three dimensional structure to a pronounceable string see Erteschik-Shir, 1987.

with the f-structure of the sentence (see section 6.1) and not from the linear position of the adverb. Let us therefore assume, as the null hypothesis, that adjuncts are phonologically linearized anywhere within the domain of the constituent to which they are attached, where the domain of a constituent includes the positions immediately to its left, immediately to its right and anywhere in between. Actual positions of adjuncts are constrained by purely phonological means (allowing for a measure of cross linguistic variation). In the case at hand, the VP adjunct can occur VP initially, VP finally and between the VP elements. Once topicalization and V-2 have occurred, we are left with the following three possibilities:

- (23) a. *i går mødte *ikke* Peter ham
 yesterday met not Peter him
 b. *i går mødte Peter *ikke* ham
 c. i går mødte Peter+ham+*ikke*

(23)a is ruled out because it violates the ID processing constraint which requires that the subject occur to the left of the adverb leaving the subject unidentified. (23)b is ruled out because the adverb is not a possible host for PI and the pronoun is therefore left stranded.¹⁴ This leaves (23)c as the only pronounceable string, with both the pronoun and the adverb incorporated as required. The three dimensional merger of adjuncts allows for the free linearization of adjuncts within the domain of the constituent to which they are attached modulo independent phonological processing and pronunciation constraints. This solves the problem of motivating adjunct movement since no motivation is needed if adjuncts are not assumed to move.¹⁵

Adopting this account of adjunct placement also leads one to question the account of the classic case of OS in (1), shown in (4). There it was assumed that the incorporated object gets a free ride with the verb and that the subject topicalizes. Yet, according to Raviv, 2005, subject topics are not pronounced with the initial stress characteristic of other fronted topics. The topicalized object in (24)b is stressed, but the subject topic in (24)a is not.

¹⁴ See section 2.4 for the phonological details of this account.

¹⁵ See Costa, 2004 for evidence from European Portuguese that the weight of adverbs determine their position in the string.

- (24) a. Peter mødte+ham ikke.
 Peter met him not
 ‘Peter didn’t meet him.’
 b. HAM mødte Peter ikke.
 him met Peter not
 ‘Peter didn’t meet him.’

This intonational difference suggests that subject topics are in fact not dislocated in the phonology. Yet the verb does occur in second position (with its incorporated pronoun), and if phonological displacement targets edges, as I claimed above, then displacing the verb without displacing the subject by topicalization will not get us the desired results. If, however, adjunct placement can be sensitized to V-2 so that the adverbs will be prevented from linearizing in second position in main clauses, neither V-2 nor topicalization need occur in this case. The adverb can only occur at the end of the clause since in initial position it has no host and if it intervenes between the verb and the pronoun, the pronoun is stranded.

The free linearization account of adjunct position however complicates the account of PI in situ invoked for the auxiliary and subordinate clause cases in (2) repeated in (25).

- (25) a. Peter har ikke mødt ham.
 Peter has not met him
 b. ... at Peter ikke mødte ham./* mødte ham ikke
 that Peter not met him

Our account so far predicts that the adverbs can occur in the positions they do. In (25)a the adverb occurs after the auxiliary (in second position) with the subject to its left. It is pronounced incorporated into the auxiliary and so no pronunciation or processing constraints are violated. Similarly, in the subordinate clause in (25)b the adverb obeys the constraints since there is no V-2 in subordinate clauses. Yet the clause final position is also licensed by these constraints but the result is ungrammatical. A last-resort type constraint must be imposed to the effect that the adverb must be linearized in the ‘earliest’ position in which it is licensed. I expect this constraint on linearization to apply only to weak adverbs which must be incorporated. The processing burden would be increased if the search for

an appropriate PI position wouldn't stop after the first hit. Strong adverbs are not expected to be subject to this constraint, since they are not limited by the need to incorporate. Whereas the positions of weak adverbs can depend on the availability of a host, the positions of strong adverbs are determined by the availability of prosodic boundaries. The constraint on weak adverb placement may not be too high a price to pay for an otherwise straightforward account based on independent phonological properties.

Double object constructions provide a test case for this approach to adverb placement. The Danish data is provided in (26)–(27).

- (26) a. Jeg gav+ikke Peter bogen.
 I gave not Peter the book
 b. Jeg gav+ham+ikke bogen.
 I gave him not the book
 c. Jeg gav+ham+den+ikke./*Peter den ikke.
 I gave him it not /*Peter it not
 d. I går gav+jeg+ham+den+ikke.
 Yesterday gave I him it not
- (27) a. Jeg gav+ikke bogen til Peter
 I gave not the book to Peter
 b. Jeg gav+den+ikke til Peter.
 I gave it not to Peter
 c. Jeg gav+den+ikke til ham/*den ham ikke til
 I gave it not to him/* it him not to
 d. I går gav+jeg+den+ikke til ham
 yesterday gave I it not to him

In (26)a the adverb is linearized in the first possible position in which it can occur. In (26)b the adverb must be placed after the pronoun in order not to disrupt PI. In (26)c the adverb must be placed after both pronouns for the same reason. With a non-pronominal indirect object, the adverb must be linearized immediately after the verb, hence OS is blocked. In (26)d the clitic cluster includes the subject.¹⁶ In (26)b–d the adverb incorporates as the final element in the cluster.¹⁷ (27)c shows that the object cannot 'shift' across the particle as expected under the view that objects don't shift. Otherwise the examples in (27) parallel those in (26).

Holmberg, 1999, (p. 17, (43)) points out that OS is possible in double object constructions in Swedish when the indirect object is removed:¹⁸

- (28) a. Vem gav du den inte?
 Who gave you it not
 ‘Who didn’t you give it to?’
 b. Henne visar jag den helst inte
 her show I it rather not
 ‘I’d rather not show it to HER.’

This follows without further ado from the restrictions on adverb linearization outlined here.

OS also occurs with pleonastic subjects of small clauses, as pointed out in Holmberg 1999, a challenge for some syntactic accounts of OS.¹⁹

- (29) a. Jeg hørte *det* ikke regne.
 I heard it not rain
 ‘I didn’t hear it rain.’
 b. Han tar *den* ikke med ro.
 He takes it not with ease.
 ‘He doesn’t take it easy.’

¹⁶ Here I adopt Josefsson’s 1993 idea that object pronouns can form ‘clitic clusters’.

¹⁷ According to Christer Platzak, Swedish also allows (i), but not (ii):

- (i) Han gav den henne inte.
 He gave it her not
 (ii) *Han har givet den henne
 he has given it her

The only way I could account for this would be by assuming phonological metathesis of the two pronouns in the cluster (positioning the longer pronoun after the short one) in order to ease the further incorporation of the adverb. This explanation would explain why metathesis doesn’t occur in (ii).

¹⁸ (28)a is ungrammatical in Danish on a par with the English ‘*Who did you give it?’ See Erteschik-Shir, 1979, for why this is so in these languages. Why Swedish allows questioning of the indirect object requires further investigation.

¹⁹ Holmberg argues, based on data such as that illustrated in (29), that OS applies to elements which are marked [–Foc]. Clearly these pleonastics are neither topics nor foci. In view of the fact that pronouns are necessarily topics, i.e., can be considered to be [–Foc], Holmberg generalizes OS of topics with the cases in (29) by labelling them all [–Foc].

The requirement on prosodic incorporation of weak pronouns argued for here, naturally carries over to pleonastics, which are phonologically identical in that they cannot be stressed. PI is therefore required of pleonastics as well. Here again the adverb is pronounced in the first position in which it is licensed. (Note that if the adverb is adjoined to the VP, its domain of linearization will include the small clause.)

The approach to adverb placement suggested here offers an interesting account of the placement of negative QPs in Scandinavian. Sells, 2001 (95–97), following Christensen, 1986, demonstrates that their placement is that of adverbs. The following Danish examples are parallel to Sells' Swedish ones:

- (30) a. Hun havde ikke sagt noget.
 she had not said anything
 b. *Hun havde sagt ingenting
 she had said nothing
 c. Hun havde ingenting sagt.
 she had nothing said

(30)b illustrates that even though it is an object, the NegQP cannot occur in canonical object position. Instead, it occurs before the verb, the canonical adverbial position. This position is also the one in which we find adjunct NegQPs as expected:

- (31) Han vil under ingen omstændigheder læse romaner.
 he will under no circumstances read novels

Accounting for the order in (30) is not simple since the object must merge VP-internally. If, however, negation like any adverbial is adjoined in a third dimension and only the nominal part of the phrase is merged in object position, we get the following string for (30)c.

- (32) hun havde ingen sagt –ting

The first position in which the adverbial can be linearized is after the auxiliary as indicated. The object '–ting' is not an independent morpheme and therefore cannot be pronounced in situ. It is therefore linearized together with its adverbial host, resulting in (30)c.²⁰ In cases in which the

nominal remnant is $\emptyset$, as in (33), no phonological displacement of the object is necessary.

(33) Hun havde ingen set.

NegQPs also occur in subject and topicalized positions:

- (34) a. Ingen så mig.
 Noone saw me
 b. Ingenting havde hun sagt.
 nothing had she said

Since the subject position in this case is identical to the adverbial position, they cannot be distinguished. Topicalizing adverbials is generally possible, therefore the occurrence of (34)b is also not surprising.

The placement of NegQPs interacts in an interesting way with OS as shown in (35).

- (35) a. Jeg gav ikke Else noget.
 I gave not Else anything
 b. *Jeg gav Else ingenting
 I gave Else nothing
 c. *Jeg gav ingenting Else
 d. Jeg gav hende ingenting.
 I gave her nothing

(35)a shows that the order of objects is IO followed by DO. (32)b is ruled out because the adverb must be linearized earlier. (32)c is ruled out by ID (extended to include both direct and indirect objects: the first object is IDed as the indirect one unless otherwise marked). (32)d, with an IO pronoun is pronounceable because in this case the final position is the first position in which the adverbial can be linearized. The NegQP data in Scandinavian therefore fall out naturally from the three-dimensional analysis of adverbials together with the free linearization approach outlined here.

²⁰ As pointed out to me by Lisa Rochman, this analysis is identical to *neg*-incorporation into indefinites in Klima, 1964, 274. There, however, the fusion was syntactic.

Section 2 started out with a free-ride approach to OS and ended up with free linearization of adverbs.²¹ Linearization of adverbs is phonologically constrained (by requirements on PI and by ID), however there is no longer any phonological displacement.

2.4. *The phonology of prosodic incorporation*

PI is a phonological process which joins elements of a phonological string to make one prosodic unit. This prosodic unit allows for no prosodic breaks to intervene between its elements and therefore cannot be broken up by the insertion of lexical elements such as adjuncts. These prosodic units make up prosodic words. Potential cases of PI include clitics and their hosts, cases of contraction (e.g., auxiliary and negation in English) and compounds. Selkirk, 1996 offers an account of the prosody of functional category words and argues that they may appear in a variety of prosodic clitic structures and that they need not have the status of prosodic words. Her analysis of weak object pronouns is particularly pertinent here.²² According to Selkirk, weak object pronouns are affixal clitics which do not themselves constitute a prosodic word. I assume the same is true of Scandinavian object pronouns. Following Selkirk, the prosodic structure of the V+object of (1) is as shown in (36) (parentheses indicate prosodic words; the pronouns are presented in their reduced incorporated form).

²¹ It remains to be seen whether this approach to adverb placement can handle the mass of related insights, prominent among them Cinque, 1999 and, for Scandinavian, Nilsen, 2003.

²² Selkirk argues that strong and weak object pronouns have different syntactic structures. Weak pronouns form a constituent with the verb, whereas strong pronouns occur in the normal object position. She notes (p. 204, fn. 9) that pronoun clusters (of double object constructions) are sisters in prosodic structure. Selkirk does not offer an account of these, yet her approach requires that such clusters together form a syntactic constituent with the V, a serious problem with the syntactic account. Selkirk also offers an account in terms of optional prosodization of the two types of pronoun. This approach, however, requires that functional categories are referred to by prosodic rules, which goes against the basic claim of the article. In view of the fact that strong pronouns are foci whereas weak ones are topics, it is possible to employ these properties to account for the different prosodic properties of the two types of pronouns. In section 6.1, I discuss the place and role of topic/focus marking in the grammar.

- (36) ((mødte) 'am)
met him

This analysis can be extended to all weak pronouns.

- (37) a. (ja' ((mødte) 'am)) or (ja' (mødte)) 'am)
I met him
b. (ja' (mødte))Peter
c. Peter ((mødte) 'am)
d. ((ga) 'am d'n)
gave him it

(37)a shows that a subject pronoun incorporates into the prosodic word as well. In principle there are two bracketing options, both of them independently necessary, as shown in (37)b and (37)c. It follows that there is no way to choose between the two options in (37)a. I conclude that both pronouns incorporate directly into the verb in the same way double object pronouns do in the 'clitic cluster' in (37)d. I therefore assume the flat prosodic structure for the incorporation of multiple pronouns as shown in (38).²³

- (38) a. (ja' +(mødte)+'am)
I met him
b. ((ga)+'am+d'n)
gave him it

Word stress rules will ignore elements that are not prosodic words within these strings so that only the verb itself is stressed and the weak pronouns are pronounced as stressless syllables. Weak pronouns can also incorporate into DPs:

- (39) Sara gav Peter+d'n.
Sara gave Peter it

Incorporation of weak pronouns into DPs is not possible in English which is why the translation of (39) cannot be pronounced. It is predicted that language variation will often depend on language particular

²³ I do not take a stand here on whether hierarchical structure is necessary on the prosodic phrase or intonational phrase levels as argued by Selkirk, 1984 and Nespor and Vogel, 1986 among others.

constraints on PI, such as which constituent types can provide hosts and which elements must PI. A careful analysis of the phonological properties of hosts, on the one hand, and elements which PI, on the other, might shed light on the source of these differences.

In section 2.2, I showed that Scandinavian adverbials also occur in strong and weak forms and that the latter incorporate. Most adverbials have at least two syllables which means that they must be analyzed as prosodic words and can in principle receive stress even when they are incorporated. Their incorporation therefore differs from that of weak pronouns as shown in (40). Here two prosodic words are combined by PI to form one complex prosodic word:²⁴

(40) ((mødte)+(ikke))

Weak adverbs, like weak pronouns, can incorporate both to the left (enclitic) and to the right (proclitic) as shown in the examples in (41).

- (41) a. Peter mødte+ikke Sara
 Peter met not Sara
 b. at Peter ikke+mødte Sara.
 that Peter not met Sara
 c. Peter har ikke+mødt Sara.
 Peter has not met Sara

Pronouncing a pause at the site of PI leads to ungrammaticality. I take this as evidence for the correctness of the attachments indicated in (41). Adverbs also PI in clusters:

- (42) Peter mødte+muligvis+ikke Sara.
 Peter met possibly not Sara
 ‘Peter possibly did not meet Sara.’

The order of the adverbs in the cluster is determined both by attachment site and by other phonological properties of the individual adverbs. In (42) *muligvis* is a sentential adverb with a higher attachment site than the

²⁴ If adverbials are lexical, then this analysis would not follow from Selkirk’s approach. It is possible that reasons can be found not to treat the result of adverb incorporation as a prosodic word, but rather a prosodic phrase. I do not take a stand on this here. What is important to me is only that PI must occur.

VP adverb *ikke*, hence the first possible attachment site of the former precedes that of the latter and these adverbs cannot be reordered.²⁵

If both pronouns and adverbs form clusters, then it is no longer obvious why weak pronouns cannot follow adverbs in an incorporated cluster, leading to the ungrammatical (43).

- (43) *Peter mødte+ikke+ham
 Peter met not him

There is actually a good reason for this. Compare the two sequences in (44):

- (44) a. (mødte)+ham+(ikke)
 b. (mødte)+(ikke)+ham

The difference between (44)a and (44)b lies in the different phonological properties of the incorporated pronoun and the incorporated adverb. The former is treated phonologically as a stressless final syllable of its host and is therefore invisible to further incorporation. The latter is itself a stressed prosodic word, therefore visible to incorporation. Since it is weak, it is not a possible host for further PI, ruling out (44)b. In the next section I discuss the reasons why the order in (44)b is in fact possible in Swedish.

An interesting upshot of this view of the pronunciation of adverbs provides a potential explanation for the different distribution of *not* and other adverbials in English. Let us hypothesize that English adverbials differ from the Scandinavian ones as follows: 1. *not* is unique among English adverbs in having only a weak form which only optionally forms a prosodic word. 2. other English adverbials have only a strong form.²⁶ The following data fall out, without requiring an additional syntactic distinction between Negation and adverbials as is customary:

- (45) a. Peter did+not/didn't meet Sara.
 b. *Peter did obviously not meet Sara.

²⁵ Erteschik-Shir and Rapoport, in preparation, propose that verbs are decomposed into meaning components with associated syntactic projections. The attachment sites of adverbs will be constrained by the meaning component modified, leading to different attachment sites with scopal consequences.

²⁶ Prosodic properties of lexical elements are to a large extent parametric, yet individual lexical items may have exceptional properties for which they must be marked.

- (46) a. Peter never met Sara.
 b. *Peter met never Sara.

(45)a and (46)a show that English VP adverbials linearize before the VP. This is possible in English which does not require V-2. (45)b shows that nothing can intervene between *not* and its host, evidence that *not* is necessarily weak and therefore must PI.²⁷ Its host cannot be the subject since this would position the adverb outside the VP domain. *Do*-support occurs to provide a verbal host and is therefore taken to be prosodically motivated.²⁸ The position of the adverb in (46)a is the first position in which the adverb can be linearized, but not the only one, since, as I suggested in the previous section, strong adverbs can occur at any prosodic boundary within their domain. Since verb phrases form prosodic phrases, adverbs cannot occur inside them, blocking (46)b. Such sequences are generally thought to be blocked by some syntactic implementation of an adjacency requirement on case assignment. This requirement would have to hold in English but not in Scandinavian, yet there is no obvious explanation for this difference. The prosodic explanation follows from the difference in V-2. We return to a discussion of V-2 and its motivation in section 6.2.

Costa, 2004 also observes that light adverbs behave in a clitic-like fashion and therefore argues that they are syntactic heads. In this way he can account for the fact that adverb position depends on weight. In my account, no syntactic distinctions need to be made between light and heavy adverbs or pronouns. Syntactically they are uniform, phonologically they differ.

PI has been employed here to explain constraints on adverb placement in Scandinavian, raising new questions about prosodic properties of adverbs in different languages. Further research should unveil what

²⁷ (i) shows that a strong (contrastive) form of *not* occurs in questions in which *do*-support is triggered by inversion and not by the negative element itself.

(i) Did Peter *nót* see Mary?

This example does not refute my claim that *do*-support with negation stems from negation being weak, but it does show that the phonology of *do*-support requires a more detailed investigation.

²⁸ In view of their different prosodic structure the two forms are also stressed differently:

i. ((*díd*)+*n't*)
 ii. ((*did*)+(*nót*))

morpho-phonological properties determine properties of PI in a particular language. What, for example, is the explanation for the different prosodic properties of adverbs in Scandinavian and English, and why do certain constituent types qualify as hosts of PI in one language but not in another. These differences should be made to follow from other language particular morpho-phonological properties. PI may also provide an explanation for more phenomena which have until now been believed to be syntactic.

3. Variation

It has often been observed (e.g., Holmberg, 1986; Josefsson, 1992, Sells, 2001), that in Swedish, object shift is optional. According to Sofie Raviv (personal communication) OS is also not required across negation in the *Ærø*/South Fyn dialect of Danish. The examples in (47) illustrate weak object pronouns following negation.

- (47) a. Jag såg +inte+den. (*Swedish*)
 I saw not it
 b. Jeg så +ikke+det (*Danish Ærø/South Fyn dialect*)
 I saw not it

Anders Holmberg (personal communication) suggests that the adverb and the pronoun in (47)a are pronounced as a clitic cluster with both PI-ed.²⁹ Since both the adverb and the pronoun must PI, and neither can host the other, this would be the only way to pronounce the sequence. Yet in the previous section I argued that in Danish (except for the *Ærø* dialect) this is not possible. Swedish differs from Danish in allowing phonological reanalysis of the verb with the incorporated adverb, with the stress on the adverb perceived as word stress on the verb+adverb sequence. In this case, then, it is the verb+adverb together which provides a host for the pronoun, not the adverb alone. Holmberg also provided me with the following Swedish example which can be viewed as evidence for this analysis (see also Holmberg 1999, 28 for the same point in Norwegian):

- (48) ?Han såg utan tvivel den.
 he saw without doubt it

²⁹ See Holmberg, 1984 and Josefsson, 2001 for an analysis of Swedish particles as clitics.

According to Holmberg acceptability is reduced with a more phonologically-complex adverb. It is well known that all things being equal, complex phonological elements are preferred after simple ones when possible.³⁰ Since the adverb is licensed both before and after the pronoun, the order which is most easy to pronounce, that with the heavier adverb at the end, will be preferred. (In the *Ærø* dialect the phenomenon is limited to negation.)

The Danish example in (35)b also corroborates this analysis. According to Sofie Raviv (pers. com.), the *Ærø* dialect is spoken with a special tune which makes it sound like Swedish and (35)b is pronounced with main stress on the incorporated negative adverb. This pronunciation of negation is completely ruled out in standard Danish. Stress shift onto the incorporated adverbial is therefore the trigger for licensing the further incorporation of the object pronoun in situ. Another piece of evidence for this analysis is the following Swedish example (Elisabet Engdahl, personal communication):

- (49) a. Där vill aldrig+nån bo.
 there wants never someone live
 ‘No one wants to live there ever.’
 b. Där vill nån aldrig bo.
 there wants someone never live
 ‘A certain person wants to never live there.’

The order in (49)a is possible in Swedish on a par with (47)a. The proform ‘nån’ can incorporate into the adverb when the adverb is incorporated into the verb and stressed. (49)b shows the shift of the pronoun, here a subject pronoun. Note that the order of the pronoun and the adverb determines the scopal relation. In (49)a the pronoun is nonspecific, in (49)b in preadverbial position it must be specific. (I return to this topic in section 6.1). Neither of the examples in (49) are grammatical in Danish. (49)a is bad because pronouns cannot PI into verb-adverbial sequences in Danish. (49)b is not possible because the parallel form in Danish *nogen* cannot be used specifically.³¹

³⁰ An obvious example is heavy NP shift. See Ross, 1975 for further examples.

³¹ Christensen, 2003 offers an OT analysis of dialectal variations in NegQP shift. I am optimistic that a careful analysis of cross-linguistic phonological differences will shed some light on the reasons for the various constraint rankings.

The possibility of incorporating both an adverb and a pronoun into the verb in Swedish also allows for the word order in (50)a (Halldór Sigurðsson, personal communication):

- (50) a. Det vet+inte+jag.
 that know not I
 ‘That I don’t know.’
 b. Det vet+jag+inte.
 that know I not

Here the object is topicalized and it is the subject which undergoes PI into the verb either before or after the adverbial.

The possibility of ‘adverbial intermingling’ in Swedish was noted in Holmberg and Platzack, 1995 (156–7). The examples in (51) are from Sells, 2001 (63, his (121)):

- (51) a. Igår läste han dem ju alltså troligen inte.
 Yesterday read he them as.you.know. thus probably not
 ‘Yesterday as you know he probably did not read them thus.’
 b. %Igår läste han ju dem alltså troligen inte.
 c. %Igår läste han ju alltså dem troligen inte.
 d. %Igår läste han ju alltså troligen dem inte.
 e. %Igår läste han ju alltså troligen inte dem.

(51)a is the normal case of OS with a string of adverbs following the object. In the rest of the examples, the pronoun follows one or more of the adverbs. This is possible only in Swedish and, as in (48), a string of adverbs preceding the object makes it less easy for the adverbs to incorporate. The oddness is therefore due to difficulty of pronunciation. This is also the explanation for the oddness of (51)b–d. These are all possible in Swedish since adverbs are not blocked from being linearized either before or after the object pronoun, yet the incorporation of a longer string of adverbs and pronouns is problematic. Holmberg and Platzack employed this data to argue against a cliticization analysis in which the pronoun cliticizes onto the verb, very much along the lines of the current approach. However, together with the free linearization of adverbs, the analysis is still viable.

‘Long Object Shift’ is another phenomenon possible only in Swedish (Josefsson, 1992). The examples in (52) are from Sells, 2001 (65, his (124)–(125)):

- (52) a. Numera gör mej alltså inte längre Helge lika irriterad
 Nowadays makes me thus not longer Helge as irritated
 som förr.
 as before
 'So nowadays Helge doesn't make me as irritated as before
 any longer.'
- b. Numera gör alltså mej inte längre Helge lika irriterad som
 förr.
- c. Numera gör alltså inte mej längre Helge lika irriterad som
 förr.
- d. Numera gör alltså inte längre mej Helge lika irriterad som
 förr.

According to Sells, the acceptability of (52)b and (52)c is reduced. This would follow from the same type of considerations as those pertaining to (51) (which would also predict the reduced acceptability of (52)d). More seriously, long OS violates the ID constraint in that the object pronoun intervenes between the verb and the subject (see (16) in section 2.2). Josefsson, 1992 considers constructions such as these highly marked and lists restrictions of various kinds on their occurrence. First of all, Long OS is restricted to cases in which the subjects are of the type which would have received quirky case in Old Swedish. If quirky case is thematically motivated, then these subjects can be IDed by their thematic role thus relaxing the prohibition on an intervening object. Another requirement is that the elements around the object pronoun must be 'heavy', this according to Josefsson to avoid stress on the pronoun, clearly a phonological restriction. Finally, the morphology of the pronoun is also important. Only object pronouns with unambiguous overt case marking are licensed in this position, clearly consistent with a processing constraint such as ID. The following example provided by Halldór Sigurðsson and Elisabet Engdahl illustrates this:

- (53) I går träffade honom hans föräldrar på stationen.
 yesterday met him his parents on the station
 'Yesterday his parents met him at the station.'

If the pronoun *honom* is replaced by the ambiguous *dom* ('they' or 'them'), then the only available reading is that in which the pronoun is interpreted as a subject, exactly as predicted by ID.

Although a counterexample to the ID ban on elements with φ -features, Long OS provides strong evidence for the phonological account of OS, in which both processing and pronunciation play a role. Yet it is not obvious how this order is derived. No obvious syntactic analysis presents itself. In view of the requirement of heaviness on either the subject or the other surrounding elements, I can only suggest that the subject is moved (to the right) beyond the object pronoun by a type of phonological metathesis akin to heavy NP shift.

Yet another difference between Swedish and the other Scandinavian languages is that OS is obligatorily blocked across particles in Swedish.

- (54) a. Vi kastade+inte+ut+den/*den inte ut. (*Swedish*)
 we threw not out it
 b. Vi smed den ikke ud/*ikke ud den. (*Danish*)
 we threw it not out

This difference follows from the difference in particle object order in these languages according to Sells, 2001, 48 as illustrated in (55):

- (55) a. Vi kastade ut boken /*boken ut. (*Swedish*)
 We threw out the book/*the book out
 b. Vi smed bogen ud /*ud bogen. (*Danish*)
 we threw the book out/*out the book

Sells, p. 68, following Holmberg, 1986, assumes that the verb-particle complex forms a unit dominated by V in Swedish. It follows that the object cannot follow the verb but must follow the whole complex.³² This complex cannot be formed in Danish. I again assume that the reason for this has to do with the possibility of pronouncing this complex in the two languages. In Swedish the incorporated particle is stressed in the same way the incorporated negative adverb is in (47)a. In Danish only phrase final particles can be stressed. This prosodic pattern may be a requirement on particle incorporation. Once this pattern is available, the pronoun can also be incorporated, again parallel to (47)a. In Swedish it is the Verb+adverb or Verb+part complexes which host PI.

³² The data is actually somewhat more complex. Here I deal only with one type of particle and ignore particles with complements. Note also that Norwegian and Icelandic allow both orders in (55), yet OS is as in Danish (see Josefsson, 1992 and Bobaljik, 2002).

The placement of the adverb must still be explained. (56) gives the three possibilities of which (56)a is the desired result:

- (56) a. vi kastade **inte** ut den
 b. *vi kastade ut **inte** den
 c. *vi kastade ut den **inte**

(56)a is pronounceable since the adverb PIs into the verb and the particle and the object pronoun follow suit. The unmarked pronunciation of the sequence has stress on the particle and not on the adverb (stress on the adverb is contrastive). This renders the appropriate contour for further incorporation of the pronoun. (56)b and c are ruled out by the last resort constraint introduced in 2.3, namely that weak adverbs must be linearized in the ‘earliest’ position in which they are licensed. A more detailed investigation of Swedish phonology may provide additional reasons.³³ The pronoun-adverb-particle order in Danish (and the other Scandinavian languages) also follows:

- (57) a. *vi smed **ikke** den ud
 b. vi smed den **ikke** ud
 c. *vi smed den ud **ikke**

In (57)a the pronoun cannot PI since Danish adverbials are not stressed the way they are in Swedish and so the V-adverb unit does not provide a host for PI, thus leaving the pronoun stranded. (57)b allows for the pronoun and adverb to PI and is therefore licensed. In (57)c the particle cannot host the incorporation of the adverb and it also violates the last resort principle.

Bobaljik, 2002 relates the lack of OS across particles in Swedish to analytic passive formation. (236, his (51a)):

- (58) *Skräpet måste bli kastat ut
 the scrap must Aux thrown out
 ‘The scrap had to be thrown out.’

In contrast the analytic passive is possible in Danish, Norwegian, and Icelandic. (59) illustrates the parallel example in Danish.

³³ According to Árnason 1996, Danish is exceptional among the Scandinavian languages in not adhering to Prokosch’s Law (i.e., that all stressed syllables are heavy). Danish therefore allows short stressed syllables. This may also turn out to be a source of differences in PI.

- (59) Affaldet blev smidt ud.
 the scrap Aux thrown out
 'The scrap was thrown out.'

Bobaljik draws the following generalization: "An object which can only appear in the post-particle position is blocked from further A-movement (i.e., object shift and passive).³⁴ This generalization would follow simply from the assumption that the particles in question cannot incorporate phrase-finally in Swedish. In Svenonius, 1996, evidence both for and against this suggestion can be found. Svenonius shows that particles incorporate obligatorily in the analytic passive in Swedish, exactly in those cases in which they cannot occur in final position (his (49)):

- (60) Mjölken blev aldrig inställd/*ställd i
 the milk was never in.put put in
 'The milk never got put away'

Unfortunately, the Swedish synthetic passive allows a final particle (Svenonius' (66)):³⁵

- (61) a. Hunden kastades ut.
 the dog threw-pass out
 'The dog was thrown out'
 b. ?? Hunden utkastades.
 the dog out.threw-pass

The distribution of sentence final particles in Swedish also shows up with Neg QPs. (62) is from Sells, 2001 (98) (used by Sells to discuss the position of the NegQP):

- (62) a. ?Jon har inga romaner läst ut.
 John has no novels read out
 b. *Jon läste inga romaner ut.
 John read no novels out

³⁴ Bobaljik, 2002 proposes an account in terms of locality of A-movement rather than the one proposed in Holmberg, 1999 in which examples such as the unacceptable string in (54)a violate the adjacency requirement between the base and shifted position of the object. See Section 5 for a comparison of Bobaljik's account with the one proposed here.

³⁵ Svenonius argues for a correlation across the Scandinavian languages between A-movement, participle agreement and particle incorporation. A discussion of this and the rich data he supplies would take us too far afield.

Here the aux-participle in (62)a is much more acceptable than the finite verb form in (62)b. The two kinds of passive also differ in this way, but the final particle is blocked in the analytic one. I expect that with more investigation into the phonology of these constructions, an account of the distribution of final particles in Swedish will be uncovered.

The ideas presented in this section are somewhat speculative. Yet, the strength of a phonological explanation for fine distinctions between otherwise highly similar languages is that it allows us to maintain the same syntax for these languages by accounting for variation in terms of morphological and phonological distinctions. Optimality theory started out as a theory of phonology. More recently, it has been applied to syntax and in particular to explain variation (e.g., Sells, 2001 who offers an OT account of OS). If variation is indeed to a large extent phonologically determined by language specific phonology and morphology, the success of OT in this area is not surprising. What is missing in OT accounts, in my opinion, is the association of the syntactic distinctions with the actual phonology and morphology of the languages in question.

4. Full DP OS in Icelandic

The free-ride approach to OS required an analysis in which DP objects could PI into a verbal host, not a very natural phonological process. Free linearization of adverbs leads to a different explanation of the difference between Icelandic and the other Scandinavian languages. An obvious difference is the fact that Icelandic has morphological case. Since overt morphological case interacts with ID, a potential explanation for the difference presents itself, if it weren't for the fact that Faroese also has overt case but doesn't allow OS of full DPs.³⁶ Here I pursue an

³⁶ Holmberg and Platzack, 1995, 172–176 argue that Faroese morphological case is weaker than that of Icelandic thus explaining the difference in OS. Bobaljik and Thráinsson 1998 argue that the presence of agreement phrases is indirectly linked to the availability of overt agreement morphology. They argue that the difference between Faroese and Icelandic is a difference in inflectional morphology and does not have to do with the availability of overt case.

examination of phonological differences between Icelandic and Faroese that predicts that only in Icelandic can the sequence DP+adverb be pronounced.

(63)a and b illustrate the optional OS of a full (definite) DP in Icelandic.

- (63) a. Jón las bókina ekki.
Jon read the books not
b. Jón las ekki bókina.
Jon read not the books

These sentences were uttered by a speaker of Icelandic, rendering the pitch contours (without the subjects) in figures 1 and 2:

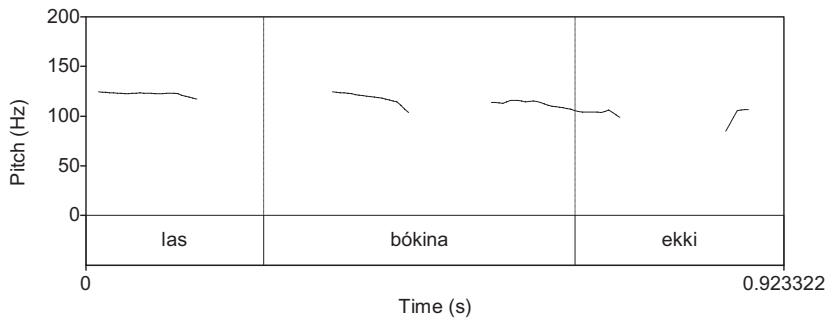


Figure 1. F_o of (63)a

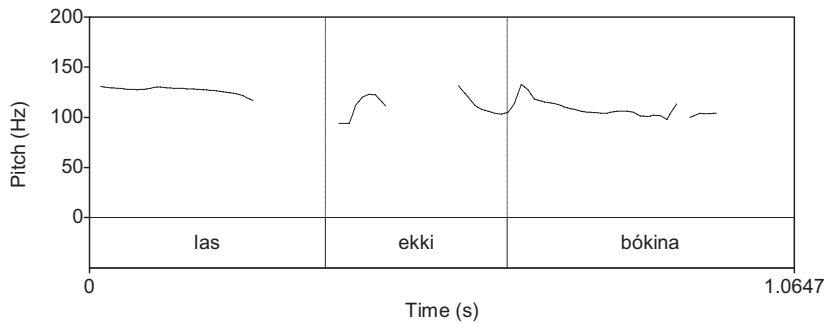


Figure 2. F_o of (63)b

In neither case is there evidence of PI into the verb. However, the negative adverb and the object form a prosodic unit in both orders. This is as clearly heard as is the lack of incorporation into the verb. Assume, therefore, that Icelandic adverbs, like all Scandinavian adverbs, are weak and must PI. Yet whereas Danish adverbs PI into the verb, Icelandic adverbs PI into the object (either pro- or enclitically) in these cases:

- (64) a. 'bókina+, ekki
books-the not
b. 'ekki+, bókina³⁷

As expected heavy noun phrases do not shift:

- (65) a. Ég sá ekki ráðherrann sem allir eru að kvarta yfir.
I saw not minister-the who all are to complain over
'I didn't see the minister who everyone was complaining about.'
b. ??Ég sá ráðherrann sem allir eru að kvarta yfir ekki.
I saw minister-the who all are to complain over not

This follows naturally from an account in terms of prosodic incorporation.

According to Árnason 1996, Icelandic and Faroese are unique among modern Western European languages in having initial stress in most of their vocabularies. Having initial stress might enable a language to tolerate a longer sequence of syllables without primary stress than a language with primary stress further towards the end of the word, thus allowing for longer prosodically incorporated strings. This reasoning should also apply to Faroese. Icelandic and Faroese differ, however, in the way they form compounds. In Icelandic, stress generally remains initial. In Faroese, stress generally shifts to the second element of the compound in borrowed compounds.³⁸ The fact that Faroese shifts the stress to the second element of the compound in new compounds indicates that this shift is productive in Faroese. A possibility worth considering is that (64)a is

³⁷ It is in fact hard to detect a difference in stress between the two. This may be due to the fact that phrasal stress is final.

³⁸ Such compound stress generally mimics the Danish stress pattern in Danish borrowings. See Árnason for details.

ruled out in Faroese because it would be unpronounceable with stress shifted onto the incorporated adverbial.³⁹

Icelandic differs from MSc in that DPs can host PI of adverbs. This may also be the source of the difference between Icelandic and Faroese. If DPs cannot host adverbial PI, then (64)a would be ruled out. Section 2.4 showed that differences in potential PI hosts can explain word order differences across languages. Further research into the phonology of both Icelandic and Faroese is required to discover which of these explanations (if any) is tenable.

Imagine a language with the stress properties of Icelandic but without overt case. I predict that full DP object shift would not be possible in such a language because ID would be violated. Overt case marking enables the identification of the object independently of its position with respect to the adverb.

If both orders are pronounceable and neither functions to ID the object, what then motivates Icelandic object shift? It is here that the interpretation of the shifted DP comes into play. It is well known that the shifted DP must be a topic (see Diesing and Jelinek, 1993, among others). Only definites shift and shifted bare plurals must be specific.⁴⁰ Shifting the object thus identifies it as a topic. The ID constraint as defined in (16) is concerned with the identification of the syntactic roles of arguments. Yet the identification of topic and focus is also critical to the interpretation of the sentence. The identification of foci is through intonation, the identification of topics is through fronting. An element fronted to the initial position in the sentence is identified as the main topic of the sentence, an element fronted within the VP is interpreted as a subordinate topic.⁴¹ Another way to mark subordinate object topics is by not stressing them. This is the strategy used in the other Scandinavian languages and in English. In other Germanic languages such as German and Dutch, topics undergo Scrambling. Scrambling has much in common with Scandinavian OS, yet

³⁹ Another possibility is that just as Danish pronunciation affects compound stress in Faroese (see Árnason 1996), it may also directly affect Faroese adverb linearization.

⁴⁰ See Cohen and Erteschik-Shir, 2002 for an analysis of specific bare plurals as topics and Erteschik-Shir, 2001; Erteschik-Shir and Strahov, 2004 for arguments that shifted objects in Icelandic must be topics.

⁴¹ See Erteschik-Shir, 1997 for properties of subordinate topics.

it occurs independently of verb movement. An examination of scrambling along the lines outlined here for OS should be fruitful.

5. A comparison with Bobaljik, 2002

Bobaljik's 2002 account is an ingenious way to allow for a purely syntactic account of OS, sensitive to phonology (adjacency), without PF filters on syntactic derivation. Since Bobaljik's proposal, in principle, could incorporate the phonological restrictions on OS proposed here, the differences between the two approaches shed light on the strengths and weaknesses of both.

Bobaljik's core idea is that LF and PF have the ability to privilege either the higher or lower position in the chain, allowing a moved element to be pronounced (and interpreted) as though it had not moved. His account of OS is as follows: All objects undergo OS to spec, AgrOP. Phonological considerations determine which copy is pronounced and semantic considerations which copy is interpreted. When verb movement does not occur, only the lower copy of the object can be pronounced, since the position of the higher copy intervenes between Infl and the verb stem blocking their PF-/morphological merger which requires adjacency.⁴² This is illustrated in (66) (Bobaljik's Icelandic example (40), p. 225).

- (66) * [_{CP/IP} Hann hefur [_{PartP} – ið [_{AgrOP} **bókina**_i [_{VP} les- ~~bókina~~_i]]]]
He has PRT read
 [Adjacency disrupted: O - - - - - * - - - - -O]

In (66) the higher copy of the object in Spec, AgrOP cannot be pronounced since it disrupts the adjacency between the main verb stem and the projection hosting participial morphology. Had the verb moved (in a non-compound tense), adjacency would not be disrupted and the object

⁴² Bobaljik rejects the syntactic account of inflection which requires syntactic raising of the verb to I in these cases. Instead he proposes post-syntactic merger of the finite verb and the inflectional features in Infl.

could be pronounced in the higher position.⁴³ Bobaljik's proposal also provides an explanation of the semantic effect of OS. According to Diesing, 1992, the object is interpreted as new information in the lower VP internal position, and as old information in the higher position. Bobaljik, (p. 251) proposes an economy condition which prefers isomorphism between PF and LF. It follows that when movement is possible (i.e., when the verb has moved), the object will be interpreted in the position in which it is pronounced.⁴⁴

Bobaljik's analysis requires that adverbs be invisible for the computation of adjacency. He achieves this effect by making a set of assumptions about the linearization of syntactic structures. Syntactic structures, according to Bobaljik (213–221), contain only hierarchical information. Linear order is determined at Spell-Out by establishing precedence among syntactic nodes. Adverbs are taken to be adjoined to VP and are therefore not dominated by VP, but only by a segment of VP. If a precedence condition requires that X precede VP, then X must precede the set of terminal nodes dominated by VP, excluding the adjoined adverb which can therefore float to the left of X in the mapping to phonology. The special status of adjunction thus follows from the segment-category distinction. I believe it to be significant that the segment-category distinction is employed by Truckenbrodt, 1999 to account for the mapping from syntax to phonological phrases for exactly the same case. He argues that in the case of adjunction to VP, the higher segment of VP that includes the adjunct is irrelevant to the syntax-prosody mapping, forcing a phonological phrase boundary between the lower VP and the adjunct. Note that if adjuncts are merged in a different dimension the segment-category distinction is not required to account for the linear position of adverbs (as in Bobaljik) nor is it required to constrain the mapping from syntax to phonological phrases (since there is no higher VP, only the lower one will

⁴³ The adjacency requirement between the verb and its inflection is immaterial to my account of OS. My account of adverb linearization, however, does predict that adverbs do not block morphological merger between the verb and Infl. This is due to the fact that linearization of adverbs presupposes lexical insertion (of both of adverbs and verbs) and the inflected verb forms must therefore be formed before adverb linearization.

⁴⁴ It is not clear to me how Bobaljik's approach can be extended to genuinely optional movement such as Topicalization, since topics can be interpreted in situ as well.

count). Instead, phonological phrasing will directly determine the possible positions of unincorporated adverbs.

There are other cases where Bobaljik must resort to rather complex considerations, diminishing the beauty of his basic idea. One illustration is the case of participle fronting which he discusses on p. 234–235. The following example of Stylistic Fronting in Swedish is from Holmberg, 1999, p. 7:

- (67) Kysst har jag henne inte.
 Kissed have I her not
 ‘Kissed her, I haven’t.’

The problem is that although the verb has moved, adjacency between the verb stem and the participial morphology is disrupted. Bobaljik follows Holmberg in assuming that the participle is fronted by long head movement which “has the effect that adjacency between the head of PartP and the lexical verb stem is not at issue and object shift should be possible.” (p. 235). In order to see this let us examine the structure Bobaljik would assign (along the lines of Bobaljik’s account of similar cases, e.g., (66)):

- (68) [CP/IP Kyss- har jag [PartP – t [AgrOP **henne**_i [VP kyss- ~~henne~~_i]]]
kissed have I PRT her
 a. [Adjacency disrupted: O - - - - - * - - - - -O]
 b. [O - - - - - * - - - - - O]

(68)a shows that the object pronoun disrupts adjacency between the participial morphology and the lower copy of the verb and (68)b shows that the subject pronoun disrupts adjacency between the higher copy of the verb and the participial morphology. Either way, merger of the verb and its morphology is blocked. The only way for Bobaljik to explain the grammaticality of the sentence is to assume that the lower copy of the object is pronounced. But then he cannot explain the position of the adverb after the object. (He suggests that in fact the pronoun can (marginally) be pronounced after the adverb, but clearly that is only one option.) This type of example does not present a problem for the current view if it is assumed that the stress on the fronted participle prevents PI of the auxiliary and the adverb so that the adverb cannot linearize after the auxiliary. The first possible position for the adverb is therefore after the object. As

seen in section 3, Swedish allows incorporation of the adverb into the verb before the object. This is why both orders are predicted to be good.⁴⁵

One of the strengths of Bobaljik's approach is his account of the difference between VO and OV languages. The fact that OS is not dependent on verb movement in OV languages follows without further stipulation. The main weakness of his account, in my opinion, is that he does not account for differences among the Scandinavian languages, which are all VO. He admits that his account sheds no new light on why OS is limited to weak pronouns except in Icelandic (beyond Bobaljik and Thráinsson 1998 mentioned in the previous section). More importantly, his approach cannot, as far as I can tell, advance the understanding of more minor differences between the Scandinavian languages such as the fact that OS is optional in Swedish, but obligatory in Danish.

6. Architecture and motivation for phonological displacement

6.1. *Topic-focus architecture*

In previous work (e.g., Erteschik-Shir, 1997) I argued for an architecture without LF-syntax, in which f(ocus)-structure, the output of syntactic computation with constituents marked for topic and focus, provides the input to interpretation.⁴⁶ In this model, f-structure is therefore responsible for scopal interpretation. Since truth value is determined by evaluating the sentence with respect to the topic, topics will always take wide scope.⁴⁷ I repeat the examples in (49) from section 3 as an illustration.

- (69) a. Där vill aldrig+nån_{foc} bo.
 there wants never someone live
 'No one wants to live there ever.'

⁴⁵ As pointed out to me by Idan Landau, Bobaljik's account also fails to account for OS in V-2 double object constructions. As shown in (26) vs. (28), OS is not possible 'across' a full DP indirect object, yet is possible when the indirect object is removed.

⁴⁶ In Erteschik-Shir, 2003, to appear, I show that top and foc features can be lexically selected on a par with φ -features. In earlier work, I assumed, in violation of the inclusiveness principle, that syntactic structures are annotated freely with these features and that these are the only features which can be inserted in the derivation.

⁴⁷ This view of topics stems from Reinhart, 1981 and is developed in Erteschik-Shir, 1997.

- b. Där vill nån_{top} aldrig bo.
 there wants someone never live
 'A certain person wants to never live there.'

In (69)a the post-adverbial quantifier is focused and hence has narrow scope, in (69)b the adverbial precedes the adverb, is interpreted as a topic and takes wide scope.

The fact that foci are marked intonationally indicates that f-structure is also spelled out phonologically. Here I explore the idea that word order is, at least in part, phonologically determined as well. In section 2.3, I proposed that the position of adverbs is due to phonological linearization. Here I explore the idea that phonology is also responsible for other word order effects. The idea that displacements can be 'stylistic' or phonological originates with Ross, 1967, who initiated the discussion of scrambling within generative grammar and viewed it as a stylistic rule of permutation. More recent accounts of phonological displacement can be found in Adger 1997, Chomsky, 2001b, Kidwai, 1999, McCloskey, 1999, and Zubizarreta, 1998, among others. Prime candidates for displacement in the phonology are heavy NP shift and other stylistic rules which are sensitive to phonological properties, e.g., constituent heaviness, adjacency, and movement to an 'edge' on the one hand, and movement motivated by focus or topic properties of the moved constituent on the other.

A common argument against phonological displacement is its sensitivity to island constraints. The violation of island constraints has traditionally been used (e.g., Chomsky, 1977) as a test for wh-movement, that is, purely syntactic A'-movement. However, island constraints, I have argued (1973, 97), are best accounted for as processing constraints sensitive to focus structure. If island constraints are formulated as constraints on PF, then rather than arguing against phonological dislocation, they provide evidence for it.⁴⁸ The constraint proposed in Erteschik-Shir, 1997 constrains not only wh-trace dependencies, but also anaphoric and negative dependencies among others. It states that such dependencies are restricted to the unmarked f-structure in (70) in which X is a dependent.

⁴⁸ The fact that resumptive pronouns 'save' islands has been employed to argue that island constraints are, in fact, a PF phenomenon (see Bošković 2001, 114–5 and the references cited therein).

- (70) SUBJECT_{top} [... X ...]_{foc}
 sTOP_t

According to this constraint, dependencies are restricted to f-structures in which either the subject or a stage is the topic and the dependent is contained in the focus.⁴⁹ The intuition behind this constraint is that the burden of processing a dependency is eased if f-structure and syntactic structure are aligned. Such alignment is disrupted when the object is the topic since the remainder of the sentence does not form a syntactic constituent. This processing constraint can be viewed as part and parcel of ID since it identifies the syntactic role of the dislocated constituent. ID should clearly be extended to include complex sentences along these lines.

6.2. *Arguments for phonological displacement*

My explanation for the linkage between OS to V-2 no longer hinges on a phonological account of V-2. It follows rather from an account of the phonological linearization of adverbs as part of spell-out: If an adverb intervenes at a required PI site, the string will not be pronounceable. In view of the fact that there is no reason for V-2 to be ordered with respect to either adverb placement or PI (as shown in section 2.1), the current proposal is compatible with both a syntactic and a phonological account of V-2. Yet there are other well-known reasons to believe that V-2 (and head movement in general) is not syntactic. Chomsky, 2001b offers several. The first reason is that head movement has no semantic effects. (Within the generally-assumed architecture only syntactic dislocation can affect interpretation since it provides the input to LF; in an architecture without LF, this argument is not decisive.) Second, overt head-raising is phonologically motivated by the phonetically affixal character of inflectional categories. Third, syntactic head-raising is a countercyclic adjunction rule with different locality constraints. Finally, there is no natural syntactic notion “second.” Even though only the last reason is theory

⁴⁹ A stage topic specifies the here-and-now of the discourse. It is contextually indexed for time and place. See Erteschik-Shir, 1997 for details.

independent, I would like to pursue what a phonological account of V-2 would look like.

According to the principles of phonological displacement outlined in section 2.1, the target must be an edge. In order for the verb to be placed in second position, topicalization must follow the fronting of the verb. A phonological account of V-2 therefore requires that topicalization also be phonological. There are independent reasons to believe that topicalization is not syntactic. Topicalization is genuinely optional, since topics can be interpreted in situ. In Danish, for example, there is a preference for topicalization, but only if there is no violation of ID. It follows that pronouns are more often topicalized than are full DPs. Topicalization is also restricted by ID, a PF processing constraint. And topicalization is motivated by processing ease: fronting facilitates the identification of the topic, as noted in section 4, by providing an initial link to previous discourse. The fact that topicalization is sensitive to island constraints now follows from the extended ID constraint. It is therefore possible to implement the phonological fronting of topics.

What, then, is the motivation for phonological V-2? The motivation must be morpho-phonological. Within minimalism the motivation for V to I movement is the affixal character of inflectional categories. Yet this approach does not distinguish overt affixes from covert inflectional features and is therefore not genuinely phonological. Under the current view, motivation for phonological processes is limited to ease of pronunciation and ease of processing where ease of processing (ID) is conditioned by the availability of overt morphology. At this point, I can only speculate as to what motivates the movement of the verb to the left edge of its string: In all cases (except for *yes-no* questions), the verb ends up in second position, since a topic, a *wh*-element, or some other element moves in front of it. Abstracting away from questions, the verbal element therefore demarcates the border between the topic and the semantic predicate with the resulting form mimicking subject-VP structure in main clauses. This is quite similar to Holmberg, 1986 who proposes that the function of V-2 is to ensure that C'/CP will be able to function as a predicate, that is, a constituent with a verbal head. Topicalization and V-2 thus combine to form an *f*-structure which is isomorphic to syntactic structure, the optimal solution to cases in which the subject-predicate structure does not match the topic-focus structure. Topicalization may

select an element from a subordinate clause, but the target to which it moves is the left edge of the main clause (or main assertion in the case of bridge verbs). V-2 is therefore also restricted to main clauses.

An interesting spin-off of this line of reasoning is that it can potentially explain why topicalization is not generally employed in English. Since English does not allow V-2, topicalization results in a non-optimal string.

Although my explanation for OS is compatible with syntactic topicalization and verb movement, I have provided some reasons that argue that these processes should be viewed as phonological. This line of thinking raises the question of how to distinguish genuinely syntactic dislocation from phonological dislocation. This is the topic of the next section.

6.3. *Is there any syntactic displacement?*

In the previous section, I argued that V-2 is a phonological process mainly in view of its interaction with topicalization. *Yes-no* questions also involve V-movement. Since it is uneconomical to assume two separate types of verb dislocation, one in the syntax and one in the phonology, I conclude that all cases of V-movement are phonological.⁵⁰ But if V-2 in questions is phonological it follows that wh-movement must be so too, since a phonological implementation of V-2 must target the left edge and any movement to 'first' position must follow it. This should not be a surprising result, since topicalization and wh-movement have both been subsumed under A'-movement, i.e., they have the same properties. Since their adherence to island constraints is no longer a problem, there is no reason to force a syntactic analysis. Wh-movement in questions is motivated by the need for the shape of questions to differ (significantly) from that of assertions. And this difference should come as early as possible in the string in order to facilitate the processing of the utterance as a question. The argument does not carry over to embedded wh-questions and relative clauses.

⁵⁰ In fact, I assume no other cases of V-movement: V-I is not needed, since handling inflection morphologically within PF is consistent with the current view. (See also Williams, 2000 for this view.) V-raising in Larsonian shells is argued against in Erteschik-Shir and Rapoport to appear; in preparation.

The discussion so far has been limited to main clauses. In order to account for the phonology of embedding I assume some version of phases à la Chomsky, 1999. Let us say that a phase is defined as the maximal extended V projection. This would allow for a flexible view of phases which can be either VP, IP or CP, depending on what has been merged. Since each such phase comes with its own edges, these will be the targets of wh-movement in embedded clauses. But what is the motivation for embedded wh-movement? First, consider again main clause questions. I have suggested that they must be phonologically signaled. I propose that a Q feature may be selected in the numeration, which can adjoin only to main clauses. The Q feature enables a question interpretation and triggers V-movement. It also functions to block topicalization of elements other than wh-constituents and it ‘occupies’ first position in *yes-no* questions leaving the verb in second position across sentence types.

Direct questions are thus signaled by inversion triggered by Q and not the fronting of wh-elements. Moreover, leaving the wh-element in situ would simplify processing in that there would be no need for ID, the existence of a variable in the string must also be marked. One way to do that would be by stress. But wh-elements are not themselves focused (except in echo questions, in which they are stressed and remain in situ), so this is not an option.⁵¹ Motivation for the movement of wh-elements (in main and embedded clauses) may simply be that this is the only way to highlight their existence in the string. In cases of multiple wh-phrases, some languages front only the first one. In Erteschik-Shir, 1997 I show that the fronted wh-phrase functions as a topic whereas the others are focused and are therefore independently highlighted. It follows that they need not front. Other languages require the fronting of all wh-phrases, or leave them all in situ. It would be interesting to explore whether the wh-parameter results from different phonological ‘high-lighting’ mechanisms across languages.

I have given a very rough outline of what a phonological account of V-movement, topicalization and wh-movement would look like. I have also suggested types of motivation for phonological dislocation. The dislocation types I have examined do not include A-movement. In

⁵¹ See Erteschik-Shir, 1986; 1997 for an analysis of the f-structure of questions.

Erteschik-Shir and Rapoport, in preparation, we argue that passives are directly merged and do not involve movement. Other cases of argument raising may also turn out to have plausible phonological analyses:

- (71) a. John/??a stranger seems to be in the garden.
 b. There seems to be a stranger in the garden.
- (72) a. John/??a stranger seems to have fallen in love with Mary.
 b. It seems that a stranger has fallen in love with Mary.

(71)a and (72)a show that a raised indefinite is degraded. This points to the topic status of raised subjects.⁵² (71)b and (72)b show that in each case an alternate unraised structure is available. Since f-structure linked optionality is characteristic of phonologically motivated dislocation, under the current proposal, it might turn out that DP displacement can also be phonologically implemented.

In this section, I have taken the phonological view of syntax to its extreme, leaving little, if anything for syntax proper. Since this exercise raises interesting questions, it may be worthwhile to pursue this line of investigation further.

7. Conclusion

Although Holmberg, 1999 casts his account of OS in terms of adjacency, he rejects a purely morpho-phonological account since it does not make reference to phonological or morphological primitives. My aim in this paper was to take Holmberg's challenge seriously and to pursue an account which exclusively makes reference to such primitives. My argument is not for or against a particular account of OS, but rather for the plausibility of accounting for word order by purely phonological means. The morpho-phonological primitives I suggest include phonological features such as prosodic properties as well as processing features such as overt case. I believe that the strength of such an approach lies in its potential to explain cross-linguistic and dialectal variation on the basis of fine

⁵² See Erteschik-Shir, 1997 for why modifying an indefinite makes it a better topic as in (i):

i. A gorgeous man seems to have captured Mary's attention.

morpho-phonological distinctions between languages, which exist anyway, while maintaining a uniform syntax.

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References

- Adger, D. (1997). VSO order and weak pronouns in Goidelic Celtic. *Canadian Journal of Linguistics* 42, 9–29.
- Áfarli, T. A. (1997). Dimensions of Phrase Structure: The Representation of Sentence Adverbials. *Motskrift*, 91–113: Institutt for nordistikk og litteraturvitenskap, NTNU.
- Árnason, K. (1996). How to Meet the European Standard: Word Stress in Faroese and Icelandic. *Nordlyd*, 24, 1–22.
- Bobaljik, J. D. (2002). A-chains at the PF interface: copies and ‘covert’ movement. *Natural Language & Linguistic Theory*, 20, 197–267.
- Bobaljik, J. D. and Thráinsson, H. (1998). Two Heads Aren’t Always Better Than One. *Syntax*, 1, 37–71.
- Bošković, E. (2001). *On the nature of the syntax-phonology interface: Cliticization and related phenomena*. Amsterdam: Elsevier.
- Chomsky, N. (1977). On Wh-Movement. In: P. W. Culicover, T. Wasow and A. Akmajian, eds. *Formal Syntax*, 71–132. New York: Academic Press.
- (1999). Derivation by phase. Cambridge, MA: MIT, ms.
- (2001a). *Beyond Explanatory Adequacy*. MIT Occasional Papers in Linguistics 20. Cambridge, MA: MITWPL.
- (2001b). Derivation by phase. In: M. Kenstowicz, ed. *Ken Hale – A Life in Language*, 1–52. Cambridge, MA: MIT Press.
- Christensen, K. K. (1986). Norwegian *ingen*: A case of post-syntactic lexicalization. In: Ö. Dahl and A. Holmberg, eds. *Scandinavian Syntax*, 21–35: University of Stockholm, Institute of Linguistics.
- Christensen, K. R. (2003). OBJ-shift, NEG-shift & Double Objects. Paper presented at Der Syntaxzirkel, Zentrum für Allgemeine Sprachwissenschaft ZAS, Berlin.
- Cinque, G. (1999). *Adverbs and Functional Heads*: Oxford Studies in Comparative Syntax. New York: Oxford University Press.
- Cohen, A. and Erteschik-Shir, N. (2002). Topic, Focus and the Interpretation of Bare Plurals. *Natural Language Semantics*, 10, 125–65.
- Costa, J. (2004). A multifactorial approach to adverb placement: assumptions, facts, and problems. *Lingua*, 114, 711–753.
- Deprez, V. (1994). Parameters of object movement. In: N. Corver and H. V. Riemsdijk, eds. *Studies on Scrambling: Movement and Non-Movement Approaches to Free Word-Order Phenomena*, 101–52. Berlin: Mouton de Gruyter.
- Diesing, M. (1992). *Indefinites*. Cambridge, MA: MIT Press.
- Diesing, M. and Jelinek, E. (1993). The Syntax and Semantics of Object Shift. *Working Papers in Scandinavian Syntax*, 51, 1–54.
- Erteschik-Shir, N. (1973). *On the Nature of Island Constraints*, MIT: Ph.D. dissertation.

- (1979). Discourse Constraints on Dative Movement. In: T. Givon, ed. *Syntax and Semantics: Discourse and Syntax*, 441–68. New York: Academic Press.
- (1986). Wh-questions and focus. *Linguistics and Philosophy*, 9, 117–49.
- (1987). Right node raising. In: M. A. Browning, E. Czaykowski-Higgins and E. Ritter, eds. *MIT Working Papers in Linguistics*.
- (1997). *The Dynamics of Focus Structure*. Cambridge: Cambridge University Press.
- (2001). P-syntactic motivation for movement: imperfect alignment in Object Shift. *Working Papers in Scandinavian Syntax*, 68, 49–73.
- (2003). On the architecture of topic and focus. GLOW workshop on Information structure in generative theory vs. pragmatics: Lund University.
- Erteschik-Shir, N. (to appear). On the architecture of topic and focus. In Molnar, V. & Winkler, S. eds. *The architecture of focus*. Berlin, Mouton de Gruyter.
- Erteschik-Shir, N. and Rapoport, T. R. (to appear). Path Predicates. In: N. Erteschik-Shir and T. R. Rapoport, eds. *The Syntax of Aspect*. Oxford, OUP.
- Erteschik-Shir, N. and Rapoport, T. R. (in preparation). *The atoms of meaning: interpreting verb projections*. Ben Gurion University.
- Erteschik-Shir, N. and Strahov, N. (2004). Focus structure architecture and P-syntax. *Lingua*, 114, 301–23.
- Hellan, L. (1994). A Note on Clitics in Norwegian. In: H. Van Riemsdijk and L. Hellan, eds. *Clitics: Their Origin Status and Position*. Tilburg: ESF-EUROTYP.
- Holmberg, A. (1984). *On certain clitic-like elements in Swedish*. Working Papers in Scandinavian Syntax, 13.
- (1986). *Word order and syntactic features in the Scandinavian languages and English*, University of Stockholm: Ph.D.
- (1999). Remarks on Holmberg's Generalization. *Studia Linguistica*, 53, 11–39.
- Holmberg, A. and Platzack, C. (1995). *The Role of Inflection in Scandinavian Syntax*: Oxford Studies in Comparative Syntax. Oxford: Oxford University Press.
- Josefsson, G. (1992). Object Shift and Weak Pronominals in Swedish. *Working Papers in Scandinavian Syntax*, 49, 59–94.
- (1993). Scandinavian Pronouns and Object Shift. *Working Papers in Scandinavian Syntax*, 52, 1–28.
- (2001). The True Nature of Homberg's Generalization Revisited – Once Again. *Working Papers in Scandinavian Syntax*, 67, 85–102.
- Kidwai, A. (1999). Word Order and Focus Positions in Universal Grammar. In: G. Rebuschi and L. Tuller, eds. *The Grammar of Focus*, 213–44. Amsterdam: John Benjamins.
- Klima, E. (1964). Negation in English. In: J. Fodor and J. Katz, eds. *The Structure of Language*, 246–323. Englewood Cliffs, N.J.: Prentice Hall.
- Lebeaux, D. (1988). *Language Acquisition and the form of the grammar*, University of Massachusetts.
- (1992). Relative Clauses, Licensing, and the Nature of the Derivation. In: S. Rothstein, ed. *Syntax and Semantics* 25, 209–40. New York: Academic Press.
- McCloskey, J. (1999). On the Right Edge in Irish. *Syntax*, 2–3, 189–209.
- Nespor, M. and Vogel, I. (1986). *Prosodic Phonology*. Dordrecht: Foris.
- Nilsen, Ö. (2003). *Eliminating Positions: Syntax and semantics of sentence modification*, Utrecht University: Ph.D.
- Raviv, S. (2005). *Identifying and processing topicalization in Danish*, Foreign Literatures and Linguistics, Ben-Gurion University: MA.

- Reinhart, T. (1981). Pragmatics and linguistics: an analysis of sentence topics. *Philosophica*, 27, 53–94.
- Ross, J. R. (1967). *Constraints on Variables in Syntax*, MIT: Ph.D. dissertation.
- (1975). World Order. In: R. E. Grossman, L. J. San and T. J. Vance, eds. *Papers from the Parasession on Functionalism*, 63–111. Chicago: Chicago Linguistics Society.
- Selkirk, E. (1984). *Phonology and Syntax*. Cambridge, MA: MIT Press.
- (1996). The Prosodic Structure of Function Words. In: J. L. Morgan and K. Demuth, eds. *Bootstrapping From Speech to Grammar in Early Acquisition*, 187–213. Mahwah, NJ: Lawrence Erlbaum Associates.
- Sells, P. (2001). *Structure, Alignment and Optimality in Swedish*. Stanford, CA: CSLI.
- Svenonius, P. (1996). The Verb-Particle Alternation in the Scandinavian Languages. University of Tromsø, ms.
- Truckenbrodt, H. (1999). On the Relationship between Syntactic Phrases and Phonological Phrases. *Linguistic Inquiry*, 30, 219–55.
- Williams, E. (2000). Three models of the morphology-syntax interface. In: W. U. Dressler, O. E. Pfeiffer, M. A. Pöchtrager and J. R. Rennison, eds. *Morphological Analysis in Comparison*, 223–47. Philadelphia, PA, USA: John Benjamins.
- Zubizarreta, M. L. (1998). *Prosody, Focus and Word Order*. vol. 33: Linguistic Inquiry Monograph. Cambridge, MA: MIT Press.

Holmberg's Generalization and Cyclic Linearization. Remarks on Fox and Pesetsky

ELENA ANAGNOSTOPOULOU

Fox and Pesetsky (henceforth F&P) propose an architecture for the mapping between syntax and phonology which relates a number of different constraints on movement to the way in which phrase structure is linearized. They investigate Object Shift (henceforth OS) and Quantifier Movement (henceforth QM) in Scandinavian and argue that the restrictions on these processes, namely Holmberg's Generalization (HG) effects on OS and what they call "the inverse Holmberg effect" on QM, reflect a requirement for preservation of the order established in the VP due to the fact that the VP is a Spell-out domain. F&P's proposal relies on Holmberg's (1999) formulation of HG which has been challenged by Anagnostopoulou (2002) on the basis of data discussed in Anagnostopoulou (2003) that directly contradict Holmberg (1999). It is my goal here to investigate how these data can be accommodated in F&P's system. I will argue that even though F&P can, in principle, account for the data in question, the attempt to unify the restrictions on OS, QM with comparable restrictions on passivization under F&P's architecture fails to express certain crosslinguistic generalizations which are straightforwardly captured in traditional locality accounts.

Holmberg's Generalization: the debate

In the literature, there is a debate concerning the scope and correct characterization of HG. According to one view (Chomsky 1993; Bobaljik 1995, 2002; Bobaljik & Jonas 1996 and others), HG is exclusively tied to V-raising. On this view, the core data to be explained are exemplified in (1) (from Swedish):

- (1) a. Jag kysste *henne* inte [_{VP} t_v t_o]
 I kissed her not
 b. *Jag har *henne* inte [_{VP} kysst t_o]
 I have her not kissed
 c. *... att jag *henne* inte [_{VP}kysste t_o]
 that I her not kissed

According to another view, defended in Holmberg (1999), the V-movement requirement is part of a more general condition preventing OS across any phonologically visible category within VP, except adjuncts. OS of a DO cannot apply across an IO in (2) and a particle in (3), just as it cannot apply across V in (1):

- (2) a. *Jag gav *den* inte Elsa t_o
 I gave it not Elsa_{IO}
 b. Jag gav inte Elsa den
 (3) a. *Dom kastade *mej* inte ut t_o
 They threw me not out
 b. Dom kastade inte ut *mej*

Importantly, OS is licit if the intervening V, IO or particle undergo movement to C leaving a trace in the VP, as in (4)–(6):

- (4) Kysst har jag *henne* inte (bara hållit henne i handen)
 Kissed have I her not (only held her by the hand)
 (5) Vem_{IO} gav du *den*_{DO} inte t_{IO} t_{DO}?
 Who gave you it not
 ‘Who didn’t you give it to?’
 (6) UT_P kastade dom *mej*_O inte t_P t_O (bara ned för trappan)
 Out threw they me not (only down the stairs)
 ‘They didn’t throw me OUT (.....)’

To account for the facts in (1)–(6), Holmberg proposes that OS takes place at PF, and therefore it can apply across a trace of any type but not across overt material. He furthermore suggests that the facts in (4)–(6) provide evidence that OS applies countercyclically: OS moves the object to a sentence medial position after V, IO and the particle move to C.

There is a third approach towards HG, which crucially relies on Holmberg’s (1999) view that HG subsumes all data in (1)–(6). According to

this approach (Kathol 2000; Müller 2000; Sells 2001 and Williams 2003), HG reflects a requirement for order preservation as a result of which OS is not allowed to revise the order of constituents in the VP. In *Cyclic Linearization of Syntactic Structure* F&P take a similar position towards HG. Crucially, though, F&P argue that ordering generalizations are established on a phase-by-phase basis; this permits them to relate the constraints that fall under Holmberg's (1999) HG to two other constraints on movement, namely successive cyclicity and "the inverse Holmberg effect".

F&P's system: an overview

F&P propose that the mapping between syntax and phonology is subject to (7):

(7) *Order Preservation*

Information about linearization, once established at the end of a given Spell-out domain, is never deleted at the course of a derivation

Certain domains created in a derivation, e.g. VP, CP and DP (roughly corresponding to *phases*), are Spell-out domains whose construction is followed by linearization. Each time the derivation constructs a Spell-out domain D, Spell-out applies linearizing D. When a new Spell-out domain D' is constructed, Spell-out linearizes the new material D' preserving the information produced by previous applications of Spell-out. Crucially, the only function of Spell-out is to *add* new information to the information already established at the previous Spell-out domain. This leads to a number of restrictions on how movement proceeds:

(a) Leftward movement of an element X from the left edge position of a Spell-out domain D to a position in a higher Spell-out domain D' is licit because X in D' continues to precede the elements that X precedes in D (and conversely for rightward movement).

(b) Movement from a non-edge position is illicit when the ordering statements established in D' are inconsistent with the ones established in D. Suppose the derivation has created the Spell-out domain D in (8):

(8) [D XYZ]

If Y moves leftward in the next Spell-out domain, as in (9), the information in (8) and the information in (9) impose conflicting requirements on linearization.

(9) $*[_{D'} \dots Y \quad [_{D} X \ t_Y Z]$

Y is pronounced after X in (8), but Y must be pronounced before X in (9) because Y precedes D (which contains X). Y must, therefore, move to the edge of D before undergoing further movement, as depicted in (10).

(10) $[_{D'} \dots Y \quad [_{D} t_Y X \ t_Y Z]$

This derives successive cyclicity.

(c) Movement from the non-edge is possible as long as the previously established linearization is not disrupted. Y can move from the non-left edge position in (8) if X moves as well and X and Y preserve their original order in the higher Spell-out domain:

(11) $[_{D'} \dots X \dots Y \dots [_{D} t_X t_Y Z]$

This is HG on F&P's view. OS is subject to HG because it is movement from a non-edge. In this analysis, there is no need to assume that OS applies counter-cyclically (contra Holmberg 1999). (4)–(6) above are licit because in the CP Spell-out domain the V, IO and particle in C continue to precede the shifted object, preserving the VP-order.

(d) In F&P's system, when Y from a non-edge position of D surfaces in a position that reverses the original order of constituents in D, this movement always proceeds through a movement step of Y to the edge of D. After moving to the edge Y precedes all material in D, and no element in D is allowed to move to a position that precedes Y in D', i.e. (12) is excluded:

(12) $*[_{D'} \dots X \dots Y \quad [_{D} t_Y X \ t_Y Z]$

This explains what F&P call “the inverse Holmberg effect”, a restriction constraining QM in Scandinavian, exemplified in (13) and (14) (Svenonius 2000):

- (13) a. Jón hefur ekkert [sagt Sveini _]
 Jon has nothing [told Sveini-DAT _]
 ‘John has told Svein nothing’
 b. ekkert [tekkert sagt Sveini tekkert]

- (14) a. *Jón sagði ekkert Sveini
 Jon said nothing Svein
 b. [CP sagði [ekkert [VP t_{ekkert} t_{sagði} Sveini t_{ekkert}]]
 ↑
 ×

In (13a) the DO undergoes QM to a position that precedes the participle and the IO. Since the DO precedes the material it follows in the VP, QM involves an intermediate step to the edge of the VP, as shown in (13b). In this analysis, (14a) is ruled out because it violates Order Preservation. After moving to the edge, the quantifier precedes V in the VP domain, but V to C movement yields a contradictory order in which V precedes the quantifier in the CP Spell-out domain, as schematized in (14b).

In conclusion, a crucial feature of F&P's theory is that linearization is relativized to particular domains, which permits unification of HG effects with successive cyclic movement. The goal of the next sections is twofold. On the one hand, I will present certain OS data which cannot be accommodated in an order preservation approach unless ordering statements are relativized to domains, arguing in favor of F&P over related theories. On the other hand, I will discuss the syntax of processes involving movement of a lower object across a higher one (OS, QM and passivization) in Scandinavian double object constructions pointing to a correlation which supports traditional locality approaches over F&P's theory.

VP vs. vP Spell-out Domains: OS across Subjects

F&P make two claims about the shape of the verb-headed phrase that counts as a Spell-out domain in Scandinavian:

(i) The Spell-out domain contains V and the objects, but not adverbs and negation. For this reason, OS is free to move across adverbs without resulting in an ordering contradiction, but is not free to move objects across V.

(ii) Since the Scandinavian languages freely raise the main verb to C over the subject, the subject is not linearized with respect to the main verb before CP is constructed, i.e. VP and not vP is a Spell-out domain.

Here I will concentrate on the second claim. I will first provide a further reason to assume that the Spell-out domain excludes the subject in Scandinavian and then I will discuss potential predictions of this hypothesis.

An argument that potentially shows that the verb phrase Spell-out domain is not vP in Scandinavian involves subject quantifier stranding (Q-stranding) under OS, as opposed to subject Q-stranding under object scrambling in Korean and Japanese.

Korean and Japanese have Q-stranding under scrambling. Kuroda (1983) and Miyagawa (1989) argue that in Q-stranding, quantifiers and their associates are generated as sisters which are separated by scrambling of the nominal to a higher position. Interestingly, object scrambling over a subject may not be followed by subject scrambling over the object, where the subject strands a quantifier, as shown in (15) with an example from Korean (Kuroda 1983; Miyagawa 1989; Ko 2004):

- (15) **Haksayng-tul-i*₁ may kcwu-lul₂ t₁ *sey-myeng* t₂
 Student-PL-NOM beer-ACC three-CL_{person}
 masi-ess-ta
 drink-Past-Dec
 ‘Three students drank beer’

Ko (2004), as cited by F&P, argues that this restriction is an instance of the “Inverse Holmberg effect”. Under the assumption that the vP is a Spell-out domain in Korean and Japanese, object scrambling over the subject involves a step by which the object moves to the edge of the vP so as to establish the “O>S” order. But then the subject is not allowed to scramble over the object stranding a quantifier as in (15) due to a conflict in linearization: O precedes S in the vP Spell-out domain but the scrambled S precedes O at a later stage in the derivation.

From the perspective of (15) it is significant that in all Scandinavian languages OS targets a position above floated quantifiers related to subjects, as discussed in Holmberg & Platzack (1995), Bobaljik (1995), Holmberg (1999) and others (data from Holmberg and Platzack 1995: 141):

- (16) *Lásu stúdentarnir greinina ekki allir?* *Icelandic*
 Read the-students the-article not all
 ‘Didn’t the students all read the article?’

- (17) *Läste studenterna den inte alla? Swedish*
 Read the-students it not all
 'Didn't the students all read it?'

Assuming that subject Q-float involves stranding of the quantifier in the vP under NP-movement of the subject, as in Sportiche (1988),¹ the vP is not a Spell-out domain in Scandinavian, or else an "Inverse Holmberg effect" is incorrectly predicted to arise in (16) and (17), on a par with (15).

The proposal that the low Spell-out domain does not include the subject in Scandinavian leads to the prediction that OS across a subject should be possible in principle. As will be seen immediately, this prediction is borne out, supporting F&P's view that the conditions on OS are limited to certain domains over competing theories where there is no domain-limitation.

As discussed in Holmberg (1999: 15), there are varieties of Swedish in which OS may apply across a subject in examples like (18):

- (18) *Därför gav mej Marit inte någon present*
 Therefore gave me Marit not any present
 'Therefore Mary did not give me any present'

The existence of (18) is predicted by F&P, while it constitutes a problem for theories that view order preservation as an absolute constraint. Moreover, (18) is a counterexample to Holmberg's (1999) formulation of HG as it shows that OS is allowed to apply across non-adjunct categories, such as subjects. To account for OS across the subject in examples like (18), Holmberg proposes that they instantiate 'long OS'² which is subject to different conditions than the type of OS illustrated in (1)–(6). Unlike (1)–(6), where the object moves to a low position close or adjoined to the vP, *mej* in (18) is high in the IP domain: it precedes the subject *Marit* which can be concluded to be vP-external (in Spec,IP) since it

¹ See Bobaljik (1995) and Holmberg (1999) for arguments against Sportiche's analysis of Q-float in Scandinavian and in favor of the view that the subject quantifiers in (16) and (17) are adjuncts. I believe that there are reasons to doubt Bobaljik's and Holmberg's analysis, but it would lead me too far to address them here. The argument in the main text relies on Sportiche's analysis.

² Long OS is possible in Swedish but not in Norwegian and Danish.

occurs before the negation, which, in turn, has been argued to be generated high in Scandinavian (the negation precedes floated quantifiers and, in non V-2 environments, auxiliaries; see Holmberg 1999, Bobaljik 2002 for discussion).

Even though it is evident that the object in (18) surfaces in a position higher than the landing site of OS in (1)–(6), it is not clear why long OS should be subject to different conditions than short OS. More importantly, there is evidence that short OS can also apply across the subject, similarly to long OS. As shown in (19) (see Jonas & Bobaljik 1993; Alexiadou & Anagnostopoulou 2001 for discussion of such examples), OS in Icelandic can take place across a vP-internal subject when this is indefinite/quantificational:

- (19) það luðu sennilega husið [vP vandlega [vP margir
 there painted probably the house carefully many
 studentar $t_{luðu}$ $t_{husið}$]
 students
 ‘Many students probably painted the house carefully’

In (19) the DO precedes the low manner adverb marking the left-edge of vP, which, in turn, precedes the subject. Expl-V-O-S orders of the type illustrated in (19) are true instances of OS as the O can precede an *in situ* S only under V-raising. Examples like (19) are expected to be well-formed if the domain relevant for linearization excludes the subject.³

On the syntax of the VP-domain: reversal of IO>DO orders in OS, QM and passives

Having seen that VP and not vP counts as a Spell-out domain in Scandinavian, I now turn to the syntax of the VP domain when two objects are present.

³ As discussed in Alexiadou and Anagnostopoulou (2001), constructions like (19) are uncommon in Scandinavian due to the independent requirement that the subject must generally raise overtly, but they are very common cross-linguistically. In pro-drop languages of the Romance/Greek type VOS orders are attested in both synthetic and periphrastic constructions. In these languages, participles move out of the vP (Alexiadou 1997), i.e. the V-raising requirement on OS is always met.

Recall that OS of a DO is generally not allowed to apply across a higher IO, as shown in (2), a fact that Holmberg (1999) takes to support his version of HG. In some varieties of Norwegian and Swedish, though, a DO pronoun is allowed to undergo OS across an *in situ* IO pronoun. Along with the grammatical IO-Neg-DO order in (20), the DO-Neg-IO order in (21) is more marked but possible (see Hellan & Platzack 1999; Anagnostopoulou, 2002, 2003, reporting judgments of Anders Holmberg, personal communication; data from Hellan and Platzack 1999: 131–132):

- (20) a. Han visade henne inte den $\sqrt{IO-Neg-DO}$
 b. Han visade 'na inte 'n
 He showed her not it
 'He did not show it to her'
- (21) a. Han gav den inte henne $\sqrt{DO-Neg-IO}$
 b. Han gav 'en inte 'na
 He gave it not her
 'He did not give it to her'

In non V-Raising, non OS environments the order of pronouns is strictly IO>DO:

- (22) a. Jag ville inte ge honom den
 I wanted not give him it
 'I didn't want to give it to him'
- b. *Jag ville inte ge den honom

Examples like (21) contradict the view that OS is not allowed to revise the order of constituents in the VP as well as the claim that OS cannot apply across higher non-adjunct categories. Just as in (18) and (19) OS applies across a subject, in (21) OS applies across an IO. OS of two objects in Norwegian and Swedish presents a further counterexample to theories that assume that the base order of arguments is always preserved under OS.⁴ When a pronominal DO and a pronominal IO shift together,

⁴ Holmberg (1999: 15 fn 10) points out that examples like (23b) can be accounted for in his system which permits OS to apply across adjuncts. Assuming that OS involves adjunction to vP, the IO *honom* adjoins to vP under OS in (23b); OS of *den* is subsequently allowed to take place because the shifted IO is an adjunct. Such a literal interpretation

DO>IO and IO>DO orders are equally possible (Hellan & Platzack 1999: 131; Anagnostopoulou, 2002, 2003):

- (23) a. Jag gav honom den inte. $\sqrt{IO-DO-adv}$
 I gave him it not
 ‘I didn’t give it to him’
 b. Jag gav den honom inte. $\sqrt{DO-IO-adv}$
 I gave it him not
 ‘I didn’t give it to him’

The DO>IO order in (23b), which reverses the order of constituents in the VP, is grammatical for all speakers, even for those that resist examples like (21).⁵

F&P can accommodate the above data due to the crucial assumption that movement to the edge of Spell-out domains may be employed to revise the base order of constituents. More specifically, (20)–(23) can be accounted for if the double object construction involves two Spell-out Domains, one containing the two objects and one containing the verb:

- (24) [Domain B V [Domain A IO DO]]

Structure (24) is consistent with decomposition approaches towards the double object construction and is most immediately compatible with a small clause analysis, as in Kayne (1984), Beck and Johnson (2004), and with a low applicative analysis as in Pylkkänen (2002). The fact that the order of IO and DO can be reversed if both are pronominal suggests that a pronominal DO may undergo OS in the presence of a pronominal IO

of ‘adjunct’, though, is unsatisfactory. The fact that OS is insensitive to phonologically visible adjuncts in itself is a problem for Holmberg which could, perhaps, be solved by postulating countercyclic insertion of adjuncts in OS constructions. If countercyclic insertion is a viable solution to the problem, then it cannot be extended to constructions with a shifted IO.

⁵ As pointed out to me by Anders Holmberg, and Øystein Nilsen (personal communication) reversal of order of pronouns under OS in Norwegian is possible only in the presence of negation or other adverbs marking the edge of the vP. In contrast, reversal of order in Swedish is licit even when there is no element marking the left edge of the vP, provided that the V-raising condition is met.

through an intermediate step by which the DO moves to the edge of Domain A, establishing a 'DO>IO' order:^{6,7}

(25) [Domain B V [Domain A DO IO t_{DO}]]

The fact that two pronominal objects never raise over the main verb furthermore suggests that movement through the edge of B is never an option under OS.

Recall at this point the Icelandic QM data in (13) and (14) where the quantificational DO precedes both the main verb and the IO and where further raising of V is blocked. Unlike OS, it is crucial to assume that QM proceeds through the edge of Domain B, the domain containing V. If Domain A is taken to be a Spell-out domain in Icelandic, as in Swedish, then *ekkert* presumably moves first to the edge of A and then to the edge of B. (13b) above must thus be re-written as (26):

(26) [Domain B *ekkert* sagt [Domain A t_{ekkert} IO t_{ekkert}]]

So far, it has been seen that Scandinavian double object constructions displaying inversion of the two objects under OS and QM can be accommodated in F&P's framework due to the assumption that movement through the edge is a strategy that reverses base orders. I will now compare these constructions to a third movement process, namely passivization.

In Anagnostopoulou (2002, 2003) I establish a correlation between OS and passives across Scandinavian. In Swedish and Norwegian where the DO may undergo OS across the IO (see (21) above) and the order of DO and IO may invert when both objects undergo OS (see (23b) above), the DO is allowed to move across the IO in passives, as shown in (27) for Norwegian:

⁶ The analysis in (25) predicts the emergence of an Inverse Holmberg effect in OS constructions with inversion of two pronominal objects. At this point, I do not know how this prediction can be tested.

⁷ Movement of the DO to the edge of A must be assumed to be impossible when the IO is a DP explaining why (2a) above and (i) with OS of two objects are ungrammatical:

(i) *Jag gav den Elsa inte
I gave it Elsa not

- (27) a. Jon ble gitt en bok $\sqrt{\text{Passivization of IO}}$
 John was given a book
 ‘John was given a book’
 b. En bok ble gitt Jon $\sqrt{\text{Passivization of DO}}$
 A book was given John

In Danish and Icelandic, where OS always preserves the base order of constituents, the DO is not allowed to undergo passivization across a higher IO. In (28)–(30), I exemplify this for Danish:⁸

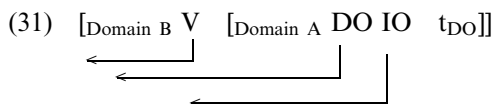
- (28) a. Peter viste hende jo den $\sqrt{\text{IO-adv-DO}}$
 Peter showed her indeed it
 ‘Peter indeed showed it to her’
 b. *Peter viste den jo hende $*\text{DO-adv-IO}$
 Peter showed it indeed her
- (29) a. Peter viste hende den jo $\sqrt{\text{IO-DO-adv}}$
 Peter showed her it indeed
 ‘Peter indeed showed it to her’
 b. *Peter viste den hende jo $*\text{DO-IO-adv}$
- (30) a. Jens blev givet bogen $\sqrt{\text{Passivization of IO}}$
 Jens was given book-the
 ‘Jens was given the book’
 b. *Bogen blev givet Jens $*\text{Passivization of DO}$

In Anagnostopoulou (2002, 2003), I argue that this correlation between OS and passivization in Scandinavian can be straightforwardly captured if movement of the DO across the IO is taken to be A-movement targeting T in passives and transitive v, v-Tr, in OS (see also Bobaljik 2002). Under the assumptions that (i) movement is constrained by the Minimal Link Condition (MLC) and (ii) apparent violations of the MLC result from the parametric availability of layered specifiers, which permit successive cyclic movement across potential interveners (cf. Ura 1996 and others), the key to the above facts is that the head introducing the IO

⁸ See Anagnostopoulou (2002, 2003) for detailed discussion of Icelandic, which has two classes of ditransitives. One in which the base order is always IO>DO and one which allows DO>IO pre-OS orders (Holmberg & Platzack 1995; Collins & Thráinsson 1996).

may or may not license a secondary specifier which permits successive cyclic A-movement (OS and passivization) of the DO across it. This specifier is licensed in Norwegian and Swedish but not in Icelandic and Danish.

Consider now how the same facts can be dealt with in F&P's terms. We have seen that OS of the DO across the IO involves the derivation in (31), where first the DO moves to the edge of Domain A, and then the V, DO and IO move further in a way that preserves their linear order:



If passivization is taken to be constrained by the same principles of linearization as OS and QM,⁹ then it is crucial that underlying objects are allowed to move across the main verb in passives, as is evident in Norwegian (27) and Danish (30) where the passive is built periphrastically. Assuming that the VP domain is a Spell-out domain in passives exactly as in actives,¹⁰ we are led to suggest that passivization of the DO across the IO in Swedish and Norwegian proceeds identically to the derivation of QM in (26). Since the DO moves across the verb as well the IO, it proceeds successive cyclically through the edge of domain A and domain B, on its way to T:



On the other hand, the DO is not allowed to move to the edge of domain A in Danish and Icelandic OS, and only the IO is allowed to move to the edge of domain B in passives. Movement of the DO across the IO is prevented.

⁹ F&P do not discuss NP movement.

¹⁰ Alternatively, Domain B is never a linearization domain in passives. This might perhaps simplify things; but it would not be clear why passive VPs differ from active VPs in this way (for vPs one could perhaps appeal to the deficiency of passive v, as opposed to active v, as in Chomsky 2000; 2001). The presence or absence of an Inverse Holmberg effect might decide between the option considered in this footnote, as opposed to the analysis presented in the main text.

- (33) a. [IO T [Domain B t_{IO} V [Domain A t_{IO} DO]]
 b. T [Domain B V [Domain A IO DO]]
- ←-----×-----|

But it is not clear why the DO is prevented from moving to the edge of the two VP Spell-out domains in Icelandic OS and passives, while it can do so in QM constructions. Is it a coincidence that OS and passives pattern together in Icelandic in not employing the movement-to-edge-strategy, as opposed to OS and passives in Swedish and Norwegian and QM in Icelandic? If not, it seems necessary to resort to independent properties of movement, for example that OS and passivization systematically pattern alike because they instantiate A movement, unlike QM, and that A movement may or may not employ the movement to edge strategy as a matter of a parameter. As far as I can see, the latter assumption would re-introduce the concept of “escape hatch” into the theory of movement, an undesirable move from F&P’s point of view. There are other questions, though, which seem to show that a general theory for when movement to the edge takes place is needed. For example, that movement across V is never possible in OS, while movement across a higher argument is licit in a number of well-defined cases. This difference is expected in approaches that treat the two as independent constraints.

The MLC analysis mentioned above groups together the conditions governing OS and passivization of an argument across a higher argument under the MLC,¹¹ and dissociates these cases from QM, which is not assumed to be A-movement, as well as from movement of an argument across a higher head (V and maybe particle, though see Bobaljik 2002), which is taken to fall under HG reflecting an independent constraint on OS. In contrast, F&P group together all instances of movement of an argument across a higher argument as well as movement of arguments across higher heads. This categorization of facts leads to an

¹¹ In an MLC analysis, OS across the subject is correctly predicted to be possible as it targets a layered specifier to v-Tr, which introduces the external argument. See also Anagnostopoulou (2002, 2003) for extensive discussion that the movement of an intervenor strategy in (5) above is not specific to OS but occurs in several NP-movement constructions, and for a locality analysis.

analysis which obscures the correlation between OS and passives across Scandinavian.

In conclusion, the more general question raised by F&P's account concerns the conditions under which different strategies are employed in order to resolve potential conflicts in linearization, and the division of labor between these strategies and independent properties specific to movement. This becomes evident once different constructions are studied and compared in detail.

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References

- Alexiadou, A. (1997). *Adverb Placement: A Case Study in Anti-symmetric Syntax*. Amsterdam: Benjamins.
- Alexiadou, A. & E. Anagnostopoulou (2001). The Subject In-situ Generalization, and the role of Case in driving computations. *Linguistic Inquiry* 32, pp. 193–231.
- Anagnostopoulou, E. (2002). Movement Across Phonologically Visible Categories. MIT Ling-Lunch talk, December 5.
- (2003). *The Syntax of Ditransitives. Evidence from Clitics*. Berlin: Mouton de Gruyter.
- Beck, S. & K. Johnson (2004). Double Objects Again. *Linguistic Inquiry* 35, pp. 97–123.
- Bobaljik, J. & D. Jonas (1996). Subject Positions and the Roles of TP. *Linguistic Inquiry* 27, pp. 195–236.
- Bobaljik, J. (1995). Morphosyntax: The Syntax of Verbal Inflection. Ph.D. Dissertation, MIT.
- (2002). A-Chains at the PF-Interface: Copies and 'Covert' Movement. *Natural Language and Linguistic Theory* 20, pp. 197–267.
- Chomsky, N. (1993). A Minimalist Program for Linguistic Theory. In: Ken Hale and Samuel Jay Keyser (eds.). *The View from Building 20* (pp. 1–52). Cambridge, Mass.: MIT Press.
- (2000). Minimalist Inquiries: The framework. In: R. Martin, D. Michaels & J. Uriagereka (eds.). *Step by step* (pp. 89–155). Cambridge, Mass.: MIT Press.
- (2001). Derivation by phase. In: M. Kenstowicz (ed.), *Ken Hale. A Life in Language* (pp. 1–52). Cambridge, Mass.: MIT Press.
- Collins, C. & H. Thráinsson (1996). VP-Internal Structure and Object Shift in Icelandic. *Linguistic Inquiry* 27, pp. 391–444.
- Fox, D. & D. Pesetsky (This Volume). Cyclic Linearization of Syntactic Structure.
- Hellan, L. & C. Platzack (1999). Pronouns in Scandinavian languages: An overview. In: H. van Riemsdijk (ed.), *Clitics in the Languages of Europe* (pp. 123–142). Language Typology, Volume III. Berlin/New York: Mouton de Gruyter.
- Holmberg, A. (1999). Remarks on 'Holmberg's Generalization'. *Studia Linguistica* 53, pp. 1–39.

- Holmberg, A. & C. Platzack (1995). *The Role of Inflection in Scandinavian Syntax*. Oxford/New York: Oxford University Press.
- Jonas, D. & J. Bobaljik (1993). Specs for Subjects: the Role of TP in Icelandic. *MIT Working Papers in Linguistics* 18, pp. 59–98.
- Kathol, A. (2000). *Linear Syntax*. Oxford: Oxford University Press.
- Kayne, R. (1984). *Connectedness and Binary Branching*. Dordrecht: Foris Publications.
- Ko, H. (2004). Asymmetries in Scrambling and Cyclic Linearization. Ms. MIT.
- Kuroda, S.-Y. (1983). What can Japanese say about Government and Binding. *Proceedings of WCCFL*: 153–164.
- Miyagawa, S. (1989). *Structure and Case Marking in Japanese*. San Diego: Academic Press.
- Müller, G. (2000). Shape Conservation and Remnant Movement. Paper presented at *NELS*, Rutgers University.
- Pylkkänen, L. (2002). Introducing Arguments. Ph.D. Dissertation. MIT.
- Sells, P. (2001). *Structure, alignment and optimality in Swedish*. Stanford, CA: CSLI Publications.
- Sportiche, D. (1988). A Theory of Floating Quantifiers and its Corollaries for Constituent Structure. *Linguistic Inquiry* 19, pp. 425–449.
- Svenonius, P. (2000). Quantifier Movement in Icelandic. Ms. Tromsø University.
- Ura, H. (1996). Multiple Feature-Checking: A Theory of Grammatical Function Splitting. Ph.D. dissertation, MIT.
- Williams, E. (2003). *Representation Theory*. Cambridge, Mass: MIT Press.

Re: CycLin and the role of PF in Object Shift*

JONATHAN DAVID BOBALJIK

This volume's two target articles explore novel approaches to word order alternations, especially Scandinavian Object Shift. They share the common perspective that aspects of linear order long considered the exclusive purview of syntax may be better understood if the burden of explanation is split between phonological and syntactic modules. The two articles differ substantially, however, in how this general hunch plays out, in particular in the amount of the explanation that is attributed to extra-syntactic factors. Fox and Pesetsky's "Cyclic Linearization" model (hereafter F&P, *CycLin*) is compatible with familiar syntactic models, and can be seen as a filter running (cyclically) on the output of syntactic derivations. F&P suggest that their proposal can explain various heretofore stipulated conditions on syntactic operations as consequences of the architecture of their system and a single axiom about linearization. Erteschik-Shir's proposal in "Sound Patterns of Syntax" (hereafter E-S) is more radical, in the sense that far less of the familiar syntax is retained; where for *CycLin* movement is still a syntactic process, on E-S's view a good deal of traditionally syntactic movement must be rethought in linear, rather than hierarchical terms. Both articles are largely exploratory and leave many of the details still to be spelled-out. To engage the ideas on specifics, then, will involve to some degree making some educated guesses about what ancillary assumptions the relevant authors might condone. I will therefore restrict myself to a few comments at a general level, though it

* I am grateful to Line Mikkelsen, Sten Vikner and especially Susi Wurmbrand for discussion of this material and to Susi in addition for pointers to the relevant processing literature. My own research on Object Shift has been in part supported by SSHRC and FQRSC.

will be impossible to do justice to these authors' ideas in the allotted space.

1. Phonology, functional motivations, parsing and grammar

One of the central hypotheses put forward by E-S is that "the motivation for [phonological displacements] is limited to easing pronunciation ... and parsing" (p. 53). While stated in functional terms, the elaboration of this idea as E-S presents it has a variety of components. On the one hand, E-S spells out this hypothesis as meaning that cross-linguistic variation in word order should be attributable to independently justifiable differences in "general phonological properties" (such as intonational phrasing), and on the other, that issues of ambiguity resolution should constrain the possibilities for word-order variation.

In practice, the former idea seems to be a promising strategy for investigation of the fine micro-variation in OS possibilities among various Scandinavian languages and dialects, sometimes glossed over in the literature. For example, it is well-known that speakers of Swedish disagree as to the obligatoriness of OS (in those environments where it is possible); for some it is obligatory, for others, apparently optional (Hellan and Platzack 1995). E-S suggests that this correlates with the possibility of realizing primary word-stress on negation; where that element may receive primary stress, it is a legitimate host for a weak (clitic-like) pronoun, and the order neg-pronoun is thus admissible, but where negation itself is weak and cannot bear word-stress, it cannot serve as a host and the pronoun must be supported by the verb or the subject. Descriptions in the literature imply that the obligatory/optional variation in Swedish (at least) is among individuals, rather than identifiable dialects; it will thus be interesting to see whether E-S's general proposal will work out at the level of individual variation in the stress patterns available to negation. Note, though, that the promise here lies in tying one point of cross-linguistic (or cross-speaker) variation to another, in this case word-order possibilities and stress, along with a theory of cliticization / prosodic incorporation that relates the two. In principle, this is independent of whether a functional motivation for the theory can (or should) be inferred.

Where E-S's proposal adopts an explicitly functional motivation is in the role of disambiguation in the account, centred around (1), her ID (16):

- (1) In a string, $X \dots Y$, where X and Y are arguments, [identify] X as subject and Y as object if neither is marked otherwise.

One observation that E-S invokes (1) to account for is the following partial paradigm. Examples (2a) and (2c) show that objects may be fronted yielding a word order consistent with the V2 constraint, but (2b), which should have a parallel object-fronting structure available to it, can, according to E-S, only be understood as 'Peter met Sara.'¹

- (2) a. Ham mødte Sara.
 him met Sara
 'Sara met him' (ES (17a))
 b. Peter mødte Sara.
 Peter met Sara
 * as 'Sara met Peter' (ES (14a))
 c. Peter mødte Sara ikke.
 Peter met Sara not
 'Sara didn't meet Peter' (ES (14b))

Since case is only marked on pronouns, and agreement and word order do not disambiguate (2b), (1) applies to exclude the object-topicalization reading. In (2a), the pronoun bears accusative case so (1) does not apply, and in (2c) word order disambiguates – the position between the finite verb and negation is only available to non-pronominal NPs that are subjects. Note that in order to block (1) from applying, that position must count as 'marking' the subject.

¹ Line Mikkelsen and Sten Vikner (personal communication 6/04) find (2b) acceptable as object topicalization, with suitable context and intonation, required also for (2c) where word order does disambiguate. (Line Mikkelsen has also drawn my attention to Ørnsnes 2002, which discusses the effects of case-marking as facilitating ambiguity resolution in certain finite subject extraction contexts in Danish – particularly relevant is his fn. 8.) E-S does not comment on whether intonation may count as 'marking' for the purposes of (1), although the discussion of her (14)–(15) implies that it doesn't. In any event, this will not affect the point to be made below.

The proposal in (1) is a venerable idea, dating at least to Jakobson (1936 [1971], p. 28).² While Jakobson apparently intended (1) as an absolute constraint, it has subsequently been argued that this represents instead a parsing *preference*, and that in suitable contexts, the object topicalization reading re-emerges (for Russian, see King 1995: 2; see also fn. 1 for Danish). There seems to be ample evidence for (1) or something like it as a parsing strategy, in particular in German where there is a demonstrable preference for initially parsing a case-ambiguous initial NP as a subject (see Schlesewsky et al. 2000, 2003 and references therein).³ Note, though, that the parsing preference is not a categorical judgment of acceptability, and is readily overcome by a variety of means, including pragmatic considerations and intonation. Thus, although feminine and neuter NPs and pronouns, including the inanimate *wh*-word *was* ‘what’, lack distinct nominative and accusative forms, speakers seem to have no trouble interpreting a question like (3) with *Maria* as the subject (presumably for reasons of pragmatic plausibility).

- (3) Was hat (die) Maria gegessen?
 What.ACC=NOM has the.ACC=NOM Maria eaten?
 ‘What did Maria eat?’ (or ‘What ate Maria?’)

This appears to hold for Danish as well: *Den her bog læste Peter* ‘this here book read Peter’ is acceptable as a object-topicalization (‘Peter read this book.’) although case is not marked, and a question like *Hvad læser Peter?* ‘What reads Peter’ is unremarkable, despite case or word

² “Wenn aber in einem derartigen Wortgefüge die Endungen der beiden Nomina ihre Kasus nicht anzeigen, darf die normale Wortfolge nicht verletzt werden.” = “If, however, in such a construction, the endings of the two nominals do not indicate their cases, the normal word order [SVO] may not be violated.” Russian has a rich case system, but certain nouns, among them ‘soft’ feminine nouns such as *mat* ‘mother’ and *doč* ‘daughter’ do not distinguish nominative from accusative; Jakobson claims that the word order freedom seen in *syn ljubit otca* ‘son-N loves father-A’ versus *syna ljubit otec* ‘son-A loves father-N’ is unavailable when *both* nouns fail to mark case and thus that *mat’ ljubit doč* ‘mother loves daughter’ is unambiguously SVO. Although Jakobson discussed only Russian, the proposal was intended to be universal, see Jakobson 1963 [1971].

³ The experimental effects are not equally strong for all constructions – see Schlesewsky et al. 2003 for a brief overview. If I have read Schlesewsky et al. correctly, the subject preference is mild, if present at all, in simple V2 constructions with the structure of (3), even when pragmatically unbiased predicates like ‘meet’ and ‘visit’ are used.

order cues to disambiguate the NPs, in violation of (1). Sten Vikner (personal communication) offers the question *Hvad spiste hesten?* ‘What ate the horse?’ as an example that is truly potentially ambiguous between subject and object and a subject question, the latter reading salient in the context of a story with monsters, for example.

That factors such as intonation and pragmatics should play a role in affecting parsing preferences should come as no surprise, of course. However, E-S proposes that (1) is responsible for a wider range of phenomena than the obvious cases of preferences in ambiguity resolution. Thus, if I have read the account correctly, (1) – and only (1) – is also responsible for excluding a variety of unwanted derivations in OS constructions, including those that yield the examples in (4).

- (4) a. *Peter ham mødte ikke.
 Peter him met not
 ‘Peter didn’t meet him.’ (E-S (6))
 b. *I går mødte ham Peter ikke.
 Yesterday met him Peter not
 * as ‘Sara met Peter’ (ES (12))

As reported in the literature, including E-S’s paper, examples such as those in (4) are apparently sharply ungrammatical despite the absence of any ambiguity.⁴ Compare (4a) with (2a). In the account of (2a), the accusative form of the pronoun serves to resolve any potential ambiguity and thus prevents (1) from applying, but in (4a), although the pronoun is the same unambiguous accusative form, the ungrammaticality is attributed to (1).⁵

⁴ Note that (4a) does not violate the V2 constraint within E-S’s system, as the sequence *Peter-ham* is a single (prosodic) constituent derived by prosodic incorporation, prior to topicalization.

⁵ The crux of the account is a requirement that no element with Φ -features intervene between the verb and the subject; I take it from E-S’s discussion below her (17) that this is intended to be derivable from (1). Even if not, my scepticism regards treating (4) as displaying the effects of a parsing constraint in the same manner as the interpretive preferences in (2b), independent of the particular parsing constraint held responsible. Note incidentally that in Icelandic, it is possible to have elements with Φ -features between the verb and the subject (and between the verb and the agreeing nominative, when the nominative is not a subject). Debates about the possibility of OS across a subject aside, indefinite subjects may extrapose to the right of the VP, as argued by Bresnan & Thráinsson 1990; yet Icelandic shares the Danish judgments on examples like (4).

To my reading then, just as it strikes me as plausible to attribute certain dispreferences to parsing strategies, it looks implausible that a pressure to resolve ambiguity between subject and object in parsing can be behind (4). Put more generally, if parsing preferences are truly responsible for the exclusion of (4) – whatever those strategies are – one might expect marked improvement in acceptability as additional factors such as intonation, case and pragmatics come into play. (Alternatively, if parsing constraints are held to be inviolable, we would expect the absence of amelioration in the well-discussed cases of ambiguity resolution.) Apparently, though, this is not what we find.⁶ In sum, my hunch is that phonology does have a role to play in the explanation of word order variation, and I feel that E-S's article has provided some intriguing new directions to explore as to how that might play out, but at this point, I remain unconvinced that the particular implementation of a functional (parsing) motivation avoids lumping together two qualitatively different classes of phenomena: those in which judgments of acceptability are readily influenced by pragmatics and other factors, and those where judgments appear to be of a more categorical nature.

2. CycLin

While E-S looks at variation within and among the Scandinavian languages, F&P present a theory which links Holmberg's Generalization (and an inverse HG effect) to the same considerations that force movement to be successive cyclic cross-linguistically, in particular in English.

⁶ As E-S notes, the construction in (4b), called Long Object Shift, is of varying acceptability in Swedish, and with more limited distribution in Icelandic. E-S treats its acceptability as a problem, and suggests an alternative direction, though one in which (1) nevertheless plays a role, as evidenced by the observation that "only object pronouns with unambiguous overt case marking are licensed in this" construction (p. 73), (but see Jónsson 1996, 53 for an Icelandic example where case is not distinguished). The important point here is that (1) alone, and the ameliorating effects of overt case-marking, cannot be the entire account of the Swedish-Danish contrast here; if Long Object Shift violates (1), it should do so in all the languages; if case-marking avoids the violation, then (4b) should be acceptable in Danish, as in Swedish. As a final note on this point, if we were to grant that the account of the Swedish-Danish difference lies elsewhere, but maintain that (1) is responsible for restrictions on Long OS in Swedish, a novel prediction emerges. Specifically, the restriction should not be precisely that the object pronoun

The goals that F&P set are quite ambitious, and include the elimination of special stipulations from the theory, arguing on the one hand that Holmberg's Generalization effects – and their distribution – fall out from the basic architecture of the theory, and on the other, that the theory need not stipulate designated escape hatches for successive cyclic movement. While the paper covers an impressive range of ground, there are some relatively tacit assumptions which bear further scrutiny.

Consider first the key parts of the analysis that are left to be filled in by more or less familiar syntactic assumptions. At the core of the CycLin proposal is the establishment of inviolable ordering statements at intermediate stages of the derivation. The two relevant phases are CP and VP. While syntactic operations beyond the phase may affect elements within the first phase, the relative order among the members of the first phase is immutable. Thus in a VO language, simple *wh*-fronting of the direct object places it in a position preceding the main verb in the CP phase; the only way for this to happen under CycLin is for the object to move through an intermediate position where it comes to precede the verb within the VP phase, indicated as *t'* in (5).

- (5) a. John has [VP read a book].
 b. What has John [VP *t'* read t]?

Without the intermediate trace, the VP in (5b) would determine the ordering statement $V < [\text{precedes}] \text{OBJ}$ at the VP phase level. Subsequent *wh*-movement of the object would yield $\text{OBJ} < \text{VP}$, which in turn gives $\text{OBJ} < V$ (i.e., since *V* is the first element of VP), a contradiction. The intermediate step avoids this, by establishing $\text{OBJ} < V$ in the first phase, an ordering which is respected by all further movements. The intermediate trace is key to allowing the long movement, but as F&P point out, “[their] proposals say nothing in themselves ... about the circumstances under which movement to these left-edge positions is allowed or prohibited” (p. 39). One must look to familiar syntax to fill in the relevant assumptions here.

bear overt case-marking, but only that either the object or the subject bear such case marking (this is explicitly claimed for V2 ambiguity resolution, see her (17b)); thus pronouns like *dom* ‘they/them’ which cannot undergo Long OS, should be able to do so when the subject is pronominal and marks case. It would be an interesting feather in this approach’s cap if that prediction could be shown to be correct.

This raises the concern that their claim to avoid syntactically designated escape hatches (p. 4) may be undermined by the admission of precisely such stipulations through the back door of an as-yet-to-be-specified syntax. I illustrate the concern here with two domains in which the concern arises, one for which the *CycLin* model might in fact suggest a fresh approach, and one in which the tacit stipulation looks (to me) less likely to fall soon to a principled analysis.

2.1. *Escape hatch asymmetries*

One aspect of the *Barriers* model (Chomsky 1986), was an asymmetry between CP and VP domains. By means of relatively unconstrained adjunction possibilities, the VP was relatively porous and allowed for multiple elements to be extracted. By contrast, the assumption that CP tolerates maximally a single specifier, along with the Adjunction Prohibition (no adjunction to CP in complement position), effectively limits extraction from a given CP to a single, designated escape hatch, namely Spec,CP. The types of facts that motivated this distinction remain to be accounted for. To start, if extraction from VP must move through the edge, then the left periphery of the VP appears to tolerate multiple landing sites for intermediate traces (positions that are not accessible as final landing sites). This is illustrated in (6); if the corresponding active clause is a reliable guide to base positions, then the adverb *derfor* ‘therefore’ and the NP have reversed their order with respect to each other and with respect to the main verb/participle; thus, both must thus have undergone intermediate movements to the left periphery of the VP.⁷

- (6) Derfor blev Marie ikke [VP t'_{ADV} t'_{NP} fyret t_{DP} t_{ADV}].
 therefore was Marie not fired
 ‘Marie was not fired for this reason.’ (Danish, Vikner 1995: 24)

By contrast, there is evidence that only a single extraction is possible from CP. For example, V2 induces an island for extraction of anything

⁷ Vikner claims that this example is ambiguous and in particular allows a reading in which ‘therefore’ is extracted from a low position below negation; this contrasts with *wh*-extraction of ‘why’, which, in a parallel sentence is unambiguous and lacks the ‘low’ reading of ‘why’.

other than the fronted topic (see Vikner 1995, 108–119).⁸ The German examples in (7) illustrate.

- (7) a. Ich glaube [_{CP} nur Bananen hat der Hans dem Peter t
I believe only bananas has Hans.N Peter.D
gegeben.]
given
'I believe that Hans gave Peter only bananas.'
- b. Wem glaubst du [_{CP} t' hat der Hans t Bananen
Who.D believe you has Hans.N bananas
gegeben?]
given
'Who do you believe Hans gave bananas to?'
- c. *Wem glaubst du [_{CP} nur Bananen hat der Hans t_{wh} t_{NP}
Who.D believe you only bananas has Hans.N
gegeben?]
given
'Who do you believe Hans gave only bananas to?'

Example (7a) shows embedded V2 topicalization (with a bridge verb), and (7b) shows the possibility of extraction from the fronted topic position; (7c) shows that extraction across a topic is impossible. Facts like this are traditionally attributed to the status of Spec,CP as the unique escape hatch at the CP phase. At first blush, then, it would seem that the kinds of assumptions (uniqueness of specifiers – or equivalently, the features that license them – and the ban on adjunction to complement CP) that effectively designate specific positions as escape hatches continue to be necessary.

In thinking more carefully about this, it has struck me that, at least for the specific case of V2 islands, *CycLin* may in fact provide a different way of looking at the puzzle. If the verb-second constraint may be stated as a restriction on ordering *at the CP phase*, this would effectively exclude all but a single position to the left of the verb (whether as an adjunct or multiple specifier). In (7c), for example, the framework requires that there be

⁸ In languages in which embedded V2 occurs under a complementizer, extraction of the fronted topic is also prohibited as (arguably) a special case of the *that-t* filter. In languages like German, embedded V2 occurs without CP recursion and only the topic is accessible for further extraction.

an intermediate trace within CP to the left of the fronted topic; without such a trace, extraction into the matrix clause (via the edge of the matrix VP) would necessarily derive an ordering contradiction.⁹ However, if the V2 requirement holds as an ordering statement at the CP phase, the structure required for subsequent extraction will necessarily violate the V2 constraint – the intermediate position does not become a trace until a subsequent cycle. These considerations will draw the right distinctions in (7), in particular admitting (7b) as V2 is satisfied at the CP phase and subsequent extraction takes the leftmost element from that phase. For the specific case of V2 islands, then, the concern about the syntactic designation of escape hatches may be avoided; the larger question will be whether there is indeed a general asymmetry between CP and VP phase, one which will require the same kinds of stipulations made in previous frameworks.¹⁰

2.2. *Is OS special?*

A second area where syntax is left unspecified lies in the account of Holmberg's Generalization, and here a way out is less readily discernable. Key to F&P's analysis is the assumption that the left-peripheral position in the VP (e.g., in (5b)) is not available for OS. This is illustrated schematically in (8) (compare F&P's (21) and (22)); (8a) shows that OS out of the VP when the verb has not raised (as in embedded clauses and compound tenses) will reverse the linear order of the object and the (main) verb, leading to an ordering contradiction.

- (8) a. *... O Adv [VP V t_O]
 b. ... O Adv [VP t'_O V t_O]

⁹ Keeping in mind that most intermediate trace positions cannot be occupied by overt elements, the fact that multiple overt positions preceding the verb are disallowed tells us nothing directly about the availability of intermediate trace positions there.

¹⁰ Susi Wurmbrand draws my attention to Zaenen 1985, where it is shown that extraction from embedded questions is freely available in Icelandic, unlike extraction from embedded topicalization structures. Zaenen relates (p. 194) the extraction asymmetry to another difference, namely that embedded *wh*-questions also fail to evidence subject-verb inversion (V2). At first blush, the account sketched in the text in which islandhood is attributed to the V2 constraint would seem to predict this difference. I do not know at this time if the account generalizes.

However, as F&P note in discussing their (18), if the left-peripheral position in VP were available (as in (8b)), movement to that position would reverse the order inside the VP phase (parallel to A'-extractions), leaving the object readily accessible to further leftwards movement, including OS, and no HG effects would be expected. Since OS is blocked in these environments, the intermediate position must be unavailable.

So why is this position unavailable for OS? The account will lie in the syntax, but what assumptions there might achieve this result? The only hint given is on p. 16, where it is implied (perhaps unintentionally) that the intermediate trace position is only available to A'-movements. At face value, though, this doesn't seem adequate. Other A-movements of underlying objects in Scandinavian, such as passive and unaccusative, do reverse the linear order of the verb and its complement. This was shown for passive in (6). Note importantly that (6) has all the ingredients for cross-phasal movement of the NP *Marie*, specifically, the movement crosses negation, which F&P explicitly take to lie outside the VP phase.

Note moreover that an appeal to a distinction between active and passive VPs, treating the latter, for example, as non-phases, seems unlikely to help.¹¹ Consider in particular double-object passives, and quirky subject constructions with a nominative object (the latter are arguably unaccusative in lacking an underlyingly external argument). In double-object passives, the dative moves across the verb, becoming a "quirky" subject, as shown in (9) (see Zaenen et al. 1985 among others).

- (9) Um veturinn voru konunginum [_{VP} t'_{PP} t'_{NP} gefnar t_{NP}
 In the.winter were.PL the.king.D given
 ambáttir t_{PP}].
 slaves.N

'In the winter, the king was given (female) slaves.'

(Zaenen et al. 1985 [1990]: 112)

Nevertheless, other order-preservation effects, including HG, are reported in DAT-NOM constructions, just as for active, transitive clauses (see Harley 1995, Jónsson 1996, 118, 147). In other words, passive and

¹¹ Since F&P explicitly argue that VP not vP constitutes the relevant domain in the Scandinavian languages, this would also look less promising than, for example, Chomsky's appeal to differences at the vP level.

unaccusative contexts allow A-movement of the (derived) subject to cross the verb, but appear to retain the other order-preserving properties attributed to VP phase-hood in F&P. It seems, then, that passive and unaccusative A-movement, and all kinds of A'-movement, have available to them an intermediate landing site in the left periphery of the VP.¹² Thus there is no general A/A' asymmetry here. Rather, it is only OS for which this position is unavailable, and it is this unavailability on which the account of HG crucially depends. While the account is quite elegant, a key component rests on an assumption that, for the moment at least, seems unfortunately *ad hoc*.

2.3. Generalizations and trade-offs

Scandinavian passives bring me to a final point regarding the scope of Holmberg's generalization and of order-preservation effects. As just mentioned, F&P use the same mechanisms to account for order-preservation restrictions holding between the verb and its internal arguments (standard HG effects) and those that obtain among direct and indirect objects, and particles (the classic HG effects). Thus, following Holmberg, F&P relate the impossibility of OS of the DO across an IO (as in (10a)) to the same mechanism that prohibits OS of the DO across the verb (ultimately, the unavailability of the left-peripheral landing site in VP). F&P's system also accounts nicely for order-preservation effects when both objects shift, as shown for Danish in (10b–c) from Müller 2001 (see also Collins & Thráinsson 1996 for Icelandic).

- (10) a. *Peter viste den jo [VP t_V Marie t_{DO}]
 Peter showed it indeed Marie
 ‘Peter indeed showed it to her.’
 b. Peter viste hende den jo [VP t_V t_{IO} t_{DO}]
 Peter showed her it indeed
 c. *Peter viste den hende jo [VP t_V t_{IO} t_{DO}]
 Peter showed it her indeed

¹² Since other A-movements appear to make use of this intermediate position, an appeal to improper movement to exclude adjunction to VP on the way to a higher A-position will not suffice.

Since, by hypothesis, order-changing movement in the VP is impossible as there are no intermediate landing sites in OS, the only admissible derivation is the one which preserves the VP-internal order as indicated by the traces. Note, though, that in Danish (and Icelandic), passive shows the same order-preservation effect between the two internal arguments: just as the DO may not undergo OS across the IO (10a), the DO is also prohibited from undergoing passive movement across the IO (11) (Holmberg & Platzack 1995: 215).

- (11) a. Jens blev [VP t'_{NP} givet t_{IO} bogen]
 Jens was given the.book
 'Jens was given the book.'
 b. *Bogen blev [VP t'_{NP} givet Jens t_{DO}]

F&P cannot provide a unified account of (10) and (11). The pair in (11) shows that the relative order of NP and verb may be changed while in the same construction, the relative order of the two objects is immutable. The acceptability of (11a) necessitates a left-peripheral intermediate trace for passive movement; it thus cannot be the unavailability of such a position per se that excludes the crossing movement in (11b).

Anagnostopoulou (2003) and Bobaljik (2002) use such contrasts to argue against Holmberg's (1999) formulation of Holmberg's Generalization (adopted by F&P) in which order preservation effects between the verb and the object have the same account as those among direct and indirect objects. The examples in (11) show that this is incorrect. F&P recognize this (fn. 21) and argue that the uniformity of their proposal across the constructions identified by Holmberg (including the V-topicalization construction, which neither Anagnostopoulou 2003 nor I account for) argues in favour of their proposal. Yet the cost is that they exclude the possibility of a unified account of (10) and (11). We are left with competing, overlapping, but ultimately incompatible empirical generalizations – on current understanding, one must be spurious.

I believe there is at least some reason to suspect that the order-preservation effects in (10b–c) should in fact be grouped together with (11) and not with the verb-movement asymmetries that form the core of (the extended) Holmberg's Generalization. This evidence comes from variation within Scandinavian. It is not the case that all Scandinavian languages show the strict order-preservation effects in double object shift

and in double object passive. Norwegian in particular is a symmetric passive language, allowing both (11a) and (b) (Holmberg & Platzack 1995: chapter 7 – they note that Swedish allows the (b) example “somewhat marginally”, p. 215). Similarly, at least some speakers of Norwegian and Swedish allow (10c) alongside (10b) (Tarald Taraldsen, personal communication 06/04, see also Anagnostopoulou 2003: 125–127 and fnn. 40–41; and E-S, n. 17). Anagnostopoulou notes moreover that at least some Swedish (and Norwegian) speakers allow constructions parallel to (10a), with OS across an unmoved IO, when both arguments are pronouns, as illustrated in (12) (judgments attributed to Holmberg, see also Hellan & Platzack 1995: 58).

- (12) Jag gav den inte [VP honom t_{DO}].
 I gave it not him.
 ‘I didn’t give it to him.’

Importantly, while there is variation reported for constructions like (10) and (11) (and for other domains in which order preservation appears to hold, including particle constructions and raising constructions), the variation in double-object constructions appears to track that in the passive (at least at a gross level). Moreover, there is no such variation reported for the canonical HG effects involving the verb. These facts have suggested to me, and to Anagnostopoulou, that there is a principled line to be drawn dividing the constructions, a line that is inconsistent with the formulation of the generalizations offered by Holmberg and by F&P. To be sure, neither Anagnostopoulou (2003) nor Bobaljik (2002) provide a complete account, nor is the empirical situation entirely clearly described.

At this stage in the research, then, the disagreements I have with E-S and F&P lie largely in educated guesses about what the proper descriptive generalizations might turn out to be. Having competing theories on the table draws attention clearly to the points of difference among perspectives, and should lead to yet another fruitful period for the development of the theory. As noted, I find the expanded role for PF in accounting for precedence relations (word order) to be an intriguing direction, and I look forward to the further elaboration of each of the theories sketched in the target articles.

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References

- Anagnostopoulou, E. (2003). *The syntax of ditransitives: evidence from clitics*. Berlin, Mouton de Gruyter.
- Bobaljik, J. D. (2002). "A-chains at the PF-interface: Copies and 'Covert' Movement." *Natural Language and Linguistic Theory* 20(2): 197–267.
- Bresnan, J. and H. Thráinsson (1990). A note on Icelandic coordination. In: J. Maling and A. Zaenen (eds.), *Modern Icelandic Syntax*. San Diego, Academic Press. 24: 355–365.
- Chomsky, N. (1986). *Barriers*. Cambridge, Mass., MIT Press.
- Collins, C. and H. Thráinsson (1996). "VP-Internal Structure and Object Shift in Icelandic." *Linguistic Inquiry* 27: 391–444.
- Harley, H. (1995). Abstracting away from abstract case. In: J. Beckman (ed.), *Proceedings of the North Eastern Linguistics Society Annual Meeting (NELS 25)*. Amherst, GLSA: 207–221.
- Hellan, L. and C. Platzack (1995). "Pronouns in Scandinavian: an overview." *Working Papers in Scandinavian Syntax* 56: 47–69.
- Holmberg, A. (1999). "Remarks on Holmberg's Generalization." *Studia Linguistica* 53(1): 1–39.
- Jakobson, R. (1936 [1971]). Beitrag zur allgemeinen Kasuslehre. Gesamtbedeutungen der russischen Kasus. *Selected Writings*. The Hague, Mouton. 2: 23–71.
- (1963 [1971]). Implications of language universals. *Selected Writings*. The Hague, Mouton. 2: 580–592.
- Jónsson, J. G. (1996). *Clausal architecture and case in Icelandic*, Ph.D. Dissertation, Amherst, University of Massachusetts.
- King, T. H. (1995). *Configuring Topic and Focus in Russian*. Stanford, CSLI.
- Müller, G. (2001). Order Preservation, Parallel Movement, and the Emergence of the Unmarked. In: G. Legendre, J. Grimshaw and S. Vikner (eds.), *Optimality-Theoretic Syntax*. Cambridge, Mass., MIT Press: 279–313.
- Ørnes, B. (2002). Case marking and subject extraction in Danish. In: M. Butt & T. H. King (eds.), *Proceedings of the LFG02 Conference*, Stanford, CSLI: 333–353.
- Schlesewsky, M., G. Fanselow, and S. Frisch. (2003). Case as a trigger for reanalysis: Some arguments from the processing of double case ungrammaticalities in German. In: S. Fischer, R. van de Vijver and R. Vogel (eds.), *Experimental Studies in Linguistics 1*. Linguistics in Potsdam 21: 31–60.
- Schlesewsky, M., G. Fanselow, R. Kliegl and J. Krems. (2000). Subject preference in the processing of locally ambiguous questions in German. In: B. Hemforth and L. Konieczny (eds.), *German Sentence Processing*. Dordrecht, Kluwer: 65–93.
- Vikner, S. (1989). "Object shift and double objects in Danish." *Working Papers in Scandinavian Syntax* 44: 141–155.
- (1995). *Verb Movement and Expletive Subjects in the Germanic Languages*. Oxford, Oxford University Press.
- Zaenen, A. (1985). *Extraction rules in Icelandic*. New York, Garland.
- Zaenen, A., J. Maling and H. Thráinsson. (1985). "Case and grammatical functions: The Icelandic passive." *Natural Language and Linguistic Theory* 3: 441–483.

Some Remarks on Fox and Pesetsky: “Cyclic Linearization of Syntactic Structure”

MOLLY DIESING

1. Introduction

Fox and Pesetsky (F&P) make the claim that certain constraints on Object Shift and Quantifier Movement in Scandinavian can be explained within their model of the mapping between syntax and phonology. Among the salient features of this model are: Spell-out occurs phase-by-phase (as in recent work by Chomsky 2000, 2001a,b), and linearization relations established at each Spell-out point must be preserved (F&P’s property of Order Preservation). As F&P demonstrate quite elegantly, this system enforces successive cyclicity in *wh*-movement, in that non-successive-cyclic derivations will result in an ordering contradiction (violating Order Preservation). Thus, the “cyclicity” of successive cyclicity is shifted to being a by-product of the cyclic spelling-out process. One might still ask why Spell-out itself is cyclic, but this is certainly a welcome simplification.

The connection with the various cases of Scandinavian object movement arises from certain novel predictions made by the F&P system. One of these is that in a domain *D* such as illustrated below:

(1) $\alpha [{}_D X Y Z]$ ordering: $X < Y$; $Y < Z$; α merges with *D*

X and *Y* should be able to move out of *D* so long as *X* and *Y* preserve their original order. As F&P note, in standard approaches to movement this would generally require some sort of “escape hatch” such as multiple specifiers. Or in the case of object shift (where a verb and an object NP move) some special assumption may come into play – such as Equidistance, as in Chomsky (1995). For F&P, these things can be

dispensed with, since linearization at the interface with phonology (Order Preservation) is all that is at stake.

For object shift, this does look like a welcome result, for as Holmberg (1999) has pointed out, the earlier Equidistance account failed to explain certain classes of examples, including those in which the main verb undergoes topicalization rather than raising to C, and cases where there are non-verb interveners (such as particles or an indirect object) blocking object shift even when the main verb moves. Indeed, Holmberg himself raised the possibility of a linearization-based account (as an alternative to Equidistance) to deal with these problems, giving up in the face of some rule-ordering puzzles (which F&P seem to have unraveled).

2. Ditransitives

In the domain of object shift there is one case that strikes me as still potentially problematic for linearization, and that is the situation with ditransitives in Icelandic. As noted by Holmberg (1999), object shift with ditransitives shows an order preservation effect in that a direct object cannot shift over an indirect object (controlling for the possibility of “inversion” of the objects with certain verbs, see Collins and Thráinsson 1996), but shifting both objects, *even while preserving order* yields a rather degraded sentence. Although Collins and Thráinsson do not indicate any ill-formedness for examples of this type, Holmberg (1999) indicates the acceptability of (2b) as “??”, and Ellert Jóhannsson (p.c.) finds it to be quite ungrammatical (hence my giving it the ?* rating):

- (2) a. Hann skilar bókasafninu aldrei bókunum.
 he returns library-the never books-the
 ‘He never returns the books to the library.’
 b. ?*Hann skilar bókasafninu bókunum aldrei.
 he returns library-the books-the never
 c. *Hann skilar bókunum/þeim aldrei bókasafninu
 he returns books-the/them never library-the

As Holmberg points out, constructions like (2b) improve if the direct object is a pronoun, so it is not simply the case that shifting two NPs is impossible. Holmberg explains this particular range of data in terms of the conditions on licensing his feature [-Foc]. But, it is not clear to me

how a simple linearization account can satisfactorily account for these differences in and of itself. Of course, an explanation may lie somewhere in “syntax proper” – or elsewhere beyond the interface of syntax and phonology that lies at the heart of the F&P proposal (though the conditions that alleviate the ungrammaticality strike me as typical of the syntax-phonology interface) – but there lies the greatest challenge in giving a full account of object shift: determining which aspects of the phenomenon are to be explained by which parts of the grammar.

3. What counts as an intervener?

Before returning to another facet of the question of object shift with ditransitives below, I wish to raise another puzzle for linearization. This concerns cross-linguistic variation in what constitutes an “intervener”, or “non-verb blocker” for object shift. The first possible case of such variation concerns particles. As is well known, particles block object shift in Swedish, though not in Danish or Icelandic (the first example illustrating this in Swedish is from F&P, the others in (3) I have taken from Vikner 2004).

- (3) a. *Dom kastade mej inte [_{VP} t_v ut t_o]
 They threw me not out
 (Swedish)
- b. Peter smed det ikke [_{VP} t_v ud t_o]
 Peter threw it not out
 ‘Peter didn’t throw it out’
 (Danish)
- c. Pétur henti henni ekki [_{VP} t_v út t_o]
 Peter threw it not out
 ‘Peter didn’t throw it out’
 (Danish)

The lack of blocking by the particle in Danish and Icelandic is easily explained within the F&P linearization account in terms of word order variation *within the VP* prior to Spell-out. Specifically, in Danish (unlike Swedish), the particle always follows its complement, regardless of whether object shift has taken place, and in Icelandic either order of particle and NP is possible inside VP (again, I use examples from Vikner 2004):

- (4) a. *Peter har ikke smidt ud tæppet
 Peter has not thrown out carpet-the
 b. Peter har ikke smidt tæppet ud
 Peter has not thrown carpet-the out
 ‘Peter has not thrown out the carpet.’
 (Danish)
- (5) a. Pétur hefur ekki hent út mottunni
 Peter has not thrown out carpet-the
 ‘Peter has not thrown out the carpet.’
 b. Pétur hefur ekki hent mottunni út
 Peter has not thrown carpet-the out
 ‘Peter has not thrown out the carpet.’

Clearly, if the particle must follow the object in the VP (as in Danish) or can follow the object (as in Icelandic), it will not create the conditions required for intervention within the F&P system – no ordering contradiction need arise when object shift takes place.

But other cross-linguistic differences in what constitutes an intervener are not so easily explained away. In addition to Object Shift, Swedish also has Long Object Shift (LOS, discussed in detail by Josefsson 1992), where a weak object pronoun can optionally appear between the finite verb and the subject in a verb second main clause with inversion.

- (6) a. Då gav honom Eva förmodligen inte några pengar.
 then gave him Eva probably not any money
 ‘Then Eva probably didn’t give him any money.’
 b. Då gav Eva honom förmodligen inte några pengar.
 then gave him Eva probably not any money
 ‘Then Eva probably didn’t give him any money.’
 (Börjars et al. 2003, 5)

This Long Object Shift is not possible in Danish or Norwegian, nor does Icelandic allow shifting of an object over a subject. Thus, a subject NP is an “intervener” in Danish, Norwegian and Icelandic, but not in Swedish. This is unexpected under a linearization account. Indeed, it’s not clear that (6a) is expected to be grammatical at all in the F&P model. The construction is admittedly somewhat marked and subject to various restrictions – for example, the subject must be a full NP, not a pronoun,

and only object pronouns that bear unambiguous morphological case marking can undergo LOS (see Josefsson 1992 for further details). But these restrictions merely suggest that LOS has the properties of a process that applies at the interface between phonology and syntax, and thus is certainly relevant here.

4. An interaction between OS and QM

One of the most interesting results of F&P is the prediction that there should be “inverse Holmberg effects” in VO languages. An inverse Holmberg effect arises when there is object movement disrupting the order of object and verb *within VP* without subsequent leftward movement of the object to a position higher than C. That is, at the point of spell-out of the VP, the object would precede the verb. In this case, verb raising to C would then yield an ordering contradiction, and should therefore be blocked. F&P claim that the phenomenon known as Quantifier Movement (QM) in Icelandic is in fact an instance of such an inverse Holmberg effect. It has been observed that QM generally fails to conform to Holmberg’s Generalization (Rögnvaldsson 1987, Jónsson 1996, Svenonius 2000). It affects negative objects obligatorily:

- (7) a. *Strákarnir höfðu hent engu grjóti í bílana.
 boys-the had thrown no rock in cars-the
 ‘The boys had thrown no rocks at the cars.’
 b. Strákarnir höfðu engu grjóti hent í bílana.
 boys-the had no rock thrown in cars-the
 (Svenonius 2000, (12))

Quantified NPs, on the other hand, move optionally, and with a slightly degraded result:

- (8) a. Strákarnir höfðu hent miklu grjóti í bílana.
 boys-the had thrown much rock in cars-the
 ‘The boys have thrown many rocks at the cars.’
 b. ?Strákarnir höfðu miklu grjóti hent í bílana.
 boys-the had much rock thrown in cars-the
 ‘The boys had thrown many rocks at the cars.’
 (Svenonius 2000, (13))

F&P use the following examples from Rögnvaldsson (1987) to illustrate the inverse Holmberg effect:

- (9) a. Jón hefur ekkert sagt Sveini.
 Jon has nothing said Svein
 ‘Jon has told Svein nothing.’
 b. *Jón sagði ekkert Sveini.
 Jon said nothing Svein

In the first example, with no verb movement out of the VP to C, the movement of the negative phrase *ekkert* ‘nothing’ to the left of the verb is perfectly grammatical (in fact, it is obligatory). But in the second example, where the verb has moved out of the VP to C, changing the ordering relation between the QP and the verb, the result is ungrammatical. F&P note (in their footnote 26) that there is an alternative explanation for the ungrammaticality here, in that the indirect object *Sveini* ‘Svein’ has not undergone object shift, even though it is possible to do so. They dismiss this alternative as unlikely, citing an example from Jónsson (1996) (with two definite NPs) in which OS of the indirect object is optional. Rögnvaldsson’s original presentation of the paradigm in (9), however, includes the example with the shifted indirect object, which is grammatical:

- (10) Jón sagði Sveini ekkert.
 Jon said Svein nothing
 ‘Jon told Svein nothing.’

And Jónsson’s (1996) characterization of the data in (9) and (10) is that object shift of the indirect object is in fact obligatory when the direct object is negative. This may seem like a mysterious stipulation, but if the semantically-conditioned nature of OS is taken into account, it is not surprising. Whether cast in terms of type-driven movement as in Diesing and Jelinek (1995), or Holmberg’s (1999) licensing of [-Foc] or Chomsky’s (2001b) INT’, the definite indirect object will be forced out of the scope of the negative quantifier, making OS obligatory in this instance (there may well be cases where OS is in fact not optional, in other words).

But we still don’t know if this has any bearing on whether or not QM is incompatible with verb movement, since we can’t really tell if the negative NP in (10) has undergone QM. Having shifting the indirect object over

the direct object, we have now lost our only “signpost” indicating the VP boundary. Stringwise, the negative NP could just as easily be down in its base position as in some position to the left of VP (where one would expect it to be if QM were outright impossible in this case). Modifying the paradigm in (9) and (10) by adding the adverb *í gær* ‘yesterday’, in (11a) we see that the result is grammatical (Ellert Jóhannsson p.c.).

- (11) a. Aðalheiður sagði Gunnlaugi ekkert í gær
 Aðalheiður said Gunnlaugur nothing yesterday
 ‘Aðalheiður told Gunnlaugur nothing yesterday.’
 b. *Aðalheiður sagði í gær Gunnlaugi ekkert
 Aðalheiður said yesterday Gunnlaugur nothing
 c. *Aðalheiður sagði Gunnlaugi í gær ekkert
 Aðalheiður said Gunnlaugur yesterday nothing

The judgments (11b) and (11c), while not conclusive (adverbs are rather mobile things), are suggestive of the possibility that both OS and QM are in fact required in this instance, as claimed by Jónsson (1996). What is surprising is (11a). Here we have a sentence with (a) verb movement, (b) object shift, and (c) quantifier movement all co-occurring. The co-occurrence of verb movement and QM is apparently dependent on (obligatory) OS, QM and verb movement being an illicit combination otherwise.

But is it an illicit combination? F&P note that QM is not *absolutely* incompatible with verb raising, but claim that it will only be permitted if the quantifier phrase moves sufficiently leftward to restore the original ordering relation established by the original application of QM, in which the QP/NegP *precedes* the verb. This situation arises with QM out of an infinitival, which is incompatible with movement of the higher verb (I repeat here F&P’s example (41)):

- (12) a. ?Hann hefur engum skipaði henni að giftast.
 he has nobody ordered her to marry
 ‘He hasn’t ordered her to marry anybody’
 b. *Hann skipaði henni engum [_{VP} t_V [_{CP} að giftast ____].
 he ordered her nobody to marry

But Johnson and Vikner (1994) have claimed that infinitival verbs raise to T in Icelandic, based on examples from Sigurðsson (1989):

- (13) a. *María lofaði að lesa ekki bókina.*
 Maria promised C to-read not book-the
 ‘Maria promised to not read the book.’
 b. *María lofaði að ekki lesa bókina.*
 Maria promised C not to-read book-the
 ‘Maria promised to not read the book.’

That is, the embedded infinitival in (12a) may raise, but QM can still take place, since it will result in the negative NP being to the left of the embedded verb in the end.

Example (11a) is not a case of this type, however. There is no subsequent leftward movement of the negative NP restoring its position to the left of the verb. Verb movement genuinely disrupts the linear order in this case – something the F&P system does not predict to be permissible. As I mentioned above, the determining factor seems to be the obligatory object shift that must take place in this configuration, and the semantically-conditioned nature of OS may be key here. Another thing worth bearing in mind is that the application of OS in this instance has the effect of returning the arguments to their base order, which leads me to suspect that linearization may in fact be at the bottom of this puzzle too, but clearly there’s still some tinkering to be done.

As a final point, I note a further complication, which is that there are other cases where QM appears to be possible with verb movement, even without the co-occurrence of OS. For example, when the quantifier is the complement of an adjective (Jónsson 1996, 92):

- (14) a. *Jón er engum líkur.*
 Jon is nobody like
 ‘Jón is like nobody else.’
 b. *María er engum háð.*
 Maria is nobody dependent
 ‘Maria is dependent on nobody.’

Another instance involves QM out of a PP (Jónsson 1996, 93); (15b) with QM is somewhat degraded, but the non-QM alternative in (15c) is even less acceptable:

- (15) a. *Ég hef engan talað við.*
 I have no one spoken with
 ‘I have spoken to no one.’

- b. ? Ég talaði engan við.
 I spoke nobody with
 ‘I spoke with nobody.’
- c. *? Ég talaði við engan.
 I spoke with nobody

These cases cast some doubt on the characterization of QM as an “inverse Holmberg effect”, at least in terms of its effects on linear order.

In summary, I find the F&P architecture to be a welcome development in that it does seem to offer the potential for both simplifications and new insights. While in the area of object shift/quantifier movement there may still be some puzzles to be solved, these strike me more as material with which to refine the model, rather than grounds for raising serious doubts.

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References

- Börjars, K., Engdahl, E., and Andréasson, M. (2003). Subject and object positions in Swedish. In: Butt, M. and King, T. H. (eds.) *Proceedings of the LFG03 Conference*, Stanford, CA: CSLI Publications.
- Chomsky, N. (1995). *The Minimalist Program*. Cambridge, Massachusetts: MIT Press.
- (2000). Minimalist inquiries. In: Martin, R., Michaels, D. and Uriagereka, J. (eds.). *Step by Step: Essays on Minimalist Syntax in Honor of Howard Lasnik*, pp. 89–156. Cambridge, Mass.: MIT Press.
- (2001a). Beyond explanatory adequacy. MITWPL, 20, pp. 1–28. Cambridge, Massachusetts.
- (2001b). Derivation by Phase. In: Kenstowicz, M. (ed). *Ken Hale: A Life in Language*, pp. 1–52. Cambridge, Mass.: MIT Press.
- Collins, C., and Thráinsson, H. (1996). VP-internal structure and object shift in Icelandic. *Linguistic Inquiry* 27, pp. 391–444.
- Diesing, M., and Jelinek, E. (1995). Distributing arguments. *Natural Language Semantics* 3, pp. 123–176.
- Holmberg, A. (1999). Remarks on Holmberg’s Generalization. *Studia Linguistica*, 53.1, pp. 1–39.
- Johnson, K., and Vikner, S. (1994). The position of the verb in Scandinavian infinitives: In V or C but not in I. *Working Papers in Scandinavian Syntax* 53, pp. 61–84. Department of Scandinavian Languages, University of Lund.
- Jónsson, J. G. (1996). *Clausal Architecture and Case in Icelandic*. Ph.D. dissertation, University of Massachusetts, Amherst.

- Josefsson, G. (1992). Object shift and weak pronominals in Swedish. *Working Papers in Scandinavian Syntax* 49, pp. 1–28. Department of Scandinavian Languages, University of Lund.
- Rögnvaldsson, E. (1987). On word order in Icelandic. In *Proceedings of the Seventh Biennial Conference of Teachers of Scandinavian Studies in Great Britain and Northern Ireland*. pp. 33–49. London: University College London.
- Sigurdsson, H. Á. (1989). *Verbal Syntax and Case in Icelandic*, Ph.D. dissertation, University of Lund.
- Svenonius, P. (2000). Quantifier movement in Icelandic. In Svenonius, P. (ed.) *The Derivation of VO and OV*. Amsterdam: John Benjamins.
- Vikner, S. (2004). Case #74: Object shift. In: *The Syntax Companion (SynCom)* (online publication). http://www.hum.au.dk/engelsk/engsv/syncom074/case_074.htm

Comments on Erteschik-Shir's article

LARS HELLAN

The article to my mind rightly emphasizes the importance of prosodic and discourse-dynamic factors. Applied to the Norwegian phenomena, these factors may be rendered as:

- (1) a. A pronoun is incorporated in a prosodic domain P only if its referent is included in the discourse-dynamic domain GIVEN relative to the proposition being expressed.¹
- b. Adverbial elements that can occur right-adjacent to a category C, can also occur right-adjacent to the cluster $C+\text{pron}^+$ when this cluster is a prosodic domain.²

In the present comment, I would like to suggest that once we exclude Verb-movement as a rule (or rule complex) in the grammar of Norwegian, the statements in (1), together with a few well known facts, cover a significant, perhaps the full, range of the phenomena commonly subsumed under the label 'Object shift' for this language. In presenting this state of affairs, I follow largely the array of examples discussed by Erteschik-Shir (ES). ES herself does not question the existence of Verb movement in the Mainland Scandinavian languages (although suggesting that it belongs in the phonological component), and my commentary in

¹ This applies to 'personal pronouns', all persons, genders and (both) numbers. Some of the pronouns have a reduced form which is used only under incorporation, but in their normal form, the pronouns are used either incorporated or not. Incorporated occurrences of pronouns are often referred to as 'weak pronouns' (many authors, possibly including the present one). The term suggests an inherent property, and as such is a misnomer, since only the reduced forms are inherently destined for incorporation.

² The superscript '+' in $C+\text{pron}^+$ is the 'Kleene '+'', meaning 'one or more'.

this respect is thus slightly orthogonal to, but perhaps not totally incompatible with, ES's argumentation.³

We first exemplify (1b). In a main clause, adverbs like *ikke* 'not' and *ofte* 'often' have as their main occurrence sites the positions stated in (2); the position stated in (2a), illustrated in (3), is perhaps the one most commonly cited, but the position in (2b), illustrated in (4) (with a topicalized object), is also frequently used:

- (2) Positions for adverbs like *ikke* and *ofte*:
- a. the position right-adjacent to the finite verb;
 - b. the position right-adjacent to an inverted subject
- (3)
- a. Jon sparket ikke/ofte ballen
Jon kicked not/ often ball-the
 - b. Jon hadde ikke/ofte sparket ballen
Jon had not/ often kicked ball-the
 - c. Jon ville ikke/ofte ha sparket ballen
Jon would not/ often have kicked ball-the
 - d. Hvorfor sparket ikke Jon ballen?⁴
why kicked not Jon ball-the
- (4)
- a. Ballen sparket Jon ikke/ofte
ball-the kicked Jon not/ often

³ Once the rule is there, possible interactions with other rules of course have to be regulated. I sympathize with ES's approach of trying to eliminate extrinsic rule ordering in accounting for these interactions, but will not comment on the over-arching principles invoked to achieve this. For a further comparison between a 'V-movement free' and a 'V-movement full' approach, which is beyond the scope of this commentary, an evaluation of these principles would in turn be necessary.

⁴ The wellformedness of (3d) perhaps does not comply with ES's judgment of her (23a) = (i),

(i) *igår mødte ikke Peter ham (Danish)
yesterday met not Peter him

as illformed. Maybe Danish and Norwegian differ at this point. ES attributes the ill-formedness of (i) in Danish to a failure of what she proposes as a processing constraint (ID) requiring a subject to occur to the left of an adverb. I'm not sure how that differs from saying that there simply is a requirement in the Danish grammar to this effect. Anyhow, as far as assignment of grammatical function goes, Danish and Norwegian are alike in requiring subject to precede object if both are postverbal; the difference with regard to adverb placement would seem a more minor difference.

- b. Ballen hadde Jon ikke/ofte sparket
ball-the had Jon not/ often kicked
- c. Ballen ville Jon ikke/ofte ha sparket
ball-the would Jon not/ often have kicked

The placements in (4) contrast with those in (5), showing that the site in (2b) does not generalize to any NP:

- (5) a. ??Vi ga Jon ikke/ofte ballen
we gave Jon not/ often ball-the
- b. *vi hadde gitt Jon ikke/ofte ballen
we had given Jon not/ often ball-the

(1b) is then illustrated by (6) for adverb site (2a), and by (7) for adverb site (2b), using '+' to indicate prosodic inclusion (in principle, also *ikke* could be incorporated in these sequences, while *ofte* could not):⁵

- (6) a. Jon sparket+den ikke/ofte
Jon kicked+it not/ often
- b. Jon gav+ham+den ikke/ofte
Jon gave+him+it not/ often

The difference seems reflected in a further difference in judgment: while for ES, both her (14a) and (15a) (= (iia,b)) are illformed, due to a lack of accompanying indicators as to what is the subject, the counterparts of these seem fine for Norwegian, albeit admittedly ambiguous:

- (ii) a. *Peter møtte Sara (Danish)
Peter met Sara
'Sara met Peter'
- b. *Peter kender Sara fra fjernsynet (Danish)
Peter knows Sara from TV-the
'Sara knows Peter from TV'

In Danish, *ikke* occurring before *Sara* in (iia) would disambiguate the latter as object, and if occurring after, *Sara* would be subject. These observations hold for Norwegian too. However, while ES from the Danish data deduces an absolute constraint to the effect that the only order between a subject and an adverb is for the former to precede the latter, such a position would be untenable for Norwegian, since the pattern in (3d) is fully acceptable.

⁵ These examples in (7) correspond to (23c) in ES's article. While the constructions (3)–(6) are all common knowledge in the literature, dating back to long before the generative tradition, constructions like (7) came to my awareness through ES's article.

- c. ga+hun+ham+den ikke/ofte?
gave-she+him+it not/ often
'didn't she give it to him?/ did she give it to him often?'
- (7) a. Hvorfor sparket Jon+den ikke/ofte?
why kicked Jon+it not/ often
- b. Hvorfor ga Jon+henne+den ikke/ofte?
why gave Jon+her+it not/ often

As has been noted by many, prosodic inclusion of pronouns is not restricted to the situation where a verb is the host, but can take place whatever occurs to the left of a pronoun, be it a head (like a preposition or adjective) or a non-head. (7a,b) illustrate prosodic pronoun incorporation into a non-verbal element. What makes the verbal cases attract the more attention, is the interaction with adverbial placement and the possibility of iteration; nevertheless, the cases in (7) illustrate the possible presence of these factors also when the host is a noun.

The condition on 'givenness' in (1a) is a rephrasing of ES's formulation in terms of 'secondary topic', or the negative 'not focus': the view in question is that behind any sentence uttered, part of its content constitutes an open proposition already assumed by speaker and hearer, and part of its content supplies 'new' information, filling in the open slots in the 'assumed' proposition. The 'GIVEN' domain referred to in (1a) is that part of the sentence whose content constitutes the non-open parts of the assumed proposition; the claim in (1a) is thus that prosodically incorporated pronouns are part of this domain.^{6,7}

The significance of prosodic incorporation in Norwegian resides not only in the factors mentioned (i.e., adverbial placement and iteration). As is commonly known, when a verb combines with any of the directional adverbs listed and exemplified in (8), it is possible to merge verb

⁶ And the view taken by ES (consonant with Holmberg 1986 and later work) is that – crosslinguistically – not only pronouns can undergo such incorporation under the discourse-dynamic condition in question, but also full NPs, instantiated in Icelandic.

⁷ The notion of 'given' used here is different from that of Gundel et al. (1993), where a proposed 'givenness' hierarchy reflects degrees to which a certain entity is known or familiar to speaker and hearer in a given discourse. It is fully possible for an entity to fill in the open slot (i.e., the slot extrinsic to the 'givenness domain') of a sentence while still being 'familiar' to speaker and hearer.

and adverb together in one prosodic domain, and the verb in this case is required to carry what is commonly referred to as ‘toneme 2’:⁸

- | | | |
|-------------------------------|-------|--|
| (8) <i>opp</i> ‘(to) up’ | as in | <i>Gutten springer opp</i> ‘(the boy jumps up)’
boy-the jumps to-up |
| <i>ned</i> ‘(to) down’ | as in | <i>Gutten springer ned</i> ‘(the boy jumps down)’
boy-the jumps to-down |
| <i>bort</i> ‘(to) away’ | as in | <i>Gutten løper bort</i> ‘(the boy runs away)’ |
| <i>vekk</i> ‘(to) away’ | as in | <i>Gutten løper vekk</i> ‘(the boy runs away)’ |
| <i>ut</i> ‘(to) out’ | as in | <i>Gutten løper ut</i> ‘(the boy runs out)’ |
| <i>inn</i> ‘(to) in’ | as in | <i>Gutten løper inn</i> ‘(the boy runs in)’ |
| <i>hjem</i> ‘(to) home’ | as in | <i>Gutten løper hjem</i> ‘(the boy runs home)’ |
| <i>frem</i> ‘(to)
forward’ | as in | <i>Gutten løper frem</i> ‘(the boy runs
forward/forth)’ |

In all the exemplifying sentences in (8), the verb form in isolation carries toneme 1. If, in these sentences, verb and adverb are read as a prosodic unit,⁹ the tone shifts to toneme 2. The same effect is seen when a verb is unisyllabic: in isolation it carries no toneme contrast, since the toneme contrast is only realized over two syllables, but in a combination like those in question, the syllabic sequence will show toneme 2. Likewise, if a verb prosodically incorporates a pronoun, which by itself has no tonal effect if the verb form already has toneme 1, and yields a toneme 1 sequence if the verb form by itself is unisyllabic, and the verb in turn incorporates with one of the adverbs in (8), then the resulting form has toneme 2. In a quasi-derivational view, this is indicated in (9):

- (9) a. ¹setter 'sets'
 b. ¹setter+han+den 'sets+he+it'
 c. ²setter+han+den+frem? 'sets-he-it-forward'

⁸ The requirement holds only when the verb form is initially stressed, and certain other conditions obtain as well which we need not go into here. Cf. Fretheim (1974).

⁹ Which is not obligatory. When it happens, I believe the givenness condition mentioned in (1a) is satisfied as well.

- d. ^{no toneme}
ga 'gave'
- e. ¹
ga+han+den 'gave+he+it'
- f. ²
ga+han+den+bort? 'gave+he+it+away'
- g. ¹
Jon ga+den 'Jon+gave+it'
- h. ²
Jon ga+den+bort 'Jon+gave+it+away'

Such an effect obtains only when the core of the cluster is a verb. Thus, in patterns like those in (7), the addition of a directional adverb has no effect on the tone of the incorporation host, as illustrated in (10) – here *gutten* 'the boy' in isolation has toneme 1, and keeps this in the environment of *inn*, and so does the corresponding sequence with *Jon*:

- (10) a. ¹
Når satte gutten+den+inn?
- b. ¹
Når satte Jon+den+inn?
when sat the boy/Jon+it+in

Examples like these corroborate the inclusion of prosodic incorporation in a grammar of Norwegian.

As expected from the formulation in (1b), repeated,

- (1b) Adverbial elements that can occur right-adjacent to a category C, can also occur right-adjacent to the cluster C+pron⁺ when this cluster is a prosodic domain.

a cluster formed by the inclusion of a directional adverb cannot host an adverb like *ikke*, the accepting sequence being constrained to consist exclusively of pronouns:

- (11) *Han setter+den+frem+ikke

Less expected from what we have said so far is that although a directional adverb can occur in the same construction as *ikke*, the presence of *ikke* 'bleeds' the toneme 2-effect of the adverb:

- (12) a. ²ga+han+den+bort? 'gave+he+it+away' (cf. (9f))
 b. ^{1 *2}ga+han+den+ikke+bort? 'gave+he+it+not+away'

These phenomena together present the grammar with some challenges, but mainly as far as the integration of discourse dynamic factors and prosody go. As far as the so-called 'Object Shift' phenomenon is concerned, there is actually nothing to be stated, once the general facts, repeated from above, are formally implemented:

- (1) a. A pronominal is incorporated in a prosodic domain P only if its referent is included in the discourse-dynamic domain GIVEN relative to the proposition being expressed.
 b. Adverbial elements that can occur right-adjacent to a category C, can also occur right-adjacent to the cluster C+pron⁺ when this cluster is a prosodic domain.
- (2) Positions for adverbs like *ikke* and *ofte*:
- a. the position right-adjacent to the finite verb;
 - b. the position right-adjacent to an inverted subject

The implementation of (1b) would appeal to the notion of *projection of C*, assuming that a sequence V+pron+pron+... is formally marked as 'V'. (The blocking of a toneme 2-effect in the presence of *ikke*, and the non-occurrence of *ikke* behind a directional adverb, will necessitate a differentiation between pure 'V+pron⁺'-projections on the one hand, and prosodic incorporation clusters headed by verbs and with *ikke* or a directional adverb as incorporated on the other; this circumstance does not significantly affect the simplicity of the picture at hand.) Phrasal combination would be assumed to proceed monotonically bottom up, with NP arguments being processed in their canonical order. The verb by itself, and its projection resulting from prosodic incorporation accompanying the regular head-complement accommodation, would by each such step be open for further incorporation. The canonical NP order is indirect object followed by direct object, preserved also if either one or both is incorporated, hence the incorporation effect is for each NP a purely local interaction between it and its immediate hosting projection. Subjects also take place in incorporation when they are inverted. If one assumes that – in

cases of inversion – the order with subject first among the postverbal NPs is the one ‘given’ to the incorporation process, then also subject incorporation is thereby accounted for. Thus, viewing the facts as suggested, the ‘Object Shift’ effect is simply one of local phrase building and adjacency-driven attraction.

Of course, the major premise for the ‘shift’ idea is that a finite verb moves from a position inside a tenseless VP to a ‘tense’ position in the tree, and then to a C position. The first part of this analysis arose in the late 70ies, replacing the view, stemming from Chomsky 1957, of having just an ‘Affix movement’ relating Tense to whatever happens to be the initial member of the verb sequence in a sentence. The second part arose about the same time, being essential to the idea of ‘V2’ as a common denominator of a group of languages. In Norwegian, its main task is to account for why verbs precede adverbs in main clauses, but succede them in subordinate clauses (and it necessitates a topicalization-like fronting of the subject in order for the normal ‘subject-verb’ order to obtain). If something like the ‘Affix movement’ view were still taken, finite verb and adverb had distinct phrasal combinatory rules in main and subordinate clauses, and the subject-verb inversion were treated as a simple local inversion, the analytic landscape would be essentially the one we have now reflected on. This is not the place for trying to assess if either of these opposing ‘packages’ of rules would, on independent grounds,¹⁰ be preferable to the other, but in the context of the reappraisal of factors governing prosodic pronoun incorporation, a ‘re-opening’ of the issue may suggest itself.

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¹⁰ I.e., independently of the Prosodic Incorporation phenomena. With the latter taken into account, one might refer to the total analyses considered as the ‘Object Shift (OS)’ analysis vs. the ‘Pronoun Incorporation (PI)’ analysis. Although a detailed presentation is beyond the scope of this exposition, the status of the ‘Pronoun Incorporation’ analysis is not exclusively programmatic: in a computational grammar of Norwegian (Hellan and Haugereid 2003), based on the platform ‘LKB’ (Copestake 2002), some of the patterns cited above (such as the patterns in (3), (5) and (6)) are correctly parsed/rejected and with semantic interpretations in the standard format of Minimal Recursion Semantics (Copestake et al. 2003). Relative to the issues presently discussed, such a grammar may serve as a useful base line for the PI position.

References

- Chomsky, N. (1957). *Syntactic Structures*. Mouton, The Hague.
- Copestake, A. (2002). *Implementing Typed Feature Structure Grammars*. CSLI Publications, Stanford.
- Copestake, A., D. Flickinger, I. Sag and C. Pollard (2003). Minimal Recursion Semantics: An Introduction (ms, Univ. of Cambridge; to appear).
- Erteschik-Shir, N. (this volume). Sound Patterns of Syntax: Object Shift.
- Fretheim, T. (1974). Some Cases of Accent Shift in East Norwegian. *Norsk Tidsskrift for Sprogvidenskap* 28: 113–129.
- Gundel, J. K., N. Hedberg and R. Zacharski (1993). Cognitive status and the form of referring expressions in discourse. *Language* 69, 274–307.
- Hellan, L., and P. Haugereid (2003). The NorSource Grammar – an exercise in the Matrix Grammar building design. In: *Proceedings of Workshop on Multilingual Grammar Engineering, ESSLLI 2003*.
- Holmberg, A. (1986). *Word order and syntactic features in the Scandinavian languages and English*. Ph.D. diss., University of Stockholm.

Linearization in two ways

ANDERS HOLMBERG

1. Introduction

What the theories of Fox & Pesetsky (F&P) and Erteschik-Shir (ES) have in common is that they both account for Holmberg's Generalization (HG) in terms of linearization. That is virtually the only thing they have in common, though, and even that is more a nominal than a substantial resemblance. For one thing, in F&P it is a matter of linearization of VP constituents as part of the narrow-syntactic derivation, in ES a matter of linearization of adverbs in relation to verb and object in a model where phonology/prosody determines word order. They can't both be right, but they can both be wrong, which, however, we can't be certain of until the set of hypotheses they propose have been thoroughly investigated and tested.

Below I will present some critical remarks on both theories, pointing out some possible false predictions in F&P in section 2, and one or two possible flaws in ES in section 3. In section 4 I will consider how the two theories fare with respect to properties of Object Shift (OS) which they do not themselves take up. In section 5 I will briefly consider the problem of inverted objects.

2. Holmberg's Generalization according to Fox & Pesetsky

F&P explain HG as the effect of a form of 'shape conservation'. In this respect the theory belongs in the same family as Sells (2001) and Williams (2003), as they note. Like Sells (2001), F&P account for just about all the facts relating to HG discussed in my 1999 paper with great elegance and

simplicity. If we accept the premises, then almost everything seems to follow with a minimum of additional, controversial hypotheses. In the case of F&P we need to accept (a) that VP is a phase, and as such is spelled out when complete (in some sense), (b) that movement of VP-internal categories is possible after spell-out, but that the linear order established at spell-out cannot be changed by subsequent movement, and (c) that OS, unlike other well-established movements, does not move stepwise, first to the edge of VP and then on to higher positions. It follows that the object can't ever shift across the verb or across any visible category which is in VP when VP is spelled out. Furthermore, it doesn't matter how the verb or the other categories are moved out of VP, as long as the linear order between them is not changed.

My first reaction was that this is so good that it must somehow be right. But as is usually the case, when you scrutinize a theory in more detail various problems and false predictions come to light.

I was initially impressed by the account of the facts in connection with 'V-topicalization', as in (1a). F&P are probably right in redefining this operation as remnant VP-fronting. (1b) is another indication that this is right (in addition to their (30)).¹

- (1) a. Sett har jag honom inte (men jag har hört hans
seen have I him not, but I have heard his
röst). (Swedish)
voice
- b. Släppt in har jag den inte (men jag har satt ut mat åt den).
let in have I it not but I have put out food for it
'I haven't let it in (but I have put out food for it).'

If so, the object pronoun must have moved out of the VP, violating HG construed as a generalization about movement. But then remnant VP-movement re-establishes the relative linear order of the VP, which in Swedish is *släppt < in < den* and the output therefore respects the linearization laws of F&P, explaining HG construed as a generalization about the output of movement.

¹ I find (1b) somewhat marginal, more so than (1a), but the contrast is not clear enough to reject (1b) while accepting (1a).

In Holmberg 1999 I rejected the idea that (1a) is derived by Object Shift followed by remnant VP-fronting, because it seemed to wrongly predict that (2), for example, should be well-formed.

- (2) *[Hört t hålla föredrag] har jag henne inte t_{VP} (Swedish)
 heard give talk have I her not

But F&P predict that (2) should not be well-formed, while (1b) should be, because only (1b) respects the linear order of the VP.

But in that case they also predict that Norwegian should contrast with Swedish with respect to (1b), since the linear order of verb, object, and particle prior to movement in Norwegian should be *sluppet < den < inn*.

- (3) a. Jeg har ikke sluppet den inn/*sluppet
 I have not let it in / let
 inn den. (Norwegian)
 in it
 b. Sluppet inn har jeg den ikke.
 let in have I it not

The prediction is false. V-topicalization/Remnant VP fronting is a marginal phenomenon in Norwegian as in Swedish, but among those Norwegians who accept it, there are those who accept (3b), just as in the case of Swedes and (1b).

The application of the theory to the 'inverse Holmberg effect' in connection with QP-fronting is also striking. This is a phenomenon which has not received the attention in the literature which it deserves, partly because no-one has had a clear idea of how to deal with it.

- (4) a. Ég hef ekkert sagt Sveini __. (Icelandic)
 I have nothing told Svein
 'I have told Svein nothing.'
 b. *Ég sagði ekkert Sveini __.
 I told nothing Svein

In (3a) the negative QP (NegQP) has moved out of the VP. As it precedes V, it must have moved via the VP-edge position (given F&P's premises), thus being spelled out in pre-V position. It follows that QP-fronting cannot then be followed by V-movement to a higher position (T or C): The linear order of the VP has to be maintained.

What about (5), though?

- (5) *Ég sagði Sveini ekkert.* (Icelandic)
 I told Svein nothing

Here OS has moved the object across the QP, and the result is fine, even though the verb precedes the QP. F&P mention (5) in a footnote, but they do not seem to consider it a problem. As far as I can see it is a problem. QP-movement is obligatory with negative QPs; compare (4a) and (6):

- (6) **Ég hef sagt Sveini ekkert.* (Icelandic)
 I have told Svein nothing

Then V-movement across the QP should be impossible whether or not the indirect object undergoes movement. In fact, (7) is a problem for the same reason.

- (7) *Ég sá ekkert.* (Icelandic)
 I saw nothing

(6) and (7) both replicate the order that the VP presumably had before anything moved – as if QP-fronting was a movement which, like OS, does not pass via the edge of VP. But in that case (4a) doesn't come out right. I imagine that some clever manipulation of spell-out domains (say, vP or VP) may provide a solution, which, however, might then lack the simplicity and elegance of F&P's present proposal.

F&P's law of order preservation allows OS across any VP-external categories. This is why OS across adverbs is possible. OS across the subject is therefore also possible, in principle. This is a welcome result, since so called Long Object Shift actually occurs in at least one variety of Scandinavian. I return to this below. OS across auxiliaries is then also allowed, in principle. Provided there is a way that the main verb can move across an auxiliary, F&P predict that an object pronoun (or in Icelandic, definite DP) should be able to move across the auxiliary, too.

Verb Topicalization/Remnant VP Fronting as in (1) is one way that a verb can move across an auxiliary. If we add another auxiliary verb, as in

(8a), and then do OS and remnant VP-fronting, can the object move across the second auxiliary verb?

- (8) a. Jag kan ha sett honom (men jag vet inte vad han
I may have seen him (but I don't know his
heter). (Swedish)
name)
b. *Sett kan jag honom ha, ...
seen may I him have

(8b) is completely ungrammatical. However, even without OS, VP-fronting is impossible from a non-finite VP, so this is not a valid test.

- c. *Sett honom kan jag ha, ...
seen him may I have

A more telling test case is provided by Stylistic Fronting, as found in Icelandic and Faroese. This operation can move a non-finite verb across an auxiliary, to what looks like a vacated subject position, as in (9a) (see Holmberg 2000). The movement is permitted by the laws of F&P as it affects a category at the edge of VP, therefore does not lead to reordering of VP-internal material. But now OS across the auxiliary is predicted to be possible, as it, too, would respect the linear order of the VP: another false prediction.

- (9) a. Hver heldur þú að stolið hefur [VP __ hjólinu]? (Icelandic)
who think you that stolen has the.bike
'Who do you think has stolen the bike?'
b. *Hver heldur þú að stolið hjólinu hefur?
who think you that stolen the.bike has

3. A phonological account of Object Shift and cross-Scandinavian variation

Erteschik-Shir (ES) is more ambitious than F&P as regards aims and coverage. Her aim, she says, is "to pursue an account /of OS/ which exclusively makes reference to /phonological and morphological/ primitives". However, the paper covers not only OS and HG, but a range of other word order phenomena in the Scandinavian languages, with the aim of accounting for some of the cross-Scandinavian variation as well. It is

embedded in a function-oriented framework where much of the syntax is a matter of facilitating production and processing of speech.²

Unlike F&P, ES discusses other properties of OS than HG. The basic idea is that there is no object shift. Instead, the position of the object in relation to the verb and other sentential categories is the effect of *prosodic incorporation* (PI) of a prosodically light object into a preceding category, usually the main verb. The illusion of object shift across adverbs in connection with verb movement, that is HG, is the effect of adverb placement: ES assumes that adverbs exist in a “third dimension” and can be linearized in a variety of positions relative to the constituent they modify, but crucially cannot intervene between an object which wants to PI with the verb. If the verb moves, as in V2 contexts in Mainland Scandinavian, the adverb is linearized after the verb+object complex. In embedded clauses it is linearized preceding the complex, following a principle according to which an adverb always linearizes in the “earliest” position available.

There is an argument against OS as a postsyntactic operation in Holmberg (1986: 168f.) and Holmberg & Platzack (1995: 150f.): It may feed topicalization. In fact, (1a,b), analyzed as remnant VP-fronting, is evidence of this. I fail to understand what position the present paper takes on topicalization, but in some previous, recent work of hers (Erteschik-Shir 2001) she argues that topicalization is a late operation, belonging in the same module as (a version of) PI. Within that model, ordering topicalization after PI may not be seen as a problem.

² I was surprised by ES's claim, following Raviv (in prep.), that (i) unambiguously means that Peter met Sara, and is ungrammatical with *Peter* a topicalized object. This claim forms part of the basis for a series of rules for the identification of grammatical relations by means of the linear order of adverbs and arguments.

- (i) Peter mødte Sara. (Danish)
Peter met Sara

I will obviously not question Raviv's or ES's judgments of Danish, but I have no doubt that the counterpart in Swedish is ambiguous, when presented out of context. There is a novel by the Swedish author Stig Claesson from the 60ies called

- (ii) Vem älskar Yngve Frej? (Swedish)
who loves Yngve Frej

I haven't read the novel, and I genuinely don't know whether it means 'Who loves Y.F.?' or 'Who does Y.F. love?', even though I suspect it is the former. It could also be an intentional ambiguity on the author's part.

I'm skeptical towards many of the claims made in the paper, but I will focus on one issue, to do with variation concerning argument placement in Mainland Scandinavian. ES believes that "the strength of /her/ approach lies in its potential to explain cross-linguistic and dialectal variation on the basis of fine morpho-phonological distinctions /.../ while maintaining a uniform syntax". The idea that linguistic/dialectal variation is really a matter of 'PF', not Narrow Syntax, is interesting and definitely worth investigating in detail. The implementation of the idea in ES is somewhat halting, though.

As discussed by ES, there is variation among the Mainland Scandinavian languages concerning the position of arguments in a range of constructions, where the arguments occur more consistently further to the left in Danish than they do in Swedish and Norwegian.

- (a) In the verb-particle construction the object always precedes the particle in Danish, always follows the particle in Swedish, while in Norwegian a pronominal object always precedes, while other objects precede or follow the particle.
- (b) Object Shift is optional in Swedish and Norwegian, but obligatory in Danish.
- (c) 'Subject Shift' is optional in Swedish and Norwegian, but obligatory in Danish. That is to say, the subject always precedes all sentence-medial adverbs in Danish, but not necessarily in Swedish or Norwegian; see Svenonius (2002).

The generalization is quite striking, but until now I have seen no serious attempt to formulate a parameter from which all of this variation would follow.

In the case of (b) and (c) ES describes the variation as a matter of allowing or not allowing a pronoun to PI into a V+adverb complex. In Danish this is not possible, so a pronominal argument cannot occur right-adjacent to an adverb.³ ES then suggests that the variation with regard to (a), (b) and (c) is due to a prosodic parameter: In Swedish the particle is always

³ This is not sufficient to account for (c), since non-pronominal subjects, too, have to precede all adverbs in Danish. In Swedish and Norwegian weak pronominal subjects strongly tend to precede any adverbs, while non-pronominal subjects need not do so. Following Raviv (in preparation) ES assumes a rule of grammatical function identification which requires a subject to precede adverbs, in Danish.

stressed in the verb-particle construction. This, according to ES, is a reflex of the verb-particle sequence in Swedish being prosodically a word-like unit into which the pronoun can PI. In Danish the verb and the particle do not form such a prosodically marked word-like unit, hence the pronoun can't PI, and is left stranded (although why it can't be left in that position when it is a lexical DP, either, is not clear; see footnote 3).

- (10) a. Jag skrev upp det. (Swedish)
 b. *Jeg skrev op det. (Danish)
 I wrote up it

ES then proposes a corresponding account of the Swedish-Danish contrast in (11): OS is optional in Swedish, obligatory in Danish.

- (11) a. Jag såg inte den. (Swedish)
 b. *Jeg såg ikke den. (Danish)
 I saw not it

The crucial difference would, again, be prosodic: ES says that the verb-adverb sequence in Swedish is pronounced with stress on the adverb, which yields a prosodically word-like unit, into which the pronoun can PI. In the Danish dialect of Ærø which ES discusses there is, apparently, a correlation between stress on the adverb and optionality of OS, which is obviously suggestive. But in Swedish or Norwegian there isn't: The stress is not on the adverb in (11a), but on the verb, in the unmarked case.

The idea that there is a prosodic parameter involved, explaining (a)–(c) is very interesting, though, and definitely worth further investigation, even if ES's present proposal turns out to be misguided. It may be the case that verb, adverb and pronoun can form a prosodic unit of some sort in Swedish and Norwegian in a way that they cannot in Danish. If that is the case, it still needs to be determined which is the chicken and which is the egg: The prosody may be a PF-adaptation to a word order derived by entirely prosody-free but parameterized narrow-syntactic rules.

4. Other properties of Object Shift

How do the theories compare with respect to other known properties of Object Shift, discussed in Holmberg & Platzack (1995) and Holmberg (1999), for example?

The clause-boundedness of Object Shift follows in both theories. In F&P it is, presumably, a consequence of the ‘non-edge property’ of OS; the movement does not land in the edge of a phase from where it could undergo edge-to-edge movement, and thus escape from the clause. In ES it is presumably an effect of the locality of PI.

How about the observation that the shifted object can’t bind a reflexive from the derived position, unlike the moved object in a passive (see Holmberg 1999)?

- (12) a. Dom ansåg honom, till hans/*sin besvikelse,
they considered him to his/REFL disappointment,
vara oduglig. (Swedish)
be incompetent
‘To his disappointment they considered him to be incompetent.’
- b. Han ansågs, till sin/*hans besvikelse,
he considered-PASS to REFL/his disappointment
vara oduglig.
be incompetent
‘He was, to his disappointment, considered to be incompetent.’

It doesn’t follow in any obvious way in F&P’s theory. It does seem to follow in ES’s theory: Since there is no object shift but only PI of the object into the verb, the object does not c-command the adverb at the relevant level of representation.

Long Object Shift (LOS), typical of Swedish, is an interesting test case. LOS is discussed in ES, who rightly points out that it provides striking confirmation for the existence of constraints on syntactic operations (or representations) dictated by processing needs: LOS is possible only when the shifted pronoun has unambiguous case marking.

- (13) Varför gör oss/*dom Helge alltid irriterad. (Swedish)
why makes us/ them Helge always irritated
‘Why does Helge always make us irritated?’

The pronoun *dom* is ambiguous between nominative and accusative, and the object reading is quite impossible. The rules of identification of grammatical relations in ES accounts for this. On the other hand, ES can’t

derive the word order in a plausible manner, as she denies the existence of object shift. The proposed solution is that the subject in the LOS construction undergoes Heavy NP Shift rightwards. This is highly implausible for several reasons, one being that the subject need not be heavy at all (ES is wrong on this point), as long as it is not a pronoun.

LOS should present no problem to F&P: Since the subject is not included in the VP at the point of spell-out (at least in Scandinavian), the theory allows free ordering of the subject and the shifted object. Crucially the shifted pronoun is still preceded by the verb. The dialectal variation (LOS possible in Swedish but impossible in Norwegian and Danish) could be a matter of a parameter which, say, allows adjunction of the shifted pronoun to TP in Swedish but not in Norwegian-Danish. The ban against ambiguous pronouns falls outside F&P's theory, though.

5. Inverted objects: a problem for shape conservation

The possibility of inversion in the double object construction presents an obvious problem particularly for F&P and any theories of OS based on shape conservation. The order in (16b) has been claimed by several scholars, including me, to be allowed in Swedish and Norwegian alongside the expected order (16a). The contrast with (16d) indicates that the inversion is an effect of OS.

- (16) a. Jag gav henne den inte. (Swedish)
 I gave her it not
 b. Jag gav den henne inte.
 I gave it her not
 c. att jag inte gav henne den.
 that I not gave her it
 d. *att jag inte gav den henne.
 that I not gave it her
 ‘(that) I didn’t give it to her.’

ES mentions this phenomenon in a footnote, and proposes a “phonological metathesis” rule. This kind of solution is not readily available for F&P: Once the VP is spelled out, it is spelled out, so any reordering must take place in VP, prior to spell-out. F&P will be encouraged,

though, by a recent investigation by Gunlög Josefsson (2003) who found that 29 out of 29 Swedish informants rejected the order (16b). This presumably licenses ignoring the inversion as an unwanted side-effect of linguistic training (this is Josefsson's explanation). I am willing to concede that my judgments are not quite as clear as indicated in (16). In fact, (16a,b,c) are all quite marginal, a prepositional indirect object being the preferred alternative. The contrast between (16b,d) seems clear enough, though. On the assumption that the inversion is part of my I-language, a late metathesis in a derived weak pronoun cluster does not seem implausible as a source of the inversion.

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References

- Erteschik-Shir, N. (2001). P-syntactic Motivation for Movement: Imperfect Alignment in Object Shift. *Working Papers in Scandinavian Syntax* 68, pp. 49–73. Department of Scandinavian Languages, Lund University.
- Holmberg, A. (1986). *Word Order and Syntactic Features in the Scandinavian Languages and English*. Ph.D. dissertation, University of Stockholm.
- (1999). Remarks on Holmberg's Generalization. *Studia Linguistica*, 53, pp. 1–39.
- (2000). Scandinavian Stylistic Fronting: How Any Category can Become an Expletive. *Linguistic Inquiry*, 31, pp. 445–484.
- Holmberg, A. & Platzack, C. (1995). *The Role of Inflection in Scandinavian Syntax*. New York and Oxford: Oxford University Press.
- Josefsson, G. (2003). Four Myths about Object Shift – and the Truth ... In: Delsing, L.-O., Falk, C., Josefsson, G., Sigurdsson, H. Å. (eds.). *Grammar in Focus vol. II. Festschrift for Christer Platzack, 18 Nov. 2003*. pp. 199–208. Department of Scandinavian Languages, Lund University.
- Sells, P. (2001). *Structure, Alignment and Optimality in Swedish*. CSLI Publications, Stanford.
- Svenonius, P. (2002). Subject Positions and the Placement of Adverbials. In: Svenonius, P. (ed.). *Subjects, Expletives and the EPP*. pp. 201–242. Oxford, New York: Oxford University Press.
- Williams, E. (2003). *Representation Theory*. Cambridge MA: MIT Press.

Remarks on cyclic linearization and order preservation

GEREON MÜLLER

1. Introduction

The goal of this paper is to compare the approach to order preservation in Fox & Pesetsky (2005) with the approach I develop in Müller (2000) on the basis of Williams (1999, 2003) and an earlier study of mine. I will proceed as follows. In section 2, I briefly sketch main features of Fox & Pesetsky's (2005) analysis. Against this background, I discuss some of the properties of the approach developed in Müller (2000) in section 3. In section 4, the two approaches are compared. Section 5 draws a conclusion.

2. Order preservation as a consequence of phase linearization

Following Chomsky (2000, 2001), Fox & Pesetsky (2005) presuppose that spell-out applies cyclically, to phases; the relevant spell-out domains are CP, VP (with vP envisaged as an alternative possibility), and DP. The crucial assumption they make is a very simple and, I think, natural one: Once a phase is completed and its elements are linearized, subsequent operations in the derivation cannot alter any of the linearization statements that have been established. This assumption can then be shown to yield a number of far-reaching, very attractive consequences. First, the generalization is derived that movement which leaves a phase must apply successive-cyclically, via the edge of the phase, with no recourse to a specific locality constraint (like Chomsky's (2001) Phase Impenetrability Condition (PIC)) being necessary. The reason is that, say, a *wh*-phrase originating in VP can only end up in a SpecC position (where it precedes

all other items of a clause) without contradicting the ordering statements for the spell-out domain VP if it first moves to the left-peripheral edge in VP. Second, the order-preservation property with both simple and multiple object shift (see Holmberg (1986, 1998), Vikner (1990, 1994), Johnson (1991), Collins & Thráinsson (1996), among many others) is derived straightforwardly, even in more subtle cases, given that object shift cannot target a phase edge position. For instance, an object pronoun that follows the verb in VP will also have to follow the verb in the next phase, after object shift to a position preceding an adverbial; similarly, an object pronoun that follows another object pronoun in the VP will have to follow it in the next phase. Third, an anti-order preservation effect with quantifier movement in the Scandinavian languages can be derived in the same way if we assume that quantifier movement, unlike object shift, proceeds via the edge of a phase: If quantifier movement first targets the edge of VP (and thereby precedes the verb), all subsequent operations will have to result in orders in which the quantifier also precedes the verbs, which blocks verb raising to a higher position after quantifier movement to a TP-internal (but VP-external) landing site. Finally, it is correctly predicted that a viable alternative to order preservation in movement is deletion.

3. Order preservation as a consequence of violable constraints

In Müller (2001), I present evidence for a Parallel Movement constraint. This constraint states that c-command relations between arguments must be preserved from one level of representation (in the sense of Chomsky (1981)) to the next one (e.g., from D-structure to S-structure). Parallel Movement was designed to capture the order preservation properties with a number of movement operations (which had in some cases previously been accounted for by construction-specific predecessors, most notably Lakoff's (1971) global rule for quantifier lowering), among them object shift and *wh*-movement in cases where there is an initial ambiguity (i.e., more than one *wh*-phrase that could in principle be moved, as in superiority contexts). Since the Parallel Movement constraint is formulated in a general way and not relativized to specific movement operations (or the features that play a part in them), it cannot possibly be surface-true.

Otherwise, there could be no movement of a *wh*-object across a non-*wh*-subject, as in (1).

- (1) What₁ did she give t₁ to John?

Consequently, the assumption was that Parallel Movement is a constraint that is minimally violable if this is the only way to respect other requirements, in particular, the triggers for movement operations like *wh*-movement. This in turn necessitates an approach that envisages minimal violability and ranking of constraints; consequently, to the extent that Parallel Movement was plausible, it could be viewed as an argument for an optimality-theoretic organization of grammar.

Parallel Movement succeeds in deriving superiority (-like) effects with *wh*-movement, and order preservation effects with multiple object shift in Scandinavian, and with multiple pronoun fronting in German. However, since the constraint is exclusively concerned with arguments, it does not yet capture the dependence of object shift on verb movement (i.e., Holmberg's generalization); as noted by Williams (1999, 2003), this can also straightforwardly be conceived of as an instance of order preservation. In Müller (2000) (written after Müller (2001), and adopting Williams's terminology), Parallel Movement is replaced by Shape Conservation:

- (2) *Shape Conservation*:

Feature checking must not change the linear order of lexical items established in vP.

The basic motivation for adopting this constraint was to force “secondary”, apparently non-feature driven movement operations as they seem to be required in approaches in which movement operations that have often been assumed to be covert are in fact treated as applying overtly. The negative XP fronting operation postulated in Kayne (1998) is a case in point. The assumption here is that the direct object DP *no books* in (3) is overtly fronted. Clearly, this necessitates additional movement operations applying to the indirect object and the remnant VP so as to restore the original order; and it does not seem to be possible to find an inherent (e.g., feature-related) trigger for these secondary movement operations (also see Koopman & Szabolcsi (2000) for this general consequence).

- (3) John [_{VP₂} gave t₁ t₃] no books₁ to Mary₃ t₂

Assuming that the general constraint prohibiting non-feature driven movement (Last Resort) is violable in favour of Shape Conservation, movement of VP₂ and DP₃ (to outer and inner specifiers of the landing site, respectively) can be forced in (3). Still, the analysis must be further restricted so as to ensure that, e.g., *wh*-movement of an object to SpecC is not wrongly predicted to trigger subsequent secondary movement operations to outer and inner specifiers of C, resulting in the order in (4) (instead of the order in (1)).

- (4) *She₂ [VP give t₁ t₃] what₁ to John₃ did t₂ t₃

The solution adopted in Müller (2000) is this: Shape Conservation is split up into two constraints, one holding for A-movement (like, by assumption, Kayne's (1998) negative XP fronting), which is ranked higher than the constraint against non-feature driven movement and may thus force secondary movement operations, and one holding for A-bar movement, which is ranked lower than this constraint. As before, this presupposes an organization of grammar in which constraints are violable and ranked. However, even a low-ranked Shape Conservation constraint for A-bar movement succeeds in deriving superiority effects: The two candidates in (5-ab) pattern in the same way with respect to all higher-ranked constraints; and Shape Conservation then prefers (5-a) over (5-b). (Note that (5-b) and (1) violate Shape Conservation in the same way, but the violation is non-fatal in the latter case because there is no alternative candidate that satisfies the featural trigger for *wh*-movement.)

- (5) a. (I wonder) who₁ t₁ bought what₂
 b. *(I wonder) what₂ who₁ bought t₂

Note finally that, even though this approach has not specifically been developed with object shift in mind, it is supposed to capture the basic facts (in particular, Holmberg's generalization and order preservation with multiple fronting; see Müller (2000, 530–531)). Consider, e.g., the Danish data in (6) (from Vikner (1994, 499)).

- (6) a. Hvorfor købte Peter den₁ ikke t₁ ?
 why bought Peter it not
 b. *Hvorfor købte Peter ikke den₁ ?
 why bought Peter not it
 c. *Hvorfor har Peter den₁ ikke købt t₁ ?
 why has Peter it not bought

- d. Hvorfor har Peter ikke købt den₁ ?
 why has Peter not bought it

Given that object shift is an A-movement operation, Shape Conservation cannot be violated by a well-formed candidate involving object shift. At this point, it is important to clarify what the local domain is in which Shape Conservation holds. The assumption in Müller (2000) is that the local domain is the phrase; this would imply that after object shift to the specifier of some vP-external projection, the verb and the subject (or a remnant verb phrase) must be fronted to an outer specifier of the same projection, and subsequently move further from these specifier positions. Alternatively, one might assume the local domain in which Shape Conservation holds to be somewhat larger (e.g., to have the size of a phrase); in that case, object shift could be saved by ordinary verb movement to T or C and subject raising to SpecT (on the potential problem of the order reversal of subject and verb in verb-second structures, see below). Thus, (6-a) (with verb movement) respects Shape Conservation, whereas (6-c) without verb movement) does not. As in other approaches, something extra needs to be said about the fact that object shift can apparently fail to apply if there is no way that it could be saved by verb movement (see (6-b) vs. (6-d); also see Erteschik-Shir (2005) on the issue of variable optionality with object shift that this observation is related to).

In the same way, Shape Conservation accounts for other order preservation effects with object shift like, e.g., the strict maintenance of vP-internal object order after multiple object shift; the Danish examples in (7-ab) are from Vikner (1990).

- (7) a. Peter viste hende₁ den₂ jo t₁ t₂
 Peter showed her it indeed
 b. *Peter viste den₂ hende₁ jo t₁ t₂
 Peter showed it her indeed

Furthermore, the fact that bare verb (participle) topicalization in the presence of an auxiliary can render object shift legitimate (see Holmberg (1998), Fox & Pesetsky (2005)) is to be expected under a Shape Conservation approach.

To conclude, in the approach developed in Müller (2000), Shape Conservation in (2) takes vP as the basic linearization domain, and states that (feature-driven) movement operations cannot change linearization

statements established for this domain. Therefore, it might at first sight look as though this approach is similar to the one developed in Fox & Pesetsky (2005). Let us now see whether this is the case.

4. Comparison

An obvious difference is that the analysis based on Shape Conservation presupposes constraint violability and constraint ranking, which the analysis in Fox & Pesetsky (2005) does not. This basic difference is responsible for the different treatment of the fact that the order preservation property cannot possibly be assumed to be surface-true in all cases. In Fox & Pesetsky's (2005) analysis, this is accounted for by postulating an intermediate step of movement to the phase edge that is available for certain movement types but not others. In contrast, in the analysis in Müller (2000), this is accounted for by splitting up the Shape Conservation constraint and ranking the version of it that is relevant for the movement types for which order preservation is not surface-true lower. Thus, where one approach attributes exceptions from strict order preservation to abstract intermediate structures, the other approach attributes them to constraint violability. This instantiates a well-known trade-off between classical derivational theories and optimality-theoretic systems (see Prince & Smolensky (1993), Halle (1995), Grimshaw (1997), and McCarthy (2002), among others). In both approaches, it is necessary to specify for which movement operations order preservation does not have to be surface-true, and for which it must be. In Müller (2000), the assumption is that the difference between movement types relevant here is the A- vs. A-bar distinction. At least as a first approximation, one can assume the same for the analysis in Fox & Pesetsky (2005) (see, e.g., their remarks in section 4): A-bar movement (which then would have to include Scandinavian quantifier movement) targets the phase edge before reaching its ultimate landing site, whereas A-movement (including object shift) does not. Of course, it remains to be shown how this result can be achieved in the general kind of approach that Fox & Pesetsky (2005) adopt. I believe that this task will eventually turn out to be far from trivial. However, Fox and Pesetsky are well aware of the problem (cf. their pertinent remark in section 6: "Our proposals say nothing in themselves, however, about the

circumstances under which movement to these left-edge positions is allowed or prohibited.”); and I think they are correct in assuming that the matter is strictly speaking orthogonal to their main claims.

So, I would like to contend that this difference in handling cases where order preservation is not a surface-true property of movement types is essentially framework-induced, and as such thus not as fundamental as it may initially look. However, closer inspection reveals that the two devices used for accounting for cases where order preservation is not surface-true yield empirical consequences that are quite different.

On the one hand, it has been argued that there is independent evidence for the use of edge positions of (vP/VP) phases as intermediate landing sites of A-bar movement. Such evidence involves reconstruction effects (see Fox (2000)), instances of stranding of parts of *wh*-phrases (see Barbiers (2002)), and occurrences of morphological markers that can be interpreted as reflexes of successive-cyclic A-bar movement (see Cole & Hermon (2000)). Thus, given that the intermediate movement steps required in the analysis that derives order preservation from cyclic spell-out of phases are independently motivated, they provide a strong argument for this approach.

On the other hand, the solution in terms of a low-ranked Shape Conservation constraint for A-bar movement makes it possible without further ado to account for superiority effects. The reason is a general property of optimality-theoretic systems: Low-ranked constraints are never switched off and may become decisive after all in contexts where higher-ranked constraints do not distinguish the competing candidates. Thus, even a low-ranked Shape Conservation constraint for A-bar movement, which is violable in contexts where there is only one *wh*-phrase that qualifies as a potential item to undergo movement, will succeed in deriving the contrast in (5), where there is an initial ambiguity. In contrast, there is no obvious way to derive superiority effects in Fox & Pesetsky’s (2005) approach: Here, order preservation becomes an issue only after the phase is completed; but at this point, it is too late to distinguish between a derivation in which the higher *wh*-phrase has been moved to the edge of the phase, and another derivation in which the lower *wh*-phrase has been moved to that position, even though the former derivation minimizes a disruption of the pre-movement order. It seems that the only possible way to make these kinds of superiority effects follow from cyclic

linearization would be to reduce the size of linearization domains further, to the sub-phase level. But then, ordinary *wh*-movement across a non-*wh*-phrase will be a problem. At the heart of this problem is the fact that different kinds of categories (in the case at hand: *wh*-phrases and non-*wh*-phrases) are not treated differently from the point of view of linearization. (Thus, one could speculate that what is called for is a concept of ‘relativized linearization domain’, but it is far from clear that this idea could be given a conceptually attractive implementation.) – That said, one might argue that there is reason not to treat superiority effects as instances of order preservation; see in particular Williams (2003, 140ff.).

Another difference between the two approaches is also directly related to the basic differences in underlying grammar design: As has been mentioned above, in the approach in Müller (2000), Shape Conservation may in fact force “secondary”, non-feature driven movement that restores the pre-movement order. In theories in which there is a constraint that demands all movement to be feature-driven (which holds, in one way or another, for both approaches currently under consideration), such secondary movement clearly presupposes that this constraint can be violated. Since Fox & Pesetsky (2005) do not envisage constraint violability, it is clear that there can be no “secondary” movement driven solely by order preservation properties. Everything then depends on how good the evidence for order-preservation driven movement then turns out to be. I take this issue to be unresolved at this point.

Next, consider Scandinavian quantifier movement. In Fox & Pesetsky’s (2005) analysis, it follows from the assumptions that (a) the quantifier phrase precedes the finite verb in VP, and (b) the ultimate landing site of the quantifier phrase follows the target position of verb-second, that quantifier movement is incompatible with verb-second movement of the main verb (see their example (38)). This result strikes me as quite attractive; and I do not see how it could be derived on the basis of the approach in Müller (2000) (the reason is that Scandinavian quantifier movement must fall under the lower-ranked Shape Conservation constraint; but then, it cannot possibly block feature-driven verb-second movement of a main verb).

A general property of optimality-theoretic systems is that of factorial typology; i.e., unless a fixed order among constraints can independently be ensured (e.g., by harmonic alignment; see Smolensky (1995)), we

expect free re-ranking. Thus, we expect languages in which Shape Conservation for A-bar movement is ranked higher than Shape Conservation for A-movement, in a system like that in Müller (2000). However, this would result in peculiar patterns which do not seem to be attested. In contrast, there is no comparable room for variation in the system assumed in Fox & Pesetsky (2005), and hence, no prediction of order preservation with A-bar movement in the absence of order preservation with A-movement.

Finally, and perhaps most importantly, in the approach developed in Müller (2000), order preservation is accounted for by postulating an explicit constraint that demands just this. In contrast, Fox & Pesetsky (2005) make do without a specific constraint; they succeed in deriving order preservation from an independently motivated basic grammatical architecture involving cyclic spell-out, enriched only by the natural assumption that linearization statements established for spell-out domains cannot subsequently be modified.

In my view, these considerations favour the analysis in Fox & Pesetsky (2005) over the one in Müller (2000). Still, I would like to end these remarks with a few related considerations that may point in the opposite direction. The first one concerns verb-second structures in an SOV language like German. The problem is how to ensure that verb-second clauses in German may exhibit a word order in which the finite main verb precedes the object (see (8-a)), whereas the object precedes a main verb which has not undergone verb-second (see (8-b)).

- (8) a. Gestern las₁ Maria [_{VP} (t₁) ein Buch₂ (t₁)]
 yesterday read Maria_{nom} a book_{acc}
 b. dass Maria gestern [_{VP} ein Buch₂ las₁ (t₂)]
 that Maria_{nom} yesterday a book_{acc} read

A standard analysis of (8-a) would involve verb-second movement from a position to the right of the object DP across VP (see Haider (1993)). However, it seems that such an analysis cannot be maintained in Fox & Pesetsky's (2005) approach: A VP spell-out domain in which the object precedes the verb prohibits a subsequent linearization in which the verb precedes the object. To solve this problem, one might assume that the verb is merged to the left of the object in VP in (8-ab). On this view, surface OV orders as in (8-b) might be derived by leftward object raising across the verb; such movement must be blocked in verb-second clauses

in which the main verb moves (unless the object ends up in SpecC). An analysis along these lines may prove viable; but it seems clear that many technical problems will need to be addressed to avoid overgeneration. Furthermore, if analyses of this type that rely on highly abstract linearization domains (that are never attested on the surface) are available for the SOV language German, one might wonder why they are not for the Scandinavian SVO languages, where a surface-oriented approach seems crucial. In contrast, verb-second movement in SOV languages is unproblematic for the Shape Conservation-based approach if it is assumed that head movement patterns with A-bar movement (rather than with A-movement) for the purposes of this constraint.

Second, given that spell-out domains are phases, that phases can be given a semantic motivation (Chomsky (2001)), and that this motivation implies that the position in which an external argument is merged is phase-internal, it strikes me as potentially problematic to assume that the verbal spell-out domain of the Scandinavian languages is VP rather than vP. It seems that Fox and Pesetsky's main reason for doing so nonetheless is theory-internal: "Since the Scandinavian languages freely raise the main verb to C over the subject, the subject is presumably not linearized in Scandinavian with respect to the main verb before CP is constructed" (section 5). In this context, it is also instructive to consider evidence from pronoun fronting in German. Fronting of unstressed pronouns is obligatory in German. In contrast, NP raising to SpecT is optional throughout. As shown in (9-ab), a fronted subject pronoun must precede a fronted object pronoun.

- (9) a. dass sie₁ es₂ wahrscheinlich t₁ t₂ gelesen hat
 that she_{nom} it_{acc} probably read has
 b. *dass es₂ sie₁ wahrscheinlich t₁ t₂ gelesen hat
 that it_{acc} she_{nom} probably read has

Given that the order preservation effect visible here cannot result from obligatory subject raising to SpecT (as it may in comparable cases in the Scandinavian languages), it seems that the most straightforward conclusion on the basis of Fox & Pesetsky's (2005) analysis is that cyclic linearization of spell-out domains is responsible. This implies that vP is the spell-out domain in German. However, this reinforces the problem with verb-second in SOV languages noted above because it suggests that the

(verbal) spell-out domain in German also includes the base position of the subject.

Note finally that German also has order-changing scrambling, even to landing sites in front of the base position of subjects (such scrambling cannot affect unstressed pronouns). But again, a finite main verb can move to C in verb-second clauses and precede scrambled objects from this position, which is again unexpected from the point of view of cyclic linearization; see (10).

- (10) Gestern gab₁ das Buch₂ wahrscheinlich niemand dem Fritz
 yesterday gave the book_{acc} probably no-one_{nom} the Fritz_{dat}
 t₂ t₁

Here, the direct object seems to have left the verbal spell-out domain (be it VP or vP); for this to be possible, it must have targetted the phase edge first. But then, we wrongly predict verb-second movement of the main verb to be ill formed, on a par with the restriction on Scandinavian quantifier movement derived in Fox & Pesetsky (2005, ex. (38)); and this effect should occur independently of whether the basic structure of OV languages is OV or VO. In contrast, neither of these cases poses a problem for the Shape Conservation-based approach: All relevant movement types involved here can be argued to fall under the lower-ranked, A-bar version of the constraint, which implies that order preservation needs only to be respected to the extent that this is possible without violating higher-ranked constraints.

5. Conclusion

These last considerations notwithstanding, I think that Fox & Pesetsky's (2005) idea that cyclic linearization of phases explains order preservation phenomena is indeed a major, innovative contribution to syntactic theory. Certain questions will ultimately have to be pursued in more depth (especially those concerning the size of spell-out domains and the OV/VO distinction). Also, I do not wish to claim that it is completely impossible to come up with theories in which order preservation effects with movement types like object shift emerge as a result of interacting constraints that are not at all related to linearization (e.g., minimality

conditions of one kind or another; see, e.g., Collins & Thráinsson (1996), Richards (2001)). However, I would like to contend that among the existing theories that do account for order preservation effects by directly invoking linearization requirements, Fox & Pesetsky's (2005) analysis is unmatched in elegance and simplicity.

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References

- Barbiers, Sjef (2002): Remnant Stranding and the Theory of Movement. In: *Dimensions of Movement*. Benjamins, Amsterdam, pp. 47–67.
- Chomsky, Noam (1981): *Lectures on Government and Binding*. Foris, Dordrecht.
- (2000): Minimalist Inquiries: The Framework. In: R. Martin, D. Michaels & J. Uriagereka, eds., *Step by Step*. MIT Press, Cambridge, Mass., pp. 89–155.
- (2001): Derivation by Phase. In: M. Kenstowicz, ed., *Ken Hale. A Life in Language*. MIT Press, Cambridge, Mass., pp. 1–52.
- Cole, Peter & Gabriella Hermon (2000): Partial Wh-Movement: Evidence from Malay. In: U. Lutz, G. Müller & A. von Stechow, eds., *Wh-Scope Marking*. Benjamins, Amsterdam, pp. 101–130.
- Collins, Chris & Höskuldur Thráinsson (1996): VP-internal Structure and Object Shift in Icelandic, *Linguistic Inquiry* 27, 391–447.
- Erteschik-Shir, Nomi (2005): Sound Patterns of Syntax: Object Shift. This volume.
- Fox, Danny (2000): *Economy and Semantic Interpretation*. MIT Press, Cambridge, Mass.
- Fox, Danny & David Pesetsky (2005): Cyclic Linearization of Syntactic Structure. This volume.
- Grimshaw, Jane (1997): Projection, Heads, and Optimality, *Linguistic Inquiry* 28, 373–422.
- Haider, Hubert (1993): *Deutsche Syntax – generativ*. Narr, Tübingen.
- Halle, Morris (1995): Comments on Luigi Burzio's 'The Rise of Optimality Theory', *GLOT International* 1(9/10), 27–28.
- Holmberg, Anders (1986): Word Order and Syntactic Features in the Scandinavian Languages and English. PhD thesis, University of Stockholm.
- (1998): Remarks on Holmberg's Generalization, *Studia Linguistica* 53, 1–39.
- Johnson, Kyle (1991): Object Positions, *Natural Language and Linguistic Theory* 9, 577–636.
- Kayne, Richard (1998): Overt vs. Covert Movement, *Syntax* 1, 128–191.
- Koopman, Hilda & Anna Szabolcsi (2000): *Verbal Complexes*. MIT Press, Cambridge, Mass.
- Lakoff, George (1971): On Generative Semantics. In: D. Steinberg & L. Jakobovits, eds., *Semantics*. Cambridge University Press, Cambridge, pp. 232–296.
- McCarthy, John (2002): *A Thematic Guide to Optimality Theory*. Cambridge University Press, Cambridge.

- Müller, Gereon (2000): Shape Conservation and Remnant Movement. In: M. Hirotani, A. Coetzee, N. Hall & J.-Y. Kim, eds., *Proceedings of NELS 30*. GLSA, Amherst, Mass., pp. 525–539. Revised and extended version appeared 2002 in Artemis Alexiadou, Elena Anagnostopoulou, Sjef Barbiers, & Hans-Martin Gärtner (eds.), *Dimensions of Movement*, 209–241. Amsterdam: Benjamins.
- (2001): Order Preservation, Parallel Movement, and the Emergence of the Unmarked. In: G. Legendre, J. Grimshaw & S. Vikner, eds., *Optimality-Theoretic Syntax*. MIT Press, Cambridge, Mass., pp. 279–313. Earlier version appeared 1997 in U. Lutz & F. d’Avis (eds.), *Zur Satzstruktur im Deutschen*. 171–214. Stuttgart/Tübingen: Arbeitspapiere des SFB 340, no. 90.
- Prince, Alan & Paul Smolensky (1993): Optimality Theory. Constraint Interaction in Generative Grammar. Book ms., Rutgers University.
- Richards, Norvin (2001): *Movement in Language*. Oxford University Press, Oxford.
- Smolensky, Paul (1995): On the Internal Structure of Con, the Constraint Component of UG. Ms., Johns Hopkins University.
- Vikner, Sten (1990): Verb Movement and the Licensing of NP-positions in the Germanic Languages. PhD thesis, University of Geneva.
- (1994): Scandinavian Object Shift and West Germanic Scrambling. In: N. Corver & H. van Riemsdijk, eds., *Studies on Scrambling*. Mouton de Gruyter, Berlin, pp. 487–517.
- Williams, Edwin (1999): Economy as Shape Conservation. Handout of talk at DGfS meeting, Universität Konstanz.
- (2003): *Representation Theory*. MIT Press, Cambridge, Mass.

Some notes on cyclic linearization

ØYSTEIN NILSEN

1. Cyclic linearization

Fox and Pesetsky (this issue, henceforth F&P) argue that the “edge effects” derived by stipulation within standard phase theory can be explained in their version, where a phase crucially triggers linearization of its constituents. Later phases may add ordering statements to an *ordering table* in a monotonic fashion, but no information can be erased or altered once it has entered the ordering table. Suppose a phase *A* contains the constituents *x*, *y*, *z*, and that they are linearized in that order. Then movement within the next phase can’t result in reordering of *x*, *y*, *z*. So *x*, being the leftmost element, can move leftwards freely within the next phase, while *y* can only move leftwards provided that *x* moves even further leftwards. This is, in essence their explanation of Holmberg’s Generalization (Holmberg, 1986, 1999) (HG). To see this, imagine that *y* in our setup is an object trying to shift, and that *x* is the verb. They also show that, if *y* moves to the left edge of *A* prior to its linearization, this may end up blocking leftwards movement of *x* in what they term the ‘inverse Holmberg’s Generalization’, and they have empirical support for the existence of that pattern. This gives a version of phase theory where phases are not entirely opaque to outside syntactic probing. Their proposal is highly innovative and elegant, and it succeeds in deriving an impressive range of facts. The following paragraphs present some relatively minor empirical problems with their treatment of HG which unfortunately seem to conspire to uncover a major one.

2. Do phases explain HG?

As F&P discuss, they sometimes need to resort to phase-internal movement to establish the required order prior to linearization, so that later movements won't create ordering contradictions. In other cases, such as Scandinavian Object Shift (OS), they need such movement to the left edge not to occur. As they put it, "[o]ur proposals say nothing in themselves, however, about the circumstances under which movement to these left-edge positions is allowed or prohibited." Constraints on phase-internal movement, it would seem, then, can't themselves be reduced to phases. I will argue that this raises non-trivial problems for F&P's explanation of HG. OS-related phenomena will require some rather complex phase-internal operations, which, unless appropriately constrained, would overgenerate.¹ A rather trivial example (which won't be discussed further here) is that of object topicalization. Such topicalization would, as F&P explain, have to proceed cyclically, through the edge of VP/vP, so that O precedes V at the VP/vP phase. But then what prevents OS from doing the same? That would produce the ordering statement $O > V$ in the first phase, and (wrongly) allow OS across the verb. Another example of the same sort is quantifier movement (QM). Since QM in Norwegian obeys HG, while in the rest of Scandinavian, it obeys the inverse HG, it seems to follow that, for F&P, Norwegian QM can't proceed cyclically, while in, say, Icelandic, it must. Icelandic QM, proceeding cyclically, will first move to the left edge of the first phase, thus yielding $Q > V$. This will then block V-to-C, unless Q moves further to spec-CP. Hence Icelandic QM can only take place when V-to-C does not. Norwegian QM, on the other hand, patterns with regular OS, i.e. it can only take place when V-to-C does take place. Hence Norwegian QM (and OS more generally) can't move through the left edge of the first phase, while Icelandic QM must do so. What governs this variation in phase-internal syntax? Do we find similar variation with respect to OS of weak pronouns?

¹ See Nilsen (2003) for an attempt to formulate such constraints with respect to HG which do not depend on phases (but are not a priori incompatible with them). Here and throughout, I use the notion of a 'phase' in F&P's sense. Hence, I am not claiming that there couldn't be an alternative conception of 'phases' which would avoid the problems to be discussed. Such an alternative remains to be formulated, however.

These complications are not devastating for F&P. They merely highlight the fact that we need to be able to distinguish between the phase internal correlate of topicalization, which is obligatory, and that of OS, which is impossible. Similar remarks could be made about the difference between Icelandic and Norwegian QM. As we shall see, however, there are less trivial cases of phase internal movement forced upon us by considering OS and HG by themselves. The problem seems to be that the phases can limit the orders in the right way only when presented with the right order to begin with.

2.1. *S-V inversion and S-O blocking*

In order to handle subject-verb inversion in Mainland Scandinavian (MSc) V2 contexts, Fox and Pesetsky (F&P) suggest that the subject is outside of the linearization domain (phase) responsible for HG. Hence, subjects are allowed to invert with V in the higher phase because they do not occur in the lower phase, so there is no ordering statement concerning the two from that phase creating potential conflict with the ordering statement given by the higher phase. This proposal runs into problems once we realize that OS can't cross subjects, just as it can't cross verbs and indirect objects. F&P allude to long OS in Swedish (Fox, p.c.), where OS does cross a subject, and conclude on that basis that subjects quite generally don't block OS. But, as is well known, long OS is limited to cases where the object bears visible case (pronouns that are syncretic along the nominative-objective dimension can't undergo long OS), and, additionally, to a small group of main verbs (the cognate of the main verb *like* allows it, while those of a.o. *kill*, *buy*, *eat*, ... don't). Furthermore, long OS is strictly ungrammatical in Norwegian and Danish, so it seems that it would be problematic to treat the Swedish exceptions as the general case. Finally, even in Swedish, shifting of both IO and DO around S is always impossible, further supporting the view that S should be internal to the relevant phase, and that long OS should be handled by some extra mechanism which is licensed in but a few cases. In fact, one can find exceptions to virtually every part of HG in different dialects, some to be discussed below, but concluding on that basis that nothing at all blocks OS seems a bit rash.

The problem with S-V inversion, then, is that V and O must be in the same phase (V blocks O); S and O must be in the same phase, too (S blocks O). By transitivity of identity, we have that S and V must be in the same phase, but then their account of S-V inversion can't be right. Müller (2004); Nilsen (2002, 2003b) argue independently that S-V inversion must take place low in V2 languages. For example, Norwegian has obligatory S-V inversion even when the verb follows sentential adverbs (v2 violations). This is triggered by the presence of focus particles associating with the verb (see Nilsen (2003b) for discussion).²

(1) Norwegian

- a. Meg vanligvis bare svarte ikke **Jens**.
me usually just answered not J
“Jens usually just didn't answer me.”
- b. Meg vanligvis bare svarte **Jens** ikke.
- c. *Meg vanligvis **Jens** bare svarte ikke.
- d. *Meg **Jens** vanligvis bare svarte ikke.

Low S-V inversion could be adopted by F&P. If inversion takes place within a low phase, all of V, O and S can be in that phase. The pattern in (1) would then follow on F&P's account because S-V inversion has taken place in the first phase, yielding the ordering statement $V > S$, so the subject would have to continue to follow V in later phases. Thus, the pattern in (1) can be taken as independent evidence that S is in the first phase and that S-V inversion must have a vP/VP-internal correlate. Scandinavian would then be closer to Korean, as F&P discuss it in their section 5 on the basis of Ko (2004). The difference with Korean, which gives rise to a sort of inverse HG effect with S and O (scrambling of O across S blocks later scrambling of S across O), would come from the assumption F&P make that Scandinavian OS is not cyclic, i.e. it doesn't proceed via the edge of vP/VP, while Korean scrambling is. They could then try to handle the Swedish exceptions (long OS) by explaining why, in precisely the relevant cases, S is outside of the low phase or, alternatively, why O is allowed to shift phase-internally to the left of S. This would amount to adopting a somewhat different clause structure than what F&P do,

² Unless there is specific indication to the contrary, all examples are Norwegian.

however, and it would involve reworking their proposal to capture lack of inversion with main verbs in English.³ Moreover, working out the details of this would necessarily involve explaining the (phase-internal) ordering possibilities of S/V and S/O, thus taking care of that part of HG without resorting to phases.

2.2. IO-DO reordering and V-IO blocking

As noted in Taraldsen (2000), the order DO > IO is impossible in Norwegian, unless IO and DO are weak pronouns followed by an adverb.

- (2) a. Hvorfor fortalte du meg det?
 why told you me it
 ‘‘Why did you tell me it?’’
 b. Hvorfor fortalte du det meg *(ikke)?
 why told you it me *(not)

Any adverb will do. If the restriction to IO > DO follows from the low phase these arguments start out from, it becomes mysterious that adverbs can have this effect. I see no easy way out for F&P here. If the phase is extended to include the adverbs, we lose the account for ordinary OS, which isn’t sensitive to adverbs. If we leave the adverb outside the relevant phase, it should have no effect on the ordering possibilities within it. Suppose, on the other hand, that we consider different positions of IO with respect to the first phase. If IO is outside of it (comparable to F&P’s problematic treatment of subjects and long OS), we run into a contradiction: V (and S) blocks IO-shift, hence they are in the same phase as IO. V (and S) also blocks O-shift, so these must also be in the same phase as O, and, by transitivity, IO and DO are in the same phase, which contradicts our assumption that they are not. Thus this seems to be a particularly nasty problem for the empirical use to which F&P put their theory. Whatever their solution would be – perhaps a VP/vP internal DO-position that

³ In the handout version of their paper, they suggest that English subjects (but not Scandinavian ones) are within the lower phase, thus disallowing S-mainV inversion in English. The discussion in the text indicates that this picture is too simple.

needs licensing from a VP/vP external adverb – it is likely to involve syntactic operations internal to the first phase, allowing DO an “escape hatch” in just the relevant cases. Explicating it would seem to suffice to explain the part of HG which restricts the ordering of IO and DO, again independently of phases. See Taraldsen (2000) for an account of this pattern in terms of remnant movement, and Anagnostopoulou (2002, 2004) for discussion of it in a somewhat different context.

2.3. *Remnant VP-topicalization*

F&P suggest that cases like (3a), where a VP lacking a pronoun has been topicalized, conform to their system, because the pronoun must be extracted from the right edge of the VP prior to VP-fronting. Thus, (3b) is bad because the order inside the VP prior to extraction of the pronoun and fronting of the participle has the pronoun preceding the embedded verb (‘give talk’), see (3c), while the surface order of (3b) has reversed this order.⁴

(3) Swedish (Holmberg, 1999)

- a. [VP Kysst t_i] har jag henne_i inte t_{VP}.
 kissed have I her not
 “I haven’t kissed her”
- b. *[VP Hört t_i hålla föredrag] har jag henne_i inte t_{VP}.
 heard give talk have I her never
- c. Jag har inte hört henne hålla föredrag.
 I have not heard her give talk

To see how this works, consider the VP of (3a) prior to OS:

[VP kissed her]

⁴ To my ears, the Norwegian (i), where the infinitival verb is unergative, is much better than (3b). This recalls the fact noted by Déprez (1988) and Kayne and Pollock (2001) that French ‘stylistic inversion’ is subject to the constraint that the fronted IP (in their analysis) can’t contain an overt argument.

(i) [Hört t_i synge]_j har jeg henne_i aldri t_j.
 [heard sing] have I her never

Linearizing this, we get the singleton linearization set $\{\langle \text{kissed}, \text{her} \rangle\}$. We now merge the negation and shift the object, yielding

[_{IP} her_i not [_{VP} kissed t_i]]

Finally, the auxilliary *have* moves to C (for simplicity, we disregard its base position as well as the introduction of the subject), and the VP topicalizes:

[CP [_{VP} kissed t_i] C+have [_{IP} I her_i not t_{VP}]]

When the CP is linearized, we still have that V precedes O, so no ordering contradiction ensues. In (3b), on the other hand, there is VP-internal material *following* the pronoun in the first phase. In particular the ECM predicate *give talk* follows it, so linearization of the first phase gives (simplifying somewhat): $\{\langle \text{heard}, \text{her} \rangle, \langle \text{her}, \text{give-talk} \rangle\}$. But after linearizing the CP in this case, we have that *give talk* linearly *precedes* the pronoun, thus contradicting the information from the first phase. In other words, remnant VP-topicalization with OS is predicted to occur only when the shifted object would have been at the *right edge* of the VP if it hadn't shifted.

Examples like the following show that things aren't this simple.⁵

(4) Norwegian

- a. % [_{VP} Lagt t_i på bordet] har jeg dem_i aldri.
 put on the-table have I them never
- b. Jeg har aldri lagt dem på bordet.
 I have never put them on the-table
- c. * Jeg har aldri lagt på bordet dem.
 I have never put on the table them

Here we have a VP-internal PP which should create the same kind of ordering contradiction as the ECM-clause in (3b). But the sentence is perfectly grammatical, apparently showing that the “right edge” isn't the relevant notion. F&P might try to handle this by saying that the weak pronoun must undergo heavy NP shift internal to the low phase prior to

⁵ Some speakers don't allow remnant VP-topicalization at all, hence the % in front of the relevant examples.

linearization (this was suggested to me by Norvin Richards, p.c.). That would get the pronoun to the right-edge of the first phase as required. One might find it slightly odd to have *weak* pronouns undergo HNPS, but I suggest we set such qualms aside for the sake of the argument. With HNPS, (4a) would be derived as follows. The VP starts out as:

[_{VP} put them on the table]

and after HNPS of the weak pronoun, we have:

[_{VP1} [_{VP2} put t_i on the table] them _{i}]

Linearizing this, we have that VP2, hence the PP contained in it, precedes the pronoun. We can now proceed to topicalize VP2 and shift the pronoun leftwards, yielding (4a) without any ordering contradiction. This would actually make sense of the contrast between (3b) and (4a), if ECM subjects can't undergo HNPS to begin with (but see footnote 4). It would also be supported by the fact that such remnant VP topics sometimes give rise to obligatory parasitic gaps: HNPS also gives rise to obligatory parasitic gaps as seen in (5c), although less strongly so than remnant VPs (see Larson (1988), and Nissenbaum (2000)). "Ordinary" OS doesn't license parasitic gaps in MSc, as illustrated in (5d), further supporting the idea that something is going on in the remnant cases which isn't going on in the "normal" cases.

- (5) a. % [Kysset t_i uten å danse med *pg* først] har jeg henne _{i}
 kissed without to dance with first have I her
 aldri _{t_{VP}}.
 never
- b. *[Kysset t_i uten å danse med henne først] har jeg henne _{i}
 kissed without to dance with her first have I her
 aldri _{t_{VP}}.
 never
- c. Jeg kysset t_i uten å danse med (??henne) først [den jenta
 I kissed without to dance with (??her) first [the girl
 du snakket om];
 you talked about]
- d. Jeg kysset henne _{i} aldri t_i uten å danse med *(henne) først.
 I kissed her never without to dance with *(her) first

So, despite its initial implausibility, Richards' HNPS idea has something to speak for it. But given the ungrammaticality of (4c), such HNPS of weak pronouns must be restricted to cases where the remnant VP subsequently topicalizes. Unless this restriction somehow follows from the spell-out cycles this looks like sneaking a notion of HG back into the theory, which is what F&P set out to derive: HNPS of a weak pronoun requires another movement, namely VP-topicalization, and such conditioning of one rule application on the application of another is precisely what HG is about. We may dub this version of HG the *backwards* HG. It's hard to see how the backwards HG *could* be made to follow from F&P's phases, because the relevant operations must take place within the first phase in order for the solution to work.⁶

Backwards HG is also operative with datives. They generally can't undergo HNPS, so, if the account under consideration is on the right track, the following sentence should be ungrammatical, contrary to fact.⁷

- (6) % [_{VP} Gitt t_i t_j tilbake] har han henne_i den_j nok ikke t_{VP}.
 given back has he her it probably not

Irrespective of the latter point, if HNPS is the relevant mechanism, the case of HPNS of a weak dative pronoun would require later HPNS of the direct object in addition to later VP-topicalization. Furthermore, the PP ('on the table') and the adjunct-like preposition 'for free' does not induce such "blocking behavior" on HNPS of weak pronouns, further strengthening the similarity with the usual "forward" HG, which quite generally disregards adjuncts.

⁶ F&P would seem to expect a backwards HG when (non-cyclic) rightwards movement targets a phase-external position, as well as an *inverse backwards* HG, i.e. cases where rightward movement of *y* blocks rightward movement of *z*. I have no idea whether these expectations are substantiated.

⁷ Another fact which would look rather mysterious on an HNPS account is that full (heavy) DPs can't give rise to such remnant VPs. F&P citing Holmberg do give some cases of stranded epithets which are reasonably OK, but regular DPs, including indefinites, are ungrammatical. This made me propose that the relevant operation is more like clitic climbing of the Romance sort (Nilsen, 2003a). That idea isn't easy to fit in with F&P's notion of phases, however:

- (i) * Lagt på bordet har jeg (ikke) den boka (jeg kjøpte igår) (ikke)
 put on the-table have I (not) that book (I bought yesterday) (not)

- (7) a. * [_{VP} Gitt t_i den gratis] har jeg henne_i ikke t_{VP}.
 given it for.free have I her not
 b. %[_{VP} Gitt henne t_i gratis] har jeg den_i ikke t_{VP}.
 given her for.free have I it not

To reiterate, it seems that F&P are going to need extensive movement to take place within their phases. These movements, in turn, will need to be constrained to apply in the right cases. These constraints (whatever their right formulation turns out to be) do not seem reducible to F&P-style phases, since the movements are internal to the relevant phases. Suppose we had such an account that told us when an (indirect) object can move phase-internally. In F&P's setup, that account would inevitably have to explain when VP-internal material can reorder and when it can't, hence it would be a phase-independent account of HG, including its empirical wrinkles. But if so, while the system of phases may be *consistent* with the facts of HG, it can't really be said to *derive* it, nor can HG be said to provide empirical support for such phases.

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References

- Anagnostopoulou, E. (2002). Movement across phonologically visible categories. Handout of talk at the MIT Ling-Lunch, December 5, 2002.
 — (2004). Holmberg's generalization and locality. Handout of talk for GLOW 2004, Thessaloniki.
 Déprez, V. (1988). Stylistic Inversion and Verb Movement. ESCOL.
 Holmberg, A. (1986). Word order and syntactic features in the Scandinavian languages and English. Doctoral Dissertation, University of Stockholm.
 — (1999). Remarks on Holmberg's generalization. *Studia Linguistica* 53, pp. 1–39.
 Kayne, R. S. & Pollock J.-Y. (2001). New thoughts on stylistic inversion. In: Hulk, A. & Pollock, J.-Y. (eds.) *Subject Inversion and the Theory of Universal Grammar*. Oxford/New York: Oxford University Press.
 Ko, H. (2004). Asymmetries in Scrambling and Cyclic Linearization. Ms. MIT.
 Larson, R. (1988). Light predicate raising. In: *Lexicon Project Working Papers*. Cambridge, MA: Center for Cognitive Science, MIT.
 Müller, G. (2004). Verb-second as vP-first. *Journal of Comparative Germanic Linguistics* 7, pp. 179–234.

- Nilsen, O. (2002). Verb second and Holmberg's generalization. In: Zwarts, J. W. & Abraham, W. (eds.). *Studies in Comparative Germanic Syntax* (pp. 153–175). Amsterdam: John Benjamins.
- (2003a). Clitics and remnants. Paper presented at the NYU workshop on antisymmetry and remnant movement.
- (2003b). Eliminating positions. Doctoral Dissertation, Utrecht University.
- Nissenbaum, J. (2000). *Investigations of covert phrase movement*. Ph.D. dissertation. Cambridge, MA: MIT.
- Taraldsen, K. T. (2000). V-movement and VP-movement in derivations leading to VO-order. In: Svenonius, P. (ed.). *The Derivation of VO and OV*. Amsterdam: John Benjamins.

Comments on Object Shift and Cyclic Linearization*

PETER SELLS

Introduction

In this short review I address certain issues and predictions that arise from the position papers by Erteschik-Shir, and by Fox and Pesetsky. The two proposed analyses are radically different, so I have made no attempt to relate them in terms of their properties. In section 1 I discuss some underlying issues in the architecture of grammar assumed by Erteschik-Shir, and in section 2 I evaluate some of the predictions generated by Fox and Pesetsky's approach. In section 3 I draw some more general conclusions about what consequences Object Shift shows us concerning the organization of grammar, and the nature of grammatical principles. While my remarks mainly address problems or drawbacks that I see with the proposals in the two position papers, they both offer a stimulating account of OS and explore the wider implications of the natures of their enterprises, and deserve fuller consideration beyond the present venue, consideration that they will surely receive.

1. Sound Patterns of Syntax: Object Shift

In her paper, Erteschik-Shir (I use 'ES' to refer to the ms. of her paper) proposes an account of clause structure and in particular of Object Shift (OS) that is driven by phonological properties, involving the mechanism

* My thanks to Brady Clark, Jong-Bok Kim and Shin-Sook Kim for useful suggestions on some of the material here.

of Prosodic Incorporation (PI). PI forms prosodic units from strings of terminal elements, and does not allow interruptability by further syntactic operations. The notion of PI is extended by Erteschik-Shir from the canonical cases of clitic pronouns, or of reduced auxiliaries in English, to forms that are phonologically considerably more substantial. Empirically, the account is very strong for the core cases of OS, ordering the shifted pronoun (right) after the verb, and then adverbs after the prosodic constituent containing the finite verb (paraphrasing the analysis a little). However, it seems to founder on Long Object Shift in Swedish (see Josefsson (1992); ES, section 2.4), which creates structures in which the object follows the finite verb but precedes the subject: for this phenomenon “[n]o obvious syntactic account suggests itself”. Conceptually, the paper makes certain assumptions about the nature of the phonology-syntax interface, and it is to these that I address my comments.

The main difficulty in evaluating Erteschik-Shir’s proposal is that the reader is given little idea of what the syntax should look like, and consequently it is hard to see if there is any advantage to the PI approach, or whether it merely recapitulates properties of structure that the syntax would have to provide anyway. The conception of syntax that Erteschik-Shir adopts appears to be broadly Minimalist – for example, in section 6, she offers some thoughts on a phonological approach to head movement and V2, citing recent Minimalist literature.

I will begin by looking at the issue of whether the phonology drives – maybe, derives – the syntax, or whether the phonological properties observed are merely consequences of the underlying syntax, assuming a standard syntax→phonology mapping. I will argue for the latter scenario: that the observed phonological properties (of OS) are a consequence of the syntax, but do not determine the syntax, in contradiction to the position adopted by Erteschik-Shir.

The first point that must be made is that there is a key difference between syntactic word-like elements which have special or restricted distributions, and clitics, which are also word-like, but which in addition are phonologically weak. In her work on X^0 -level categories which do not project a phrase, Toivonen (2003, 45ff.) argues for the typology of words in (1), suggesting instances as shown. This extends the perhaps more familiar 3-way typology of Cardinaletti and Starke (1996), by providing two categories of X^0 element.

(1)

	NON-PROJECTING	PROJECTING
PHONOLOGICALLY DEPENDENT	Romance clitics	English reduced auxiliaries
PHONOLOGICALLY INDEPENDENT	Swedish pronouns and particles	most words

The table shows two distinct dimensions: one, of being phonologically weak – that is, dependent – or not and the other, of being projecting or not – whether X^0 projects up to XP or not. Non-projecting words often show special syntax, but they need not have any special phonological properties.¹ This means that words may have special syntax without having special phonology – falling in the bottom left cell of (1). To illustrate, I first consider negation in English, and the well-known fact that *not* has a different distribution from a negative adverb such as *never*:

- (2) a. Sam always sat in the front row, but Max did not.
b. *Sam always sat in the front row, but Max did never.
- (3) a. I asked her to go first, but she would NOT.
b. *I asked her to go first, but she would NEVER.

Erteschik-Shir proposes that all adverbs in English have phonologically strong forms, except for *not*, which only has a weak form, and which optionally forms a prosodic word. In this final case, *not* must PI with a preceding verb, and this could be used as the basis of an account of the different ellipsis facts in (2). However, exactly the same facts hold in (3), where clearly *not* is not prosodically weak (this problem may be recognized in ES, footnote 27). In the literature, there is a variety of analyses available for these facts, attributing a special syntax to *not* – as being in a privileged relation to INFL, as in Bresnan (2000), or as being a complement of an auxiliary verb, as in Kim and Sag (2002), or as being in a privileged position in NegP, as in Zanuttini (1997). All of these are syntactic

¹ See e.g., the many examples in Toivonen (2003), Asudeh and Mikkelsen (2000) on Danish, Sells (1994) on Korean.

accounts, generating strings with stranded *not* in them, which may interface with the phonology as a strong or a weak form.

In the case of Scandinavian OS of pronouns, it is not because pronouns are prosodically weak that they undergo/allow OS, but because they are X^0 s in the syntax, and have special positions where they are licensed as such. In Sells (2001) I developed an account of OS which provides special positions for ‘shifted’ pronouns, which are syntactically X^0 s but not XPs, but where exactly the same principles of linear ordering apply to all clauses, regardless of whether their objects are shifted or not, or whether the objects are pronominal or not – which derives ‘Holmberg’s Generalization’ (Holmberg (1986)) as a necessary consequence (Sells (2001, 5, 53)). The special positioning of non-projecting pronouns is a property found throughout Germanic both synchronically and diachronically (see e.g., Cardinaletti and Roberts (2002, 133ff.) on German, and Clark (2004) on earlier stages of English).

The motivation for a ‘phonological’ account of OS is undermined by the fact that OS in Swedish shows the same pattern of facts as those of *not* in (2)–(3): unstressed but non-weak pronouns and phonologically weak clitic forms of those pronouns have exactly the same syntactic distribution, so it would appear to be problematic to attribute the special syntax to special phonological properties of the pronouns – the non-weak pronouns have no special properties but do have special syntax (see Sells (2001, 90ff.)).²

A second motivation for a ‘phonological’ account is that there may be interpretive effects of OS, and the interpretive differences may be signalled phonologically. However, the relation between OS and interpretation is indirect. While Erteschik-Shir takes it that OS has interpretive effects (ES, p. 2, 41) in that “only topics or non-foci shift”, this is not a good basis for an analysis. First, it is not clear what it means to be “non-focus” (though possibly what is intended is the [-Foc] feature of Holmberg (1999)). Second, relatedly, the fact that expletive pronouns must shift (Holmberg (1999, (288)), Sells (2001, 168)) seems rather problematic for an interpretation-driven view of OS. And third, as Vikner (2001) shows,

² There is only a partial correlation in the other direction: an accented (therefore strong) pronoun does not shift (Sells (2001, 45)).

when OS is blocked by syntactic constraints, whatever interpretation of the object that would be available if it had shifted is available in situ (a fact that figures prominently in the proposal about OS in Chomsky (2001)). In other words, no interpretations are blocked even if OS is blocked; so it cannot be that interpretation drives OS.

Looking now in more detail at Erteschik-Shir's account of OS, it is this in barest outline: a pronominal object PIs with a preceding verb; adverbs are represented on a separated dimension and are linearized into the clausal structure respecting the object's requirement to PI, and as far to the left as possible (following the subject; (ES, p. 17, 18)). This gives the effect of medial adverbs being generated in positions just to the left of VP, which has been the usual approach in syntactic analyses of Scandinavian (e.g., Holmberg and Platzack (1995, 49)). Erteschik-Shir assumes that some adverbs, such as Swedish *inte* ('not') also PI into the V(+Pro) unit, though I do not fully understand the motivation for this. As the examples in (4) show, there may be several medial adverbs, which come in a fixed order, of which the middle ones are phonologically rather bulky. (4)b also shows OS, with the shifted pronoun underlined.

- (4) a. De har ju därför faktiskt aldrig rest
 they have you.know therefore actually never travelled
 utomlands
 abroad
 'They have really actually never travelled abroad, as you know.'
 (Sells (2001, 57))
- b. Igår läste han dem ju alltså troligen inte.
 yesterday read he them as.you.know thus probably not
 'Yesterday as you know he probably did not read them thus.'
 (Holmberg and Platzack (1995, 56))

It is hard to imagine that these adverbs are phonologically weak and therefore require PI in order to integrate into the syntax. Additionally, the simple negative marker *inte* in Swedish may appear in the initial position in a V2 clause (unlike its counterpart in Danish (cf. (19)a in Erteschik-Shir's paper), suggesting that it is not phonologically weak.

In summary, it seems that Erteschik-Shir's account relies on assumptions about the phonological nature of pronouns and adverbs that may not find independent support outside of the analysis of OS.

2. Cyclic Linearization of Syntactic Structure

Fox and Pesetsky (I use ‘FP’ both for the authors and their ms.) propose an account of OS as part of a general theory of linearization. At various cyclic points in the derivation, structures are ordered by familiar principles, such as Spec precedes Head, and Head precedes Complement, and the categories within them are then related by ‘ordering statements’, which accrue incrementally throughout the derivation. Suppose at a given point, a head *V* precedes a complement *DP*, then the ordering statement *V* < *DP* will be generated. (Using my own terminology for convenience, I will say below that “*V* S-precedes *DP*”, where the “S” is mnemonic for the ‘ordering Statement’. Ordering statements are determined at each Spell-out Domain, which, for my purposes here, is the same domain as the notion of ‘phase’ that is standard within Minimalist syntax.) If, later on, *DP* moves to the left of *V* by the point at which another ordering statement is generated (say, in the next phase up), the statement *DP* < *V* will conflict with the earlier statement, and the derivation will be ruled out. FP motivate their system as a general approach to cyclic movement, as opposed to one-fell-swoop movement, and attempt to use their linearization analysis of various kinds of facts as evidence for a system that involves an intermediate phase at the *VP* or *vP* level. I will argue that instead, the facts motivate only a system that can refer to the initial ‘base’ order, and the final clause-level order. The fact that ordering statements may not conflict is called ‘Order Preservation’ (FP, p. 5) and it is a plausible approach to linearization, though not a necessary one – in fact, the standard copy theory of movement of Chomsky (1995) assumes the opposite. Consider the abstract structure in (5):

- (5) ... [*YP* *XP*_i^α ... [*ZP* ... *V* ... *XP*_i^β]]

Imagine that *XP* moves from its post-verbal to the pre-verbal position as shown in (5). If ordering statements were added to the copy theory of movement, then *XP*^α S-precedes *V*, but *V* S-precedes the copy *XP*^α. This conflict of statements is effectively avoided on the assumption that only one copy of *XP* is pronounced (usually the higher one).

In contrast, in FP’s approach, statements that create conflict cause derivations to crash. Suppose that *YP* and *ZP* in (5) are Spell-out Domains: then in the inner *ZP* Spell-out Domain, *V* S-precedes *XP*; after

that, the YP Spell-out Domain is built, XP moves, and therefore in YP, XP S-precedes V. Such a situation cannot be resolved and simply leads to a crash of the derivation.³

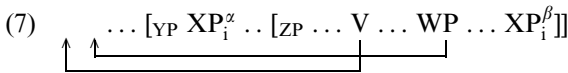
Now we consider the case where (5) represents a single Spell-out Domain, YP – in other words, ZP is a maximal projection, but not a Spell-out Domain. Then (5) represents the standard case of movement to a phase edge, the edge of YP, prior to subsequent movement. If both occurrences of XP ‘count’, then the ordering statements will say that XP S-precedes V and that V S-precedes XP, which the grammar cannot tolerate. As for the mechanism of movement, FP assume that it is not the creation of copies, but rather is a ‘remerge’ operation (FP, p. 39). Reremerge associates a single syntactic object with more than one position in the structure, rather than associating copies and positions in a one-one fashion. Only the most recent Merge of a category counts as the one which defines its mother, and ordering statements are determined with respect to a mother (specifically, the lowest mother which is also a Spell-out Domain). So if ZP is not a Spell-out Domain, the most recent Merge is XP^α , and XP has ordering statements for it determined (only) within YP. In this case, the only ordering statement is that XP S-precedes V. On the other hand, if ZP is a Spell-out Domain, the most recent Merge of XP is XP^β , and ZP is its mother within which ordering statements are determined; further statements will be determined within YP. Now we move to a variant of (5), as a way of understanding FP’s account of OS. Assume that XP is an object which is shifting, and that YP and ZP are Spell-out Domains. (6) is just like (5) except an extra VP-internal category, WP, has been added.

(6) ... [_{YP} XP_1^α ... [_{ZP} ... V ... WP ... XP_1^β]]

ZP is a Spell-out Domain, and clearly V S-precedes XP and WP, and WP S-precedes XP. A key aspect of FP’s proposal for OS is that it is what I will call “delayed-cyclic” movement, in the sense that XP does not move until the higher YP phase; it does not first move to the left edge of ZP. In other words, by the time XP moves, the ordering statements just mentioned have already been generated. Suppose now in the YP phase that

³ This is not quite accurate; the conflict can be avoided if the constituent containing V, namely ZP, is elided – “Salvation by Deletion” (FP, p. 14).

XP moves as shown in (6) – then in the YP ordering domain, XP will S-precede V and WP, in violation of the earlier statements. So, how could XP ever move? – It could do so only if V and WP also moved, to positions higher than XP^α – as indicated by the arrows in (7).



This represents ‘Holmberg’s Generalization’ (Holmberg (1986)) – the final surface order of constituents must not alter the underlying relative order of constituents.

In the case where YP is a Spell-out Domain but ZP is not, then XP in (7) will S-precede V and WP. FP dub this the “Inverse Holmberg Effect” – for XP must precede V and WP at every later Spell-out Domain in the derivation, requiring the opposite of the underlying relative order.

FP’s proposal accounts for all of the core cases of OS and seems to extend straightforwardly to Long Object Shift (on the assumption that the subject is not part of the inner Spell-out Domain). However, there appear to be several empirical problems that appear when the interaction of different movements is considered. As I have just described, a VP-internal constituent XP can adjoin to the edge of VP during the VP phase, thereby S-preceding V, and then XP will undergo further movement. Otherwise, XP remains VP-internal during the VP phase and therefore V will S-precede XP in that Spell-out Domain. The first problem I will discuss concerns passive:

(8) Max_i has been [_{VP} interviewed t_i].

In the VP Spell-out Domain, the V *interviewed* S-precedes the DP *Max* (which is in the position of the trace t_i in that domain). However, at the clause level, in the higher Spell-out Domain, clearly the DP S-precedes V. In other words, while OS obeys Holmberg’s Generalization, passive does not. This problem with passive can be avoided by assuming that the moving DP moves to the left edge of the VP first, even though it is not undergoing A’-movement. This move has the consequence that OS cannot be either A’- or A-movement, raising the issue of exactly what kind of movement it is supposed to be.

The Inverse Holmberg effects argued for by FP create empirical problems, in Swedish. In section 5 they argue that VP-internal quantifica-

tional XPs should undergo ‘Quantifier Movement’ to a position external to VP; they first move to the edge of VP during the VP phase, thereby S-preceding V and any other nonmoved VP-internal XPs. I will illustrate with Negative Quantifiers (NegQPs) in Swedish. Following the foundational work by Christensen (1991), Svenonius (2000) and Sells (2000, 2001) argue that NegQPs in Norwegian and Swedish are obligatorily external to VP in the surface analysis. To derive such structures, FP assume that an object NegQP is first merged within VP and moves to a VP-external position, proceeding via the edge of VP. (9) shows the movement within the VP phase:

- (9) [_{VP} NegQP_i [_{VP} V XP t_i]]

In (9), even though V and XP precede NegQP in its original base position (indicated by t_i), as NegQP moves to the edge of VP during the VP phase, when ordering is read off the structure, NegQP S-precedes both V and XP. Hence, the Inverse Holmberg Effect predicts that in a higher Spell-out Domain, it will not be grammatical for V or XP to precede NegQP. However, both of these possibilities exist as grammatical structures in Swedish.

Even a simple example like (10) runs afoul of the ordering statements.⁴

- (10) Jan såg ingenting.
Jan saw nothing

As a main clause, this is a V2 structure, and the V must have moved out of VP to some higher head position, maybe Fin⁰ or C⁰. Whatever its position, V S-precedes NegQP in the clausal Spell-out Domain, in conflict to what is determined by (9). To account for this example, FP would have to assume that the movement of NegQP is optional, and that *ingenting* has not moved in (10). While the general operation of Quantifier Movement may be optional in Icelandic, it appears to be obligatory for NegQPs in the mainland Scandinavian languages, as noted above.

Returning to (9), what if XP undergoes OS? By FP’s assumptions, a shifting object does not move in the lower phase, but only delayed-

⁴ FP present an Icelandic example with a similar structure to (10) which is ungrammatical, and use it to argue for their proposal. However, the data I present here indicates that the distribution of NegQPs in Swedish shows no evidence of the Inverse Holmberg Effect.

cyclically in the phase higher than VP. The proposal leads us to expect that XP cannot precede NegQP in that Spell-out Domain, as that would violate the S-precedence relations determined for (9). However, as shown in Sells (2001, 97), the facts are exactly the opposite; a structure with the underlying order XP–NegQP is only grammatical under two conditions: if (i) XP undergoes OS and (ii) the base order as just given is preserved. The two bare objects of a ditransitive verb in Swedish come in the order IO–DO, and if the NegQP is the DO, the only grammatical examples involve a pronominal (and preceding) IO, as seen in the contrast between (11)a and (11)b. Sells argues that the NegQP *inga pengar* is only grammatical if it is VP-external, and this can only be the case if everything that would have preceded it in VP is also VP-external. As the non-pronominal *Sven* cannot undergo OS in (11)b, it must be VP-internal, and therefore so must *inga pengar* – hence the example is ungrammatical.

- (11) a. Jag lånade dig inga pengar.
 I lent you no money
 (*dig* can shift to be outside VP, so *inga pengar* can be VP-external)
- b. *Jag lånade Sven inga pengar.
 I lent Sven no money
 (*Sven* must be in VP, so *inga pengar* must be VP-internal)
- c. Jag lånade inte Sven några pengar.
 I lent not Sven any money

The acceptable ‘version’ of (11)b is shown in (11)c – using a periphrastic combination of negation plus an existential quantifier for the DO, which indeed remains in situ in the VP. The fact that the underlying order IO–DO has to be maintained in these examples suggests that the account of the distribution of NegQPs in terms of (9) and the Inverse Holmberg Effect is incorrect, because the data in (11) indicate that the DO (the NegQP) shows a ‘regular Holmberg effect’ with respect to the IO.

For Swedish, at least, we see that the generalizations about order emerge at the clause level (CP), but that there is no evidence for a CP-internal Spell-out Domain such as vP or VP, and no evidence for the role of S-precedence statements that are generated at intermediate points in the derivation.

As more evidence for these intermediate ordering statements, FP offer the account of Ko (2004) regarding the distribution of floated quantifiers in Korean and Japanese, exemplified in the Korean data paradigm in (12). Ko's argument for cyclic linearization is based on her proposal that subject scrambling is possible (revising the long-standing view that subject scrambling is not possible, based on the arguments in e.g., Saito (1985)).

- (12) a. maykcwu-lul John-i sey-pyeng masi-ess-ta
 beer-Acc John-Nom three-bottle drink-Past-Decl
 'John drank three bottles of beer.'
- b. *haksayng-tul-i maykcwu-lul sey-myeng masi-ess-ta
 student-Pl-Nom beer-Acc three-person drink-Past-Decl
 'Three students drank beer.'

The apparent generalization is that the object may not intervene between the subject and a floated quantifier (FQ) related to the subject, while the subject may intervene between the object and an FQ related to the object.

It is assumed that (12)b is derived as follows: the underlying structure has the order: subject–FQ–object–V; first, the object scrambles in front of the subject, and then the subject scrambles in front of the scrambled object. The analysis developed by Ko is that vP is a Spell-out Domain in Korean, a domain which includes the subject. Within the vP phase, the object must scramble to the left edge, the first movement just mentioned. Hence in the vP Spell-out Domain, the object S-precedes the subject. In the next domain up, scrambling of the subject to derive (12)b also creates an ordering conflict, as now the subject S-precedes the object, and the example is out. This is an Inverse Holmberg effect: the O>S order (an 'inverse' order) must be maintained through the derivation, as it is in (12)a, which only involves ordering domains where the statement O<S holds.

However, it seems unlikely that the paradigm in (12) is to be explained in terms of the interaction of S-precedence statements. There are examples with the same structure as (12)b which are acceptable, such as (13), as observed by Kuno and Kim (2004).⁵ Kuno and Kim argue that the linear order in examples such as those in (12) is determined by information

⁵ See Gunji and Hasida (1998) and Hoji and Ishii (2004) for similar observations on Japanese.

structural properties such as topic and focus, but that there are no specific structural bans on the structure.

(13) A: Have the students in your school applied to our college?

B: ney, cehuy pan haksayng-tul-i ku tayhak-ey
 yes my class student-Pl-Nom that college-to
 wense-lul sey-myeng nay-ass-eyo
 application-Acc three-person submit-Past-Decl
 ‘Yes, three students in my class submitted an application to
 your college.’
 (slightly modified from Kuno and Kim (2004, 5))

Ko (2004) acknowledges that examples of this form with a focussed FQ may be acceptable (footnote 2). As the effect in (12)b is attributed to the relative order of the subject and the object, it is not immediately clear why properties of the FQ would interact with this. Looking beyond Scandinavian, then, the positive evidence for the Inverse Holmberg Effect does not seem to be as strong as FP might hope.

3. Summary

The idea that OS is in some sense ‘phonological’ seems to be a response to two observations: first, that OS is not like the standard types of A- or A’-movement (as observed by Holmberg and Platzack (1995)), and second, that it is constrained only by overt elements in syntax. However, OS is not ‘phonological’ – it has no special phonological properties – but it does involve non-canonical syntactic structures (either special positions for objects external to VP, as in the account of Sells (2001), or delayed-cyclic movement, as in FP).

Rather, OS is constrained by some principles of linear order, however they are characterized and enforced. The evidence that FP present for ordering statements which may conflict (rather than resolving by one requirement overruling the other) is only strong with respect to data which obeys Holmberg’s Generalization, which always refers to the base order of the elements in question. As OS is taken to be delayed-cyclic, this also means that Holmberg’s Generalization provides no evidence for a Spell-out Domain at the vP or VP level – the ordering statements could simply

refer to the base order and the clause-level Spell-out Domain, without any loss of empirical coverage. This puts FP's proposal in the class of order-preservation approaches, along with Müller (2001) and Williams (2003), among others.

The evidence for the Inverse Holmberg Effect, which requires ordering statements that are generated during the derivation (and which crucially ignores the underlying base order), is much weaker, and in some cases leads to incorrect predictions. It is this evidence which motivates the existence of a Spell-out Domain smaller than CP; if the evidence is not strong, again we fall back to a theory that refers only to base order and clause-level order.

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References

- Asudeh, A. & Mikkelsen, L. H. (2000). Incorporation in Danish: Implications for interfaces. In: Cann, R., Grover, C. & Miller, P. (eds.), *Grammatical Interfaces in HPSG*. (pp. 1–15). Stanford: CSLI Publications.
- Bresnan, J. (2000). Optimal syntax. In: Dekkers, J., Leeuw, F. van der & Weijer, J. van de (eds.), *Optimality Theory: Phonology, Syntax and Acquisition*. (pp. 334–385). Oxford: Oxford University Press.
- Cardinaletti, A. & Roberts, I. (2002). Clause structure and x-second. In: Cinque, G. (ed.), *Functional Structure in DP and IP: The Cartography of Syntactic Structures, Volume 1*. (pp. 123–166). New York: Oxford University Press.
- Cardinaletti, A. & Starke, M. (1996). Deficient pronouns: A view from Germanic. In: Thráinsson, H., Epstein, S. & Peter, S. (eds.), *Comparative Studies in Germanic Syntax II*. (pp. 21–65). Dordrecht: Kluwer.
- Chomsky, N. (1995). *The Minimalist Program*. Cambridge: MIT Press.
- (2001). Derivation by phase. In: Kenstowicz, M. (ed.), *Ken Hale – A Life in Language*. (pp. 1–52). Cambridge: MIT Press.
- Christensen, K. K. (1991). Complex passives reanalyzed. *Working Papers in Scandinavian Syntax* 48, pp. 45–75.
- Clark, B. Z. (2004). *A Stochastic Optimality Theory Approach to Syntactic Change*. Doctoral dissertation, Stanford University.
- Gunji, T. & Hasida, K. (1998). Measurement and quantification. In: Takao Gunji and Kôiti Hasida (eds.), *Topics in Constraint-Based Grammar of Japanese*. (pp. 39–79). Dordrecht: Kluwer Academic.
- Hoji, H. & Ishii, Y. (2004). What gets mapped to the tripartite structure of quantification in Japanese. In: Schmeiser, B. et al. (ed.), *Proceedings of WCCFL*, Vol. 23. Somerville, MA: Cascadilla Press.

- Holmberg, A. (1986). *Word Order and Syntactic Features in the Scandinavian Languages and English*. Stockholm, University of Stockholm: Department of Linguistics.
- (1999). Remarks on Holmberg's generalization. *Studia Linguistica* 53, pp. 1–39.
- Holmberg, A. & Platzack, C. (1995). *The Role of Inflection in Scandinavian Syntax*. New York: Oxford University Press.
- Josefsson, G. (1992). Object shift and weak pronominals in Swedish. *Working Papers in Scandinavian Syntax* 49, pp. 59–92.
- Kim, J.-B. & Sag, I. A. (2002). Negation without head movement. *Natural Language and Linguistic Theory* 20, pp. 339–412.
- Ko, H. (2004). Constraining scrambling: Cyclic linearization and subject movement. In: Schmeiser, B. et al. (ed.), *Proceedings of WCCFL*, Vol. 23. Somerville, MA: Cascadilla Press.
- Kuno, S. & Kim, S.-Y. (2004). On syntactic intervention. In: Kuno, S. et al. (ed.), *Harvard Studies in Korean Linguistics*, Vol. 10. Harvard University: Dept. of Linguistics, pp. 3–31.
- Müller, G. (2001). Order preservation, parallel movement, and the emergence of the unmarked. In: Legendre, G., Grimshaw, J. & Vikner, S. (eds.), *Optimality-Theoretic Syntax*. (pp. 279–313). Cambridge: MIT Press.
- Saito, M. (1985). *Some Asymmetries in Japanese and Their Theoretical Implications*. Doctoral dissertation, MIT.
- Sells, P. (1994). Sub-phrasal syntax in Korean. *Language Research* 30, pp. 351–386.
- (2000). Negation in Swedish: Where it's not at. In: Butt, M. & King, T. H. (eds.), *Online Proceedings of the LFG-00 Conference*. Stanford: CSLI Publications. (At <http://csli-publications.stanford.edu/LFG/5/lfg00.html>).
- (2001). *Structure, Alignment and Optimality in Swedish*. Stanford: CSLI Publications.
- Svenonius, P. (2000). Quantifier movement in Icelandic. In: Svenonius, P. (ed.), *The Derivation of VO and OV*. (pp. 255–292). Amsterdam, Philadelphia: John Benjamins.
- Toivonen, I. (2003). *Non-Projecting Words: A Case Study of Swedish Particles*. Dordrecht: Kluwer.
- Vikner, S. (2001). The interpretation of Object Shift and Optimality Theory. In: Müller, G. & Sternefeld, W. (eds.), *Competition in Syntax*. (pp. 321–340). Berlin, New York: Mouton de Gruyter.
- Williams, E. (2003). *Representation Theory*. Cambridge: MIT Press.
- Zanuttini, R. (1997). *Negation and Clausal Structure: A Comparative Study of Romance Languages*. New York: Oxford University Press.

Object Shift and linearization at the PF interface

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Abstract

Both target articles, in conformity with minimalist ideals, aim to discover to what extent syntax itself can be relieved of descriptive burden, and just how much of word order can be attributed to mechanisms of linearization at the PF interface. I point out two paradoxes the treatment of V2 gives rise to in Erteshik-Shir's paper, and dispute the status and effectiveness of processing constraints in accounting for (un)grammaticalities in various word order patterns. I discuss the relation of Fox and Pesetsky's cyclic linearization algorithm to 'tucking in' as well as to the 'Attract Closest' property of movement. Qualms are raised regarding several apparently stipulative and/or conceptually unattractive properties of the linearization mechanism, and ostensible violations of order preservation are examined.

1. Introduction

What makes the juxtaposition of the two target articles in this volume particularly thought-provoking is that both are directly concerned with issues of linearization at the PF interface of the grammar, aiming to discover to what extent syntax itself can be relieved of descriptive burden, and just how much of word order can be attributed to PF mechanisms. In addition to the shared interest in architectural considerations of division of labor, Scandinavian Object Shift (OS) plays a key role in both argumentations.

The role of linear order and PF is central to a number of current theories of OS, most notably Holmberg (1999) (where in fact OS is treated essentially as a PF operation) and Chomsky (2001)¹, as well as

¹ Interpretive conditions and effects (cf. specificity, focus) figure both in Holmberg (1999) and in Chomsky (2001). The LF interface is left to the side in both target articles.

Bobaljik (1995, 2002) (where PF-adjacency functions prominently). The two target papers can be considered members of the ‘PF’ family of accounts of OS. Both studies make assumptions with intriguing consequences for the analysis of OS and far-reaching implications beyond. In what follows, I elucidate my main concerns about each of the two proposals, and also spell out several of their apparently desirable consequences.

2. Linearization and PF operations

Erteshik-Shir (ES) argues in favor of a radical position according to which a number of displacements that are commonly viewed as syntactic are to be viewed as resulting from operations at PF. The essential claim about OS defended by ES is that it does not exist: it is the result of the PF operation of Prosodic Incorporation (PI) of prosodically weak elements (pronominal objects and weak adverbs like negation) and adverb linearization (adverbs are assumed to be merged to the phrase they modify in a third dimension, cf. e.g. Áfarli 1997), interacting with independently necessary displacements. Basing herself on these fundamentals, ES provides an engagingly detailed treatment of OS phenomena across Scandinavian.

One significant advantage of ES’s approach, beyond its impressive coverage of variation across Scandinavian, is that it provides a straightforward explanation of why OS has ‘no effect’ (it has neither A- nor A-bar properties: it does not affect binding configurations, does not license parasitic gaps, or give rise to weak crossover effects). The reason is simple: no independent displacement of OS exists. Another welcome consequence of this idea is order preservation in constructions with multiple objects and multiple adverbs: since objects do not shift, their order will remain unaffected even if adverbs get linearized in between or after them.² The approach is also designed to furnish a fairly direct answer to the question of why weak pronouns are the typical elements undergoing OS. This is

² As far as ‘free ride’ to V2 is concerned, given the order inside VP, only the IO, or both IO and DO, can be incorporated and be taken along to V2, but DO alone cannot.

due to the assumption that it is weak pronouns (as opposed to full-fledged DPs) that have a need to be PI-ed. Finally, ES's system effectively rules out an inadmissible VP-topicalization construction that would cause trouble for many accounts of OS. In this construction the fronted VP contains the verb and the direct object, but not the indirect object (schematically: [V — DO] ... IO ...; cf. Fox and Pesetsky's (30b)). Such a structure is excluded on ES's assumptions once again owing to the fact that there is no OS per se: the indirect object cannot be removed by OS from the VP prior to VP-topicalization.

If viable, ES's account is bound to emerge as an extremely competitive alternative on the OS scene. However, in order to really get the approach off the ground, several complications need to be resolved and a number of points must be clarified.

2.1. *PF-operations and adverb linearization*

One aspect of the account that needs clarification is the analysis of verb displacement to the V2 position, which gives rise to two intricate paradoxes with regard to V2.

On the one hand, it appears that V-movement to V2 must be a PF phenomenon (as suggested by ES), for two reasons. First, the PF operation of PI can feed it. Second, in ES's model V-movement to V2 needs to be made sensitive to targeting a linearly second PF position (the V2 restriction applies at PF). This latter requirement is necessary: otherwise several contrary-to-fact word orders would be generated, for instance orders schematically represented in (1a) and (1b). (1a) is the result of PI of DO into IO, which in turn gets topicalized; in (1b) it is the subject pronoun that is PI-ed into V proclitically, feeding movement to V2.

- (1) a. *IO+DO V ... Adv ...³
 b. *Obj Subj+V ... Adv [...]

³ IO = indirect object (a lexical DP in (1a)), DO = direct object (a weak pronoun in (1a)). ES's example (6d) would also be left without an account without maintaining a phonological V2 constraint: the proposed phi-intervention constraint cannot apply there, since the object pronoun is morphologically marked (cf. §2.2).

Paradoxically, on the other hand, V-movement to the clause periphery appears to be a syntactic, rather than a PF operation. First, if it is construed as PF displacement, yes-no questions turn out to pose a complication. The reason is that in this clause type a phonological V2 restriction remains inapplicable, hence (2), where the subject pronoun *PI*-s into *V* prior to V-movement, is ruled in, contrary to fact.

(2) *Subj+V ... Adv [... Obj]

Another indication that V-movement to the periphery is syntactic is that it can have scopal effects when the moved verbal item is scope-sensitive (this is true of head movement more generally, cf. Benedicto 1997, Müller 2002, Note 2 and references cited there; see Zwart 2001 for arguments against a PF-treatment of head movement).

A further significant shortcoming of ES's proposal to treat V-movement to V2 as a PF displacement, which again speaks in favor of a syntactic analysis, is that the motivation of this operation is ill-founded. On the one hand, a PF V2 restriction is doubtful in view of systematic exceptions to such a phonological constraint in Mainland Scandinavian (which even *PI* into the *V* cannot explain away), as discussed in Egerland (1998) and Nilsen (2003: 79–80). On the other hand, the proposed discourse-functional trigger, namely that *V* in the V2 position serves to demarcate the topic and the semantic predicate expression (p. 19), is doubtful for several reasons. For one, the restriction that topics in Scandinavian are unique should suffice to identify the border of topic and predicate phrase. Also, the idea involves a clear lookahead property: at the point where *V* is moved to the edge in the derivation, no topic exists to its left. Further, if V2 is to mark topicalized material, then the question is why topicalization is not invariably linked to V2 across languages.

The apparent, and apparently contradictory, conclusions are that within ES's model, V-displacement to the V2 slot must be a PF operation, and at the same time, it must be a syntactic movement.

A second paradox concerns the order of operations. Consider the sentence pattern in (3) below, where an indirect object has been topicalized/*wh*-moved, while the direct object pronoun has undergone OS, being given a free ride by the verb.

- (3) IO_{Top/Wh} V+DO_{Pron} ... Adv [...] (Swedish)
 ?Honom gav + dig Kleopatra inte
 he-dat gave you-acc K. not
 'Cleopatra did not give you to HIM'⁴

It is clear that V-movement to the edge (=V2) must precede topicalization (=TOP) in the derivation generally (whether V-displacement is phonological or syntactic): V2 < TOP. However, at the point where movement of the verb applies, V and DO are separated by IO, hence PI could not have applied. The only way to derive (3) is if TOP < V2, hence an operation ordering paradox is derived.⁵

2.2. *Linearization and processing*

The other major component of the system that, in my view, gives cause for concern is the status and effectiveness of applying processing motivations in accounting for (un)grammaticality of linearization patterns. The first issue is the status of restrictions derived from processing. For instance, if ID (=ES's (16)) is a processing constraint, then it is unclear (i) why it should apply to production too, and (ii) why it should be inviolable. As regards question (ii), although it is true that in the absence of any other disambiguating factor ID is frequently taken to be the processing strategy applied, various pragmatic clues, including animacy or previous discourse, can trigger a violation of ID, even if neither the subject nor the

⁴ In the interest of an animate DO pronoun, the example involves a scenario where slaves are talking about who Cleopatra didn't give one of them to as a present. Note that this pattern cannot be generated even by a rule of 'metathesis' postposing the subject (cf. ES's discussion of long OS). Thanks to Jonas Winnerlov and Kristina Marosi for help with Swedish data.

⁵ Regarding the issue of the ordering of operations, even though ES claims that no (extrinsic) ordering of her PF operations is necessary, this contention is challenged as far as the operation of adverb linearization (AdvLin) is concerned. This is because AdvLin needs to obligatorily follow both V2, subject raising, as well as clause-internal subject movement (which operation is still required) if the account is to be prevented from over-generating by allowing linearized and PI-ed weak adverbs a 'free ride' indiscriminately, thereby giving rise to reordering of adverb sequences.

object is grammatically marked (morphologically, or through their position). This should mean that ID cannot be made responsible for outright and exceptionless ungrammaticalities.

ES's model also relies on a processing constraint which blocks elements with phi-features from intervening between subject and verb (or verb and subject) (p. 3). ES claims that this constraint is derived from ID (though it is unclear how exactly this is done). Indeed, from her discussion of 'long OS' in Swedish (p. 13) it appears that the phi-intervention constraint, as a corollary of ID, applies only in cases where neither the subject nor the object is grammatically marked as such. However, then, contrary to ES's intentions, examples like her (12) (given below as (4a)) are not ruled out: the object pronoun is morphologically marked.⁶ The same type of problem arises for ID itself: ES's (23a)(=4b) should not come under the effect of ID, since the object is marked as such. Thus, the example is (falsely) predicted to be well-formed.

- (4) a. *I går mødte + ham Peter ikke (Danish)
 yesterday met him P. not
 b. *I går mødte ikke Peter ham

Further interesting questions arise regarding VOS and SOV languages. For it is far from obvious whether ID and especially its corollary, the phi-intervention constraint, wouldn't prove much too restrictive in view of various word order patterns available in these language types (even if 'positional clues' are taken into account). Clearly, if they are to be effectively exploited, the status and scope of processing constraints in the account should need further elaboration.

3. Linearization and derivational cycles

Fox and Pesetsky (F&P) explore to what extent combining a particular algorithm of linearization with a phase-based derivational theory (Chomsky 2000, 2001) can derive constraints on the computation of word order.

⁶ Her (6d) is also left without an account, unless a phonological V2 constraint is introduced, cf. Note 3.

In particular, according to their proposal relations of precedence are computed cyclically at the completion of each phase, and precedence relations once established in the course of a derivation must be conserved.⁷ F&P's account of HG qua order preservation straightforwardly extends to derive the order conservation property of constructions involving multiple objects and multiple adverbs as well.

The main appeal of the proposal lies in the fact that it exploits a property of the current multiple Spell Out model of minimalist theory that is independently necessary (i.e. some linearization algorithm must be part of Spell Out) in order to provide a unified explanation of a set of apparently unrelated word order restrictions (including HG), as well as a deep property of (long) syntactic movement in general: its successive cyclic nature.

3.1. *Tucking in and Attract Closest*

Given the order preserving character of cyclic linearization, as far as I can see, the approach can potentially derive 'tucking in' patterns involved in certain multiple movement constructions (e.g. multiple *wh*-movements) discussed in Richards (1997, 1998) and in much recent work. According to Richards, 'tucking in' involves multiple movements to a projection where the element to be moved later, instead of extending the root node, 'tucks in' countercyclically below the elements that have already been

⁷ This conception places F&P's approach in the family of approaches to OS/HG which crucially draw on an 'order preserving' property of syntax, cf. e.g. Müller (2001), Sells (2001), Williams (2003). An area that could furnish supporting evidence for F&P is apparent rightward movements. Providing that such movements exist, the prediction is that rightward movement is restricted to phase-final elements (a potential case in point is Right Node Raising), unless rightward movements touch down at the right edge of the phase, or more than one element gets right-moved, in which cases their relative reordering should be possible. If rightward movement is in reality movement to the left (call it M_1) followed by movement of the remnant (call it M_2), then it must be noted that F&P's particular implementation of 'order preservation' is unique in that a shape conserving motivation for remnant movement is absent if (i) M_1 has moved from one phase to another via an 'edge' position, or (ii) if M_1 is a phase-internal movement, and in that M_2 is not triggered immediately after M_1 : M_2 can be delayed up until the completion of the phase.

moved.⁸ ‘Tucking in’ appears necessary in the context of the Attract Closest property of movement, which is built into Attract (Chomsky 1995) / Agree (Chomsky 2000, 2001) by definition. This is because if the closest relevant item has to be moved first (by Attract Closest), then the original hierarchical relations (and hence, order) are preserved only if the next moved item lands below the one that has already been moved. Importantly, F&P’s system can potentially derive tucking in patterns without actually ‘tucking in’, i.e. without introducing any countercyclic movement operations.

It can do so, because it may well allow us to drop the Attract Closest property from the definition of Move (i.e. Attract/Agree). In fact, such an amendment would constitute a significant improvement in the minimalist theory of movement. For one thing, the Attract Closest property is a plain stipulation as it stands, even though it is commonly claimed to be motivated by the need to minimize operational complexity, which is a corollary of the Strong Minimalist Thesis (cf. Chomsky 2000, 2001). The problem with this claim is twofold. On the one hand, the Attract Closest property reduces operational complexity only if the ‘search for Goal’ operation is taken literally to be a search operation. As far as I can see, however, there is no compelling reason to assume that movement needs to involve a search operation in the model of competence, and indeed the ‘search for Goal’ operation itself is left undefined: if it were defined, it would very likely involve selecting syntactic items one after the other based on a given ordering of those items as determined by an algorithm operating on the syntactic hierarchy. If such an incremental selection operation and an independent order-determining algorithm were indeed part of the competence model, then that would clearly add to the complexity (and operational complexity) of the model itself, making it less optimal. On the other hand, the Attract Closest property is a major hindrance to efforts aiming to eliminate the remaining representational residue of the derivational minimalist approach: insofar as it involves a search operation carried out on a (partial) representation, the model continues to remain

⁸ ‘Tucking in’ movement may well be the only remaining non-root-extending transformation if head movement is not to be analyzed as involving head-adjunction, as is held by a variety of recent proposals.

a mixed derivational–representational theory (cf. Brody 2002). Significantly, Attract Closest in fact becomes to a large extent redundant if F&P’s proposal is correct: if elements E_1 and E_2 are of the same type [E] and E_1 c-commands E_2 at some phase PH, they will be ordered as $E_1 < E_2$. As this order must be preserved, if a probe attracting type [E] is introduced in the derivation, only E_1 can be moved, while E_2 cannot, because that would reverse the established order.⁹

If Attract Closest turns out to be eliminable, that is a significant and welcome consequence in itself, as I have argued.¹⁰ Returning to ‘tucking in’ patterns now, if Attract Closest can be dropped, they are clearly derivable without countercyclic movements. This is because we are free to move to the current projection a ‘more distant’ element (say, an object *wh*-phrase) first, and a ‘closer’ element (say, a subject *wh*-phrase) next, extending the root at each step, as long as at the completion of the phase (say, CP), the order of the moved elements is unchanged compared to their original order. This is a highly attractive perspective, one that evidently opens up an array of interesting novel research questions.¹¹

⁹ The question of cases where E_1 does not c-command E_2 , but is contained in a phrase that c-commands E_2 , must also be addressed. However, a model incorporating Attract Closest is no different: the same issue arises there too.

¹⁰ In fact, if the cyclic linearization algorithm and Attract Closest exhibit large-scale redundancy, then ideally, one of the two must go. I have suggested that optimally it is Attract Closest that is to be dispensed with. However, if there still remains independent motivation/need for Attract Closest (that is not covered by the linearization algorithm), then that can be held against the linearization algorithm itself. Potential scenarios where Attract Closest can, but cyclic linearization cannot act to constrain derivations include cases where E_1 and E_2 are both contained within the same phase (or rather, the same ‘slice’, cf. §3.2), or where E_1 and E_2 are in adjacent phases. For instance, in languages where VP functions as a Spell Out domain, cyclic linearization does not effectively derive superiority of subject over object either in *wh*-movement to CP or in A-movement to TP.

¹¹ Chomsky (2004) suggests that A-movements may not alter previously existing hierarchical relations that A-bar movements are sensitive to. This would neatly follow from an account of ‘closeness’ along the lines laid out above (whereas it potentially poses a problem for ‘Attract Closest’; Chomsky’s (2004) proposal is that A-bar movements to CP remerge occurrences located within the ν P, instead of those in the TP/AgrP-domain). This is because ordering relations established at the ν P phase remain unaffected in the derivation up until the CP node even if the hierarchical relations introduced above the ν P level reverse those that have existed within ν P.

3.2. *Cyclic linearization*

One aspect of the model that needs some further elaboration, in my view, is the linearization algorithm itself. Two qualms relate to the status of notions figuring in the Laws of Precedence. The first concerns the fact that in the Laws themselves (cf. F&P's (48)) the notions specifier and complement play the key role. Linearization can rely on such notions at no theoretical cost only if they are, or can be derived in a non-stipulative fashion from, primitives of the theory – an issue that remains elusive in the paper (see Epstein (1999), and especially Brody (1997), for relevant discussion).¹²

Second, the fact that non-last (i.e. trace) occurrences of an element (taking movement to be re-merge, as assumed by F&P) do not count for linearization is given more or less directly as a stipulation: the 'X is the mother of α and β ' condition seems to be inserted in the Laws of Precedence merely to then define 'mother of α ' as "the constituent formed by the most recent Merge of α " (F&P's (49)), thereby restricting the scope of the Laws of Precedence previously given to exclude non-last, i.e. trace, occurrences.¹³

A final comment concerns the proposed execution of the cyclic property of linearization. F&P suggest that the choice of linearization mechanism they put forward is in part "dictated by a consideration of the degree of redundancy that is allowed to exist" in the final set of ordering statements. One aspect of this linearization mechanism is that at each phase level *all* nodes created up to that point are ordered. Crucially, contents of the current phase are ordered not only with respect to each other,

¹² In Brody's Mirror Theory, where the (sole) rule of linearization orders specifiers to precede heads, 'specifier' is a primitive of the theory (while 'complement' is not).

¹³ A closely related issue is whether copies/occurrences created by covert movement matter for Laws of Precedence. On F&P's definition at least covert category movement (argued elsewhere by Pesetsky to exist) should count, as it creates a new occurrence. If, however, covert movement is not relevant for some reason (for instance, if covert movement does not create new occurrences, e.g. because it is Agree or FF-movement), then in so far as cyclic linearization is assumed to be responsible for successive cyclicity, covert movements are not expected to proceed successive cyclically. Indeed, this is an assumption that was frequently made in GB theory to account for the lack of *wh*-island and other Subadjacency effects with certain *in situ* elements, especially *wh*-*in situ* (cf. Watanabe (1992) and references cited there).

but also they are determined to precede material of the next lower phase (call this ‘on-line’ ordering). The point to emphasize here is that this is by no means an inherent property of a phase-by-phase (cyclic) linearization; it is an additional feature. This additional feature is required, however, quite apart from considerations of (non-)redundancy, if one is to effectively capture the successive cyclic nature of (long) movement.

It is the operation Linearize (52) that plays the key role in achieving ‘on-line’ ordering. Linearize orders with respect to each other first and last elements of such complex nodes that are already in a precedence relation, and this applies also to phasal nodes themselves. For instance, a clause-embedding head *V* and the (leftmost element in the leftmost) *XP* in the edge of *V*’s phasal complement *CP* will be ordered upon the completion of the next phase above *V*.¹⁴ If *V* = *wonder* and *XP* = *what*, *wonder* < *what* is generated, correctly. Consider the case when there is yet another clause-embedding predicate above *V* = *think* (as in [_{CP1} *What did you say* [_{CP2} *they think* [_{CP3} *we bought?*]]]), and the derivation is at stage *CP*₁. Now it crucially matters at this point whether (i) *what* itself (when in the edge of *CP*₃) was ordered with respect to *think* at the *CP*₂ phase level (and derivatively with respect to other elements of the *CP*₂ phase), as F&P assume, as part of Linearize, or (ii) only the phasal category *CP*₃ was (and the final set of ordering statements applying to terminal nodes is computed at some later stage, e.g. when the derivation is completed). Assume (ii), contra F&P, for the sake of the argument. In the case of (ii), *what* would be ordered at the *CP*₃ phase level to precede all other material of *CP*₃, and at the *CP*₂ phase level *what* would not be directly concerned in new ordering statements. Continuing this derivation, at the *CP*₁ phase level *what* is moved to [*Spec*,*CP*₁] for *wh*-feature-checking reasons. *What* now still precedes material in *CP*₃, and importantly, *what* itself has not been ordered up to the *CP*₁ phase level with respect to anything other than material in *CP*₃, hence no ordering contradiction ensues by moving *what* from [*Spec*,*CP*₃] to [*Spec*,*CP*₁] in one fell swoop. In short, the successive cyclic property of (long) movement is not derived. Assuming (i) (i.e. ‘on-line’ ordering), in contrast,

¹⁴ I disregard here *vP* phases in order to keep the presentation simple; the point of the argument is not affected by this idealization.

effectively enforces successive cyclic movement: *what* must be moved to the left edge of CP₂ at the CP₂ phase level, otherwise it would be fatally trapped in [Spec,CP₃] by virtue of being ordered after *think* (and hence all material belonging to the CP₂ phase). Then, apparently, an ‘on-line’ execution of cyclic linearization is required to be able to derive successive cyclicity of movement.

3.3. *Spell Out domains*

In their account of HG and the ‘Inverse Holmberg Effect’ obtaining in Scandinavian Quantifier Movement, F&P propose that (i) OS does not proceed through a verb phrase edge, and (ii) in Scandinavian languages the relevant Spell Out domain (which for F&P equals (strong) phase) is VP instead of *v*P. The first assumption is stipulative as it stands – if compliance with subsequent ordering statements can drive movements to phase edges in general, it should also be able to do so in OS. Assumption (ii) is crucial, because if the subject was included in the first Spell Out domain, then the verb would never be able to surface to its left at a later stage, which is contrary to fact.¹⁵ Clearly, identifying VP instead of *v*P as the first (strong) phase is a departure from standard assumptions. It is not obvious whether the set of evidence that has been taken to motivate movements that target the edge of *v*P (e.g. stranding, reconstruction facts) can entirely be reinterpreted as being in conformity to a VP-edge based analysis – this remains to be demonstrated.

More interestingly, intriguing predictions are made for OV V2 languages like Dutch or German. Given their V2 character, here too VP must act as a Spell Out domain instead of *v*P.¹⁶ Assuming a VO base and O-to-[Spec,AgrOP]/[Spec,*v*P] to derive surface OV order, the expectation is that O must raise to [edge,VP] in OV patterns, but must not in V2 clauses (except if it ends up in [Spec,CP]). On an OV base hypothesis, the VP itself must minimally contain a further XP that dominates the basic constituent of O+V. This is necessary in order to allow V to precede

¹⁵ Another factor that enforces (ii) is the existence of constructions involving OS across *in situ* subjects in Icelandic (cf. e.g. Jonas 1996), as well as ‘long OS’ in Swedish.

¹⁶ The same applies to VSO languages: otherwise VSO orders would not be able to arise.

O at the VP-phase level, which is necessary in V2 clauses. It would be interesting to see what independent support can be found for these predictions on either the OV or the VO theory of the base.

3.4. *Conflicts in linearization and crash*

In some varieties of Norwegian and Swedish direct object pronouns can be shifted across a VP-internal indirect object pronoun (Hellan and Platzack 1999). In addition, in all dialects of these languages the direct object may precede the indirect object when both pronouns are shifted. These patterns involve apparent violations of order preservation. The strategy for F&P is to argue that they are indeed merely apparent exceptions. To do so, they need to assume some VP-internal movement of DO above IO. Another set of cases that seem to disobey order preservation are instances of passivization of a DO across an IO, which is permitted in Norwegian and Swedish.

- (5) En bok ble gitt Jens (Norw.)
a book was given J.

Importantly, VP-*internal* movement of DO above IO will not save such examples, since that would yield an ordering where DO follows the verb at the Spell Out of VP, which in turn would effectively prevent the passive transformation itself. Then here, it seems, some other movement must be at work, placing DO to the left of the verb at the VP-level.

Beyond the need for auxiliary movements, this crucially means that F&P's system does not appear to be capable of capturing a prominent correlation within Scandinavian (cf. Anagnostopoulou 2004 for this general point): Danish and Icelandic do not allow the crossing of IO by DO either in OS or in passives, while Norwegian and Swedish permit both in both types of constructions (cf. Anagnostopoulou 2003).¹⁷

A final note on the status of linearization conflicts is in order. F&P's model has a prominent property that may well be considered a weakness: it relies heavily on crash. Movements to phase edges are not locally

¹⁷ Another case of apparent conflict in the linearization process is presented by floated object quantifiers in OS; but see Holmberg (1999, Note 11).

triggered (or locally determined to be inevitable), but are enforced if the derivation is to avoid a final crash (e.g. unless a *wh*-phrase has moved to (all) the intermediate phase edge(s), it will be unable to move to [+wh] C). This property is problematic if syntax is to be crash-proof on minimalist assumptions, as convincingly argued by Frampton and Gutmann (2002).

4. Concluding remarks

Despite these criticisms, I am confident that both approaches can be further developed to meet the challenges highlighted here. Although OS is on center stage in both papers, the scope of the proposals being put forward goes well beyond OS: albeit in radically different ways, both theories exploit linearization at the PF interface, opening up intriguing new perspectives in accounting for word order in general. This appears a fruitful direction of research which is fully in line with minimalist principles: the burden of explanation is being pushed from within syntax to the interfaces.

Acknowledgements

I gratefully acknowledge the support provided by postdoctoral grant No. D-048454 of OTKA (Hungarian Scientific Research Fund).

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References

- Áfarli, T. A. 1997. Dimensions of phrase structure: The representation of sentence adverbials. *Motskrift*. Institutt for Nordistikk og Litteraturvitenskap, NTNU. 91–113.
 Anagnostopoulou, E. 2003. *The Syntax of Ditransitives: evidence from clitics*. Studies in generative grammar 54. Moton de Gruyter: Berlin/New York.
 —. 2004. Holmberg's Generalization and locality. GLOW abstract.

- Benedicto, E. 1997. V-movement and its interpretational effects. *GLOW Newsletter* 38, pp. 14–15.
- Bobaljik, J. D. 1995. Morphosyntax: The Syntax of Verbal Inflection. Doctoral dissertation. MIT.
- . 2002. A-chains at the PF-interface: Copies and ‘covert’ movement. *Natural Language and Linguistic Theory* 20.2, pp. 197–267.
- Brody, M. 1997. Mirror theory. Ms. UCL, London.
- . 2002. On the status of representations and derivations. In: Epstein, S. D. & Seely, T. D. (eds.) *Derivation and Explanation in the Minimalist Program* (pp. 19–41). Oxford: Blackwell.
- Chomsky, N. 1995. *The Minimalist Program*. Cambridge, MA: MIT Press.
- . 2000. Minimalist inquiries: the framework. In: Martin, R., Michaels, D. & Uriagereka, J. (eds.) *Step by Step. Essays on minimalist syntax in honor of Howard Lasnik* (pp. 89–155). Cambridge, MA: MIT Press.
- . 2001. Derivation by phase. In: Kenstowicz, M. (ed.) *Ken Hale: a life in language* (pp. 1–52). Cambridge, MA: MIT Press.
- . 2004. Questions about phases. Invited talk at GLOW 27, Thessaloniki.
- Egerland, V. 1998. On Verb-Second violations in Swedish and the hierarchical ordering of adverbs. *Working Papers in Scandinavian Syntax* 61.
- Epstein, S. 1999. Un-principled syntax and the derivation of syntactic relations. In: Epstein, S. & Hornstein, N. (eds.) *Working Minimalism* (pp. 317–345). Cambridge, MA: MIT Press.
- Frampton, J. & Gutmann, S. 2002. Crash-proof syntax. In: Epstein, S. D. & Seely, T. D. (eds.) *Derivation and Explanation in the Minimalist Program* (pp. 90–105). Oxford: Blackwell.
- Hellan, L. & Platzack, C. 1999. Pronouns in Scandinavian languages: an overview. In: Riemsdijk, H. van (ed.) *Clitics in the Languages of Europe* (pp. 123–142). Berlin: Mouton de Gruyter.
- Holmberg, A. 1999. Remarks on Holmberg’s Generalization. *Studia Linguistica* 53.1, pp. 1–39.
- Jonas, D. 1996. Subject Positions and the Roles of TP. *Linguistic Inquiry* 27, pp. 195–236.
- Müller, G. 2001. Order preservation, parallel movement, and the emergence of the unmarked. In: Legendre, G., Grimshaw J. & Vikner, S. (eds.) *Optimality-Theoretic Syntax* (pp. 279–313). Cambridge, MA: MIT Press.
- . 2002. Verb-Second as vP-first. Ms., IDS, Mannheim. To appear in *Journal of Comparative Germanic Linguistics*.
- Nilsen, Ø. 2003. Eliminating Positions: Syntax and semantics of sentence modification. PhD dissertation. University of Utrecht.
- Richards, N. 1997. What Moves Where When in Which Language? PhD dissertation, MIT.
- . 1998. The principle of minimal compliance. *Linguistic Inquiry* 29, 599–629.
- Sells, P. 2001. *Structure, Alignment and Optimality in Swedish*. Stanford monographs in linguistics. Stanford, CA: CSLI Publications.
- Williams, E. 2003. *Representation Theory*. Cambridge, MA: MIT Press.
- Zwart, J.-W. 2001. Syntactic and phonological verb movement. *Syntax* 4.1, pp. 34–62.

How Phonological is Object Shift?

PETER SVENONIUS

1. Introduction

The papers here represent very different analyses of Object Shift (OS), and take very different positions on the interaction of phonology and syntax, an area in which there are many unsettled questions.

Mainstream work tends to hold that syntax is blind to phonological content, with certain exceptions, for example sometimes phonetically null elements require special syntactic licensing (Chomsky 1981), or certain syntactic rules only apply to nodes with phonetically visible features (Holmberg 2001). Basically falling within the mainstream are proposals that syntactic movement can be blocked by or driven by requirements that have phonological effect at the output, such as adjacency (Bobaljik 1995, Kidwai 1999) or rules matching prosodic structure with focus structure (Zubizarreta 1998). Such accounts generally describe movements in the syntactic terms of specifiers and feature checking and so on, and do not rely on the visibility to syntax of strictly phonological features. We can call these all Syntactic accounts.¹ Most accounts of OS are Syntactic in these terms, for example those of Holmberg (1986), Holmberg and Platzack (1995), and Bobaljik (1995, 2002).

In other work, a distinction is countenanced between Syntactic and Postsyntactic movement. Such accounts typically argue that Postsyntactic movement has properties that syntactic movement doesn't have; for example violating the Coordinate Structure Constraint (Adger 1997),

¹ The non-movement account of scrambling in Neeleman and Reinhart (1998) is essentially Syntactic in these terms, though their analysis of English case assignment makes crucial reference to prosodic structure and so is Postsyntactic.

violating the Head Movement Constraint (Adger 2000), violating islands (McCloskey 1999), lowering rather than raising (Embick & Noyer 2001),² or not feeding A-bar movement (Chomsky 2001 on Th/Ex). A Postsyntactic account of OS is Holmberg (1999); on that account, OS violates the Extension Condition, which Holmberg assumes is valid for syntactic movement. Another is Chomsky (2001).³

Both of the accounts presented in this volume are highly original and neither fits neatly into the familiar range of Syntactic and Postsyntactic analyses. Erteschik-Shir promotes a new extreme of phonologizing syntax, proposing that OS and other apparently syntactic phenomena should be handled Postsyntactically, though I suggest that in the end she accomplishes this by syntacticizing phonology. Pesetsky and Fox, for their part, propose that a phonological property (linear order) constrains syntax in a way not explored before. I point out some empirical challenges for their proposal.

2. Erteschik-Shir

Erteschik-Shir (henceforth ES) hopes to develop what she calls “a purely phonological account of object shift.” This might lead one to expect that

² The assumptions of Bobaljik (1995, 2002) are very similar to those of Embick & Noyer (2001), but I place them in these different groups because Bobaljik’s focus is on OS, which he assumes to be a syntactic movement, facilitating the adjacency of morphologically related heads, while Embick & Noyer’s focus is on morphological merger, the lowering of affixal material from one head to another.

³ This is partly obscured by the fact that Chomsky (2001) uses the abbreviation OS only for the first, syntactic step of a two-stage movement. On his account, the object first moves to the EPP position of ν^* ; if it does not, then it is assigned a feature *Int'* which is incompatible with an unfocused definite object. This is a Syntactic step in my terms because phonology is invoked only in the statement of the rule assigning *Int'*, which refers to the “phonological border of ν^*P ” (p. 35). The identification of the “phonological border” is sensitive not to details of phonological structure but only to whether there is phonological material to the left of the object or not; a moved verb is invisible at the border of ν^*P , leaving the object to get the feature *Int'*, whereas an unmoved verb harmlessly absorbs *Int'*, preventing the object from shifting. The second step of object shift in Chomsky’s account is called *Disl[ocation]* (p. 30), and takes the object across adverbs. Chomsky does not *Disl* in detail, but suggests that it is a phonological rule like Th/Ex. Since the two steps together are OS in my terms, and since the second one has properties crucially distinguishing it from syntactic movement, Chomsky’s account is Postsyntactic in my terms.

phonological features such as foot structure or the presence of onsets would be visible to syntax, but instead ES's account relies on what most accounts would take to be syntactic, rather than phonological features. For example, where previous accounts have assumed that pronouns are syntactically special, stipulating a syntactic [+pronoun] feature, ES assumes that pronouns are prosodically special, essentially introducing a [+pronoun] feature in the phonology. The only surface phonetic manifestation of such features, I believe, is in determining word order. However, since these features do not interact with syntax, in ES's system, she achieves a separation of syntax and phonology that is highly original. Syntax proper, on her account, builds base structures on which phonology operates to linearize constituents.

A central tool in ES's account of OS is Åfarli's (1997) multi-dimensional analysis of adverbs, whereby adverbial adjuncts to an XP may linearize to the left or right of XP, or between the specifier of XP and its head. The linearization process for adjuncts is called 'bending.'

ES assumes that adverbs adjoin to VP (§2.3). Linearization of an adverb to the peripheries of VP would put it in the positions indicated in (1), where the actually ungrammatical linearizations are marked with stars (I illustrate throughout with Norwegian, unless otherwise noted).

- (1) a. Han så [_{VP} (aldri) t_V [_{SC} analysen løse oppgaven]
 he saw never the.analysis solve the.assignment
 (*aldri)]
 never
 'He (never) saw the analysis solve the assignment'
- b. Han så [_{VP} (*aldri) t_V [_{SC} den løse oppgaven] (*aldri)]
 he saw never it solve the.assignment never
 'He saw (*never) it solve the assignment'

The badness of the rightmost positions in (1a–b) is characterized by ES by a rule that an adverb must linearize "in the 'earliest' position in which it is licensed" (§2.3). As long as the adverb cannot bend upwards to precede the finite verb, this gets the facts in (1a).

The badness of the VP-initial position in (1b) leads to two additional assumptions – that the pronoun must incorporate prosodically, and that the adverb cannot host it (§2.4). But no evidence is presented that these properties are phonological; adverbs do not have audibly distinctive

phonology (cf. Nilsen's 2003: 96 minimal pair *Han har det fortsatt* 'He has it still_{Adv}' vs. *Han har fortsatt det* 'He has continued_V it'). ES suggests a distinction between weak and strong adverbs (§2.2), but all adverbs are crossed by OS, if they are between the finite verb and its trace, not just the ones which cannot be VP-final.

Furthermore, the pronoun is phonologically indistinguishable from a definite article, which cannot undergo OS. Consider the ambiguous string in (2a), disambiguated by OS in (2b) or its failure to apply in (2c).

- (2) a. Han så den løse oppgaven.
 he saw it/the solve/loose the.assignment
 'He saw it solve the assignment' or 'He saw the loose assignment'
 b. Han så den aldri løse oppgaven.
 he saw it never solve the.assignment
 'He never saw it solve the assignment'
 c. Han så aldri den løse oppgaven.
 he saw never the loose the.assignment
 'He never saw the loose assignment'

Thus, it seems that on ES's account, special diacritics have to be assigned to pronouns indicating that they must incorporate, and special diacritics have to be associated with adverbs indicating that they are not hosts for prosodic incorporation; these diacritics are not realized phonologically at the surface, so the relationship of word order rules to phonology is abstract.

Of course, the prosodic structure built on the DP in (2c) might be assumed to allow or even force the determiner to incorporate prosodically to the right at some abstract level, and there might be some reason that this cannot happen in (1b)/(2b).

But note that ES has to assume that linearization rules violate syntactic constituency. In order to get the actual order in (2b) the adverb must bend in between the small clause subject and predicate.⁴ ES provides no

⁴ Here ES departs from Áfarli's (1997) assumptions about adverb placement. Áfarli argues that adverbs adjoin to AgrP, assuming that Icelandic full DP objects move to SpecAgrP, and that weak pronouns in Scandinavian more generally undergo clitic movement.

reason that an adverb should not be able to bend in between a determiner and an adjective.

Again, an abstract prosodic structure could be postulated that would make a DP an inviolable domain, but just as with the special marking of adverbs as non-hosts, this would seem to be a relabeling of traditionally syntactic properties as abstractly phonological.

Given the freedom of adverb bending in ES's system, there are many unattested positions in which we might incorrectly expect to see adverbs. The constraint that only the earliest (i.e. leftmost) possible position is acceptable does important work in ruling out most of the unattested options, but it is not clear how the constraint actually operates. For example, in extending her analysis to Negative Movement (§2.3), ES discusses the Danish examples in (3a–b).

- (3) a. * Jeg gav Else ingenting.
 I gave Else nothing
 b. * Jeg gav ingenting Else.
 I gave nothing Else
 c. * Jeg gav ingen Else -ting.
 I gave no Else thing

ES's suggestion is that *ingen* is an adverb, and *-ting* is a bound morpheme. (3a) is ruled out because the adverb *ingen* has been bent into VP after the indirect object, violating the 'earliest possible' rule. But (3b) is bad as well – according to ES, because of a processing constraint blocking the reordering of arguments – as is (3c), presumably because *-ting* is a bound morpheme. It seems that the 'earliest possible' condition should pick out (3a), which should therefore be grammatical. I return to this construction in the next section.

There is a heavy empirical burden on the bending mechanism, but it is not explicated fully enough to evaluate. ES assumes that the rather rigid order of multiple adverbs is determined by their attachment sites (§2.4). This would easily get the right order when adverbs are to the left of VP, but when they are anywhere else it is unclear why the same left-right order should be preserved (see Nilsen 1997 for a critique of Áfarli's multi-dimensional analysis).

The fact that the mechanisms are not made fully explicit does not mean that they cannot be made to work, but I suspect that when

they are fleshed out, they will look more like syntax than like phonology. However, this may be because I have the wrong preconceptions about what syntax looks like; daringly, ES extends her notion of phonological movement even to classic syntactician's staples like Topicalization.

ES suggests that V2 is derived by the interaction of two phonological processes, one moving the verb to the left edge, and the other moving a topical element X to the left edge. To account for the impossibility of applying these rules in the wrong order (yielding *VXSO), ES suggests a 'processing constraint' that prevents elements with phi-features from coming between V and S (§2.1); but the order is blocked even if X has no phi-features (e.g. **Så da jeg han* 'saw then I him'). On the impossibility of topicalizing the subject itself, followed by verb fronting (yielding *VS(Adv)O, e.g. **Så jeg (da) han* 'saw I (then) him') ES seems to invoke a requirement that the verb be in second position (§2.1); but surely this cannot be a phonological constraint on the verb itself, which is not second in V1 or in embedded clauses.

Quite generally, the characteristics of topicalization seem to be more syntactic than phonological. For example, the non-restrictive relative clause in (4a) can be moved along with the object to the initial position, even though it defines a separate prosodic phrase; it cannot be stranded, as shown in (4b), unless it is extraposed; and it can only be extraposed, as shown in (4c), if it is attached at the level of the highest DP – thus the relative clause in (4a) ambiguously modifies 'a picture' or 'the child,' whereas that in (4c) can only modify 'a picture.'

- (4) a. Et bilde av barnet, som Ragnhild syntes var pent,
 a picture of the.child as Ragnhild thought was pretty
 tok jeg på festen.
 took I at the.party
 'I took a picture of the child, which Ragnhild thought was
 pretty, at the party' (*which* = picture or child)
- b. * Et bilde av barnet tok jeg, som Ragnhild syntes var
 a picture of the.child took I as Ragnhild thought was
 pent, på festen.
 pretty at the.party
- c. Et bilde av barnet tok jeg på festen,
 a picture of the.child took I at the.party

som Ragnhild syntes var pent.
as Ragnhild thought was pretty
'I took a picture of the child at the party, which Ragnhild
thought was pretty' (*which* = picture)

Of course, an abstract prosodic domain could be postulated to mimic the ban on extraction from an embedded DP, but again this would simply make abstract phonology emulate syntax, without phonetic expression.

Thus, it seems to me that to the extent that ES's account handles word order variation, it can do so only by making reference to the kinds of features and categories that we know from syntax; once the account is fully fleshed out, it may not look so radical after all.

3. Pesetsky & Fox

Pesetsky & Fox's paper (henceforth P&F) is quite different in its approach to the PF side of syntax. The provocative proposal is that the linear order of VP-internal material is fixed at an early point in the derivation and cannot be undone by later movements. Holmberg's Generalization facts fall out because the relative order of V and O matters in a way that the relative order of adverbs and V and of adverbs and O doesn't. The account is straightforwardly Syntactic in the terms sketched in §1 here, but the introduction of Order Preservation as a constraint on syntax represents a novel conception of a way that syntax can make reference to a property with phonologically audible output. This approach needs to be tested on a much wider range of data, but here I simply point out a couple of problems for the specifics of the account of the Scandinavian facts.

P&F include only VP-internal material from the initial Spell-Out domain, excluding the subject and auxiliaries. However, OS also fails to cross auxiliaries, as noted by Holmberg (this volume). To include auxiliaries in the first Spell-Out domain, without also counting the subject, would run counter to the assumption, standard since Koopman and Sportiche (1991), that the subject originates in a projection associated with the verb from which it receives its thematic role.

However, it is not completely clear that the subject should be excluded; OS is quite generally prevented from crossing the subject (cf. Nilsen 1997), apart from a restricted class of exceptions known as 'long object

shift' (cf. Josefsson 1992). For most speakers, the only possible orders are those in (5) (cf. Nilsen 2003: 113; as indicated, the first pronoun following the adverb must be focused).

- (5) a. Vanligvis gir ikke HAN meg den.
 b. Vanligvis gir han ikke MEG den.
 c. Vanligvis gir han meg ikke DEN.
 d. Vanligvis gir han meg den ikke.
 usually gives he me it not he me it
 'Usually he doesn't give me it'

All alternatives are excluded, for most speakers (again setting aside the long-OS construction). P&F might invoke Relativized Minimality to rule out objects crossing the subject, but this will also rule out *IO<DO (cf. Josefsson 2003), so the real empirical burden of the novel part of P&F's proposal is the relative ordering of the verb to the internal arguments.

However, S<V is arguably just as strong as V<O preservation, in the Middle Field. In Icelandic, Swedish, and Norwegian, the subject can appear in any of several positions relative to adverbs (Sigurðsson 1990, Holmberg 1993, Nilsen 1997).⁵

- (6) a. Sânt roter gutten sannsynligvis vanligvis bort.
 b. Sânt roter sannsynligvis gutten vanligvis bort.
 c. Sânt roter sannsynligvis vanligvis gutten bort.
 such messes the.boy probably the.boy usually the.boy away
 'Stuff like that, the boy probably usually loses'

The low position requires a certain degree of focus (the necessary degree varies across the Scandinavian languages, cf. Svenonius 2002a). The pattern in (6) is also found in Northern Norwegian dialects, which have in addition optional short verb movement in many kinds of embedded clauses (Bentzen 2003); this movement crosses only lower adverbs, and not higher ones ((7) is Northern Norwegian, from Bentzen 2003: 581).

⁵ I use these adverbs to keep the data minimally different from Bentzen's original examples in (7); however, the example is awkward with these two adverbs in a row, much like the effect in English when two *-ly* adverbs succeed each other. See Nilsen (2003) for many more examples, including some taken from natural contexts.

- (7) a. Ho kan ha nøkkelen, med mindre ho rote vanligvis
 she can have the.key with less she messes usually
 bort sånt.
 away such
 ‘She can have the key, unless she usually loses things like
 that’
- b. * Ho kan ha nøkkelen, med mindre ho rote
 she can have the.key with less she messes
 sannsynligvis bort sånt.
 probably away such

Bentzen demonstrates that this is not embedded V2; topicalization is impossible in clauses introduced by ‘unless,’ and high adverbs, if present, precede the short-moved verb. Bentzen has pointed out (personal communication) that short verb movement never combines with the low subject position ((8) is Northern Norwegian, provided by Kristine Bentzen).

- (8) a. ... med mindre ho Ida vanligvis rote bort sånt
 b. ... med mindre vanligvis ho Ida rote bort sånt
 c. ... med mindre ho Ida rote vanligvis bort sånt
 d. *... med mindre rote vanligvis ho Ida bort sånt
 with less she Ida messes usually she Ida messes away such
 ‘unless Ida usually loses things like that’

(8a) is the standard order, for all Mainland Scandinavian; (8b) is possible with focus on the subject, with much dialectal variation. (8c) is peculiar to those dialects with verb movement in embedded clauses, including Icelandic and Northern Norwegian; but (8d) is systematically impossible.

So, arguably, the order that should be respected is not just $V<O$, or $V<IO<DO$, but $S<V<O$, and $S<V<IO<DO$. The problem is that if anything becomes a topic, even something that does not originate within VP, then the verb precedes the subject, which is why P&F leave the subject out of the first Spell-Out domain.

There are several ways that P&F could preserve the heart of their analysis while including the subject in the first Spell-Out domain, it seems to me. Lacking the space to develop such a proposal, I turn to the case of Quantifier Movement (QM) in Icelandic.

P&F propose an intriguing account of certain QM patterns by appealing to a sort of ‘Inverse Holmberg Effect.’ Here, the fact that a quantified expression has crossed the verb within VP means that the verb cannot cross it again to undergo V2. However, it seems the analysis cannot deal with the basic data from a subcase of QM, namely Negative Movement (NM) (Christensen 1986, 1988, 1995, Rögnvaldsson 1987, Jónsson 1996, Svenonius 2000).

NM is obligatory, that is, a negatively quantified argument must move out of VP, in all Scandinavian languages (except for narrow scope negation, cf. Svenonius 2002b). In contexts where NM is blocked, negatively quantified arguments in VP are ungrammatical; this can be seen in the example in (3a) above, from Danish, in which it is difficult to OS a full DP indirect object. When the indirect object is a pronoun, it undergoes OS, permitting NM of the direct object, and *Jeg gav hende ingenting* ‘I gave her nothing’ is grammatical in Danish.

In Icelandic, to demonstrate the same fact requires an adverb, to show when the IO has undergone OS, as in (9) (based on Rögnvaldsson 1987 and Jónsson 1996), where I include VP brackets in the gloss to indicate the structural positions of the arguments.

- (9) a. * Jón sagði (hreinlega) ekkert Maríu.
 b. Jón sagði Maríu (hreinlega) ekkert.
 c. * Jón sagði hreinlega Maríu ekkert.
 Jon told Maria simply nothing [_{VP} _{TV} Maria nothing]
 ‘Jon (simply) told Maria nothing’

As Rögnvaldsson (1987) demonstrates, NM adjoins to the highest VP shell. This means that sentences with negatively quantified objects have the structures sketched in (10) before verb movement, showing the structures with and without an auxiliary but omitting the IO, which is optional with *segja* ‘say, tell.’

- (10) a. ... ekkert [_{VP} sagði _{TP}]
 nothing said
 b. ... ekkert [_{VP} hefur [_{VP} sagt _{TP}]]
 nothing has said

After verb movement from (10a), the order V<O will be reinstated, whereas in (10b), the surface order will be Aux<O<V. The odd fact that

P&F want to account for is that in (10b), NM may cross an unmoved indirect object, if present (cf. P&F's (36a)), whereas in (10a) it may not, as illustrated in (9a) (cf. P&F's (38)).

P&F assume that what is wrong with (9a) is that verb movement disrupts the first Spell-Out domain's $DO < V < IO$, but then they have to assume that (10a) is also impossible, since there, verb movement would disrupt $DO < V$. However, (10a) is a good (intermediate) structure, since NM is obligatory and yet *Jón sagði ekkert* 'Jon said nothing' is grammatical. P&F would have to deny that NM even occurs in clauses with a simple verb, complicating the account of NM and leaving unexplained data like that in (9).

Jónsson (1996) showed that NM in (10b) is A-bar movement. In Svenonius (2000), I demonstrated that NM in (10a) shows the characteristics of A-movement, including failure to strand prepositions (pseudopassives are impossible in Icelandic) and failure to license parasitic gaps. I argued that the failure to cross the indirect object was another A-movement characteristic (under Relativized Minimality), explained if adjunction to the VP headed by a theta-assigning lexical V, as in (10a), is A-movement, whereas adjunction to the VP headed by a non-theta-assigning, functional auxiliary verb, as in (10b), is A-bar movement.

The account in Svenonius (2000), then, allows a simple and general characterization of obligatory NM, whereas it seems to me that the P&F account does not.

4. Conclusion

Both of these papers make important contributions to our thinking about the syntax-phonology interface, in ways that I think reach far beyond the specifics of their accounts of OS and NM.

ES's account suggests that we have been thinking about displacement phenomena in fundamentally the wrong way. My challenges for the approach focus on the ways in which the rules proposed do not refer to actual phonology, but some abstract phonological representation that does not necessarily have phonetic significance independently of the way it affects word order. This might be an acceptable result, but there is a lot of work to be done to show that such an account can be made to bear fruit.

P&F's account suggests that Order Preservation is an important part of grammar, and promises to explain a whole range of edge effects. Edge phenomena have been observed in many domains (see Svenonius 2004 for some), and it will be interesting to see if the P&F model can be extended to them. My challenges for the account suggest to me that Order Preservation might be a violable constraint, perhaps making it even more similar to William's (2003) notion of Shape Conservation.

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References

- Adger, David. 1997. VSO order and weak pronominals in Celtic. *Canadian Journal of Linguistics* 42.1: 9–31.
- . 2000. First position and the Syntax/Prosody interface: Old Irish preverbs. *WCCFL 19 Proceedings*, 101–114. Somerville, Ma.: Cascadia Press.
- Bentzen, Kristine. 2003. V-to-I movement in the absence of morphological cues: Evidence from adult and child Northern Norwegian. *Nordlyd* 31.3: 573–588 (<http://www.ub.uit.no/munin/nordlyd/>).
- Bobaljik, Jonathan David. 1995. Morphosyntax: The Syntax of Verbal Inflection. Unpublished PhD dissertation, Massachusetts Institute of Technology, Cambridge, Ma.
- . 2002. A-chains at the PF-interface: Copies and 'covert' movement. *Natural Language and Linguistic Theory* 20.2: 197–267.
- Chomsky, Noam. 1981. *Lectures on Government and Binding*. Dordrecht: Foris.
- . 2001. Derivation by phase. In *Ken Hale: A Life in Language*, Michael Kenstowicz (ed.), 1–52. Cambridge, Ma.: MIT Press.
- Christensen, Kirsti Koch. 1986. Norwegian *ingen*: A case of post-syntactic lexicalization. In *Scandinavian Syntax*, Östen Dahl and Anders Holmberg (eds.), 21–35. Institute of Linguistics, University of Stockholm.
- . 1988. Modern Norwegian *ingen* and the ghost of an Old Norse particle. In *Proceedings from the Seventh Biennial Conference of Teachers of Scandinavian Studies*, J. Allen and M. Barnes (eds.).
- . 1995. Quantifier-movement and its syntactic reflexes. Ms. University of Bergen.
- Embick, David and Ralf Noyer. 2001. Movement operations after syntax. *Linguistic Inquiry* 32.4: 555–595.
- Erteschik-Shir, Nomi. This volume. Sound patterns of syntax: Object shift. *Theoretical Linguistics*.
- Fox, Danny and David Pesetsky. This volume. Cyclic linearization of syntactic structure. *Theoretical Linguistics*.
- Holmberg, Anders. 1986. *Word order and syntactic features*. Unpublished Ph.D. dissertation, University of Stockholm.
- . 1993. Two subject positions in IP in Mainland Scandinavian. *Working Papers in Scandinavian Syntax* 52: 29–41.

- . 1999. Remarks on Holmberg's Generalization. *Studia Linguistica* 53.1: 1–39.
- . 2001. Scandinavian Stylistic Fronting: How any category can become an expletive. *Linguistic Inquiry* 31.3: 445–483.
- . This volume. Linearization in two ways: Comments on Fox & Pesetsky and Erteschik-Shir. *Theoretical Linguistics*.
- Holmberg, Anders and Christer Platzack. 1995. *The Role of Inflection in Scandinavian Syntax*. New York: Oxford University Press.
- Josefsson, Gunlög. 1992. Object shift and weak pronominals in Swedish. *Working Papers in Scandinavian Syntax* 49: 59–94.
- . 2003. Four myths about Object Shift in Swedish – and the Truth ... In *Grammar in Focus: Festschrift for Christer Platzack 18 November 2003*, vol. 2, Lars-Olof Delsing, Cecilia Falk, Gunlög Josefsson, and Halldór Á. Sigurðsson (eds.), 199–207. Lund: Department of Scandinavian Languages, Lund University.
- Jónsson, Jóhannes Gísli. 1996. Clausal Architecture and Case in Icelandic. Unpublished Ph.D. dissertation, University of Massachusetts at Amherst.
- Kidwai, Ayesha. 1999. Word order and focus positions in Universal Grammar. In *The Grammar of Focus*, Georges Rebuschi and Laurice Tuller (eds.), 213–244. Amsterdam: John Benjamins.
- Koopman, Hilda and Dominique Sportiche. 1991. The position of subjects. *Lingua* 85: 211–258.
- McCloskey, James. 1999. On the right edge in Irish. *Syntax* 2.3: 189–209.
- Neeleman, Ad and Tanya Reinhart. 1998. Scrambling and the PF interface. In *The Projection of Arguments*, Miriam Butt and Wilhelm Geuder (eds.), 309–353. Stanford, Ca.: CSLI.
- Nilsen, Øystein. 1997. Adverbs and A-shift. *Working Papers in Scandinavian Syntax* 59: 1–31.
- . 2003. Eliminating Positions: Syntax and Semantics of Sentential Modification. Unpublished PhD dissertation, University of Utrecht.
- Rögnvaldsson, Eiríkur. 1987. On word order in Icelandic. In *Proceedings of the Seventh Biennial Conference of Teachers of Scandinavian Studies in Great Britain and Northern Ireland*, 33–49. London: University College London.
- Sigurðsson, Halldór Ármann. 1990. V1 declaratives and verb raising in Icelandic. In *Modern Icelandic Syntax*, Joan Maling and Annie Zaenen (eds.), 41–69. New York: Academic Press.
- Svenonius, Peter. 2000. Quantifier movement in Icelandic. In *The Derivation of VO and OV*, Peter Svenonius (ed.), 255–292. John Benjamins, Amsterdam.
- . 2002a. Subject positions and the placement of adverbials. In *Subjects, Expletives, and the EPP*, Peter Svenonius (ed.), 201–242. New York: Oxford University Press.
- . 2002b. Strains of negation in Norwegian. *Working Papers in Scandinavian Syntax* 69: 121–146.
- . 2004. On the edge. In *Peripheries: Syntactic Edges and their Effects*, David Adger, Cécile de Cat, and George Tsoulas (eds.), 259–287. Dordrecht: Kluwer.
- Williams, Edwin. 2003. *Representation Theory*. Cambridge, Ma.: MIT Press.
- Zubizarreta, Maria Luisa. 1998. *Prosody, Focus, and Word Order*. MIT Press, Cambridge, Ma.
- Áfarli, Tor. 1997. Dimensions of phrase structure: The representation of sentence adverbials. *Motskrift*, 91–113. Trondheim: NTNU.

A Note on “Cyclic Linearization”

EDWIN WILLIAMS

“Cyclic Linearization” (CL) is similar in its intent to my own “Representation Theory” (Williams 1998, 2003) (RT) in that it seeks to explain some parallelism constraints in terms of general architecture, and in fact both were inspired by Holmberg’s generalization and the general properties of Object Shift in the formulation of central tenets.¹ But here, I will explore a difference in the models that has more to do with WH movement.

In CL Fox and Pesetsky present a scheme for phase derivation in which all movement is subject to a parallelism constraint – the linear order established in earlier phases must be maintained in later phases. Like Chomsky’s (2001) “Derivation by Phase” theory, all movements are bounded by the same set of items, or phases; thus if vP (or VP in CL) is a phase, then WH movement is bounded by it, and so WH movement must proceed via the specV ‘escape hatch’ on its way to specC. CL rationalizes Chomsky’s specV escape hatch, but it suffers, I think, from submitting all movement to the same bounding, as attractive as that may seem at the start.

RT is a model of grammar in which much of what is done with movement in Minimalist practice (MP) in general (including CL) is rather done by mapping one structure into another ‘holistically’; thus, for example, the structure (1a) is mapped onto the structure (1b) by matching up parts:

¹ In fact, neither really explains Holmberg’s generalization in the sense of making it inevitable – in RT, deviations of parallelism are always allowed, so why not [V NP] ← [NP V]? And in CL, it must be stipulated that OS, unlike WH, cannot access specV.

- (1) a. [Agent [predicate Theme]]
- b. [Nom [V Acc]]

thus identifying Agent and Nom, etc. The program is extended to other relations, like Topic, etc., in a ramified model, in which the mapping relations are, approximately,

- (2) Theta $\rightarrow$ Case $\rightarrow$ Q/WH $\rightarrow$ Information structure.

The overwhelming fact about such mappings is that they are ‘shape conserving’ in a sense made somewhat, but not fully, precise in Williams (2004).

Locality in the RT model derives from the timing of embedding and movement operations. Embedding types (VP, Small Clause, CP) are embedded at the level at which they are defined (Theta, Case, and Q/WH, respectively); movement applies only to what is already embedded at the level at which it applies. Thus case-driven movement, for example, can extract from Small Clauses and directly embedded VPs, but not from CPs, since CPs are not embedded at the time that case-driven movement takes place.

The extractive dimension of various types of movement is correlated with interpretive effects which hold at different levels, thus explaining reconstruction effects – if binding theory and WCO hold, for example, at Case Structure, then the relatively ‘short’ movements which precede it will not show reconstruction effects, whereas the relatively ‘long’ movements that follow it will show reconstruction effects. Locality and reconstruction are correlated throughout the model, as detailed in Williams 2003, ch. 3; and they together correlate with a third dimension, a generalization of the A/A’ distinction.

As CL is built from Minimalist practice with Phase theory by adding preservation constraints, it is inevitable that CL and RT will resemble each other to some degree. So, for example, the structural correspondences that must hold under Shape Conservation in RT between Theta structure and Case structure will be mirrored by order correspondences between the interior and exterior of VP in CL. Also, CL and RT are similar in that the preservations are ‘subcyclic’ (or ‘phasic’): whatever is established in VP in CL (or, in Theta Structure in RT) must be preserved in TP in CL (or, in Case Structure in RT), etc.

But CL and RT differ in some important ways. First, what is preserved in the two accounts is different; in CL linear order is preserved, and in RT, whatever structure is present at one level is preserved in the next level; in principle that could be either linear order, hierarchical structure, or both, or some further aspect of structure, like coindexing. Fox and Pesetsky, in a discussion of cases involving remnant movement, make it clear that it is only linear order that is preserved, and not, in particular, c-command relations.

CL and RT differ in a further crucial way that I will explore here. In CL, order is *always* preserved. Thus, WH movement from VP must always begin by moving to specV, otherwise a later movement will produce an ordering conflict. But in RT, structure is conserved only in the unmarked case; Shape Conservation is violated to achieve otherwise unreachable structures or meanings; Level A+1 may not have a structure that matches a structure *a* in A, but only one that is ‘close’; or, Level A+1 may have two structures *a1* and *a2* which match *a* in A, with *a1* matching better than *a2*; but if *a2* has a semantics that *a1* does not have, then the structure in A can be matched with either *a1* or *a2*.

A further related difference is that there is no such thing as ‘long distance movement’ in CL (like DBP) – all movement is bounded by phases, but with chaining for long distance effects. In RT there is a grade of length of movement – in early levels, movement is bounded relative to movement in later structures. The bounding is a consequence of what structures are developed at the different levels – early levels have smaller parts of clause structure, and have less embedding.

There is a class of problems for a theory like CL where all movements are bounded in the same way, which the following is meant to illustrate. Suppose a language allows more than one thing to escape the VP, leaving the V behind. The things in question will not be OS type things, but rather WH type things, and in fact we might have in mind languages which allow multiple WH movement, such as Bulgarian, Serbo-Croatian, etc. Whatever the details of the analysis of such phenomena, we can conclude at least the following about their treatment in CL: if two WHs (say DO and IO) in the same VP both escape from that VP without the verb, then they both moved to the front of that VP in the VP phase:

- (3) [CP ... [VP WH_{DO} WH_{IO} V t_{DO} t_{IO}]]

Furthermore, the order of the two WHs at the end of the VP phase will be preserved in the CP phase, so (3) yields (4):

- (4) [CP WH_{DO} WH_{IO} ... [VP V t_{DO} t_{IO}]]

But in fact there is another derivation – one in which the two WHs switch order in the derivation of the VP phase; that order is then preserved in the CP phase:

- (5) a. [CP ... [VP WH_{IO} WH_{DO} V t_{DO} t_{IO}]]
 b. [CP WH_{IO} WH_{DO} ... [VP V t_{DO} t_{IO}]]

Serbo-Croatian WH movement provides a puzzle for the CL theory. In multiple questions in that language more than one WH word may be moved to the beginning of the clause, and the two WHs can appear in either order:

- (6) a. Ko je koga vidjeo
 ‘who AUX whom saw,
 b. Koga je ko vidjeo

Serbo-Croatian thus shows no Superiority effects here; however, for long multiple WH movement (not grammatical for all speakers) Superiority effects show up again:

- (7) a. Ko je koga vidjeo
 ‘who AUX whom saw,
 b. Koga je ko vidjeo
 c. Ko si koga tvrdio da je istukao
 ‘who AUX whom claimed that t AUX beaten t,
 d. *Koga si ko tvrdio da je istukao
 (Richards (p. 32))

In CL, then, we must allow both orders to be derived on the lowest phase that contains both WHs – in the case of (6–7), that means IP or CP; where both start out as members of VP, it means VP. However, in long-distance multiple WH, multiple orders are not derivable – only the order that matches the base order is allowed. The problem I think goes to the heart of CL – CL says that any reordering between two items must occur in the lowest phase in which they both occur; in the simple case, we see that the reordering is possible, but in the long distance case, which should ‘contain’ the simple case as a stage of development, the reordering is impossible. Various simple kinds of solution will not suffice.

For example, there is no structure that one can put above the lowest phase in the simple case that will allow both orders for that case only, and not the more complex case. One would have to posit that the lowest phases themselves were differently constituted in the simple and complex case, which is to give the game away.

RT resolves this problem with two different movements, one prior to Superiority, one subsequent to it; the prior scrambling provides the input to Superiority, and thus is not subject to it, whereas the later one must respect the relation between the WHs that Superiority establishes². Further, it is a consequence of the architecture that the prior one will be bounded relative to the second, so it is not arbitrary that it is the local one that allows both orders. CL cannot easily borrow such an account, as any scrambling of two elements must already have occurred in the smallest possible phase.

We might envision a modification of CL which would accommodate such things, a revision more thinkable in CL than its ancestor, “Derivation by Phase”. In the present version of CL, evading Holmberg’s generalization requires movement to specV, and different kinds of movement are arbitrarily assigned this ability – OS not, WH yes. In the CL mode of analysis, violating Holmberg’s constraint comes down to asserting orderings that conflict with earlier ordering assertions. In a revised version of the theory, we could assign the ability to *revise* or *supersede* earlier ordering assertions directly (but no less arbitrarily) to the class of relevant processes – OS not, WH yes, just as before. In this reconception, we retain the notion that phases are the locus of ordering assertions – an ordering assertion holds unless a certified process revises it later. CL thus revised retains cyclic order assertions, but eliminates the need to use specV as an escape hatch, and so would escape the problem with multiple WH discussed earlier.

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² Superiority is argued in Williams (1994, 2003) to be the component of Binding Theory which establishes the ‘dependent/independent variable’ relation, and thus to be part of the A system, and thus to intervene between these two scramblings or movements discussed in the text.

References

- Chomsky, N. (2001). Derivation by Phase. In: Kenstowicz, M. (ed.). *Ken Hale: a life in language*. Cambridge, MA: MIT Press.
- Richards, N. (1997). *What Moves Where When in Which Language?* PhD Dissertation, MIT.
- Williams, E. (1998). Economy as Shape Conservation. In: Keyser, J. & Hale, K. (eds.). *N. Chomsky's 70th Birthday Celebration. Web Page*. MIT Press.
- (2003). *Representation Theory*. Cambridge, MA: MIT Press.

Cyclic Linearization and its interaction with other aspects of grammar: a reply

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Our proposal is concerned with the relation between an aspect of phonology (linearization) and syntax.¹ In the picture that we had in mind, the syntax is autonomous – “it does what it does” – but sometimes the result maps to an unusable phonological representation. In this sense, linearization acts logically as a filter on derivations. We know of no evidence that the syntax can predict which syntactic objects will be usable by the phonology, and we know of no clear evidence that the phonology communicates this information to the syntax. In this sense, our proposal fits squarely into the tradition that Svenonius characterizes as the “mainstream”.² We thus attempted to identify certain deviant configurations that are not plausibly excluded for syntax-internal reasons, but are filtered out in the linearization process.

¹ Our reply to the ten commentaries on our paper is, of necessity, very far from comprehensive – given the wide range of topics, ideas and criticisms raised. We have chosen to focus most of our attention on those concerns that appear to be shared by several commentators – in particular because this set of concerns has a clear internal coherence to it, as we discuss below. Unfortunately, a number of other important issues could not be addressed here. We hope to return to them in future work.

² We thus disagree with Svenonius’s claim that the overall architecture of our proposal is particularly novel. One might ask whether this is a positive result. Possible evidence that the phonology does indeed send information back to the syntax might come from the observation that weak pronouns in Mainland Scandinavian seem to undergo Object Shift obligatorily when possible, but are allowed to stay in situ (at least in the overt syntax) when Object Shift would violate Holmberg’s Generalization. If we are correct that the effects of Holmberg’s Generalization arise from linearization contradictions, this might be a case in which the obligatoriness of an overt syntactic process is conditioned by the actions of the phonology. See Bobaljik (2002), however, for a “mainstream” approach that could be adapted to our system with minor modifications.

Occupying center stage in our proposal is our view of Spell-out – in particular, the role played by the monotonic mapping between the syntactic derivation and the Ordering Table. (As the structure grows, the ordering table grows.) No account of actual linguistic phenomena, however, can rely solely on these abstract properties of Spell-out. Two other issues must be clarified and resolved in every case – or there is no explanation of anything to be had:

1. How does syntactic structure grow? What laws govern external merge and internal merge (movement)? In what ways does the class of possible merge operations differ from language to language? More concretely: In what positions are arguments and adjuncts externally merged? What triggers movement, and what are the possible landing sites for movement operations?
2. What is the “periodicity” of the mapping between syntactic structure and phonology? More concretely: Where in the derivation does Spell-out take place, i.e. what are the “phases” or Spell-out domains, and why are they what they are?

If we are right, the system as a whole may be thought of as having three components: a monotonic mapping between syntax and phonology (Order Preservation), rules for external and internal merge, and a specification of Spell-out domains.

Faced with a particular phenomenon, it is a complex task to determine which of its aspects bear on which components of the overall system. Our strategy was therefore practical and fairly unambitious. We took for granted a general proposal concerning the laws of syntax that invokes rules of Merge in the sense of Chomsky (1995a, 2000, 2001a) and thus constructs structures “bottom-to-top”. This is one example of a proposal for syntax that is compatible with our notion of “periodicity”. Our goal was to see what happens when we take this view of syntax as a framework for provisional answers to the questions in 1, and combine these answers with a particular set of hypotheses relevant to the questions in 2.

We claimed that the result was a simplification of the syntactic system itself. We suggested first that the requirement of successive cyclicity did not need to be stipulated, and continued with arguments that certain locality properties of Object Shift, Quantifier Movement and other constructions follow directly from our answers to the questions in 1 and 2 above.

In general, if our system is correct, we expect to find a characteristic pattern of interaction among movement operations that place an element *E* outside its original Spell-out domain *D*.

$$(1) \quad \dots E \dots \left[\begin{array}{c} \overbrace{} \\ D \ X \dots t_E \dots \end{array} \right]$$

In particular, when an element *E* exits *D* but cannot stop first at the edge of *D* – and the edge of *D* is occupied by some other element *X* – this movement will yield a Holmberg or Inverse Holmberg Effect. In the Object Shift construction, for example, *E* is the object and *X* may be an unmoved verb or other element. In the Quantifier Movement construction, *E* is the verb and *X* is the moved QP. In both constructions, the presence of *X* blocks the movement of *E* in ways discussed in our paper. The fact that Holmberg and Inverse Holmberg Effects are found is thus an argument in favor of our overall system.

What we did *not* attempt was a comprehensive explanation of which movements do or do not target the edge of *D*. In some cases, our assumptions reflect the results of a substantial literature. For example, when QM places a quantified phrase in Spec,VP, verb movement is blocked for linearization reasons, but would not be blocked if *V* could adjoin to the edge of its own maximal projection. We assumed that movement of this type is excluded for more general reasons, following the literature on the nature of head movement. In other cases, we made assertions concerning the presence or absence of movement through the edge of Spell-out domains which – though not outlandish – still await independent confirmation. We agree with our commentators that these issues provide an important agenda for future research, and that the results of this research are very likely to bear on the evaluation of our proposals. One step towards further progress might be a more precise characterization of the properties of movement required to derive the Holmberg and Inverse Holmberg Effects within our system. In the next section, we offer some clarifications and speculations on this topic.

1. Movement through the edge: OS vs. QM

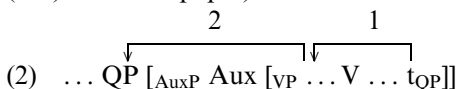
Acute questions arise when we compare our account of Holmberg's Generalization (HG) in OS with the account of the Inverse Holmberg Effect

(IHE) in QM. Opposite assumptions seem to be needed for the two constructions. Movement through the edge of VP is prohibited for OS, but is found with QM. One would like to understand why. First, however, we should clarify what assumptions are needed, especially in light of a variety of observations and suggestions offered by the commentators.

We agree with Sells that a key difference between our proposal and the larger family of order preservation/shape conservation accounts in the literature is its interaction with successive cyclicity, as evidenced principally by the IHE that we discussed in connection with Scandinavian QM and Korean scrambling (following Ko (2004)). When we observe apparent violations of shape conservation, one might imagine, in agreement with Williams's commentary, that we are observing the effects of deletion of ordering statements (a possibility also discussed in a slightly different form by Müller) – thereby sidestepping an ordering contradiction. It is the IHE that teaches us (we suggest) that the apparent violation of shape conservation does not result from deletion of an ordering statement, but instead results from an intermediate, phase-internal, successive-cyclic step.

In QM, the apparent violation of shape conservation compels us to posit successive cyclic movement through the edge of VP. It is the quantified phrase in this intermediate VP-initial position, in turn, that blocks V-movement to C. If Williams were correct, and it is deletion of ordering statements, rather than successive-cyclic movement, that yields the apparent violation of shape conservation, the interaction with V-movement would be unexpected (without further assumptions).

Since the IHE plays a central role in distinguishing our proposal from possible alternatives, it is important to discuss considerations offered by our commentators that might call into question the existence of the effect. As we noted in our paper (following Rögnvaldsson (1987)), the QM construction, even when it first targets the left edge of VP, appears to involve a second step of movement to the left of auxiliary verbs (see example (36c) from our paper):



If step 1 occurs before spell-out of VP, V-raising is impossible. (This is the IHE). Step 2 does not block Aux raising to C because Spell-out does not re-occur until CP has been constructed.

Diesing, Holmberg, Nilsen, Sells and Svenonius present examples like (3) as a problem for our proposal. In this example, a negative QP (NegQP) co-occurs with V-raising to C:³

- (3) Jan såg ingenting. (Swedish; Sells' (10))
 Jan saw nothing

An example of this type is surprising, given the instances of the IHE that we discuss, if it involves QM of the NegQP in addition to V-raising. In fact, as our commentators note, contrasts like (4) below have been taken to suggest that QM for NegQPs (unlike other QPs) is obligatory:

- (4) a. *Jag lånade Sven inga pengar. (Swedish; Sells' (11b))
 I lent Sven no money
 b. Jón hefur ekkert [sagt Sveini ____] (Icelandic)
 Jon has nothing said Svein-DAT
 'Jon has told Svein nothing.'

More precisely, (3) has the force of a counterexample to our proposals if both of the following claims are true:⁴

- A. A NegQP must always undergo QM.
 B. QM always proceeds through the left edge of VP.

³ Sells also presents an apparent problem for Ko's IHE, which we do not have the space to discuss. The issue is discussed at length by Ko (2004).

⁴ Diesing responds to this example with an alternative view of part of our IHE paradigm – a proposal similar to one that we (following Jónsson (1994)) argued against in note 26 of our paper. Diesing suggests that it is not the case that when the verb raises, a quantificational DO may not undergo QM. Rather, when the verb raises, an IO can and must undergo OS. This in turn removes the order-reversing possibility for QM. Thus in Diesing's (10), we see both OS and QM, under Diesing's analysis.

Diesing argues that her (9b) (an example of the IHE for us) is excluded because the indirect object *Svein* does not undergo OS. Thus (10), with IO-DO order, is acceptable, and allegedly shows OS of *Sveini* and QM of *ekkert* coexisting with V-to-C movement. Diesing's suggestion raises many interesting questions and possibilities, which we lack the space to address here. (Her proposals are uniformly consistent with our system in the variant discussed as "Alternative 2" below.) It is, however, worth noting that our account extends to cases where the considerations raised by Diesing are irrelevant, e.g. cases in which QM – if it could occur – would cross an element that cannot itself undergo OS. See also discussion of this point in connection with QM stranding prepositions by Svenonius (2000).

Claim A means that a NegQP must undergo movement to the landing site of step 2 in (2), and Claim B means that all movement of this sort also involves step 1 in (2).

Both A and B are reasonable assumptions – but they are not the only natural assumptions compatible with the data. As we will argue, it is possible to reject either A or B and still maintain our explanation for the IHE without running afoul of the facts in (3) and (4).⁵ We first discuss a solution that rejects A (in favor of a well-motivated alternative), and then discuss a different solution that maintains A but rejects B.

Alternative 1: a NegQP need not undergo QM

The fact that the NegQP in (4) must undergo QM does not necessarily teach us that QM is formally obligatory for NegQPs (claim A). A plausible alternative, we think, is provided by Christensen (1986).⁶ If her proposal is correct, QM of a NegQP is not obligatory in *all* situations. It is forced for principled reasons in exactly those constructions like (4) in which we seem to detect its obligatoriness – and is *not* required elsewhere.

Christensen suggests that a NegQP is licensed when it is string-adjacent to (phonologically null) sentential negation. This “licensing” could be thought of as a well-formedness requirement on NegQPs, e.g. (5):

(5) *[… Neg X NegQP …], where X is overt.

– or else as a condition on the morphological formation of the negative word from its semantic subparts (e.g. the formation of *no book* from *not + any book*):⁷

(6) [… Neg X [*any* NP]] → X [*no* NP], where X is *not* overt.

If Christensen’s proposal is correct, the natural conclusion is that QM is formally optional for all QPs. When the absence of QM leaves overt

⁵ In footnote 22 of our paper we accepted assumption A, which would force us to reject B, as in Alternative 2 below. But we know of no actual reason to prefer this choice.

⁶ Kayne’s (1998) alternative development of Christensen’s proposal arises in part from a commitment to claim A, and thus would require us to adopt the view discussed below that maintains A.

⁷ The phenomenon of “scope splitting” (discussed by Jacobs (1980), Kratzer (1995), Potts (2000), Swart (2000) and Penka (2001) – among others) provides independent

material intervening between sentential negation and the quantified expression, a negative word cannot be licensed (or formed).

Consider now a circumstance in which a (potential) NegQP α is separated from negation by VP-internal overt material that does not later move out of VP, as in (4a):

(7) Neg [_{VP} ... X ... α ...] (*X* overt)

Here, if α does not move to the left of X (as it does in (4b)), it will never have a chance to appear adjacent to Neg. It is in this situation that QM will appear to be obligatory, even though it is formally optional. On the other hand, consider a circumstance in which X is an element that itself moves out of VP to a position to the left of Neg, as is the case when X is a V that moves to C. If no other material intervenes between α and Neg, no QM will be necessary in such a case. For this reason, when a NegQP is the leftmost element in VP except for the verb itself, V-to-C movement is possible, as in example (3) – the case of concern to our commentators. If this is the correct view of the apparent obligatoriness of QM for NegQPs, then the co-occurrence of a NegQP object with V-to-C movement seen in (3) is not a problem.

This approach provides a simple perspective on the variation observed among the Scandinavian languages in the distribution of NegQPs (Svenonius (2000); Nilsen (this issue)). As Christensen notes, a NegQP object in Norwegian is acceptable *only* in a clause in which V-to-C movement has taken place, and furthermore may not surface to the left of other VP-internal material. Thus, examples like (3) have an acceptable Norwegian counterpart, but neither example in (4) is acceptable in Norwegian.

It should be obvious how such a language might fit naturally into the alternative sketched here. Norwegian is a language in which NegQPs are licensed or formed in accordance with the adjacency requirement discussed above – but QM is entirely unavailable. (Alternatively, QM is available but covert.) The adjacency requirement entails that structures in which Neg and the relevant QP are not adjacent will be

arguments for the latter approach (and certain versions of the former). Scope splitting is a situation in which a quantifier takes scope between the *not* and *any* components of a NegQP.

unacceptable,⁸ just as they are in the other Scandinavian languages, but the situation cannot be mitigated by movement of the NegQP.⁹

Alternative 2: QM need not proceed through the edge of VP

Suppose now that we learn somehow that Alternative 1 is wrong and that QM is in fact formally obligatory for NegQPs across the Scandinavian languages. We would then naturally take examples such as (3) to call into question claim B – that QM always proceeds in two steps via the left edge of VP, as in (2). There is in fact nothing intrinsic to our system that forces QM to uniformly proceed through the edge of VP. Instead, it is crucial that QM proceeds through the edge of VP only in those cases in which we observe successful fronting of a QP over overt VP-internal material. It is entirely possible to reject claim B in favor of the hypothesis that step 1 of QM is optional in the Scandinavian languages (other than Norwegian, to which we return below).

Consider a situation in which we observe a QP preceding VP-internal material that it would otherwise follow (as in (4b)). If we can be sure that this order reversal is not the result of phase-internal movement to a position lower than the edge (as discussed in Anagnostopoulou's commentary; see section 3 below and in particular footnote 25), it must be movement *through* the edge. Such movement will have exactly the consequences we discussed in our paper under the rubric of IHE. Consider a case in which the verb in VP is finite, and moves to C. If QM still proceeds through the edge, an ordering contradiction will result, just as discussed in our paper. If, on the other hand, QM does not proceed through the edge, no ordering contradiction will result, and the structure will correctly be allowed.

More specifically, the paradigms that are relevant for testing IHE involve cases in which we know that a QP could precede an element *a*

⁸ If NegQPs are morphologically composed under adjacency, the result will not be unacceptable, but will yield an overt *not ... any* rather than a NegQP. (See Sells example (11c)).

⁹ This view of the variation internal to Scandinavian invokes none of the "complications" seen by Nilsen in his discussion of how our proposal might handle the syntax of Norwegian NegQPs.

internal to the VP only if the first step in (2) has applied. In such cases, we predict that V-movement to C would be impossible. We thus predict the incompatibility of V-movement with non-string vacuous QM (relative to VP-internal material; see examples (38), (40d) and (41b) from our paper) – but do not exclude the co-occurrence of V-movement with string-vacuous QM as in (3), even under the assumption that Claim A is correct and Alternative 1 is wrong.

What is crucial is that the first step in (2) is optional. If QM proceeds through the edge of VP, it allows reversal of order among a QP and other overt VP internal material, but blocks V-movement to C. If QM does not proceed through the edge of VP, it may not produce a reversal of order among the QP and other overt VP-internal material – which is why V-to-C movement is possible – and in fact required – in such configurations.

This approach, like Alternative 1, allows a simple statement of the difference in this domain between Norwegian and the other Scandinavian languages. In essence, the only type of QM allowed in Norwegian is the kind seen in (3), where in the current approach, QM has not proceeded through the left edge of VP, but moves the QP in one step to its final landing site. What is excluded in Norwegian are those cases of QM that crucially involve an initial step through the left edge of VP. Consequently, as we noted in footnote 22 of our paper, and as stressed by Nilsen in his commentary, we would characterize Norwegian as a language in which step 1 in (2) is impossible (or obligatorily covert), with the result that QM must move in one step from inside the VP Spell-out domain to its final landing site.

It thus seems to us that examples such as (3) do not undermine the evidence that we provide for the IHE, and with it, the strong evidence for a central claim of our approach: that what is involved in order preservation is not faithfulness to an underlying order (as in many other approaches) but faithfulness to the order established at particular stages of the derivation.

Cross-linguistic variation:

It is also important, of course, to determine (if possible) which of the two alternatives just sketched is correct. Alternative 2 makes a cross-linguistic

prediction. If Alternative 2 is correct, the availability of step 1 of QM (movement to the edge of VP) is subject to parametric variation. It is impossible in Norwegian and optional in the other Scandinavian languages. All things being equal, we might conceivably expect to find languages in which this step is obligatory. Imagine that Alternative 1 is false, and that claim A (obligatory QM for NegQP) is thus true. In a language in which the first step of QM is obligatory, examples like (3) should be impossible, but the contrast in (4) should still hold.¹⁰ We do not know of any such language at present, but investigation of the prediction should help decide among the competing proposals.¹¹

More generally, as our commentators note, one wants to ask whether there is a larger theory that determines which movements proceed through the edge of VP and which do not.¹² Why should it be the case, in some languages at least, that QM may proceed through the edge of

¹⁰ This is the pattern that many of our commentators thought we were committed to, as a description of Scandinavian more generally (because of their assumption that we were committed to both Claim A and Claim B).

¹¹ It would be useful in this connection to have an independent way of diagnosing movement through the edge of VP. If Nissenbaum (2000) is correct, the licensing of parasitic gaps by a moved element should provide such a test. In this context, it will be interesting to examine Svenonius's (2000) argument that parasitic gap licensing requires QP>V order, though the judgments are controversial (Jónsson (1994)). See also Svenonius (this issue). As examples (3)–(5) in Nilsen's commentary make clear, the distribution of parasitic gaps might diagnose movement through the edge of VP even in environments in which OS takes place – in particular under the analysis attributed by Nilsen to Norvin Richards (personal communication). As Nilsen points out, the data can be interpreted as indicating that whenever linearization considerations would need movement through the edge, parasitic gaps are obligatory in adjuncts, in obedience to what Nissenbaum calls Larson's Generalization (Larson (1988)).

The main problem raised by Nilsen for this analysis is the need to posit a Heavy NP Shift-type rule applying to a weak pronoun that usually would not undergo such a process. (A similar objection is made by Holmberg, in connection with his example (3).) We might suggest, however, (as Richards points out to us) that the heaviness requirement otherwise found with Heavy NP Shift might be a by-product of a particular phonological configuration disrupted in Nilsen's examples by remnant VP fronting.

¹² For example, Bobaljik suggests that the grammar might impose an ordering requirement on the output of a given phase that would in turn force phase-internal movement to take place. He points out that if we take V2 to be a requirement of this sort, imposed when CP is spelled out, cyclic linearization might account for the special strength of *wh*-island effects in V2 languages. The proposal is ingenious and we hope to return to it in future work. If Bobaljik is correct, the "larger theory" mentioned in the text will be diverse.

VP, but OS cannot? It is possible, of course, that there is no interesting answer to this question, and that it is an accident of history that the Germanic languages do not contain an unequivocal example of OS proceeding through the left edge of VP.¹³

Suppose we could convince ourselves, however, that OS across languages never involves an initial step of movement to the edge of VP. A familiar strategy for explaining such a fact would propose that movement is always blocked unless required by some independent factor. If this strategy is productive, we will discover factors that motivate movement through the edge of VP in exactly those cases where HG is not observed, and we will discover no such factor in the case of OS.

One possible motivation for movement through the edge of VP is the interaction between semantics and economy principles within syntax. Assume that quantificational elements may not be interpreted in argument positions (Heim (1997); Larson and Segal (1995); among others). If this is true, a QP may not be interpreted in situ, and must undergo movement. (Similar considerations might apply to *wh*-phrases and to phrases with topic/focus features.) It is possible that movement to the edge of VP is motivated by this semantic factor, and that there is no independent feature of *v* that could motivate such movement. In the case of quantificational phrases, it is the semantics that motivates the movement, and in the case of OS there is no motivation.¹⁴

Things cannot be quite this simple, given various observations made by our commentators (and in particular by Bobaljik) concerning passive and unaccusative VPs. Bobaljik makes it clear (based on the interaction of passive and OS) that the surface subject of a passive or unaccusative

¹³ As Bobaljik (1995) points out, in the OV Germanic languages (German and Dutch), OS-like fronting can apply in the absence of verb movement to C. This observation, however, does not necessarily provide us with examples of OS through the edge of VP, since (as Bobaljik argued), the OV character of the languages itself predicts this pattern of data if (as Bobaljik argued, like us) linear order is relevant to HG.

¹⁴ Note that these considerations would require us to assume that all QPs move through the edge of VP in all languages. In particular, in Norwegian (and whenever a QP is pronounced in situ), we would posit covert movement through the edge of VP. The linearization consequences of covert movement are, of course, the same as the consequences of failure to move in the first place. See Fox and Pesetsky (forthcoming) for a theory of covert movement, based on ideas of Nissenbaum (2000), consistent with our approach to Spell-out.

clause must move through a left-edge position within the lower Spell-out domain. We might view this movement as a response to the case requirement of the noun-phrase, much like movement through the specifier of a “defective” infinitival TP in a raising construction, or alternatively as a response to a feature and an EPP requirement on νP . (See Pesetsky and Torrego (2004) for one possible approach to such constructions.) In this context, we agree with our commentators that more work must be done to distinguish the possibility of movement through the edge of VP in passive and unaccusative constructions from its apparent impossibility in OS.

2. Periodicity

As many of our commentators note, the precise identity of the Spell-out domains (i.e. the *periodicity* of Spell-out) is just as crucial to our accounts of HG and IHE as is the question of which movements target the edge. As always, one wants to determine whether there is a hypothesis about the periodicity of Spell-out that is coherent and in accordance with the facts of individual languages, and also whether there is a plausible general theory that predicts the array of Spell-out domains cross-linguistically.

If detailed investigation uncovers Spell-out domain *paradoxes*, we would of course have reason to be suspicious of the approach. One salient question raised in our paper concerns the inclusion (or exclusion) of the external argument in the Spell-out domain that includes the verb and object. A similar question can be asked about auxiliary verbs. Both questions are taken up by some of our commentators. We offer a few comments on the issue here.

The status of the external argument:

As we noted in our paper, the ability of a finite verb to move to the left of the subject in Scandinavian V2 constructions indicates that the subject and the verb are not ordered until the domain in which V2 is established is Spelled-out. We assumed that this domain is CP. We therefore concluded that the subject in Scandinavian is not part of the lower Spell-out domain – i.e. not part of the domain that is responsible for HG effects and IHE. We thus proposed that νP is not a Spell-out domain in Scandinavian, but that VP is.

Certain observations by our commentators and others can be taken to reinforce the view that the subject does not form part of the lower Spell-out domain. The “scorecard” that we might assemble from our paper, the commentators’ remarks, and relevant other literature includes the following considerations:

- (8) *Arguments that the external argument is not part of lower Spell-out domain*
- a. The finite verb may move over the external argument in Scandinavian V2 (as just discussed).
 - b. OS may move an object over the external argument under certain circumstances in Swedish and Icelandic (Anagnostopoulou (this volume), examples (18) and (19); as well as remarks by Nilsen, Sells, Suranyi and Svenonius).
 - c. A floated quantifier associated with the subject may appear to the right of an object that has undergone OS in Icelandic, while the subject itself appears to its left (Anagnostopoulou (this volume), examples (16) and (17)).

At the same time, other considerations point in the opposite direction:

- (9) *Arguments that the external argument is part of lower Spell-out domain*
- a. Short V-movement in Northern Norwegian (independently attested) may not apply over a ν P-internal left-peripheral external argument (a position for external arguments that is also independently attested) (Svenonius (this issue), citing Bentzen (2003)).
 - b. OS over the subject, of the type mentioned in (8b), is *impossible* in Norwegian and Danish – and under certain circumstances, in Swedish as well (Nilsen (this issue); see also Anagnostopoulou (this issue, note 3)).
 - c. A floated quantifier associated with the subject may *not* appear to the right of an object that has undergone scrambling in Korean and Japanese, while the subject itself appears to its left (Ko (2004), citing Haig (1980); Kuroda (1983); Saito (1985), Miyagawa (1989); Fujita (1994), among others) – in contrast to the situation in Icelandic presented in (8c), as observed by Anagnostopoulou (this issue).

To the extent that the apparently conflicting evidence comes from distinct languages and dialects, one might account for the diversity of relevant data by proposing that there is cross-language variation in Spell-out domains, i.e. variation in periodicity. In some cases, we suspect that this approach is on the right track.¹⁵

As noted in (9b), for example, Ko (2004) shows that the interaction of subject-related floated quantifiers with object scrambling suggests that *v*P is a Spell-out domain in Korean and Japanese (in addition to VP). Anagnostopoulou (this issue), noting the argument from (8a) against *v*P as a Spell-out domain in Scandinavian, pointed out the correct prediction in (8b). In agreement with Anagnostopoulou, we do not believe that the contradictory data from Korean/Japanese and Scandinavian present a “periodicity paradox”. Instead, we suggest that they argue for cross-linguistic variation in periodicity.¹⁶

Other cases of apparent differences among languages, on the other hand, might very well be due to differences in syntax proper, rather than periodicity differences. Consider, for example, the question of object shift over the subject – that is, (8b) vs. (9b).

Following Sells (2001), we might assume that the subject is occupying its normal surface position in such cases and that OS is thus moving the object higher than usual, yielding a case of “Long Object Shift”. The argument that *v*P is not a Spell-out domain is straightforward. The existence of HG effects with OS makes it clear that, if *v*P is a Spell-out

¹⁵ In Fox and Pesetsky (in preparation), we actually argue that Spell-out domains do not vary across languages, but that external merge (basic phrase structure building) may occur before or after spellout, i.e. overtly or covertly. In a language in which *v*P appears not to be a Spell-out domain, specifiers and adjuncts of *v*P are merged after Spell-out of *v*P. As a consequence, they do not figure in the ordering statements supplied when *v*P is spelled out. In order to be linearized under the algorithm offered in the appendix to our paper, the subject will have to raise further.

¹⁶ Ko (2004) notes that certain apparent floated quantifiers in Korean – principally those that have case markers suffixed to them – do not show the effects that argue for *v*P (and VP) as a Spell-out domain. The existence of such elements in Korean raises a caution concerning the proper interpretation of the Scandinavian data cited by Anagnostopoulou. We must determine whether the failure of Scandinavian floated quantifiers to show the effects found in Korean is due to periodicity differences or whether the Scandinavian quantifiers are actually counterparts of the Korean quantifiers that also fail to show the relevant effects. Anagnostopoulou suggests that there are arguments that settle the matter in favor of her interpretation of the data.

domain, OS does not move through its edge. Given this fact, if νP is a Spell-out domain, Subject<Object ordering is established when it is spelled out. If the syntax allows the object to undergo OS that places it to the left of the subject, an ordering paradox results, which should exclude the construction. If, on the other hand, νP is not a Spell-out domain, the subject will not be ordered with respect to the object until Spell-out applies again, presumably at the CP level. As a consequence, the object is free to undergo movement that places it to the left of the subject, without any danger of an ordering contradiction. Thus, we might take the possibility of “Long Object Shift” in Icelandic and Swedish as an argument that νP is not a Spell-out domain – a conclusion consistent with the other evidence in (8).

How might we explain those Scandinavian languages and syntactic environments that disallow Object Shift over the subject? In these cases, we must assume that some other property of the grammar prevents the object from preceding the subject. We might assume that in these cases, νP *is* a Spell-out domain, with the consequence that Subject<Object order is established early in the derivation. While this view correctly excludes “Long Object Shift”, it now becomes a puzzle that the verb may precede the subject in V2 (and in VP-fronting constructions with and without remnant movement, as discussed in our paper). One might conclude that we have found a “periodicity paradox”, as suggested by several commentators.

It is possible, however, that syntax itself – not periodicity issues connected with the syntax-phonology interface – distinguishes these languages. (See the commentaries by Anagnostopoulou, Bobaljik and Diesing.) To allow Long Object Shift, under an analysis of the sort we have been considering, there must be a landing site for the object above Spec,TP. In a feature-driven theory of movement, for example, there must be an appropriately high head triggering the movement. It is quite possible that such a head is simply absent in the languages that disallow the construction (or triggers covert movement only), in which case one might suggest that νP is not a Spell-out domain in any of the Scandinavian languages.

Following suggestions of Anagnostopoulou (this issue) and Alexiadou and Anagnostopoulou (2001), one might also question the assumption that the subject in these constructions actually does occupy its usual

high, ν P-external position. In languages that allow apparent Long Object Shift, on this view, OS is actually “short” – targeting its usual position. It is the external argument that shows special behavior; it remains in a low position, perhaps Spec, ν P, instead of raising to Spec,TP. If this analysis is correct, what distinguishes languages that allow the construction from those that do not is the possibility of a low position for the external argument. Once again, ν P is crucially not a Spell-out domain. Crucially, once again, there is no periodicity paradox.

Some cases may pose more difficult problems, especially cases in which a single language appears to present conflicting evidence about the periodicity of Spell-out. For example, Svenonius points out, in connection with (9a) above, that though short V-movement in Northern Norwegian may not reorder the verb and external argument, long V-movement in the V2 construction may do so.¹⁷ In this connection, it might be fruitful to re-examine the extent to which V2 presents a clear argument that ν P is not a Spell-out domain.

As we noted above, it is crucial that the ordering of the subject and verb take place in the same Spell-out domain as that which establishes V2 order. In his commentary, Nilsen suggests that this domain might be lower than CP (Müller (2004); Nilsen (2003)), contrary to what we had assumed. Call this lower domain μ P.¹⁸ In effect, the movement of V over the subject in V2 would be a process akin (for Linearization

¹⁷ Svenonius’s and Bentzen’s arguments rest on the following premises: (1) that the subject may occur to the right of adverbs like *usually*; (2) that short V-movement may apply over the same set of adverbs; and (3) that these adverbs have a fixed position. The fact that the order *V-usually-Subject* is not possible is thus a surprise, which Svenonius suggests may indicate that Subject-V order is fixed within a lower Spell-out domain. Note, however, that premise (3) might be questioned, in which case premise (1) would not necessarily argue that the subject may remain internal to the ν P. If the subject must raise in Northern Norwegian, the impossible *V-usually-Subject* order could be explained straightforwardly as a consequence of the non-application of obligatory subject raising. Bentzen’s observations also suggest that the denial of premise (3) might have to be extended to negation, which behaves like *usually* in this paradigm. That might be a problematic move.

¹⁸ A suggestion of this sort would lead to rather different proposals about cross-linguistic variation of the sort studied by Ko and extended by Anagnostopoulou, which we have not yet investigated. At the same time, this type of analysis might eliminate some of the questions noted by Suranyi in his commentary that arise in connection with our proposal in OV languages with Verb Second.

purposes) to the movement of a QP in the QM construction across VP- (and ν P-)internal material prior to movement from μ P into the higher Spell-out domain CP.

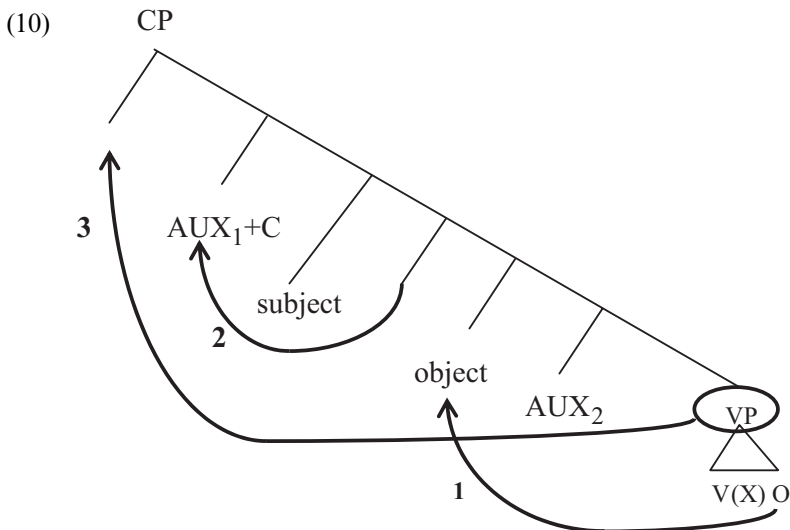
As Shoichi Takahashi (personal communication) has pointed out to us, however, constructions with remnant [V(+X)] fronting can also yield V(+X)-S-O order. Since V moves over S here too, the reasoning just sketched would entail that the remnant movement takes place first within μ P, like V-movement in V2 – which in turn means that OS over the verb took place within μ P as well. Since these remnant movement constructions appear to obey HG (one of our central points), there must have been a Spell-out domain smaller than μ P in which the relative order of the verb and VP-internal material was fixed. So if Nilsen's suggestion is on the right track, there must be three Spell-out domains in the clause: CP, μ P and something like VP. It is interesting to investigate this possibility in light of Ko's (2004) arguments that ν P and VP are both Spell-out domains in Korean. Specifically, one might suggest that universally the Spell-out domains are VP, CP and a projection in between – either ν P or μ P, depending on whether or not the language has V2.

The status of auxiliary verbs:

The fact that auxiliary verbs are outside the lower Spell-out domains of a clause is perhaps less controversial than the periodicity status of the external argument and ν P. Here too, important issues arise, as noted in particular by Holmberg in his commentary.

As Holmberg points out, if the first Spell-out domain above an auxiliary verb *Aux* is CP, we expect OS to be able to cross *Aux* as long as $V < O$ order is preserved. If, on the other hand (as pointed out by Svenonius in his commentary), there is a Spell-out domain between VP and CP that includes *Aux* but does not include the landing site of OS, then we do not expect OS across *Aux* to be possible. These questions interact crucially with the question of the external argument just discussed – since if *Aux* is contained within a Spell-out domain lower than CP, *a fortiori* the external argument is also contained within that Spell-out domain.

As it happens, we have been conducting a pilot study on these very questions.¹⁹ A relevant configuration, as Holmberg notes, is one in which OS occurs in conjunction with remnant VP movement to Spec,CP – and enough auxiliary verbs are present to allow one to see OS moving over one of them, while (of necessity) another auxiliary verb has moved to C, as part of the V2 phenomenon.²⁰



VP ordering: $V (< X) < O$

CP ordering: $VP < AUX_1 \rightarrow V \text{ (or } X) < AUX_1$
 $AUX_1 < \text{subject}$
 $\text{subject} < \text{object}$
 $\text{object} < AUX_2$

¹⁹ We wish to thank Alevtina Asarina for conducting these investigations under MIT's Undergraduate Research Opportunities Program. This research was made possible by a grant from the Dean's Fund of the School of Humanities, Arts and Social Sciences, MIT.

²⁰ Holmberg suggests that another relevant configuration might be found in the domain of Icelandic Stylistic Fronting, where (as in remnant VP movement), the main verb of a sentence may come to precede the auxiliary. In this construction, the auxiliary verb occupies the "second position" relevant to V2 – either C or perhaps I in Icelandic – and the next highest verb moves to the left of the auxiliary. This second instance of V-fronting should license OS in our approach, and possibly permit us to see OS move the

If there is no Spell-out domain that contains Aux₂ in (10) while excluding the landing site of OS, our approach predicts that the result should be acceptable.

This prediction can only be tested if VP topicalization in a complex structure of this sort is acceptable independent of OS. In his commentary, Holmberg offers the judgment (his (8c)) that VP topicalization is in fact excluded in these environments, independent of OS, rendering the prediction untestable.

Our three consultants presented a more mixed picture. One of them showed a pattern that can be interpreted as fitting the prediction under the assumption stated, though the best examples were still degraded. Another consultant provided very similar judgments, but only if VP-fronting stranded the light verb ‘do’ – a phenomenon more general in Swedish, whose analysis is somewhat open (Anders Holmberg, personal communication). Finally, a third consultant presented judgments much closer to Holmberg’s.²¹ We hope to explore these issues in greater detail in ongoing research.

3. OS of IO over DO: just linearization?

As many of our commentators have pointed out, some of the facts that are explained by Cyclic Linearization (CL) might also follow from

direct object over the auxiliary. Unfortunately, since the auxiliary verb has raised to I or C, we only predict the possibility of OS over the auxiliary if there is a landing site for OS that is higher than that usually observed in Icelandic.

²¹ The examples tested were:

- (i) Jag skulle ha öppnat dörren/den.
I shall have opened the-door/it.
- (ii) Öppnat dörren/den skulle jag ha.
Opened the-door/it should I have.
- (iii) Öppnat skulle jag ha dörren/den.
Opened should I have the-door/it.
- (iv) Öppnat skulle jag dörren/den ha.
Opened should I the-door/it have.

Because of the possibility of various confounds (e.g. causative vs. aspectual uses of ‘have’), more careful study will be necessary before we can reach firm conclusions, of course.

syntax-internal principles such as Attract Closest (AC; see Chomsky (1995b, 2000), Richards (1997, 1999) among others). As we will see in the next section, Superiority phenomena do not follow from CL. We thus will need both AC and CL. This raises immediate questions about the division of labor between these two constraints. The questions are particularly pertinent to the investigation of the distribution of IO-DO reorderings.

In a range of cases, IO blocks movement of DO. One would like to know whether this is an effect of AC or CL, or both.²² One reason to think that CL is the governing factor in regulating OS is the apparent fact that DO can undergo object shift over IO when subsequent movements restore IO<DO order. This, if our data are correct, is the case when OS of DO is followed by remnant movement of the verb and IO.

- (11) ?[Gett henne t_{DO}] har jag den_{DO} inte ...
 given her have I it not (Holmberg, p.c.)

In (11) ((30a) from our paper) the environment for remnant movement was first created when DO exited the VP by OS, crossing over IO. This movement would constitute a counterexample to an AC account of those cases in which IO does successfully block OS of DO. However, the availability of the initial movement in this context is predicted by the CL account.

The facts are also consistent with a theory that posits AC as an explanation for other phenomena (e.g. the blocking effect of IO in passive), so long as OS is sensitive to a feature that can be present on some DPs and not on others. Thus, for example, in (11), the DO pronoun would be taken to bear a feature not present on IO.²³

²² Note that this type of overlap is not necessarily an indication of a redundancy. Two constraints can overlap in empirical coverage without either being redundant. See also section 4 below.

²³ There are other examples in which DO undergoes OS over IO and a later instance of movement “legalizes” this otherwise impossible instance of IO. For example, if IO undergoes *wh*-movement, OS of DO becomes possible, as noted by Holmberg. Here, one might claim, with Anagnostopoulou (2003), that the trace of *wh*-movement fails to count as an intervener, with Attract Closest considerations determined phase-finally (i.e. after both OS and *wh*-movement have taken place), as proposed in Chomsky (2001a) and Chomsky (2001b). Such a solution is not immediately available for our examples with remnant movement, since the intervening IO has not itself moved (though the containing *vP* has).

We must still ask, of course, whether we can detect a role for AC in regulating OS, and especially whether IO can be shown to act as an intervener for OS of DO. If OS involves a feature that might be present on DO but not on IO in a given derivation, as (11) seems to indicate, we do not expect to be able to find easy evidence that AC might play such a role.²⁴

Anagnostopoulou and Bobaljik argue, however, that OS is not sensitive to a feature that might distinguish an IO from a DO. They call attention to an important cross-linguistic correlation discovered by Anagnostopoulou (2003). Anagnostopoulou discovered that the Scandinavian languages and dialects that allow long passive of DO over IO are precisely those languages that allow a similar long instance of OS. She furthermore suggests that this correlation reflects variation in the availability of a landing site for DO lower than the landing sites for OS and passive – i.e. an initial movement step that can reorder IO and DO. If DO undergoes this “short” initial step, it will be closer to the triggers for OS and passivization than IO. If this initial step is unavailable in a language, DO will remain farther from these triggers than IO.

The correlation discovered by Anagnostopoulou is important, and her proposed account in terms of an initial reordering seems to us likely to prove correct. (See Doggett (2004) for other arguments for this reordering.) Note, however, that what is responsible for the correlation is not any claim about the nature of constraints that restrict OS or passive, but rather the fact that the output of the initial reordering feeds these operations. We propose, therefore, to adopt Anagnostopoulou’s suggestion, without, however, disrupting in any way our account of (11) and similar facts.

Anagnostopoulou argues that a particular variant of her proposal, if transferred into our overall framework, would entail undesirable complications. The variant she considers is one in which initial reordering takes place within a spell-out domain (“Domain A”) lower than VP.²⁵ It

²⁴ If the relevant feature were overtly morphologically marked, then in an example in which both IO and DO bore such a feature, we would expect AC to block movement of DO over IO. We are not aware of such a case.

²⁵ The main complication that she notes relates to IHE. If DO could be moved to the left of IO within a Spell-out domain lower than VP, we might expect that DO < IO order would be available in the QM paradigm in Icelandic without simultaneously blocking verb movement. That is, we might incorrectly fail to predict certain instances of the IHE.

is not necessary, however, to assume that the domain in which initial reordering takes place (in those languages that allow it) is itself a Spell-out domain. Thus, we see no obstacle to adapting Anagnostopoulou's proposal *in toto* within our framework.²⁶

4. Superiority: beyond just linearization

In his commentary, Müller notes a certain "order preservation" character to standard Superiority effects in multiple questions, and observes (correctly) that these effects do not follow from our system. *Wh*-movement clearly has the capacity to reverse the order of a *wh*-phrase and its phase-mates. For this reason, we assume that it proceeds successive-cyclically through the edges of Spell-out domains. At the same time, one might view Superiority effects in multiple questions as a case in which *wh*-movement lacks the capacity to reverse order with respect to a particular phase-mate – and thus as an order preservation effect.

In our system, if we observe order preservation between elements X and Y, but not between X and Z or Y and Z, we might suppose that X and Y are contained in a Spell-out domain D that does not contain Z. Thus, the order of X and Y will be fixed within D, and must be maintained thereafter, but X and Y are free to move over Z in a later Spell-out domain. This strategy cannot explain, as Müller notes, the phenomena of Superiority relevant to such contrasts as *Who saw what.* vs. **What did who see.* There is clearly no order preservation effect preventing *what* from moving over the verb, but there is at the same time no plausible domain D that contains *who* and *what* but not the verb. Consequently, in the framework of our paper, we must probably conclude that Superiority facts are not order preservation effects, but arise from some other factor, e.g. AC, discussed above.

Müller points out that this failure of unification could be viewed as a potential argument in favor of the alternative, Optimality Theoretic perspective advanced by Müller (2000), building on ideas of Williams (2003)

²⁶ We inherit, of course, Anagnostopoulou's own unanswered questions, including (a) why initial reordering *must* feed OS or passive, i.e. why **I gave the book Bill* is impossible; and (b) why, in certain dialects of Norwegian and Swedish, initial reordering can proceed across a pronoun, but not a full DP (Anagnostopoulou's (20)–(21)).

that are also reflected in Williams' own commentary. In these frameworks, all instances of movement are governed by a very general Order Preservation (Shape Conservation) constraint – which is, however, out-ranked by certain other constraints that require it to be violated from time to time. One such constraint is the EPP requirement on C that at least one instance of overt *wh*-movement take place in an English interrogative. This requirement will motivate a violation of Order Preservation, allowing (in fact, forcing) a *wh*-phrase to move over such elements as the verb in order to satisfy the EPP requirement. At the same time, the system will pick the derivation that minimizes the violation of Order Preservation. Thus, when there are two *wh*-phrases that in principle could move leftward to satisfy the relevant EPP requirement, picking the leftmost of these phrases will violate Order Preservation with respect to fewer elements than would the choice of the rightmost.²⁷

In contrast, Object Shift in Scandinavian is viewed as the product of an EPP requirement on a head other than C (e.g. AgrO) that is ranked lower than Order Preservation. As a consequence, violations of Order Preservation in service of EPP satisfaction will not be found. Instead, EPP satisfaction will be violated in service of Order Preservation. Thus, when OS would violate Order Preservation with respect to the relative position of V and O, it simply does not take place.

²⁷ If one were to adopt this more complex model, the “tucking in” character of multiple-specifier formation posited by (Richards (1997, 1999)) might also be explained as anticipated in Suranyi's commentary on our paper. A non-tucking in derivation requires deletion of an ordering statement relating the two *wh*-phrases. This deletion can be avoided by the tucking-in derivation which otherwise involves the same amount of deletion.

This suggestion presupposes the countercyclic tucking-in of Richards. Suranyi suggests that our proposal could be adapted to avoid such countercyclic movement entirely. Expanding on his suggestions, one might imagine that the “Laws of Precedence” leave multiple specifiers of relevant categories unordered among themselves – determining only that specifiers of XP precede non-specifier dependents of XP. The task of ordering multiple specifiers would then fall to statements on the “Provisional Ordering Table” that reflect their earlier positions. This proposal would predict, however, that in a multiple specifier configuration the linear order of the specifiers might not reflect their relative scope (c-command relations) after movement. Evidence against this conclusion is provided by Nissenbaum (2000, appendix to chapter 3), who shows that the relative order of *wh*-specifiers in Bulgarian multiple questions does reflect their relative structural prominence for binding and related phenomena. If Nissenbaum is correct, tucking in is simply a fact. If so, the alternative sketched in this paragraph is not viable.

In our paper, we argued that Order Preservation (i.e. monotonicity) is irrelevant to phase-internal operations. We implemented this idea with the proposal that Ordering Statements are produced by the grammar only at the point of Spell-out for each phase (along with the assumption that the Ordering Table has to grow as the derivation proceeds). We could have assumed a slightly more complex implementation in which Ordering Statements are produced at every step of the derivation but could be deleted to resolve contradiction before the next Spell-out takes place. For example, we could imagine a “Provisional Ordering Table” for which monotonicity doesn’t hold (i.e. statements can be deleted from the Provisional Ordering Table to resolve contradiction). Phase-finally, the contents of the Provisional Ordering Table are placed on the “Definitive Ordering Table” – a set from which ordering statements are *never* deleted (which renders contradictions fatal and unresolvable).

In this more complex variant of our system, we could now imagine that the syntax, whenever it is faced with a choice among possible next steps – e.g. a choice between moving *who* or *what* in (1) – will make the choice that requires deletion of the fewest Ordering Statements from the Provisional Ordering table to resolve contradiction. This proposal, in turn, if rephrased in OT terms, amounts to the claim that the grammar includes a constraint *forbidding* deletion of Ordering Statements from the Provisional Ordering Table (“NoDELETE”). The NoDELETE constraint is soft – outranked by another factor (EPP for the *wh*-feature) that requires violations of Order Preservation within the Provisional Ordering Table in certain circumstances. The Definitive Ordering Table, which stands outside this system, is not subject to deletion of the sort just described, and is uninterpretable if it contains contradictions.

This variant of Müller’s proposal leaves unchanged the fundamental properties of our system, since it continues to assume that Ordering Statements are sent to the (Definitive) Ordering Table as each phase is spelled out, with the consequences suggested in our paper. Thus, this variant preserves our account of the IHE, which relies on the observation that long-distance movement with a particular set of predicted properties proceeds successive-cyclically through the edge.

Williams, discussing Superiority Effects in his commentary, presents an interesting challenge to even this aspect of our approach – in particular, the idea that there are Ordering Statements that reflect intermediate

positions occupied in the course of successive-cyclic movement. He notes (correctly) that we predict that orders established when any given domain is spelled out will be inherited by later domains. He offers as a possible counterexample the behavior of multiple *wh*-movement in Serbo-Croatian. Specifically, he offers examples of short-distance *wh*-movement in which Superiority is apparently violated, and then notes that Superiority may not be violated in long-distance questions, even when both *wh*-phrases started as phase-mates.

This is problematic for our approach, Williams argues, in that a local instance of multiple *wh*-movement that violates Superiority (e.g. his (6b)) should be able to serve as input to further *wh*-movement. This would incorrectly permit apparent Superiority violations in long-distance *wh*-movement – inherited from violations first incurred in an earlier Spell-out domain. That is, the acceptability of his (6b) should entail the acceptability of his (7b).

It is not clear, however, that this contrast poses a problem for our approach. Williams' discussion presupposes that the relevant distinction between cases in which Superiority can be violated in Serbo-Croatian and cases in which it cannot is the distinction between local and long-distance *wh*-movement. If Bošković (1997) is correct, however, the relevant distinction is not between local and long-distance *wh*-movement, but between *wh*-movement in matrix vs. embedded clauses. Bošković notes that the possibility of violating Superiority (as in Williams' (6b)) is in fact limited to matrix questions. In embedded questions (and in multiple-*wh* correlatives), Superiority effects resurface:

- (12) a. Jovan i Marko ne znaju [ko je koga istukao].
 Jovan and Marko not know who AUX whom beaten
 'Jovan and Marko do not know who beat whom.'
 b. ?*Jovan i Marko ne znaju [koga je ko istukao].
 Jovan and Marko not know whom AUX who beaten
 (Bošković (1997), example (12))

If the possibility of violating Superiority is limited to matrix clauses, we do not expect to find any legal Superiority violations in an embedded clause, and thus such violations should not be inherited by higher clauses. Thus, we do not, in fact, predict examples like (7b) to be acceptable. Williams' (6b) is not expected to serve as input to further *wh*-movement,

and there is no special problem for our overall approach. Of course, we must hope that the correct account of the matrix/subordinate asymmetry will not conflict with this line of reasoning. For example, it will be profitable to ask whether Bošković's own account of these phenomena (and related constructions in French and other languages) is consistent – or can be made consistent – with our approach.

We thus see again, as we have seen throughout this reply, that evaluation of our proposal can only be made in the context of the syntax whose results are being linearized. This is a complex task, of course, and many relevant questions remain open. In this reply, we have attempted to clarify a few of the most important issues raised by our commentators in this connection.

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References

- Alexiadou, A. & Anagnostopoulou, E. (2001). The Subject-in-Situ Generalization and the Role of Case in Driving Computations. *Linguistic Inquiry*, 32, 193–231.
- Anagnostopoulou, E. (2003). *The Syntax of Ditransitives: Evidence from Clitics*. Berlin: Mouton de Gruyter.
- Bentzen, K. (2003). V-to-I Movement in the Absence of Morphological Cues: Evidence from Adult and Child Northern Norwegian. *Nordlyd*, 31, 573–588.
- Bobaljik, J. (1995). *Morphosyntax: The Syntax of Verbal Inflection*, Massachusetts Institute of Technology: Doctoral Dissertation.
- (2002). A-Chains at the PF-Interface: Copies and 'Covert' Movement. *Natural Language and Linguistic Theory*, 20, 197–267.
- Bošković, Ž. (1997). Superiority Effects with Multiple *Wh*-Fronting in Serbo-Croatian. *Lingua*, 102, 1–20.
- Chomsky, N. (1995a). *The Minimalist Program*. Cambridge, Mass.: MIT Press.
- (1995b). Categories and Transformations. In: *The Minimalist Program* (pp. 219–394). Cambridge, Mass.: MIT Press.
- (2000). Minimalist Inquiries. In: Martin, R., Michaels, D. & Uriagereka, J. (eds.) *Step by Step: Essays on Minimalist Syntax in Honor of Howard Lasnik* (pp. 89–156). Cambridge, Mass.: MIT Press.
- (2001a). Derivation by Phase. In: Kenstowicz, M. (ed.) *Ken Hale: A Life in Language* (pp. 1–52). Cambridge, Mass.: MIT Press.
- (2001b). *Beyond Explanatory Adequacy*. Cambridge, Mass.: MIT Working Papers in Linguistics.
- Christensen, K. K. (1986). Norwegian *Ingen*: A Case of Post-Syntactic Lexicalization. In: Östen, D. & Holmberg, A. (eds.) *Workshop at the Ninth Scandinavian Conference* (pp. 21–35). Stockholm: University of Stockholm.

- Doggett, T. B. (2004). *All Things Being Unequal*, Massachusetts Institute of Technology: PhD dissertation.
- Fujita, N. (1994). *On the Nature of Modification: A Study of Floating Quantifiers and Related Constructions*, University of Rochester: Doctoral Dissertation.
- Haig, J. H. (1980). Some Observations on Quantifier Float in Japanese. *Linguistics*, 18, 1065–1083.
- Heim, I. (1997). Predicates or Formulas? Evidence from Ellipsis. In: Lawson, A. & Cho, E. (eds.) *Proceedings of Salt VII*. Cornell University: CLC Publications.
- Jacobs, J. (1980). Lexical Decomposition in Montague Grammar. *Theoretical Linguistics*, 7.
- Jónsson, J. (1994). On Case and Agreement in Icelandic. In: Benedicto, E. and Runner, J. (eds.) *Functional Projections* (pp. 85–102). Amherst: Graduate Linguistic Student Association.
- Kayne, R. (1998). Overt vs. Covert Movement. *Syntax*, 1, 128–191.
- Ko, H. (2004). Asymmetries in Scrambling and Cyclic Linearization. Ms., MIT.
- Kratzer, A. (1995). Stage-Level and Individual-Level Predicates. In: Carlson, G. N. & Pelletier, F. J. (eds.) *The Generic Book* (pp. 125–175). Chicago: The University of Chicago Press.
- Kuroda, S.-Y. (1983). What Can Japanese Say About Government and Binding. In: Barlow, M. (ed.) *Proceedings of the West Coast Conference on Formal Linguistics* (pp. 153–164). Stanford: Stanford Linguistics Association.
- Larson, R. (1988). Light Predicate Raising. vol. 27. Lexicon Project Working Papers. Cambridge, MA: Center for Cognitive Science, MIT.
- Larson, R. K. & Segal, G. (1995). *Knowledge of Meaning: An Introduction to Semantic Theory*. Cambridge, Mass.: MIT Press.
- Miyagawa, S. (1989). *Structure and Case Marking in Japanese*. San Diego: Academic Press.
- Müller, G. (2000). Shape Conservation and Remnant Movement. Paper presented at Proceedings of the North East Linguistic Society, Rutgers University.
- (2004). Verb-Second as VP-First. *Journal of Comparative Germanic Linguistics*, 7, 179–234.
- Nilsen, Ø. (2003). *Eliminating Positions: Syntax and Semantics of Sentence Modification*, University of Utrecht.
- Nissenbaum, J. (2000). *Investigations of Covert Phrase Movement*, Linguistics, Massachusetts Institute of Technology: Doctoral Dissertation.
- Penka, D. (2001). *Kein Muß Kein Rätsel Sein. Zur Semantik der Negativen Indefinita im Deutschen*, University of Tübingen: Master's thesis.
- Pesetsky, D. & Torrego, E. (2004). The Syntax of Valuation and the Interpretability of Features. Ms., MIT and University of Massachusetts, Boston.
- Potts, C. (2000). When Even *No*'s Neg Is Splitsville. Paper presented at Jorge Hankamer's Web Fest.
- Richards, N. (1997). *What Moves Where When in Which Language?*, Massachusetts Institute of Technology: Doctoral Dissertation.
- (1999). Featural Cyclicity and the Ordering of Multiple Specifiers. In: Epstein, S. & Hornstein, N. (eds.) *Working Minimalism* (pp. 127–158). Cambridge, Mass.: MIT Press.
- Rögnvaldsson, E. (1987). On Word Order in Icelandic. In: *Proceedings of the Seventh Biennial Conference of Teachers of Scandinavian Studies in Great Britain and Northern Ireland* (pp. 33–49). London: University College London.

- Saito, M. (1985). *Some Asymmetries in Japanese and Their Theoretical Implications*, Massachusetts Institute of Technology: Doctoral Dissertation.
- Sells, P. (2001). *Structure, Alignment and Optimality in Swedish*. Stanford, CA: CSLI Publications.
- Svenonius, P. (2000). Quantifier Movement in Icelandic. In: P. Svenonius (ed.) *The Derivation of VO and OV*. Amsterdam: John Benjamins.
- Swart, H. D. (2000). Scope Ambiguities with Negative Quantifiers. In: Heusinger, K. V. & Egli, U. (eds.) *Reference and Anaphoric Relations* (pp. 118–142). Dordrecht: Kluwer.
- Williams, E. (2003). *Representation Theory*. Cambridge, Mass.: MIT Press.

What is Syntax?

NOMI ERTESCHIK-SHIR

Abstract

In this response, I address the main points of my article raised by the reviews in this volume. I argue that the properties responsible for word order are in fact phonological. I clarify the properties of the identification constraint, revise my account of V-2, and elaborate on some of the phonological properties involved in incorporation.

1. Introduction

Syntactic accounts are often rejected when the phenomenon under examination violates well-established syntactic constraints such as the Head Movement Constraint, the ban on rightward movement, and island constraints. Object shift is such a phenomenon (re: A/A-bar, Extension condition) and has therefore been relegated to a post-syntactic component. (Svenonius, this volume). Taking such a tack has the advantage of leaving syntax unperturbed by ‘messy’ phenomena, but it has the disadvantage of creating a new component, post-syntax, which is undefined and therefore merely provides a dumping ground for unaccounted for data. A more conservative tack is to attempt to reexamine the validity of the constraints. This may lead one to question basic tenets of the theory and end up with a radically changed model of grammar. This is the spirit in which my paper was written.

Starting anew in this way means that nothing can be taken for granted except the most basic premises. I propose the following (uncontroversial) initial division of labor between syntactic and phonological computation:

(1) <u>Syntax</u>	<u>Phonology</u>
lexically determined	morphologically determined
hierarchical-	linear-
c-command	adjacency, left/right edges
syntactic constituents	prosodic constituents

Sells and Svenonius question the need for the phonological computation of word order. What I argue is that the factors listed under Phonology in (1) are not properties of syntactic computation and should therefore not play a role in syntax. To the extent that factors such as these do play a role in the account of word order, my account is therefore not syntactic. To the extent that these factors belong in phonology, my account is indeed phonological.¹

2. Processing

Processing involves the identification of at least subject/object, topic/focus, and the force of the sentence. This is achieved by language specific means, such as word order, morphology and intonation. In my article (henceforth SPS), I introduce a constraint (ID) which accounts for subject/object identification and show how it works in Danish.

2.1. *Identification (ID)*²

Several of the reviewers point out both theoretical and empirical problems with the ID as I presented it. In particular Bobaljik, Holmberg, and Suranyi argue that my claim that the position of adverbs, auxiliaries and modals plays a role in ID must be misguided since in these cases identification is absolute, whereas it is only a preference in the remaining cases. Bobaljik quotes King's 1995: 2 criticism of Jakobson's 1936/1971: 28 classic proposal that word order remains fixed when the case of the

¹ An interesting task would be to tease apart those phonological features that are relevant to the pronunciation of sentences (e.g. the ones listed under phonology in (1)) and those that are not (e.g., feet, onsets).

² See section 2.2 (16) of Erteschik-Shir (this volume) for the introduction of the ID constraint.

nominal is unclear. Her criticism is that in context the constraint doesn't hold. Bobaljik concludes that ID must be taken as "a parsing *preference*."

I agree that context (not to mention thematic roles) 'marks' arguments, just as case does, yet the ID constraint should still be viewed as an absolute constraint, since the arguments are marked unambiguously in each individual case:³

- (2) Den her bog læste Peter. (Bobaljik)
This here book read Peter
'Peter read this book.'
- (3) Hvem var det Sarah mødte, Peter eller Thomas? Peter mødte Sarah.
Who was it Sarah met, Peter or Thomas? Peter met Sarah
'Sarah met Peter.'

In (2) both arguments are lexically marked since the verb selects a human subject and a 'readable' object, leaving no room for ambiguity in any context. In (3), the best context possible to induce the reading with 'Sarah' as subject has been set up: The context forces a contrastive reading of the topicalized object and still allows for a topic reading of the subject.⁴ Yet Danish informants reject the sentence in this context because they still want to assign the reading in which *Peter* is the subject. (Note that the contrastive stress on the topic does not function to disambiguate the sentence since the same intonation would be used if *Peter* were a contrastive subject.) It seems that Danish is particularly rigid in this respect. A disambiguating context can serve to identify subjects and objects more easily in Swedish and Norwegian.⁵ Sofie Raviv (pers. com.) suggests a possible reason for this difference. In Norwegian (4) is ambiguous because the pronoun *han* is unmarked for case:

³ See Erteschik-Shir, to appear for a more extensive discussion of the ID constraint.

⁴ For definitions of topic, focus and contrast and the contexts which promote readings with each, see Erteschik-Shir, 1997.

⁵ Ørsnes, 2002, fn. 8 makes a similar point. In his discussion of subject extraction from embedded that-clauses in Danish, he points out that the potentially ambiguous

- (i) Peter tror Louise kommer.
a. Peter believes Louise is coming.
b. Peter Louise believes is coming.

is interpreted as in a. out of context and that instances of b. are "strikingly rare".

- (4) Han så Peter.

He saw Peter/Peter he saw

Since Norwegian (Nynorsk) has a number of case-ambiguous pronouns, different ways of disambiguating are required for many more cases in Norwegian than in Danish, especially since the most common kind of topicalization is the topicalization of pronouns. This is very likely the reason that Norwegian is less rigid than Danish in allowing contextual identification. It is possible that this argument carries over to colloquial Swedish, in which, according to Josefsson, 1992: 69, third person singular masculine, feminine and plural pronouns are case-neutral. Further investigation will show whether this is the reason that some Scandinavian languages are more open to contextual identification than others.⁶

There is no difference among the Scandinavian languages in the processing of ambiguous wh-questions. They are ambiguous in Danish as well (as pointed out by Holmberg and Bobaljik):

- (5) Hvem mødte Sarah?

Who met Sarah

'Who did Sarah meet?'/ 'Who met Sarah?'

The reason wh-questions are processed differently than declaratives with respect to ID is that the initial wh-phrase triggers a search for a trace and is therefore not automatically identified as a subject. The wh-marking therefore also serves a purpose in identification. Further, since wh-questions never occur out-of-the blue, their interpretation is necessarily linked to a disambiguating context.⁷ Finally, ambiguous wh-phrases

⁶ Hellan notes that Norwegian subjects are not restricted to a position preceding the adverb as shown in (i):

(i) Hvorfor sparket ikke Jon ballen?

why kicked not Jon ball-the

According to Westergaard, 2003, a subject which follows the adverb in a wh-questions is focussed. In Norwegian this type of generalization interacts with verb position as well (V-2 vs. V-3) (see Nilsen, 1997, 2003, Vangsnes, 2004 for discussion). These facts call for an analysis in terms of focus identification, particular to Norwegian. It is possible that intonation takes over in those Scandinavian dialects in which negation does not play a role in ID.

⁷ It makes sense that such ambiguity is used for poetic purposes, as Holmberg says (fn. 2).

are rarely found in Danish. They are replaced by circumlocutions such as that in (6):

- (6) Hvem er det (som) Sarah mødte?/ Hvem er det som/der mødte
 Who is it (that) Sarah met? / Who is it that met
 Sarah?
 Sarah?

From this discussion, I conclude that even when identification is contextually determined, it is still absolute, and no less absolute than identification by overt case marking, agreement, and adverb position. I therefore believe that the application of ID to word order phenomena should be pursued further, allowing for an explanation of 'syntactic' phenomena in terms of a processing constraint.

2.2. Long Object Shift (LOS)⁸

LOS involves object shift across a subject. It provides strong evidence for an account in terms of ID, since it is blocked with case-ambiguous pronouns and is only licensed with marked thematic roles. Yet, as SPS section 2.2 points out, it violates the ban on elements with φ -features intervening between subject and verb. (7) illustrates a case in which the φ -intervention constraint applies (the object pronoun intervenes between the verb and the subject) and (8) an example of LOS in which it does not.

- (7) *I går mødte +ham Peter ikke (Danish)
 yesterday met him Peter not
 (8) I hallen mötte honom en hemsk syn.
 In hall.the met him a horrible view
 (Swedish, Josefsson, 2003: 204)
 'In the hall he met a terrible sight.'

(7), as well as other similar cases to which the constraint applies, can be derived under the free-ride account of OS in which the object pronoun incorporates and moves along with the verb. Such strings are not, however, derivable under my current view in which such word orders are not derived by object movement, but rather by the linearization (of adverbs, for example) which, obviously, must precede incorporation

⁸ See SPS, section 3.

(pronunciation). The intervention constraint is therefore unnecessary and no longer a problem in the context of LOS.⁹

According to Josefsson, 1992, elements surrounding the object pronoun in LOS constructions must be heavy, aligning the object pronoun before the heavy stressed elements. SPS therefore concludes that LOS constructions are derived by metathesis between the subject and the object in order to achieve the optimal alignment just as in cases of heavy NP shift. Holmberg finds this derivation implausible since “the subject need not be heavy at all as long as it is not a pronoun.” Indeed it is not the heaviness of the subject itself which triggers the inversion, but rather the phonological alignment of the complete structure. I employ the notion of phonological metathesis which applies within words to ease pronunciation and argue that this process can also apply to ease the pronunciation of sentences. The available data is rather limited, but from the examples I have examined, the object is followed either by a rather heavy subject or else a less heavy subject combines with other elements to form a heavy string. The idea that metathesis applies for informational and/or phonological reasons is to my mind still plausible, especially in view of the fact (Josefsson, 2003) that no syntactic analysis has been proposed that accounts fully for the distribution of LOS.

2.2. *V-2*

Hellan, Surányi, and Svenonius point out problems with my account of V-2.¹⁰ One of the problems is that it was not well-motivated. Here I adopt a recent proposal by Brandner, 2004: 1 who extends Cheng’s 1991 notion of typing to declaratives and argues “that V2 is a strategy to specify a FORCE-value in an explicit way” in languages which do not have typing particles. According to Brandner, V-2 languages mark every FORCE-value, including the declarative, whereas in other languages the declarative is the default case. In such languages the verb moves because of the need for FORCE-specification and the value is delivered by a subsequently moved constituent. If the moved constituent is a *wh*-phrase, the

⁹ The intervention constraint is criticized with good reason by Surányi and Svenonius.

¹⁰ Surányi mistakenly argues that PI can feed V-movement. This was the case under the free ride account which I outline and reject in SPS.

FORCE value will be interrogative; if a DP, it will be declarative. Brandner treats verb-initial structures as underspecified for FORCE. Their interpretation depends on intonation and discourse strategies, yielding yes-no questions, conditionals and exclamation.

Brandner offers an ingenuous crosslinguistic syntactic analysis which avoids both the pitfalls of V-2 as head movement and the need for designated positions for *wh*-phrases and topics. Since her analysis does not depend on *c*-command relations, but rather on adjacency between the verb and the element contributing the FORCE specification, her proposal could easily be rearticulated within a linear phonological theory.

The motivation for V-2 can be seen as another aspect of identification. Just as arguments must be identified, so must the force of a sentence.

2.3. *Top-foc processing*

Yet another aspect of the sentence which requires identification is the information structure of the sentence, i.e., topic and the focus. These, as is well known, are marked positionally, morphologically and/or by intonation.¹¹ In Scandinavian, topics (unless contrastive) are unstressed and may be fronted to sentence initial position or scrambled in front of an adverb, as in Icelandic. This is not a requirement for topic interpretation, however. Topics may remain in situ (as long as they are unstressed), as evidenced by the lack of OS in subordinate clauses. This does not contradict the idea that object-adverb order is motivated by topic interpretation, as Sells suggests, since this is not the only way of marking topics.

Pronouns are topics (see Erteschik-Shir, 1997) but the reason they precede adverbs is their need to prosodically incorporate. Sells' point that expletives undergo OS (i.e., they occur preverbally), is therefore not a problem under my account, but follows naturally.

In the framework presented in Erteschik-Shir, 1997, focus structures are syntactic in that topic and focus status is assigned to syntactic constituents. Focus structure must be phonologically visible in view of the fact that it determines intonation. Some syntactic bracketing therefore remains in the phonology. This explains why topicalization may apply to

¹¹ See SPS, section 4.

a complex constituent and is not restricted to prosodic constituents as Svenonius assumes.¹²

3. Phonology

One of the main points of my analysis of the OS data is that pronouns are prosodically weak and therefore must be incorporated. Svenonius and Sells maintain that since pronouns are distinguished syntactically anyway, it is the syntactic feature which figures in OS, and it is this feature which I have disguised as a phonological one. This is not at all the case. Both strong (stressed) pronouns and weak (unstressed) pronouns (including clitic forms) would be syntactically [+pronoun], but only weak pronouns are incorporated.¹³

Under my analysis, weak adverbs must also incorporate. The idea that adverbs incorporate is criticized by several of my reviewers yet, as I have belatedly discovered, morphological adverb incorporation is well-known in languages as different as Greek, Chukchi and Eskimo (Rivero, 1992, Spencer, 1995, Alexiadou, 1997). My claim is that adverbs can occur in strong and weak forms. Whether an adverb is strong or weak is language specific and may also be a lexical property of the adverb or adverb type. The phonology of the adverb (e.g. its length) may also be a factor. Adverbs, just like pronouns, may also alternate between strong and weak forms. Scandinavian adverbs, I claim, all have a weak form and many also have a strong form. Strong adverbs are distinguished by their ability to occur at the sentence peripheries, separated from the sentence with comma intonation. Weak adverbs cannot occur in these positions, and must be pronounced as a prosodic unit together with some host. Adverbs which follow object pronouns in Scandinavian are incorporated in the verb-pronoun cluster if they are weak. If they have a strong form, they are not required to incorporate and can occur in final position. The Danish adverb *igår* ('yesterday') has a strong form and can occur either

¹² Svenonius' example (4) deserves more attention than I can give it here, but see Erteschik-Shir, 1981, 1997 and Guéron, 1980 for explanations of related data in terms of focus structure.

¹³ I am grateful to Hellan for more evidence supporting this view of pronoun and adverb incorporation.

incorporated or independently as in (9), but *ikke* ('not') only has a weak form and can only be pronounced incorporated:¹⁴

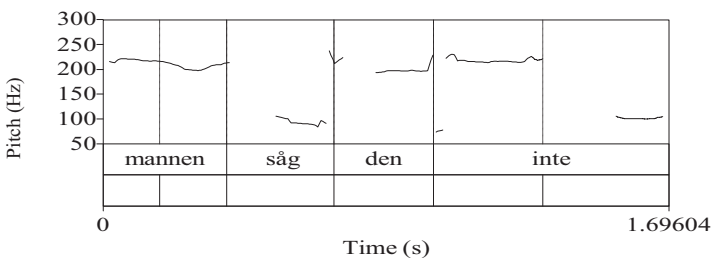
- (9) Jeg så+ham+igår. /Jeg så ham igår.
I saw him yesterday.
(10) Jeg så+ham+ikke. /*Jeg så ham ikke.
I saw him not

3.1. Optionality of OS in Swedish

In SPS, I suggest that when OS is optional (in Swedish and in the Danish dialect of Ærø) it is due to language-particular prosodic properties.¹⁵ According to Holmberg the particular prosody I assume, with stress on the adverb, is wrong in Swedish and in Norwegian. There the stress is on the verb in the unmarked case. In order to discover whether the dialects which allow the pronoun to follow the adverb have distinct prosodic properties, I have begun an examination of the prosody of the two word orders in Swedish. I am able here to present only initial results. The sentences in (11) were uttered by a young Swedish woman from Malmö rendering the pitch contours in (12).

- (11) Mannen såg den inte/ inte den. (Swedish)
the man saw it not

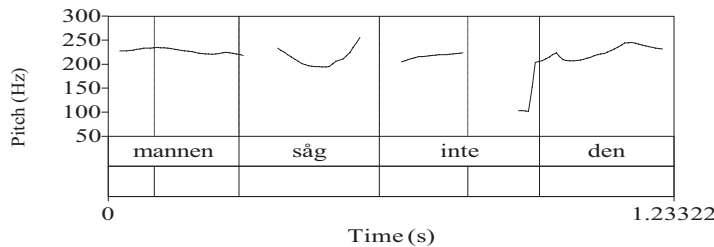
- (12) a.



¹⁴ According to Sells, Swedish *inte* ('not') can occur sentence initially, indicating that it has a strong form. I predict, then, that it should also be able to be left unincorporated in final position.

¹⁵ According to Josefsson, from Basbøll (1986: 173), OS is also optional on the Danish island Femø. It would be interesting to discover how widespread this phenomenon is in southern Denmark.

b.



The pronoun and adverb form a prosodic unit with the verb in both orders, yet the two contours are very different. The most conspicuous difference is the HLH contour on the verb in (12)b which is absent in (12)a. I wonder whether this contour is associated with accent 2 in Swedish. This speculation is obviously premature, and I only introduce it here because of Hellan’s inspiring discussion of the role of accent (or toneme) in defining the prosodic domain of a verb combined with a directional adverb in Norwegian.¹⁶ The idea that accent 2 identifies prosodic units can be found already in Haugen, 1967. According to Haugen “tone serves to join successive elements more closely than would otherwise be the case” (p. 198) and it helps the hearer identify the morphemic structure (p. 201). Interestingly, he points out that “tone is added to a stressed monosyllabic verb form ... in rapid speech with certain following adverbs, especially *så* (‘so’) and *ikke* (‘not’)” (p. 198). In view of these observations it is plausible that a correlation can be found between speakers who integrate adverbs by using accent 2, and those who are able to pronounce the order in (12)b.¹⁷

4. Conclusion

I agree with my reviewers that the program outlined in SPS leaves many details to be worked out. Here I’ve just attempted to clarify some of the

¹⁶ Such data originates with Fretheim, 1974, Fretheim and Halvorsen, 1975.
¹⁷ Interesting observations concerning the choice between the two orders can be found in Josefsson, 2003.

more basic points of SPS in order to encourage further research into the role of phonology in determining word order.

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References

- Alexiadou, A. (1997). *Adverb placement: a case study in antisymmetric syntax*, Amsterdam, John Benjamins.
- Brandner, E. (2004). Head-movement in minimalism, and V/2 as Force-marking. In Lohnstein, H. & Trissler, S. (Eds.) *Syntax and Semantics of the Left Periphery*. Berlin, de Gruyter.
- Cheng, L. (1991). *On the Typology of wh-Questions*, Cambridge, MA, MIT Working Papers in Linguistics.
- Erteschik-Shir, N. (1981). On extraction from noun phrases (picture noun phrases). In Belletti, A., Brandi, L. & Rizzi, L. (Eds.) *Theory of Markedness in Generative Grammar: Proceedings of the 1979 GLOW Conference*. Pisa, Scuola Normale Superiore di Pisa.
- (1997). *The Dynamics of Focus Structure*, Cambridge, Cambridge University Press.
- (to appear). On the architecture of topic and focus. In Molnar, V. & Winkler, S. (Eds.) *The architecture of focus*. Berlin, Mouton de Gruyter.
- Fretheim, T. (1974). Some cases of Accent Shift in East Norwegian. *Norwegian Journal of Linguistics*, 28, 113–129.
- Fretheim, T. & Halvorsen, P.-K. (1975). Norwegian cliticization. In Dahlstedt, K.-H. (Ed.) *The Nordic Languages and Modern Linguistics 2*. Stockholm, Almqvist & Wiksell International.
- Guéron, J. (1980). On the Syntax and Semantics of PP Extraposition. *Linguistic Inquiry*, 11, 637–678.
- Haugen, E. (1967). On the rules of Norwegian tonality. *Language*, 43, 185–202.
- Jakobson, R. (1971). Beitrag zur allgemeinen Kasuslehre: Gesamtbedeutungen der Russischen Kasus. *Selected Writings II*. The Hague, Mouton.
- Josefsson, G. (1992). Object Shift and Weak Pronominals in Swedish. *Working Papers in Scandinavian Syntax*, 49, 59–94.
- (2003). Four myths about Object Shift in Swedish – and the Truth. In Delsing, L.-O., Falk, C., Josefsson, G. & Sigurdsson, H. Á. (Eds.) *Grammar in Focus vol. II. Festschrift for Christer Platzack, 18 Nov. 2003*. Lund, Department of Scandinavian Languages, Lund University.
- King, T. H. (1995). *Configuring Topic and Focus in Russian*, Stanford, CSLI.
- Nilsen, Ø. (1997). Adverbs and A-shift. *Working Papers in Scandinavian Syntax*, 59, 1–31.
- (2003). Eliminating Positions: Syntax and semantics of sentence modification. Published by LOT, Utrecht, the Netherlands, Utrecht University.
- Ørnes, B. (2002). Case Marking and Subject Extraction in Danish. In Butt, M. & Hollo-way, T. K. (Eds.) *LFG02 Conference*. National Technical University of Athens, Athens, CSLI.

- Rivero, M. L. (1992). Adverb incorporation and the syntax of adverbs in Modern Greek. *Linguistics and Philosophy*, 15, 289–331.
- Spencer, A. (1995). Incorporation in Chukchi. *Language*, 71, 439–489.
- Vangsnes, Ø. A. (2004). On *wh*-questions and V2 across Norwegian dialects. A survey and some speculations. *Working Papers in Scandinavian Syntax*, 73, 1–59.
- Westergaard, M. R. (2003). Word order in *wh*-questions in a North Norwegian dialect: some evidence from an acquisition study. *Journal of Nordic Linguistics*, 26, 81–109.